1

Introduction to Category Theory

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Notation

$\mathcal{M}_+(X, \mathcal{A})$	finite measures on $(X, \mathcal{A})$
v	velocity

Preface

cc-by-nc-sa

Chapter 1

Preliminary Material

1.1 First Order Logic

maybe reorganize and move to logic notes?

1.2 von Neumann–Bernays–Gödel (NBG) Set Theory

maybe just use iterated extensions to not worry about if stuff is well defined or small, etc, see iterated NBG math overflow

Definition 1.2.0.1. Let x be a class. Then x is said to be a set iff there exists a class A such that $x \in A$.

Definition 1.2.0.2. Let x and y be classes. Then x is said to be a **subclass** of y, denoted $x \subset y$, if for each set a, $a \in x$ implies that $a \in y$.

Definition 1.2.0.3. Let x and y be classes. Then x is said to be **equal** to y if $x \subset y$ and $y \subset x$.

Axiom 1.2.0.4. Axiom of Extensionality:
Let x and y be classes. If for each set a, $a \in x$ iff $a \in y$, then $x = y$.

Axiom 1.2.0.5. Axiom of Pairing:
Let a, b be sets. Then there exists a set p such that for each for each set x, $x \in p$ iff $x = a$ or $x = b$.

Definition 1.2.0.6. product of two classes

Definition 1.2.0.7. Let A, B be classes and $R \subset A \times B$. elation from A to B.

Note 1.2.0.8. We can define cartesion products, relations, and functions for classes just like for sets.

Exercise 1.2.0.9. Let a, b be sets. Then there exists a unique set p such that for each for each set x, $x \in p$ iff $x = a$ or $x = b$.

Proof. By Axiom 1.2.0.5 implies that there exists a set p such that for each for each set x, $x \in p$ iff $x = a$ or $x = b$. Let q be a set. Suppose that for each for each set x, $x \in q$ iff $x = a$ or $x = b$. Then $\square$

Definition 1.2.0.10. Let x and y be sets. We define $(x, y) = \{\}$, denoted

Axiom 1.2.0.11. Axiom of Replacement:
Let A, B be classes and $f : A \to B$. If A is a set, then $f(A)$ is a set.

Axiom 1.2.0.12. Schema of Specification:
Let ϕ a propositional function on sets. Then there exists a class A such that for each set x, $x \in A$ iff $\phi(x)$.

Exercise 1.2.0.13. There exists a class A such that for each class x, $x \in A$ iff x is a set.

Proof. Define ϕ by

$$\phi(x) : x = x$$

Axiom 1.2.0.12 implies that there exists a class A such that for each set x, $x \in A$ iff $x = x$. Let x be a class. If $x \in A$, then by definition, x is a set.
Conversely, if x is a set, then by construction, $x \in A$. □

Exercise 1.2.0.14. There exists a class A such that for each class G and $* : G \times G \to G$, $(G, *) \in A$ iff $(G, *)$ is a group.

Proof. Define ϕ_1, ϕ_2 and ϕ_3 by

- $\phi_1(G, *) : * : G \times G \to G$ is associative
- $\phi_2(G, *)$: there exists $e \in G$ such that for each $g \in G$, $e * g = g * e = g$
- $\phi_3(G, *)$: for each $g \in G$ there exists $h \in G$ such that $g * h = h * g = e$

Define ϕ by

$$\phi(G, *) : \phi_1(G, *) \text{ and } \phi_2(G, *) \text{ and } \phi_3(G, *)$$

Then there exists a class A such that for each set G and $* : G \times G \to G$, $(G, *) \in A$ iff $\phi(G, *)$ $(G, *)$ "is a group". Therefore, for each group $(G, *)$, $(G, *) \in A$. **FINISH!!!** □

1.2.1 TO DO

1. cover existence of subclasses, products of classes to be able to define class relations and subsequently class functions
2. need to merge this with set theory notes and into to math notes, can first have naive set theory and then the NBG set theory or something

1.2.2 Relations

can I define relations on classes instead of just sets? do i need to move this stuff before to prelim material since we need class functions which I guess are class relations?
apparently yes class,set theory notes (link, but already downloaded)

Definition 1.2.2.1. Let A, B be classes and $R \subset A \times B$. Then R is said to be a **relation between** A **and** B. write out 1st order logic definition like in notes class,set theory notes (link, but already downloaded)

Definition 1.2.2.2. Let A, B, C be classes, R a relation between A and B, S a relation between B and C. We define the **composition relation of** R **with** S **between** A **and** C, denoted $R \circ S$, by $R \circ S := \{(a, c) \in A \times C$: there exists $b \in B$ such that $(a, b) \in R$ and $(b, c) \in S$

Exercise 1.2.2.3. show composition of class relations exists

Exercise 1.2.2.4. Let A, B, C, D be classes, R a relation between A and B, S relation between B and C and T relation between C and D. Then $(T \circ S) \circ R = T \circ (S \circ R)$.

Proof.

- Let $(a, d) \in (T \circ S) \circ R$. Then there exists $b \in B$ such that $(a, b) \in R$ and $(b, d) \in T \circ S$. Since $(b, d) \in T \circ S$, there exists $c \in C$ such that $(b, c) \in S$ and $(c, d) \in T$. Since $(a, b) \in R$ and $(b, c) \in S$, $(a, c) \in S \circ R$. Since $(a, c) \in S \circ R$ and $(c, d) \in T$, we have that $(a, d) \in T \circ (S \circ R)$. Since $(a, d) \in (T \circ S) \circ R$ is arbitrary, we have that for each $(a, d) \in (T \circ S) \circ R$, $(a, d) \in T \circ (S \circ R)$. Hence $(T \circ S) \circ R \subset T \circ (S \circ R)$.
- Similarly, $T \circ (S \circ R) \subset (T \circ S) \circ R$.

Since $(T \circ S) \circ R \subset T \circ (S \circ R)$ and $T \circ (S \circ R) \subset (T \circ S) \circ R$, we have that $(T \circ S) \circ R = T \circ (S \circ R)$.
review, input any necessary axioms for classes □

Definition 1.2.2.5. Let A be a class. We define the **diagonal relation on** A, denoted Δ_A, by $\Delta_A := \{(a, a) : a \in A\}$. make and show exists in a previous ex

Exercise 1.2.2.6. Let A, B be classes and R a relation between A and B. Then

1. $R \circ \Delta_A = R$
2. $\Delta_B \circ R = R$

Proof.

1.
   - Let $(a, b) \in R \circ \Delta_A$. Then there exists $a' \in A$ such that $(a, a') \in \Delta_A$ and $(a', b) \in R$. By definition of Δ_A, $a = a'$. Therefore $(a, b) \in R$. Since $(a, b) \in R \circ \Delta_A$ is arbitrary, we have that for each $(a, b) \in R \circ \Delta_A$, $(a, b) \in R$. Hence $R \circ \Delta_A \subset R$.
   - Let $(a, b) \in R$. Then $a \in A$. Therefore $(a, a) \in \Delta_A$. Since $(a, a) \in \Delta_A$ and $(a, b) \in R$, we have that $(a, b) \in R \circ \Delta_A$. Since $(a, b) \in R$ is arbitrary, we have that for each $(a, b) \in R$, $(a, b) \in R \circ \Delta_A$. Hence $R \circ \Delta_A \subset R$.

   Since $R \circ \Delta_A \subset R$ and $R \circ \Delta_A \subset R$, we have that $R \circ \Delta_A = R$.
2. Similar to (1).

□

1.2.3 Functions

1.2.4 Digraphs

Definition 1.2.4.1. talk about what it means for two diagrams to commute, talk about parallel morphisms being the only exception

Definition 1.2.4.2.
- We say a digraph (G, E) **commutes** if for each $x, y \in G$ and paths p, q, if p, q start at x and end at y, then $p = q$ need to fix this definition

Note 1.2.4.3. We make the following exception about parallel arrows. if two ore more edges are equal, we draw them as curved arrows, not parallel.

We define parallel arrows FINISH!!!

Chapter 2

Basic Concepts

2.1 Categories

2.1.1 Introduction

Definition 2.1.1.1. Let $\mathcal{C}_0$, $\mathcal{C}_1$ be classes and $\mathrm{dom}, \mathrm{cod} : \mathcal{C}_1 \to \mathcal{C}_0$. Define $\mathcal{C}_1^{(2)} \subset \mathcal{C}_1 \times \mathcal{C}_1$ by $\mathcal{C}_1^{(2)} := \{(g, f) \in \mathcal{C}_1 \times \mathcal{C}_1 : \mathrm{dom}(g) = \mathrm{cod}(f)\}$. Let $\circ : \mathcal{C}_1^{(2)} \to \mathcal{C}_1$. Then $(\mathcal{C}_0, \mathcal{C}_1, \mathrm{dom}, \mathrm{cod}, \circ)$ is said to be a **category** if

1. **(Composition):**
   for each $(g, f) \in \mathcal{C}_1^{(2)}$, $\mathrm{dom}(g \circ f) = \mathrm{dom}(f)$ and $\mathrm{cod}(g \circ f) = \mathrm{cod}(g)$
2. **(Associativity):**
   for each $f, g, h \in \mathcal{C}_1$, if $(g, f), (h, g) \in \mathcal{C}_1^{(2)}$, then
   $$(h \circ g) \circ f = h \circ (g \circ f)$$
3. **(Identity):**
   for each $X \in \mathcal{C}_0$, there exists $e \in \mathcal{C}_1$ such that $\mathrm{dom}(e) = X$, $\mathrm{cod}(e) = X$ and for each $f, g \in \mathcal{C}_1$, if $\mathrm{dom}(f) = X$ and $\mathrm{cod}(g) = X$, then
   $$f \circ e = f \text{ and } e \circ g = g$$

Definition 2.1.1.2. Let $\mathcal{C}$ be a category. Write $\mathcal{C} = (\mathcal{C}_0, \mathcal{C}_1, \mathrm{dom}, \mathrm{cod}, \circ)$.

- We define the
  - **objects of** $\mathcal{C}$, denoted $\mathrm{Obj}(\mathcal{C})$, by $\mathrm{Obj}(\mathcal{C}) := \mathcal{C}_0$
  - **morphisms of** $\mathcal{C}$, denoted Hom, by $\mathrm{Hom}_{\mathcal{C}} := \mathcal{C}_1$
  - **domain function of** $\mathcal{C}$, denoted $\mathrm{dom}_{\mathcal{C}}$, by $\mathrm{dom}_{\mathcal{C}} := \mathrm{dom}$
  - **codomain function of** $\mathcal{C}$, denoted $\mathrm{cod}_{\mathcal{C}}$, by $\mathrm{cod}_{\mathcal{C}} := \mathrm{cod}$
  - **composition operator of** $\mathcal{C}$, denoted $\circ_{\mathcal{C}}$, by $\circ_{\mathcal{C}} := \circ$
- For $X, Y \in \mathrm{Obj}(\mathcal{C})$, we define the **Morphisms of** $\mathcal{C}$ **from** X **to** Y, denoted $\mathrm{Hom}_{\mathcal{C}}(X, Y)$, by $\mathrm{Hom}_{\mathcal{C}}(X, Y) := \{f \in \mathrm{Hom}(\mathcal{C}) : \mathrm{dom}_{\mathcal{C}}(f) = X \text{ and } \mathrm{cod}_{\mathcal{C}}(f) = Y\}$.

Note 2.1.1.3. When the context is clear, we write dom, cod and $g \circ f$ in place of $\mathrm{dom}_{\mathcal{C}}$, $\mathrm{cod}_{\mathcal{C}}$ and $g \circ_{\mathcal{C}} f$ respectively.

Note 2.1.1.4. We typically define a category $\mathcal{C}$ by specifying

- $\mathrm{Obj}(\mathcal{C})$
- for $X, Y \in \mathrm{Obj}(\mathcal{C})$, the class $\mathrm{Hom}_{\mathcal{C}}(X, Y)$
- for $X, Y, Z \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(Y, Z)$, the composite morphism $g \circ f \in \mathrm{Hom}_{\mathcal{C}}(X, Z)$.

and then show

- well-definedness of composition
- associativity of composition
- existence of identities

Exercise 2.1.1.5. Uniqueness of Identities:
Let $\mathcal{C}$ be a category. Then for each $A \in \mathrm{Obj}(\mathcal{C})$, there exists a unique $e \in \mathrm{End}_{\mathcal{C}}(A)$ such that for each $B \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(B, A)$, $f \circ e = f$ and $e \circ g = g$.

Proof. Let $A \in \mathrm{Obj}(\mathcal{C})$.

- **Existence:**
  Since $\mathcal{C}$ is a category, by definition there exists $e \in \mathrm{Hom}_{\mathcal{C}}(A, A)$ such that for each $B \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(B, A)$, $f \circ e = f$ and $e \circ g = g$.
- **Uniqueness:**
  Let $e_1, e_2 \in \mathrm{Hom}_{\mathcal{C}}(A, A)$. Suppose that for each $B \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(B, A)$, $f \circ e_1 = f$, $f \circ e_2 = f$, $e_1 \circ g = g$ and $e_2 \circ g = g$. Then

$$\begin{aligned} e_1 &= e_2 \circ e_1 \\ &= e_2. \end{aligned}$$

□

Definition 2.1.1.6. Let $\mathcal{C}$ be a category and $A \in \mathrm{Obj}(\mathcal{C})$. We define the **identity morphism on** A, denoted $\mathrm{id}_A^{\mathcal{C}} \in \mathrm{Hom}_{\mathcal{C}}(A, A)$, to be the unique $e \in \mathrm{Hom}_{\mathcal{C}}(A, A)$ such that for each $B \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(B, A)$, $f \circ e = f$ and $e \circ g = g$.

Note 2.1.1.7. When the context is clear, we write id_A in place of $\mathrm{id}_A^{\mathcal{C}}$.

Definition 2.1.1.8. Let $\mathcal{C}$ be a category. Then $\mathcal{C}$ is said to be

- **small** if $\mathrm{Obj}(\mathcal{C})$ and $\mathrm{Hom}_{\mathcal{C}}$ are sets
- **locally small** if for each $A, B \in \mathrm{Obj}(\mathcal{C})$, $\mathrm{Hom}_{\mathcal{C}}(A, B)$ is a set

Exercise 2.1.1.9. Let $\mathcal{C}$ be a category. If $\mathcal{C}$ is small, then $\mathcal{C}$ is a set.

Proof. Suppose that $\mathcal{C}$ is small. Then $\mathrm{Obj}(\mathcal{C})$ and $\mathrm{Hom}_{\mathcal{C}}$ are sets. Then $\mathcal{P}(\mathrm{Obj}(\mathcal{C}))$, $\mathcal{P}(\mathrm{Hom}_{\mathcal{C}})$ and $\mathrm{Obj}(\mathcal{C})^{\mathrm{Hom}_{\mathcal{C}}}$ are sets. Hence $\mathrm{Obj}(\mathcal{C}) \times \mathrm{Hom}_{\mathcal{C}} \times \mathrm{Obj}(\mathcal{C})^{\mathrm{Hom}_{\mathcal{C}}} \times \mathrm{Obj}(\mathcal{C})^{\mathrm{Hom}_{\mathcal{C}}}$ is a set. By definition, $\mathcal{C} = (\mathrm{Obj}(\mathcal{C}), \mathrm{Hom}_{\mathcal{C}}, \mathrm{dom}, \mathrm{cod}) \in \mathrm{Obj}(\mathcal{C}) \times \mathrm{Hom}_{\mathcal{C}} \times \mathrm{Obj}(\mathcal{C})^{\mathrm{Hom}_{\mathcal{C}}} \times \mathrm{Obj}(\mathcal{C})^{\mathrm{Hom}_{\mathcal{C}}}$. By definition, $\mathcal{C}$ is a set. □

Exercise 2.1.1.10. There exists a class A such that $\mathcal{C} \in A$ iff $\mathcal{C}$ is a small category.

Proof. Exercise 2.1.1.9 implies that for each category $\mathcal{C}$, $\mathcal{C}$ is small implies that $\mathcal{C}$ is a set. Define ϕ by

$$\phi(\mathcal{C}) : \mathcal{C} \text{ is a small category}$$

Then Axiom 1.2.0.12 implies that there exists a class A such that $\mathcal{C} \in A$ iff $\mathcal{C}$ is a small category. □

prosets

Definition 2.1.1.11. Let $\mathcal{P}$ be a category. Then $\mathcal{P}$ is said to be a

1. **proset category** if for each $a, b \in \mathrm{Obj}(\mathcal{P})$, $\#\,\mathrm{Hom}_{\mathcal{P}}(a, b) \leq 1$
2. **poset category** if $\mathcal{P}$ is a proset category and for each $a, b \in \mathrm{Obj}(\mathcal{P})$, $\#\,\mathrm{Hom}_{\mathcal{P}}(a, b) = 1$ and $\#\,\mathrm{Hom}_{\mathcal{P}}(b, a) = 1$ implies that $a = b$.

Definition 2.1.1.12. Let $\mathcal{P}$ be a proset. Set $P := \mathrm{Obj}(\mathcal{P})$. We define the **less-than-or-equal-to relation** on P, denoted $\leq \subset P \times P$, by

$$\leq := \{(a, b) \in P \times P : \mathrm{Hom}_{\mathcal{P}}(a, b) \neq \varnothing\}.$$

Definition 2.1.1.13. Let $\mathcal{C}$ be a category. Then $\mathcal{C}$ is said to be **discrete** if $\mathrm{Hom}_{\mathcal{C}} = \{\mathrm{id}_X : X \in \mathrm{Obj}(\mathcal{C})\}$.

2.1.2 Fundamental Examples

Definition 2.1.2.1. We define the **empty category**, denoted $\mathbf{0}$, by

- $\mathrm{Obj}(\mathbf{0}) = \varnothing$
- $\mathrm{Hom}_{\mathbf{0}} = \varnothing$

Exercise 2.1.2.2. We have that $\mathbf{0}$ is a category.

Proof. Vacuously true. □

Definition 2.1.2.3. We define the **trivial category**, denoted $\mathbf{1}$, by

- $\mathrm{Obj}(\mathbf{1}) = \{*\}$
- $\mathrm{Hom}_{\mathbf{1}} = \{\mathrm{id}_*\}$

Exercise 2.1.2.4. We have that $\mathbf{1}$ is a category.

Proof. Clear. □

Definition 2.1.2.5. We define **Set** by

- $\mathrm{Obj}(\mathbf{Set}) = \{A : A \text{ is a set}\}$
- for each $A, B \in \mathrm{Obj}(\mathbf{Set})$, $\mathrm{Hom}_{\mathbf{Set}}(A, B) = \{f : f : A \to B\}$
- for $A, B, C \in \mathbf{Set}$, $f \in \mathrm{Hom}_{\mathbf{Set}}(A, B)$ and $g \in \mathrm{Hom}_{\mathbf{Set}}(B, C)$, $g \circ_{\mathbf{Set}} f = g \circ f$.

Exercise 2.1.2.6. We have that **Set** is a category.

Proof.

- **well-definedness of composition:**
- **associativity of composition:**
- **existence of identities:**

FINISH!!! □

Exercise 2.1.2.7. Let $A, B \in \mathrm{Obj}(\mathbf{Set})$, show $\mathcal{R}(A, B)$ is a category, (category of relations)

Exercise 2.1.2.8. preordered set is a category

FIX!!!! need to decide what to do about relations, on the one hand, we can define relations as classes like in the previous section, on another hand, we can define the category of relations based on **Set**, on the third hand, looking at 'Elementary categories, elementary toposes' by mcLarty who shows you cand define relations in terms of subobjects, i.e. morphisms, does this more general idea also form a category, of which the category of relations based on set can be considered a subobject(i.e. a subcategory)?

Definition 2.1.2.9. We define the **category of relations**, denoted **Rel**, by

- $\mathrm{Obj}(\mathbf{Rel}) := \mathrm{Obj}(\mathbf{Set})$
- for each $A, B \in \mathrm{Obj}(\mathbf{Rel})$, $\mathrm{Hom}_{\mathbf{Rel}}(A, B) := \mathcal{P}(A \times B)$
- for each $A, B, C \in \mathrm{Obj}(\mathbf{Rel})$, $R \in \mathrm{Hom}_{\mathbf{Rel}}(A, B)$ and $\mathrm{Hom}_{\mathbf{Rel}}(B, C)$, we define $S \circ_{\mathbf{Rel}} R := \{(a, c) \in A \times C :$ there exists $b \in B$ such th

Proof.

- **well-definedness of composition:**
  By definition, for each $A, B, C \in \mathrm{Obj}(\mathbf{Rel})$, $R \in \mathrm{Hom}_{\mathbf{Rel}}(A, B)$ and $S \in \mathrm{Hom}_{\mathbf{Rel}}(B, C)$, we have that
  $$\begin{aligned} S \circ_{\mathbf{Rel}} R &\in \mathcal{P}(A \times C) \\ &= \mathrm{Hom}_{\mathbf{Rel}}(A, C). \end{aligned}$$
  So $\circ_{\mathbf{Rel}}$ is well-defined.
- **associativity of composition:**
  ref ex in basic concepts chapter
- **existence of identities:**
  ref ex in basic concepts chapter

FINISH!!! □

maybe discuss a quiver as a category whos objects are vertices and morphisms are and composition is given by concatenation (i.e. I guess a quiver equipped with the path algebra), maybe this is not appropriate quiver wikipedia

Definition 2.1.2.10. We define the **category of quivers**, denoted **Quiv**, by

- $\mathrm{Obj}(\mathbf{Quiv})\}$
- for $\mathcal{Q}$, $\mathcal{L} \in \mathrm{Obj}(\mathbf{Quiv})$,
  $$\mathrm{Hom}_{\mathbf{Quiv}}(\mathcal{Q}, \mathcal{L}) := \{\phi : \phi \text{ is a quiver}\}$$

2.1.3 Opposite Categories

Definition 2.1.3.1. Let $\mathcal{C}$ be a category, we define the dual of $\mathcal{C}$ or the **opposite of** $\mathcal{C}$, denoted $\mathcal{C}^{\mathrm{op}}$, by

- $\mathrm{Obj}(\mathcal{C}^{\mathrm{op}}) := \mathrm{Obj}(\mathcal{C})$
- for $X, Y \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$, $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, Y) := \mathrm{Hom}_{\mathcal{C}}(Y, X)$
- for $X, Y, Z \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ and $f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, Y), g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(Y, Z)$, $g \circ_{\mathcal{C}^{\mathrm{op}}} f := f \circ_{\mathcal{C}} g$

Exercise 2.1.3.2. Let $\mathcal{C}$ be a category. Then $\mathcal{C}^{\mathrm{op}}$ is a category.

Proof.

- for $W, X, Y, Z \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(W, X)$ and $g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, Y)$ and $h \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(Y, Z)$. Then
  $$\begin{aligned} (h \circ_{\mathcal{C}^{\mathrm{op}}} g) \circ_{\mathcal{C}^{\mathrm{op}}} f &= f \circ_{\mathcal{C}} (h \circ_{\mathcal{C}^{\mathrm{op}}} g) \\ &= f \circ_{\mathcal{C}} (g \circ_{\mathcal{C}} h) \\ &= (f \circ_{\mathcal{C}} g) \circ_{\mathcal{C}} h \\ &= h \circ_{\mathcal{C}^{\mathrm{op}}} (f \circ_{\mathcal{C}} g) \\ &= h \circ_{\mathcal{C}^{\mathrm{op}}} (g \circ_{\mathcal{C}^{\mathrm{op}}} f) \end{aligned}$$
  So composition is associative.
- Let $X \in \mathrm{Obj}(\mathcal{C})$ and $f, g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}$. Suppose that $\mathrm{dom}(f) = X$ and $\mathrm{cod}(g) = X$ Then
  $$\begin{aligned} f \circ_{\mathcal{C}^{\mathrm{op}}} \mathrm{id}_X &= \mathrm{id}_X \circ_{\mathcal{C}} f \\ &= f \end{aligned}$$
  and
  $$\begin{aligned} \mathrm{id}_X \circ_{\mathcal{C}^{\mathrm{op}}} g &= g \circ_{\mathcal{C}} \mathrm{id}_X \\ &= g \end{aligned}$$
  So $(\mathrm{id}_X)_{\mathcal{C}^{\mathrm{op}}} = (\mathrm{id}_X)_{\mathcal{C}}$.

□

Exercise 2.1.3.3. Let $\mathcal{C}$ be a category. Then $(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} = \mathcal{C}$.

Proof. Set $\mathcal{D} := \mathcal{C}^{\mathrm{op}}$.

- We have that

$$\begin{aligned}\mathrm{Obj}((\mathcal{C}^{\mathrm{op}})^{\mathrm{op}}) &= \mathrm{Obj}(\mathcal{D}^{\mathrm{op}}) \\ &= \mathrm{Obj}(\mathcal{D}), \quad \text{(by definition)} \\ &= \mathrm{Obj}(\mathcal{C}^{\mathrm{op}}) \\ &= \mathrm{Obj}(\mathcal{C}), \quad \text{(by definition)}.\end{aligned}$$

- Let $X, Y \in \mathrm{Obj}((\mathcal{C}^{\mathrm{op}})^{\mathrm{op}})$. Then

$$\begin{aligned}\mathrm{Hom}_{(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}}}(X, Y) &= \mathrm{Hom}_{\mathcal{D}^{\mathrm{op}}}(X, Y) \\ &= \mathrm{Hom}_{\mathcal{D}}(Y, X), \quad \text{(by definition)} \\ &= \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(Y, X) \\ &= \mathrm{Hom}_{\mathcal{C}}(X, Y), \quad \text{(by definition)}.\end{aligned}$$

□

Exercise 2.1.3.4. Let $\mathcal{P}$ be a category. Then

1. $\mathcal{P}$ is a proset iff $\mathcal{P}^{\mathrm{op}}$ is a proset
2. $\mathcal{P}$ is a poset iff $\mathcal{P}^{\mathrm{op}}$ is a poset

Proof. FINISH!!! □

Definition 2.1.3.5. Let $\mathcal{P}$ be a proset. Set $P := \mathrm{Obj}(\mathcal{P})$. We define the **less-than-or-equal-to relation** on P, denoted $\leq \subset P \times P$, by

$$\leq := \{(a, b) \in P \times P : \mathrm{Hom}_{\mathcal{P}^{\mathrm{op}}}(a, b) \neq \varnothing\}.$$

2.2 Functors

2.2.1 Introduction

Definition 2.2.1.1. Let $\mathcal{C}$ and $\mathcal{D}$ be categories and $F_0 : \mathrm{Obj}(\mathcal{C}) \to \mathrm{Obj}(\mathcal{D})$, $F_1 : \mathrm{Hom}_{\mathcal{C}} \to \mathrm{Hom}_{\mathcal{D}}$ class functions. Set $F = (F_0, F_1)$. Then F is said to be a $(\mathcal{C}, \mathcal{D})$-functor, denoted $F : \mathcal{C} \to \mathcal{D}$, if

1. for each $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$, $F_1(f) \in \mathrm{Hom}_{\mathcal{D}}(F_0(A), F_0(B))$
2. for each $A, B, C \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(B, C)$, $F_1(g \circ f) = F_1(g) \circ F_1(f)$
3. for each $A \in \mathrm{Obj}(\mathcal{C})$, $F_1(\mathrm{id}_A) = \mathrm{id}_{F_0(A)}$

Note 2.2.1.2. When the context is clear, we may say "functor" instead of "$(\mathcal{C}, \mathcal{D})$-functor".

Note 2.2.1.3. For $A \in \mathrm{Obj}(C)$ and $f \in \mathrm{Hom}_{\mathcal{C}}$, we typically write $F(A)$ and $F(f)$ instead of $F_0(A)$ and $F_1(f)$ respectively. maybe discuss how this is just a quiver homomorphism

Definition 2.2.1.4. Let $\mathcal{C}$ be a category. We define the **empty functor** from $\mathbf{0}$ to $\mathcal{C}$, denoted $E_{\mathcal{C}} : \mathbf{0} \to \mathcal{C}$ by $(E_{\mathcal{C}})_0 := \varnothing$ and $= (E_{\mathcal{C}})_1 := \varnothing$.

Exercise 2.2.1.5. Let $\mathcal{C}$ be a category. Then $E_{\mathcal{C}} : \mathbf{0} \to \mathcal{C}$ is a functor.

Proof. Since $\mathrm{Obj}(\mathbf{0}) = \varnothing$ and $\mathrm{Hom}_{\mathbf{0}} = \varnothing$, this is vacuously true. □

Definition 2.2.1.6. Let $\mathcal{C}, \mathcal{D}$ be categories and $X \in \mathrm{Obj}(\mathcal{D})$. We define the **constant functor** from $\mathcal{C}$ onto X, denoted $\Delta_X^{\mathcal{C}} : \mathcal{C} \to \mathcal{D}$ by

- $\Delta_X^{\mathcal{C}}(A) = X$
- $\Delta_X^{\mathcal{C}}(f) = \mathrm{id}_X$

Exercise 2.2.1.7. Let $\mathcal{C}, \mathcal{D}$ be categories and $X \in \mathrm{Obj}(\mathcal{D})$. Then $\Delta_X^{\mathcal{C}} : \mathcal{C} \to \mathcal{D}$ is a functor.

Proof.

1. Let $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Then
$$\begin{aligned}\Delta_X^{\mathcal{C}}(f) &= \mathrm{id}_X \\ &\in \mathrm{Hom}_{\mathcal{D}}(X, X) \\ &= \mathrm{Hom}_{\mathcal{D}}(\Delta_X^{\mathcal{C}}(A), \Delta_X^{\mathcal{C}}(B))\end{aligned}$$
2. Let $A, B, C \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(B, C)$. Then
$$\begin{aligned}\Delta_X^{\mathcal{C}}(g \circ f) &= \mathrm{id}_X \\ &= \mathrm{id}_X \circ \mathrm{id}_X \\ &= \Delta_X^{\mathcal{C}}(g) \circ \Delta_X^{\mathcal{C}}(f)\end{aligned}$$
3. Let $A \in \mathrm{Obj}(\mathcal{C})$. Then
$$\begin{aligned}\Delta_X^{\mathcal{C}}(\mathrm{id}_A) &= \mathrm{id}_X \\ &= \mathrm{id}_{\Delta_X^{\mathcal{C}}(A)}\end{aligned}$$

So $\Delta_X^{\mathcal{C}} : \mathcal{C} \to \mathcal{D}$ is a functor. □

2.2.2 Fundamental Examples

discuss a quiver as a functor from the free quiver (a category) to **Set** or other category quiver wikipedia

Note 2.2.2.1. We recall for $X, Y \in \mathrm{Obj}(\mathbf{Set})$, $f \in \mathrm{Hom}_{\mathbf{Set}}(X, Y)$, the image and preimage maps induced by f, denoted $f_* \in \mathrm{Hom}_{\mathbf{Set}}(\mathcal{P}(X), \mathcal{P}(Y))$ and $f^* \in \mathrm{Hom}_{\mathbf{Set}}(\mathcal{P}(Y), \mathcal{P}(X))$ are defined by $f_*(A) := f(A)$ and $f^*(B) := f^{-1}(B)$. make exercise in set theory notes showing $(g \circ f)_* = g + * \circ f_*$ and similar for contravariant version

Definition 2.2.2.2. We define the

- **image power set functor on Set**, denoted $\mathcal{P}_* : \mathbf{Set} \to \mathbf{Set}$ by
  - $\mathcal{P}_*(X) := \mathcal{P}(X), \quad X \in \mathrm{Obj}(\mathbf{Set})$
  - $\mathcal{P}_*(f) := f_*, \quad f \in \mathrm{Hom}_{\mathbf{Set}}$
- **preimage power set functor on Set**, denoted $\mathcal{P}^* : \mathbf{Set}^{\mathrm{op}} \to \mathbf{Set}$ by
  - $\mathcal{P}^*(X) := \mathcal{P}(X), \quad X \in \mathrm{Obj}(\mathbf{Set})$
  - $\mathcal{P}^*(f) := f^*, \quad f \in \mathrm{Hom}_{\mathbf{Set}}$

Exercise 2.2.2.3. We have that

1. $\mathcal{P}_* : \mathbf{Set} \to \mathbf{Set}$ is a functor.
2. $\mathcal{P}^* : \mathbf{Set}^{\mathrm{op}} \to \mathbf{Set}$ is a functor.

Proof.

1. (a) For each $X, Y \in \mathrm{Obj}(\mathbf{Set})$ and $f \in \mathrm{Hom}_{\mathbf{Set}}(X, Y)$,

$$\begin{aligned}\mathcal{P}_*(f) &= f_* \\ &\in \mathrm{Hom}_{\mathbf{Set}}(\mathcal{P}(X), \mathcal{P}(Y)) \quad (refexinsettheorysectionintrotomathnotes) \\ &= \mathrm{Hom}_{\mathbf{Set}}(\mathcal{P}_*(X), \mathcal{P}_*(Y)).\end{aligned}$$

   (b) For each $X, Y, Z \in \mathrm{Obj}(\mathbf{Set})$, $f \in \mathrm{Hom}_{\mathbf{Set}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathbf{Set}}(Y, Z)$,

$$\begin{aligned}\mathcal{P}_*(g \circ_{\mathbf{Set}} f) &= (g \circ_{\mathbf{Set}} f)_* \\ &= g_* \circ_{\mathbf{Set}} f_* \quad \text{(ref ex in set theory section intro to math notes)} \\ &= \mathcal{P}_*(g) \circ_{\mathbf{Set}} \mathcal{P}_*(f).\end{aligned}$$

   (c) For each $X \in \mathrm{Obj}(\mathbf{Set})$,

$$\begin{aligned}\mathcal{P}_*(\mathrm{id}_X^{\mathbf{Set}}) &= \mathrm{id}_{\mathcal{P}(X)}^{\mathbf{Set}} \quad \text{(ref ex in set theory section intro to math notes)} \\ &= \mathrm{id}_{\mathcal{P}_*(X)}^{\mathbf{Set}}.\end{aligned}$$

   Thus $\mathcal{P}_* : \mathbf{Set} \to \mathbf{Set}$ is a functor.

2. (a) Let $X, Y \in \mathrm{Obj}(\mathbf{Set}^{\mathrm{op}})$ and $f \in \mathrm{Hom}_{\mathbf{Set}^{\mathrm{op}}}(X, Y)$. Then $X \in \mathrm{Obj}(\mathbf{Set})$ and $f \in \mathrm{Hom}_{\mathbf{Set}}(Y, X)$. Therefore

$$\begin{aligned}\mathcal{P}^*(f) &= f^* \\ &\in \mathrm{Hom}_{\mathbf{Set}}(\mathcal{P}(X), \mathcal{P}(Y)) \quad (refexinsettheorysectionintrotomathnotes) \\ &= \mathrm{Hom}_{\mathbf{Set}}(\mathcal{P}^*(X), \mathcal{P}^*(Y)).\end{aligned}$$

   Since $X, Y \in \mathrm{Obj}(\mathbf{Set}^{\mathrm{op}})$ and $f \in \mathrm{Hom}_{\mathbf{Set}^{\mathrm{op}}}(X, Y)$ are arbitrary, we have that for each $X, Y \in \mathrm{Obj}(\mathbf{Set}^{\mathrm{op}})$ and $f \in \mathrm{Hom}_{\mathbf{Set}^{\mathrm{op}}}(X, Y)$, $\mathcal{P}^*(f) \in \mathrm{Hom}_{\mathbf{Set}^{\mathrm{op}}}(\mathcal{P}^*(X), \mathcal{P}^*(Y))$.

(b) Let $X, Y, Z \in \mathrm{Obj}(\mathbf{Set}^{\mathrm{op}})$, $f \in \mathrm{Hom}_{\mathbf{Set}^{\mathrm{op}}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathbf{Set}^{\mathrm{op}}}(Y, Z)$. Then $X, Y, Z \in \mathrm{Obj}(\mathbf{Set})$, $f \in \mathrm{Hom}_{\mathbf{Set}}(Y, X)$ and $g \in \mathrm{Hom}_{\mathbf{Set}}(Z, Y)$. Therefore

$$\begin{aligned}
\mathcal{P}^*(g \circ_{\mathbf{Set}^{\mathrm{op}}} f) &= (g \circ_{\mathbf{Set}^{\mathrm{op}}} f)^* \\
&= (f \circ_{\mathbf{Set}} g)^* \\
&= g^* \circ_{\mathbf{Set}} f^* \quad \text{(ref ex in set theory section intro to math notes)} \\
&= \mathcal{P}^*(g) \circ_{\mathbf{Set}} \mathcal{P}^*(f).
\end{aligned}$$

(c) For each $X \in \mathrm{Obj}(\mathbf{Set}^{\mathrm{op}})$,

$$\begin{aligned}
\mathcal{P}^*(\mathrm{id}_X^{\mathbf{Set}^{\mathrm{op}}}) &= (\mathrm{id}_X^{\mathbf{Set}^{\mathrm{op}}})^* \\
&= (\mathrm{id}_X^{\mathbf{Set}}) \\
&= \mathrm{id}_{\mathcal{P}(X)}^{\mathbf{Set}^{\mathrm{op}}} \quad \text{(ref ex in set theory section intro to math notes)} \\
&= \mathrm{id}_{\mathcal{P}^*(X)}^{\mathbf{Set}^{\mathrm{op}}}.
\end{aligned}$$

Thus $\mathcal{P}^* : \mathbf{Set}^{\mathrm{op}} \to \mathbf{Set}$ is a functor.

□

2.2.3 Category of Small Categories

Definition 2.2.3.1. Let $\mathcal{C}$, $\mathcal{D}$ and $\mathcal{E}$ be categories and $F : \mathcal{C} \to \mathcal{D}$, $G : \mathcal{D} \to \mathcal{E}$ functors. We define the **composition of G with F**, denoted $G \circ F : \mathcal{C} \to \mathcal{E}$, by

- $G \circ F(A) := G(F(A))$
- $G \circ F(f) := G(F(f))$

Note 2.2.3.2. Writing $G \circ F = (H_0, H_1)$, we have that $H_0 = G_0 \circ F_0$ and $H_1 = G_1 \circ F_1$.

Exercise 2.2.3.3. Let $\mathcal{C}$, $\mathcal{D}$ and $\mathcal{E}$ be categories and $F : \mathcal{C} \to \mathcal{D}$, $G : \mathcal{D} \to \mathcal{E}$ functors. Then $G \circ F : \mathcal{C} \to \mathcal{E}$ is a functor.

Proof.

1. Let $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Since $F(f) \in \mathrm{Hom}_{\mathcal{D}}(F(A), F(B))$, we have that $G(F(f)) \in \mathrm{Hom}_{\mathcal{E}}(G(F(A)), G(F(B)))$. Then

$$\begin{aligned}
G \circ F(f) &= G(F(f)) \\
&\in \mathrm{Hom}_{\mathcal{E}}(G(F(A)), G(F(B))) \\
&= \mathrm{Hom}_{\mathcal{E}}(G \circ F(A), G \circ F(B))
\end{aligned}$$

2. Let $A, B, C \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(B, C)$. Then

$$\begin{aligned}
G \circ F(g \circ f) &= G(F(g \circ f)) \\
&= G(F(g) \circ F(f)) \\
&= G(F(g)) \circ G(F(f)) \\
&= G \circ F(g) \circ G \circ F(f)
\end{aligned}$$

3. Let $A \in \mathrm{Obj}(\mathcal{C})$. Then

$$\begin{aligned}
G \circ F(\mathrm{id}_A) &= G(F(\mathrm{id}_A)) \\
&= G(\mathrm{id}_{F(A)}) \\
&= \mathrm{id}_{G(F(A))} \\
&= \mathrm{id}_{G \circ F(A)}
\end{aligned}$$

So $G \circ F : \mathcal{C} \to \mathcal{E}$ is a functor. $\square$

Exercise 2.2.3.4. Let $\mathcal{C}$, $\mathcal{D}$, $\mathcal{E}$, $\mathcal{F}$ be categories and $F : \mathcal{C} \to \mathcal{D}$, $G : \mathcal{D} \to \mathcal{E}$, $H : \mathcal{E} \to \mathcal{F}$ functors. Then $(H \circ G) \circ F = H \circ (G \circ F)$.

Proof. Let $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Then

- $$\begin{aligned}(H \circ G) \circ F(A) &= H \circ G(F(A)) \\ &= H(G(F(A))) \\ &= H(G \circ F(A)) \\ &= H \circ (G \circ F)(A)\end{aligned}$$

- $$\begin{aligned}(H \circ G) \circ F(f) &= H \circ G(F(f)) \\ &= H(G(F(f))) \\ &= H(G \circ F(f)) \\ &= H \circ (G \circ F)(f)\end{aligned}$$

Hence $(H \circ G) \circ F = H \circ (G \circ F)$. $\square$

Definition 2.2.3.5. Let $\mathcal{C}$ be a category. We define the **identity functor from $\mathcal{C}$ to $\mathcal{C}$**, denoted $\mathrm{id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$, by

- $\mathrm{id}_{\mathcal{C}}(A) := A, \quad (A \in \mathrm{Obj}(\mathcal{C}))$
- $\mathrm{id}_{\mathcal{C}}(f) := f, \quad (f \in \mathrm{Hom}_{\mathcal{C}})$

Exercise 2.2.3.6. Let $\mathcal{C}$ be a category. Then $\mathrm{id}_{\mathcal{C}} : \mathcal{C} \to \mathcal{C}$ is a functor.

Proof.

1. Let $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Then
$$\begin{aligned}\mathrm{id}_{\mathcal{C}}(f) &= f \\ &\in \mathrm{Hom}_{\mathcal{C}}(A, B) \\ &= \mathrm{Hom}_{\mathcal{C}}(\mathrm{id}_{\mathcal{C}}(A), \mathrm{id}_{\mathcal{C}}(B))\end{aligned}$$

2. Let $A, B, C \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(B, C)$. Then
$$\begin{aligned}\mathrm{id}_{\mathcal{C}}(g \circ f) &= g \circ f \\ &= \mathrm{id}_{\mathcal{C}}(g) \circ \mathrm{id}_{\mathcal{C}}(f)\end{aligned}$$

3. Let $A \in \mathrm{Obj}(\mathcal{C})$. Then
$$\begin{aligned}\mathrm{id}_{\mathcal{C}}(\mathrm{id}_A) &= \mathrm{id}_A \\ &= \mathrm{id}_{\mathrm{id}_{\mathcal{C}}(A)}\end{aligned}$$

$\square$

Exercise 2.2.3.7. Let $\mathcal{C}$ and $\mathcal{D}$ be categories and $F : \mathcal{C} \to \mathcal{D}$. Then

1. $\mathrm{id}_{\mathcal{D}} \circ F = F$
2. $F \circ \mathrm{id}_{\mathcal{C}} = F$

Proof.

1. Let $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Then

$$\begin{aligned} \mathrm{id}_{\mathcal{D}} \circ F(A) &= \mathrm{id}_{\mathcal{D}}(F(A)) \\ &= F(A) \end{aligned}$$

and

$$\begin{aligned} \mathrm{id}_{\mathcal{D}} \circ F(f) &= \mathrm{id}_{\mathcal{D}}(F(f)) \\ &= F(f) \end{aligned}$$

Since $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ are arbitrary, $\mathrm{id}_{\mathcal{D}} \circ F = F$.

2. Let $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Then

$$\begin{aligned} F \circ \mathrm{id}_{\mathcal{C}}(A) &= F(\mathrm{id}_{\mathcal{C}}(A)) \\ &= F(A) \end{aligned}$$

and

$$\begin{aligned} F \circ \mathrm{id}_{\mathcal{C}}(f) &= F(\mathrm{id}_{\mathcal{C}}(f)) \\ &= F(f) \end{aligned}$$

Since $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ are arbitrary, $F \circ \mathrm{id}_{\mathcal{C}} = F$.

□

Exercise 2.2.3.8. Let $\mathcal{C}$ and $\mathcal{D}$ be categories and $F : \mathcal{C} \to \mathcal{D}$. If $\mathcal{C}$ is small, then F is a set.

Proof. Suppose that $\mathcal{C}$ is small. Then $\mathrm{Obj}(\mathcal{C})$ and $\mathrm{Hom}_{\mathcal{C}}$ are sets. By definition, there exist $F_0 : \mathrm{Obj}(\mathcal{C}) \to \mathrm{Obj}(\mathcal{D})$ and $F_1 : \mathrm{Hom}_{\mathcal{C}} \to \mathrm{Hom}_{\mathcal{D}}$ such that $F = (F_0, F_1)$. Axiom 1.2.0.11 implies that $F_0(\mathrm{Obj}(\mathcal{C}))$ and $F_1(\mathrm{Hom}_{\mathcal{C}})$ are sets. Therefore, $\mathrm{Obj}(\mathcal{C}) \times F_0(\mathrm{Obj}(C))$ and $\mathrm{Hom}_{\mathcal{C}} \times F_1(\mathrm{Hom}_{\mathcal{C}})$ are sets. Hence $\mathcal{P}(\mathrm{Obj}(\mathcal{C}) \times F_0(\mathrm{Obj}(C)))$ and $\mathcal{P}(\mathrm{Hom}_{\mathcal{C}} \times F_1(\mathrm{Hom}_{\mathcal{C}}))$ are sets. Since $F_0 \subset \mathrm{Obj}(\mathcal{C}) \times F_0(\mathrm{Obj}(C))$ and $F_1 \subset \mathrm{Hom}_{\mathcal{C}} \times F_1(\mathrm{Hom}_{\mathcal{C}})$, we have that $F_0 \in \mathcal{P}(\mathrm{Obj}(\mathcal{C}) \times F_0(\mathrm{Obj}(C)))$ and $F_1 \in \mathcal{P}(\mathrm{Hom}_{\mathcal{C}} \times F_1(\mathrm{Hom}_{\mathcal{C}}))$. Hence F_0 and F_1 are sets. Thus $F = (F_0, F_1)$ is a set. □

Exercise 2.2.3.9. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Suppose that $\mathcal{C}$ is small. Then there exists a class A such that for each class F, $F \in A$ iff $F : \mathcal{C} \to \mathcal{D}$.

Proof. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Suppose that $\mathcal{C}$ is small. Define ϕ by

$$\phi(F) : F : \mathcal{C} \to \mathcal{D}$$

Then there exists a class A such that for each set F, $F \in A$ iff $\phi(F)$. Let F be a class. Suppose that $F \in A$. By Definition 1.2.0.1, F is a set. Since F is a set and $F \in A$, we have that $\phi(F)$. Hence $F : \mathcal{C} \to \mathcal{D}$.
Conversely, suppose that $F : \mathcal{C} \to \mathcal{D}$. Exercise 2.2.3.8 implies that F is a set. Since F is a set and $\phi(F)$ is true, we have that $F \in A$. □

Definition 2.2.3.10. We define **Cat** by

- $\mathrm{Obj}(\mathbf{Cat}) = \{\mathcal{C} : \mathcal{C} \text{ is a small category}\}$.
- for $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathbf{Cat})$,
$$\mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D}) = \{F : F : \mathcal{C} \to \mathcal{D}\}$$
- for $\mathcal{C}, \mathcal{D}, \mathcal{E} \in \mathrm{Obj}(\mathbf{Cat})$, $F \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D})$ and $G \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{D}, \mathcal{E})$,
$$G \circ_{\mathbf{Cat}} F = G \circ F$$

Exercise 2.2.3.11. We have that **Cat** is

1. a category
2. locally small

Proof.

1. Exercise 2.2.3.3 implies that composition is well defined. Exercise 2.2.3.4 implies that composition is associative. Exercise 2.2.3.6 and Exercise 2.2.3.7 imply the existence of identities.
2. Let $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathbf{Cat})$ and $F \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D})$. Definition 2.1.1.8 implies that $\mathrm{Obj}(\mathcal{C})$, $\mathrm{Obj}(\mathcal{D})$, $\mathrm{Hom}_{\mathcal{C}}$ and $\mathrm{Hom}_{\mathcal{D}}$ are sets. Then $\mathrm{Obj}(\mathcal{D})^{\mathrm{Obj}(\mathcal{C})}$ and $\mathrm{Hom}_{\mathcal{D}}^{\mathrm{Hom}_{\mathcal{C}}}$ are sets. Hence $\mathrm{Obj}(\mathcal{D})^{\mathrm{Obj}(\mathcal{C})} \times \mathrm{Hom}_{\mathcal{D}}^{\mathrm{Hom}_{\mathcal{C}}}$ is a set. Let $F \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D})$. Then there exist $F_0 \in \mathrm{Obj}(\mathcal{D})^{\mathrm{Obj}(\mathcal{C})}$ and $F_1 \in \mathrm{Hom}_{\mathcal{D}}^{\mathrm{Hom}_{\mathcal{C}}}$ such that $F = (F_0, F_1)$. Therefore $F \in \mathrm{Obj}(\mathcal{D})^{\mathrm{Obj}(\mathcal{C})} \times \mathrm{Hom}_{\mathcal{D}}^{\mathrm{Hom}_{\mathcal{C}}}$. Since $F \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D})$ is arbitrary,
$$\mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D}) \subset \mathrm{Obj}(\mathcal{D})^{\mathrm{Obj}(\mathcal{C})} \times \mathrm{Hom}_{\mathcal{D}}^{\mathrm{Hom}_{\mathcal{C}}}$$
which implies that $\mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D})$ is a set. Therefore, **Cat** is locally small.

□

2.2.4 Comma Categories

Definition 2.2.4.1. Let $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ be a categories and $S : \mathcal{A} \to \mathcal{C}$, $T : \mathcal{B} \to \mathcal{C}$ functors. We define the **comma category of S to T**, denoted $(S{\downarrow}T)$, by

- $\mathrm{Obj}(S{\downarrow}T) = \{(A, B, h) : A \in \mathrm{Obj}(\mathcal{A}), B \in \mathrm{Obj}(\mathcal{B}), \text{ and } h \in \mathrm{Hom}_{\mathcal{C}}(S(A), T(B))\}$
- For $(A_1, B_1, h_1), (A_2, B_2, h_2) \in \mathrm{Obj}(S{\downarrow}T)$,
$$\begin{aligned}&\mathrm{Hom}_{(S\downarrow T)}((A_1, B_1, h_1), (A_2, B_2, h_2)) = \\ &\quad \{(\alpha, \beta) : \alpha \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2),\ \beta \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2) \text{ and } T(\beta) \circ_{\mathcal{C}} h_1 = h_2 \circ_{\mathcal{C}} S(\alpha)\}\end{aligned}$$
i.e. for $(A_1, B_1, h_1), (A_2, B_2, h_2) \in \mathrm{Obj}(S{\downarrow}T)$, $\alpha \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$ and $\beta \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$, $(\alpha, \beta) \in \mathrm{Hom}_{(S\downarrow T)}((A_1, B_1, h_1), (A_2, B_2,$ iff the following diagram commutes:
$$\begin{tikzcd} S(A_1) \arrow[r, "S(\alpha)"] \arrow[d, "h_1"'] & S(A_2) \arrow[d, "h_2"] \\ T(B_1) \arrow[r, "T(\beta)"'] & T(B_2) \end{tikzcd}$$
- For
  - $(A_1, B_1, h_1), (A_2, B_2, h_2), (A_3, B_3, h_3) \in \mathrm{Obj}(S{\downarrow}T)$
  - $(\alpha_{12}, \beta_{12}) \in \mathrm{Hom}_{(S\downarrow T)}((A_1, B_1, h_1), (A_2, B_2, h_2))$
  - $(\alpha_{23}, \beta_{23}) \in \mathrm{Hom}_{(S\downarrow T)}((A_2, B_2, h_2), (A_3, B_3, h_3))$

  we define
$$(\alpha_{23}, \beta_{23}) \circ_{(S\downarrow T)} (\alpha_{12}, \beta_{12}) = (\alpha_{23} \circ_{\mathcal{A}} \alpha_{12}, \beta_{23} \circ_{\mathcal{B}} \beta_{12})$$

Exercise 2.2.4.2. Let $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ be a categories and $S : \mathcal{A} \to \mathcal{C}$, $T : \mathcal{B} \to \mathcal{C}$ functors. Then $(S{\downarrow}T)$ is a category.

Proof.

- **well-definedness of composition:**
  Let
  - $(A_1, B_1, h_1), (A_2, B_2, h_2), (A_3, B_3, h_3) \in \mathrm{Obj}(S{\downarrow}T)$

- $(\alpha_{12}, \beta_{12}) \in \mathrm{Hom}_{(S\downarrow T)}((A_1, B_1, h_1), (A_2, B_2, h_2))$
- $(\alpha_{23}, \beta_{23}) \in \mathrm{Hom}_{(S\downarrow T)}((A_2, B_2, h_2), (A_3, B_3, h_3))$

By definition, $\alpha_{12} \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$, $\alpha_{23} \in \mathrm{Hom}_{\mathcal{A}}(A_2, A_3)$, $\beta_{12} \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$, $\beta_{23} \in \mathrm{Hom}_{\mathcal{B}}(B_2, B_3)$, $T(\beta_{12}) \circ_{\mathcal{C}} h_1 = h_2 \circ S(\alpha_{12})$ and $T(\beta_{23}) \circ_{\mathcal{C}} h_2 = h_3 \circ_{\mathcal{C}} S(\alpha_{23})$,
i.e. the following diagram commutes:

$$\begin{tikzcd}
S(A_1) \arrow[r, "S(\alpha_{12})"] \arrow[d, "h_1"'] & S(A_2) \arrow[r, "S(\alpha_{23})"] \arrow[d, "h_2"] & S(A_3) \arrow[d, "h_3"] \\
T(B_1) \arrow[r, "T(\beta_{12})"'] & T(B_2) \arrow[r, "T(\beta_{23})"'] & T(B_3)
\end{tikzcd}$$

Then $\alpha_{23} \circ_{\mathcal{A}} \alpha_{12} \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_3)$, $\beta_{23} \circ_{\mathcal{B}} \beta_{12} \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_3)$ and

$$\begin{aligned}
T(\beta_{23} \circ_{\mathcal{B}} \beta_{12}) \circ_{\mathcal{C}} h_1 &= (T(\beta_{23}) \circ_{\mathcal{C}} T(\beta_{12})) \circ_{\mathcal{C}} h_1 \\
&= T(\beta_{23}) \circ_{\mathcal{C}} (T(\beta_{12}) \circ_{\mathcal{C}} h_1) \\
&= T(\beta_{23}) \circ_{\mathcal{C}} (h_2 \circ_{\mathcal{C}} S(\alpha_{12})) \\
&= (T(\beta_{23}) \circ_{\mathcal{C}} h_2) \circ_{\mathcal{C}} S(\alpha_{12}) \\
&= (h_3 \circ_{\mathcal{C}} S(\alpha_{23})) \circ_{\mathcal{C}} S(\alpha_{12}) \\
&= h_3 \circ_{\mathcal{C}} (S(\alpha_{23}) \circ_{\mathcal{C}} S(\alpha_{12})) \\
&= h_3 \circ_{\mathcal{C}} S(\alpha_{23} \circ_{\mathcal{A}} \alpha_{12})
\end{aligned}$$

i.e. the following diagram commutes:

$$\begin{tikzcd}
S(A_1) \arrow[r, "S(\alpha_{23} \circ_{\mathcal{A}} \alpha_{12})"] \arrow[d, "h_1"'] & S(A_3) \arrow[d, "h_3"] \\
T(B_1) \arrow[r, "T(\beta_{23} \circ_{\mathcal{B}} \beta_{12})"'] & T(B_3)
\end{tikzcd}$$

Hence $(\alpha_{23} \circ_{\mathcal{A}} \alpha_{12}, \beta_{23} \circ_{\mathcal{B}} \beta_{12}) \in \mathrm{Hom}_{(S\downarrow T)}((A_1, B_1, h_1), (A_3, B_3, h_3))$ and composition is well defined.

- **associativity of composition:**
  Let
  - $(A_1, B_1, h_1), (A_2, B_2, h_2), (A_3, B_3, h_3), (A_4, B_4, h_4) \in \mathrm{Obj}(S\downarrow T)$
  - $(\alpha_{12}, \beta_{12}) \in \mathrm{Hom}_{(S\downarrow T)}((A_1, B_1, h_1), (A_2, B_2, h_2))$
  - $(\alpha_{23}, \beta_{23}) \in \mathrm{Hom}_{(S\downarrow T)}((A_2, B_2, h_2), (A_3, B_3, h_3))$
  - $(\alpha_{34}, \beta_{34}) \in \mathrm{Hom}_{(S\downarrow T)}((A_3, B_3, h_3), (A_4, B_4, h_4))$

  Then

$$\begin{aligned}
[(\alpha_{34}, \beta_{34}) \circ_{(S\downarrow T)} (\alpha_{23}, \beta_{23})] \circ_{(S\downarrow T)} (\alpha_{12}, \beta_{12}) &= (\alpha_{34} \circ_{\mathcal{A}} \alpha_{23}, \beta_{34} \circ_{\mathcal{B}} \beta_{23}) \circ_{(S\downarrow T)} (\alpha_{12}, \beta_{12}) \\
&= ([\alpha_{34} \circ_{\mathcal{A}} \alpha_{23}] \circ_{\mathcal{A}} \alpha_{12}, [\beta_{34} \circ_{\mathcal{B}} \beta_{23}] \circ_{\mathcal{B}} \beta_{12}) \\
&= (\alpha_{34} \circ_{\mathcal{A}} [\alpha_{23} \circ_{\mathcal{A}} \alpha_{12}], \beta_{34} \circ_{\mathcal{B}} [\beta_{23} \circ_{\mathcal{B}} \beta_{12}]) \\
&= (\alpha_{34}, \beta_{34}) \circ_{(S\downarrow T)} (\alpha_{23} \circ_{\mathcal{A}} \alpha_{12}, \beta_{23} \circ_{\mathcal{B}} \beta_{12}) \\
&= (\alpha_{34}, \beta_{34}) \circ_{(S\downarrow T)} [(\alpha_{23}, \beta_{23}) \circ_{(S\downarrow T)} (\alpha_{12}, \beta_{12})]
\end{aligned}$$

  So composition is associative.

- **existence of identities:**
  Let
  - $(A_1, B_1, h_1), (A_2, B_2, h_2), \in \mathrm{Obj}(S\downarrow T)$

- $(\alpha, \beta) \in \mathrm{Hom}_{(S \downarrow T)}((A_1, B_1, h_1), (A_2, B_2, h_2))$

By definition,

- $\alpha \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$, $\beta \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$
- $h_1 \in \mathrm{Hom}_{\mathcal{C}}(S(A_1), T(B_1))$, $h_2 \in \mathrm{Hom}_{\mathcal{C}}(S(A_2), T(B_2))$
- $T(\beta) \circ h_1 = h_2 \circ S(\alpha)$

Since $\mathrm{id}_{A_1} \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_1)$, $\mathrm{id}_{B_1} \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_1)$, and

$$\begin{aligned} T(\mathrm{id}_{B_1}) \circ_{\mathcal{C}} h_1 &= \mathrm{id}_{T(B_1)} \circ_{\mathcal{C}} h_1 \\ &= h_1 \\ &= h_1 \circ_{\mathcal{C}} \mathrm{id}_{S(A_1)} \\ &= h_1 \circ_{\mathcal{C}} S(\mathrm{id}_{A_1}) \end{aligned}$$

i.e. the following diagram commutes:

$$\begin{tikzcd} S(A_1) \arrow[r, "S(\mathrm{id}_{A_1})"] \arrow[d, "h_1"'] & S(A_1) \arrow[d, "h_1"] \\ T(B_1) \arrow[r, "T(\mathrm{id}_{B_1})"'] & T(B_1) \end{tikzcd}$$

we have that $(\mathrm{id}_{A_1}, \mathrm{id}_{B_1}) \in \mathrm{Hom}_{(S \downarrow T)}((A_1, B_1, h_1), (A_1, B_1, h_1))$. Similarly $(\mathrm{id}_{A_2}, \mathrm{id}_{B_2}) \in \mathrm{Hom}_{(S \downarrow T)}((A_2, B_2, h_2), (A_2, B_2, h_2))$. Therefore

$$\begin{aligned} (\alpha, \beta) \circ_{(S \downarrow T)} (\mathrm{id}_{A_1}, \mathrm{id}_{B_1}) &= (\alpha \circ_{\mathcal{A}} \mathrm{id}_{A_1}, \beta \circ_{\mathcal{B}} \mathrm{id}_{B_1}) \\ &= (\alpha, \beta) \end{aligned}$$

and

$$\begin{aligned} (\mathrm{id}_{A_2}, \mathrm{id}_{B_2}) \circ_{(S \downarrow T)} (\alpha, \beta) &= (\mathrm{id}_{A_2} \circ_{\mathcal{A}} \alpha, \mathrm{id}_{B_2} \circ_{\mathcal{B}} \beta) \\ &= (\alpha, \beta) \end{aligned}$$

Since $(A_1, B_1, h_1), (A_2, B_2, h_2), \in \mathrm{Obj}(S \downarrow T)$ and
$(\alpha, \beta) \in \mathrm{Hom}_{(S \downarrow T)}((A_1, B_1, h_1), (A_2, B_2, h_2))$ are arbitrary, we have that for each $(A, B, h) \in \mathrm{Obj}(S \downarrow T)$, $\mathrm{id}_{(A,B,h)} = (\mathrm{id}_A, \mathrm{id}_B)$.

□

Definition 2.2.4.3. define the arrow category of $\mathcal{C}$ text

Note 2.2.4.4. explain with diagram how in the case $\alpha = \Delta^{\mathbf{1}}_X$ and how we can contract one edge of the rectangle diagram to get a triangle

Definition 2.2.4.5. Let $\mathcal{C}, \mathcal{D}$ be a categories, $X \in \mathrm{Obj}(\mathcal{D})$ and $F : \mathcal{C} \to \mathcal{D}$. We define the **comma category from X to F**, denoted $(X \downarrow F)$, by $(X \downarrow F) := (\Delta^{\mathbf{1}}_X \downarrow F)$.
We may make the following identification:

- $\mathrm{Obj}(X \downarrow F) = \{(A, f) : A \in \mathrm{Obj}(\mathcal{C}) \text{ and } f \in \mathrm{Hom}_{\mathcal{D}}(X, F(A))\}$
- For $(A_1, f_1), (A_2, f_2) \in \mathrm{Obj}(X \downarrow F)$,

$$\mathrm{Hom}_{(X \downarrow F)}((A_1, f_1), (A_2, f_2)) = \{\alpha \in \mathrm{Hom}_{\mathcal{C}}(A_1, A_2) \text{ and } F(\alpha) \circ f_1 = f_2\}$$

i.e. for $(A_1, f_1), (A_2, f_2) \in \mathrm{Obj}(X \downarrow F)$ and $\alpha \in \mathrm{Hom}_{A_1, A_2}$, $\alpha \in \mathrm{Hom}_{(X \downarrow F)}((A_1, f_1), (A_2, f_2))$ iff the following diagram commutes:

$$\begin{tikzcd} & X \arrow[dl, "f_1"'] \arrow[dr, "f_2"] & \\ F(A_1) \arrow[rr, "F(\alpha)"'] & & F(A_2) \end{tikzcd}$$

- For
  - $(A_1, f_1), (A_2, f_2), (A_3, f_3) \in \mathrm{Obj}(X \downarrow F)$
  - $\alpha \in \mathrm{Hom}_{(X \downarrow F)}((A_1, f_1), (A_2, f_2))$
  - $\beta \in \mathrm{Hom}_{(X \downarrow F)}((A_2, f_2), (A_3, f_3))$

  we define
$$\beta \circ_{(X \downarrow F)} \alpha = \beta \circ_{\mathcal{C}} \alpha$$

Note 2.2.4.6. note that the coslice category under X is a comma categroy from X to $\mathrm{id}_{\mathcal{C}}$

Definition 2.2.4.7. Let $\mathcal{C}, \mathcal{D}$ be a categories, $X \in \mathrm{Obj}(\mathcal{D})$ and $F : \mathcal{C} \to \mathcal{D}$. We define the **comma category from F to X**, denoted $(F \downarrow X)$, by $(F \downarrow X) := (F \downarrow \Delta^{\mathbf{1}}_X)$.
We may make the following identification:

- $\mathrm{Obj}(F \downarrow X) = \{(A, f) : A \in \mathrm{Obj}(\mathcal{C}) \text{ and } f \in \mathrm{Hom}_{\mathcal{D}}(F(A), X)\}$
- For $(A_1, f_1), (A_2, f_2) \in \mathrm{Obj}(F \downarrow X)$,
$$\mathrm{Hom}_{(X \downarrow F)}((A_1, f_1), (A_2, f_2)) = \{\alpha \in \mathrm{Hom}_{\mathcal{C}}(A_1, A_2) \text{ and } f_2 \circ F(\alpha) = f_1\}$$
  i.e. for $(A_1, f_1), (A_2, f_2) \in \mathrm{Obj}(F \downarrow X)$ and $\alpha \in \mathrm{Hom}_{A_1, A_2}$, $\alpha \in \mathrm{Hom}_{(F \downarrow X)}((A_1, f_1), (A_2, f_2))$ iff the following diagram commutes:
$$\begin{tikzcd} F(A_1) \arrow[rr, "F(\alpha)"] \arrow[dr, "f_1"'] & & F(A_2) \arrow[dl, "f_2"] \\ & X & \end{tikzcd}$$
- For
  - $(A_1, f_1), (A_2, f_2), (A_3, f_3) \in \mathrm{Obj}(F \downarrow X)$
  - $\alpha \in \mathrm{Hom}_{(F \downarrow X)}((A_1, f_1), (A_2, f_2))$
  - $\beta \in \mathrm{Hom}_{(F \downarrow X)}((A_2, f_2), (A_3, f_3))$

  we define
$$\beta \circ_{(F \downarrow X)} \alpha = \beta \circ_{\mathcal{C}} \alpha$$

Definition 2.2.4.8. Let $\mathcal{C}$ be a category and $X \in \mathrm{Obj}(\mathcal{C})$. We define the **arrow category of $\mathcal{C}$**, denoted $\mathrm{Arr}(\mathcal{C})$, by $\mathrm{Arr}(\mathcal{C}) := (\mathrm{id}_{\mathcal{C}} \downarrow \mathrm{id}_{\mathcal{C}})$.

Definition 2.2.4.9. Let $\mathcal{C}$ be a category and $X \in \mathrm{Obj}(\mathcal{C})$. We define the **slice category of $\mathcal{C}$ over X** or the **over category of $\mathcal{C}$ over X**, denoted $\mathcal{C} \downarrow X$, by $\mathcal{C} \downarrow X := (\mathrm{id}_{\mathcal{C}} \downarrow X)$.

Definition 2.2.4.10. Let $\mathcal{C}$ be a category and $X \in \mathrm{Obj}(\mathcal{C})$. We define the **coslice category of $\mathcal{C}$ over X** or the **under category of $\mathcal{C}$ under X**, denoted $X \downarrow \mathcal{C}$, by $X \downarrow \mathcal{C} := (X \downarrow \mathrm{id}_{\mathcal{C}})$.

give examples in posets, and other

2.3 Natural Transformations

2.3.1 Introduction

Definition 2.3.1.1. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, $F, G : \mathcal{C} \to \mathcal{D}$ and $\alpha : \mathrm{Obj}(\mathcal{C}) \to \mathrm{Hom}_{\mathcal{D}}$. Then α is said to be a **natural transformation from** F **to** G, denoted $\alpha : F \Rightarrow G$, if

1. for each $A \in \mathrm{Obj}(\mathcal{C})$, $\alpha_A \in \mathrm{Hom}_{\mathcal{D}}(F(A), G(A))$
2. for each $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$, $G(f) \circ \alpha_A = \alpha_B \circ F(f)$, i.e. the following diagram commutes:

$$\begin{tikzcd} F(A) \arrow[r, "\alpha_A"] \arrow[d, "F(f)"'] & G(A) \arrow[d, "G(f)"] \\ F(B) \arrow[r, "\alpha_B"'] & G(B) \end{tikzcd}$$

Definition 2.3.1.2. Horizontal Composition:
Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be categories, $F, G : \mathcal{A} \to \mathcal{B}$, $F', G' : \mathcal{B} \to \mathcal{C}$ functors and $\alpha : F \Rightarrow G$, $\alpha' : F' \Rightarrow G'$ natural transformations. We define the **horizontal composition of** β **with** α, denoted $\beta * \alpha : F' \circ F \Rightarrow G' \circ G$, by

$$(\alpha' * \alpha)_A = \alpha'_{G(A)} \circ F'(\alpha_A)$$

Exercise 2.3.1.3. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be categories, $F, G : \mathcal{A} \to \mathcal{B}$, $F', G' : \mathcal{B} \to \mathcal{C}$ functors and $\alpha : F \Rightarrow G$, $\alpha' : F' \Rightarrow G'$ natural transformations. Then $\alpha' \circ \alpha : F' \circ F \Rightarrow G' \circ G$ is a natural transformation.

Proof.

1. Let $A \in \mathrm{Obj}(\mathcal{A})$. Since $\alpha : F \Rightarrow G$, we have that $\alpha_A \in \mathrm{Hom}_{\mathcal{B}}(F(A), G(A))$. Since $F' : \mathcal{B} \to \mathcal{C}$, we have that

$$\begin{aligned} F'(\alpha_A) &\in \mathrm{Hom}_{\mathcal{C}}(F'(F(A)), F'(G(A))) \\ &= \mathrm{Hom}_{\mathcal{C}}(F' \circ F(A), F' \circ G(A)). \end{aligned}$$

   Since $G : \mathcal{A} \to \mathcal{B}$, we have that $G(A) \in \mathrm{Obj}(\mathcal{B})$. Since $\alpha' : F' \Rightarrow G'$, we have that

$$\begin{aligned} \alpha'_{G(A)} &\in \mathrm{Hom}_{\mathcal{C}}(F'(G(A)), G'(G(A))) \\ &= \mathrm{Hom}_{\mathcal{C}}(F' \circ G(A), G' \circ G(A)). \end{aligned}$$

   Therefore

$$\begin{aligned} (\alpha' * \alpha)_A &= \alpha'_{G(A)} \circ F'(\alpha_A) \\ &\in \mathrm{Hom}_{\mathcal{C}}(F' \circ F(A), G' \circ G(A)). \end{aligned}$$

   Since $A \in \mathrm{Obj}(\mathcal{A})$ is arbitrary, we have that for each $A \in \mathrm{Obj}(\mathcal{A})$, $(\alpha' * \alpha)_A \in \mathrm{Hom}_{\mathcal{C}}(F' \circ F(A), G' \circ G(A))$.

2. Let $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$ and $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$. Since $F, G : \mathcal{A} \to \mathcal{B}$, we have that $F(f) \in \mathrm{Hom}_{\mathcal{B}}(F(A_1), F(A_2))$ and $G(f) \in \mathrm{Hom}_{\mathcal{B}}(G(A_1), G(A_2))$. Since $\alpha : F \Rightarrow G$, we have that for each $k \in [2]$, $\alpha_{A_k} \in \mathrm{Hom}_{\mathcal{B}}(F(A_k), G(A_k))$ and $G(f) \circ \alpha_{A_1} = \alpha_{A_2} \circ F(f)$, i.e. the following diagram commutes:

$$\begin{tikzcd} F(A_1) \arrow[r, "\alpha_{A_1}"] \arrow[d, "F(f)"'] & G(A_1) \arrow[d, "G(f)"] \\ F(A_2) \arrow[r, "\alpha_{A_2}"'] & G(A_2) \end{tikzcd}$$

   Since $F', G' : \mathcal{B} \to \mathcal{C}$, we have that

$$\begin{aligned} F' \circ F(f) &= F'(F(f)) \\ &\in \mathrm{Hom}_{\mathcal{C}}(F'(F(A_1)), F'(F(A_2))) \\ &= \mathrm{Hom}_{\mathcal{C}}(F' \circ F(A_1), F' \circ F(A_2)), \end{aligned}$$

$$\begin{aligned} F' \circ G(f) &= F'(G(f)) \\ &\in \mathrm{Hom}_{\mathcal{C}}(F'(G(A_1)), F'(G(A_2))) \\ &= \mathrm{Hom}_{\mathcal{C}}(F' \circ G(A_1), F' \circ G(A_2)), \end{aligned}$$

$$\begin{aligned} G' \circ G(f) &= G'(G(f)) \\ &\in \mathrm{Hom}_{\mathcal{C}}(G'(G(A_1)), G'(G(A_2))) \\ &= \mathrm{Hom}_{\mathcal{C}}(G' \circ G(A_1), G' \circ G(A_2)), \end{aligned}$$

and for each $k \in [2]$,

$$\begin{aligned} F'(\alpha_{A_k}) &\in \mathrm{Hom}_{\mathcal{C}}(F'(F(A_k)), F'(G(A_k))) \\ &= \mathrm{Hom}_{\mathcal{C}}(F' \circ F(A_k), F' \circ G(A_k)). \end{aligned}$$

Since $\alpha' : F' \Rightarrow G'$, we have that for each $k \in [2]$,

$$\begin{aligned} \alpha'_{G(A_k)} &\in \mathrm{Hom}_{\mathcal{C}}(F'(G(A_k)), G'(G(A_k))) \\ &= \mathrm{Hom}_{\mathcal{C}}(F' \circ G(A_k), G' \circ G(A_k)). \end{aligned}$$

Therefore

$$\begin{aligned} [F' \circ G(f)] \circ F'(\alpha_{A_1}) &= F'(G(f)) \circ F'(\alpha_{A_1}) \\ &= F'(G(f) \circ \alpha_{A_1}) \\ &= F'(\alpha_{A_2} \circ F(f)) \\ &= F'(\alpha_{A_2}) \circ F'(F(f)) \\ &= F'(\alpha_{A_2}) \circ [F' \circ F(f)], \end{aligned}$$

and

$$\begin{aligned} [G' \circ G(f)] \circ \alpha'_{G(A_1)} &= G'(G(f)) \circ \alpha'_{G(A_1)} \\ &= \alpha'_{G(A_2)} \circ F'(G(f)) \\ &= \alpha'_{G(A_2)} \circ [F' \circ G(f)], \end{aligned}$$

i.e. the following diagram commutes:

$$\begin{tikzcd} F' \circ F(A_1) \arrow[r, "F'(\alpha_{A_1})"] \arrow[d, "F' \circ F(f)"'] & F' \circ G(A_1) \arrow[r, "\alpha'_{G(A_1)}"] \arrow[d, "F' \circ G(f)"] & G' \circ G(A_1) \arrow[d, "G' \circ G(f)"] \\ F' \circ F(A_2) \arrow[r, "F'(\alpha_{A_2})"'] & F' \circ G(A_2) \arrow[r, "\alpha'_{G(A_2)}"'] & G' \circ G(A_2) \end{tikzcd}$$

Hence

$$\begin{aligned} [G' \circ G(f)] \circ (\alpha' * \alpha)_{A_1} &= [G' \circ G(f)] \circ [\alpha'_{G(A_1)} \circ F'(\alpha_{A_1})] \\ &= ([G' \circ G(f)] \circ \alpha'_{G(A_1)}) \circ F'(\alpha_{A_1}) \\ &= (\alpha'_{G(A_2)} \circ [F' \circ G(f)]) \circ F'(\alpha_{A_1}) \\ &= \alpha'_{G(A_2)} \circ ([F' \circ G(f)] \circ F'(\alpha_{A_1})) \\ &= \alpha'_{G(A_2)} \circ (F'(\alpha_{A_2}) \circ [F' \circ F(f)]) \\ &= [\alpha'_{G(A_2)} \circ F'(\alpha_{A_2})] \circ [F' \circ F(f)] \\ &= (\alpha' * \alpha)_{A_2} \circ [F' \circ F(f)], \end{aligned}$$

i.e. the following diagram commutes:

$$\begin{tikzcd} F' \circ F(A_1) \arrow[r, "(\alpha' * \alpha)_{A_1}"] \arrow[d, "F' \circ F(f)"'] & G' \circ G(A_1) \arrow[d, "G' \circ G(f)"] \\ F' \circ F(A_2) \arrow[r, "(\alpha' * \alpha)_{A_2}"'] & G' \circ G(A_2) \end{tikzcd}$$

Since $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$ and $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$ are arbitrary, we have that for each $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$ and $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$, $[G' \circ G(f)] \circ (\alpha' * \alpha)_{A_1} = (\alpha' * \alpha)_{A_2} \circ [F' \circ F(f)]$.

Thus $\alpha' * \alpha : F' \circ F \Rightarrow G' \circ G$. □

Definition 2.3.1.4. Whiskering:

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-

FINISH!!!

talk about following example preimages and set comprehension vs post composition and predicates will need to talk about contravariant functors from preim, pred, denoted $F, G : \mathbf{Set} \to \mathbf{Set}$ respectively given by $F(X) := \mathcal{P}(X)$, $F(f) := f^{-1}$ and $G(X) := \{0,1\}^X$, $G(f) := - \circ f$. This post/functor is interesting because it highlights the associativity of $(\mathrm{NOT} \circ \phi) \circ f = \mathrm{NOT} \circ (\phi \circ f)$ as a commutativity $f^{-1}(A^c) = (f^{-1}(A))^c$. That is if $X = f^{-1}$ and $Y = -^c$, then $XYA = YXA$ so that $XY = YX$. It begs the question whether or not commutativity can always be recast as associativity in some other context.

below is wild speculation and off topic question The author writes that associativity shows you that for any appropriate f, ϕ, post composing with NOT first and then pre composing with f vs precomposing with f first then poste composing with NOT are the same thing and in a sense orthogonal and independent, as in they do not affect the outcome. So is there a general view in which associativity (or via natural transformation commutativity) is a measure of independence?

Can we connect this with probability/physics in an information theoretic sense using the motivation/intuition that the kl divergence (or other notion of info loss) is a functor? i.e. $XY = YX$ or $x \circ (y \circ a) = (x \circ y) \circ a$ iff $\mathrm{info_loss}[\mathrm{joint}(X,Y), \otimes(X,Y)] = 0$ or some generalization? We would need a notion of joint for X and Y and a (not product because category of prob dists doesnt have products) tensor product, so maybe we need to consider monoidal cats)

what X, Y would correspond to classical/quantum probability? In the classical/quantum case, X, Y are marginal systems and $\mathrm{joint}(X,Y)$ is the given bipartite system from which we isolate the marginals. In the classical case, $\mathrm{info_loss}[\mathrm{joint}(X,Y), \otimes(X,Y)] = 0$ iff $\mathrm{joint}(X,Y) = \otimes(X,Y)]$, that is, there is the two marginal states of the bipartite system contain as much information as the bipartite system itself, i.e. there is no interaction. Im not sure what the associated commuting/associative operations would be, but maybe we can get a clue from the above example or by looking further into similar examples, maybe toposes or other tools cover this?

If it were possible to find commutative/associative operations that capture two subsystems being independent and not having an interaction in the classical/quantum probabilistic sense, then this would seem to suggest that we would be able to recast interaction between subsystems in terms of some associative/commutative operations. How would this compare/contrast with the characterization of interaction via homological algebra using exactness?

2.3.2 Category of Functors

Definition 2.3.2.1. Vertical Composition:
Let $\mathcal{C}$, $\mathcal{D}$ be categories, $F, G, H : \mathcal{C} \to \mathcal{D}$ functors and $\alpha : F \Rightarrow G$, $\beta : G \Rightarrow H$ natural transformations. We define the **vertical composition of β with α**, denoted $\beta \circ \alpha : F \Rightarrow H$, by

$$(\beta \circ \alpha)_A = \beta_A \circ \alpha_A$$

Exercise 2.3.2.2. Let $\mathcal{C}$, $\mathcal{D}$ be categories, $F, G, H : \mathcal{C} \to \mathcal{D}$ functors and $\alpha : F \Rightarrow G$, $\beta : G \Rightarrow H$ natural transformations. Then $\beta \circ \alpha : F \Rightarrow H$ is a natural transformation.

Proof.

1. Let $A \in \mathrm{Obj}(\mathcal{C})$. Since $\alpha_A \in \mathrm{Hom}_{\mathcal{D}}(F(A), G(A))$ and $\beta_A \in \mathrm{Hom}_{\mathcal{D}}(G(A), H(A))$, we have that
$$\begin{aligned}(\beta \circ \alpha)_A &= \beta_A \circ \alpha_A \\ &\in \mathrm{Hom}_{\mathcal{D}}(F(A), H(A))\end{aligned}$$

2. Let $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Since $\alpha : F \Rightarrow G$ and $\beta : G \Rightarrow H$, $G(f) \circ \alpha_A = \alpha_B \circ F(f)$ and $H(f) \circ \beta_A = \beta_B \circ G(f)$. Therefore
$$\begin{aligned}H(f) \circ (\beta \circ \alpha)_A &= H(f) \circ (\beta_A \circ \alpha_A) \\ &= (H(f) \circ \beta_A) \circ \alpha_A \\ &= (\beta_B \circ G(f)) \circ \alpha_A \\ &= \beta_B \circ (G(f) \circ \alpha_A) \\ &= \beta_B \circ (\alpha_B \circ F(f)) \\ &= (\beta_B \circ \alpha_B) \circ F(f) \\ &= (\beta \circ \alpha)_B \circ F(f)\end{aligned}$$

So $\beta \circ \alpha : F \Rightarrow H$ is a natural transformation. □

Exercise 2.3.2.3. Let $\mathcal{C}$, $\mathcal{D}$ be categories, $F, G, H, K : \mathcal{C} \to \mathcal{D}$ functors and $\alpha : F \Rightarrow G$, $\beta : G \Rightarrow H$ and $\gamma : H \Rightarrow K$ natural transformations. Then
$$(\gamma \circ \beta) \circ \alpha = \gamma \circ (\beta \circ \alpha)$$

Proof. Let $A \in \mathrm{Obj}(\mathcal{C})$. By definition,
$$\begin{aligned}[(\gamma \circ \beta) \circ \alpha]_A &= (\gamma \circ \beta)_A \circ \alpha_A \\ &= (\gamma_A \circ \beta_A) \circ \alpha_A \\ &= \gamma_A \circ (\beta_A \circ \alpha_A) \\ &= \gamma_A \circ (\beta \circ \alpha)_A \\ &= [\gamma \circ (\beta \circ \alpha)]_A\end{aligned}$$

Since $A \in \mathrm{Obj}(\mathcal{C})$ is arbitrary,
$$(\gamma \circ \beta) \circ \alpha = \gamma \circ (\beta \circ \alpha)$$

□

Definition 2.3.2.4. Let $\mathcal{C}$, $\mathcal{D}$ be categories and $F : \mathcal{C} \to \mathcal{D}$. We define the **identity natural transformation from F to F**, denoted $\mathrm{id}_F : F \Rightarrow F$, by
$$(\mathrm{id}_F)_A = \mathrm{id}^{\mathcal{D}}_{F(A)}$$

Exercise 2.3.2.5. Let $\mathcal{C}$, $\mathcal{D}$ be categories and $F : \mathcal{C} \to \mathcal{D}$. Then $\mathrm{id}_F : F \Rightarrow F$ is a natural transformation.

Proof.

1. Let $A \in \mathrm{Obj}(\mathcal{C})$. Then
$$\begin{aligned}(\mathrm{id}_F)_A &= \mathrm{id}^{\mathcal{D}}_{F(A)} \\ &\in \mathrm{Hom}_{\mathcal{D}}(F(A), F(A))\end{aligned}$$

2. Let $A, B \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Then
$$\begin{aligned}F(f) \circ_{\mathcal{D}} (\mathrm{id}_F)_A &= F(f) \circ_{\mathcal{D}} \mathrm{id}^{\mathcal{D}}_{F(A)} \\ &= F(f) \\ &= \mathrm{id}^{\mathcal{D}}_{F(B)} \circ_{\mathcal{D}} F(f) \\ &= (\mathrm{id}_F)_B \circ_{\mathcal{D}} F(f),\end{aligned}$$

i.e. the following diagram commutes:

$$\begin{tikzcd} F(A) \arrow[r, "(\mathrm{id}_F)_A"] \arrow[d, "F(f)"'] & F(A) \arrow[d, "F(f)"] \\ F(B) \arrow[r, "(\mathrm{id}_F)_B"] & F(B). \end{tikzcd}$$

□

Exercise 2.3.2.6. Let $\mathcal{C}$, $\mathcal{D}$ be categories, $F, G : \mathcal{C} \to \mathcal{D}$ and $\alpha : F \Rightarrow G$. Then

1. $\mathrm{id}_G \circ \alpha = \alpha$
2. $\alpha \circ \mathrm{id}_F = \alpha$

Proof.

1. Let $A \in \mathrm{Obj}(\mathcal{C})$. Then

$$\begin{aligned} (\mathrm{id}_G \circ \alpha)_A &= (\mathrm{id}_G)_A \circ \alpha_A \\ &= \mathrm{id}_{G(A)} \circ \alpha_A \\ &= \alpha_A \end{aligned}$$

   Since $A \in \mathrm{Obj}(C)$ is arbitrary, $\mathrm{id}_G \circ \alpha = \alpha$

2. Let $A \in \mathrm{Obj}(\mathcal{C})$. Then

$$\begin{aligned} (\alpha \circ \mathrm{id}_F)_A &= \alpha_A \circ (\mathrm{id}_F)_A \\ &= \alpha_A \circ \mathrm{id}_{F(A)} \\ &= \alpha_A \end{aligned}$$

   Since $A \in \mathrm{Obj}(C)$ is arbitrary, $\alpha \circ \mathrm{id}_F = \alpha$.

□

Exercise 2.3.2.7. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, $F, G : \mathcal{C} \to \mathcal{D}$ and $\alpha : F \Rightarrow G$. If $\mathcal{C}$ is small, then α is a set.

Proof. Suppose that $\mathcal{C}$ is small. Then $\mathrm{Obj}(\mathcal{C})$ is a set. Since $\alpha : \mathrm{Obj}(\mathcal{C}) \to \mathrm{Hom}_{\mathcal{D}}$, Axiom 1.2.0.11 implies that $\alpha(\mathrm{Obj}(\mathcal{C}))$ is a set. Then $\mathrm{Obj}(\mathcal{C}) \times \alpha(\mathrm{Obj}(\mathcal{C}))$ is a set. Therefore $\mathcal{P}(\mathrm{Obj}(\mathcal{C}) \times \alpha(\mathrm{Obj}(\mathcal{C})))$ is a set. Since $\alpha \subset \mathrm{Obj}(\mathcal{C}) \times \alpha(\mathrm{Obj}(\mathcal{C}))$, we have that $\alpha \in \mathcal{P}(\mathrm{Obj}(\mathcal{C}) \times \alpha(\mathrm{Obj}(\mathcal{C})))$ which implies that α is a set. □

Exercise 2.3.2.8. Let $\mathcal{C}$ and $\mathcal{D}$ be categories and $F, G : \mathcal{C} \to \mathcal{D}$. If $\mathcal{C}$ is small, then there exists a class A such that for each class α, $\alpha \in A$ iff $\alpha : F \Rightarrow G$.

Proof. Suppose that $\mathcal{C}$ is small. Define ϕ by

$$\phi(\alpha) : \alpha : F \Rightarrow G$$

Axiom 1.2.0.12 implies that there exists a class A such that for each set α, $\alpha \in A$ iff $\phi(\alpha)$. Let α be a class. Suppose that $\alpha \in A$. By Definition 1.2.0.1, α is a set. Since α is a set and $\alpha \in A$, we have that $\phi(\alpha)$. Hence $\alpha : F \Rightarrow G$.
Conversely, suppose that $\alpha : F \Rightarrow G$. Since $\mathcal{C}$ is small, Exercise 2.3.2.7 implies that α is a set. Since $\phi(\alpha)$, we have that $\alpha \in A$. □

Definition 2.3.2.9. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Suppose that $\mathcal{C}$ is small. We define the **functor category from $\mathcal{C}$ to $\mathcal{D}$**, denoted $\mathcal{D}^{\mathcal{C}}$, by

- $\mathrm{Obj}(\mathcal{D}^{\mathcal{C}}) = \{F : F : \mathcal{C} \to \mathcal{D}\}$
- For $F, G \in \mathrm{Obj}(\mathcal{D}^{\mathcal{C}})$, $\mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(F, G) = \{\alpha : \alpha : F \Rightarrow G\}$
- For $F, G, H \in \mathrm{Obj}(\mathcal{D}^{\mathcal{C}})$, $\alpha \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(F, G)$ and $\beta \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(G, H)$, $\beta \circ_{\mathcal{D}^{\mathcal{C}}} \alpha = \beta \circ \alpha$

Exercise 2.3.2.10. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Suppose that $\mathcal{C}$ is small. Then $\mathcal{D}^{\mathcal{C}}$ is a category.

Proof. Exercise 2.3.2.2 implies that composition is well-defined. Exercise 2.3.2.3 implies that composition is associative. Exercise 2.3.2.5 and Exercise 2.3.2.6 imply the existence of identities. □

Exercise 2.3.2.11. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Suppose that $\mathcal{C}$ and $\mathcal{D}$ are small. Then $\mathcal{D}^{\mathcal{C}}$ is small.

Proof. We have that

$$\begin{aligned}
\mathcal{C} \text{ and } \mathcal{D} \text{ are small} &\iff \mathrm{Obj}(\mathcal{C}), \mathrm{Obj}(\mathcal{D}), \mathrm{Hom}_{\mathcal{C}}, \mathrm{Hom}_{\mathcal{D}} \in \mathrm{Obj}(\mathbf{Set}) \\
&\iff \mathrm{Obj}(\mathcal{D})^{\mathrm{Obj}(\mathcal{C})}, \mathrm{Hom}_{\mathcal{D}}^{\mathrm{Hom}_{\mathcal{C}}} \in \mathrm{Obj}(\mathbf{Set}) \\
&\iff \mathrm{Obj}(\mathcal{D})^{\mathrm{Obj}(\mathcal{C})} \times \mathrm{Hom}_{\mathcal{D}}^{\mathrm{Hom}_{\mathcal{C}}} \in \mathrm{Obj}(\mathbf{Set}).
\end{aligned}$$

Since $\mathcal{D}^{\mathcal{C}} \subset \mathrm{Obj}(\mathcal{D})^{\mathrm{Obj}(\mathcal{C})} \times \mathrm{Hom}_{\mathcal{D}}^{\mathrm{Hom}_{\mathcal{C}}}$ and $\mathcal{C}$, $\mathcal{D}$ are small, we have that $\mathcal{D}^{\mathcal{C}} \in \mathrm{Obj}(\mathbf{Set})$. Thus $\mathcal{D}^{\mathcal{C}}$ is small. □

2.3.3 Diagonal Functor

Definition 2.3.3.1. Let $\mathcal{C}$, $\mathcal{D}$ be categories, $X, Y \in \mathrm{Obj}(\mathcal{D})$ and $f \in \mathrm{Hom}_{\mathcal{D}}(X, Y)$. We define the **constant natural transformation on $\mathcal{C}$ at f**, denoted $\delta_f^{\mathcal{C}} : \Delta_X^{\mathcal{C}} \Rightarrow \Delta_Y^{\mathcal{C}}$, by

$$(\delta_f^{\mathcal{C}})_A = f$$

Exercise 2.3.3.2. Let $\mathcal{C}$, $\mathcal{D}$ be categories, $X, Y \in \mathrm{Obj}(\mathcal{D})$ and $f \in \mathrm{Hom}_{\mathcal{D}}(X, Y)$. Then $\delta_f^{\mathcal{C}} : \Delta_X^{\mathcal{C}} \Rightarrow \Delta_Y^{\mathcal{C}}$ is a natural transformation.

Proof.

1. By definition, for each $A \in \mathrm{Obj}(\mathcal{C})$ $(\delta_f^{\mathcal{C}})_A \in \mathrm{Hom}_{\mathcal{D}}(\Delta_X^{\mathcal{C}}(A), \Delta_Y^{\mathcal{C}}(A))$.
2. Let $A, B \in \mathrm{Obj}(\mathcal{C})$ and $g \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Then

$$\begin{aligned}
\Delta_Y^{\mathcal{C}}(g) \circ (\delta_f^{\mathcal{C}})_A &= \mathrm{id}_Y \circ f \\
&= f \\
&= f \circ \mathrm{id}_X \\
&= (\delta_f^{\mathcal{C}})_B \circ \Delta_X^{\mathcal{C}}(g)
\end{aligned}$$

i.e. the following diagram commutes:

$$\begin{tikzcd}
\Delta_X^{\mathcal{C}}(A) \arrow[r, "(\delta_f^{\mathcal{C}})_A"] \arrow[d, "\Delta_X^{\mathcal{C}}(g)"'] & \Delta_Y^{\mathcal{C}}(A) \arrow[d, "\Delta_Y^{\mathcal{C}}(g)"] \\
\Delta_X^{\mathcal{C}}(B) \arrow[r, "(\delta_f^{\mathcal{C}})_B"] & \Delta_Y^{\mathcal{C}}(B)
\end{tikzcd}
=
\begin{tikzcd}
X \arrow[r, "f"] \arrow[d, "\mathrm{id}_X"'] & Y \arrow[d, "\mathrm{id}_Y"] \\
X \arrow[r, "f"] & Y
\end{tikzcd}$$

So $\delta_f^{\mathcal{C}} : \Delta_X^{\mathcal{C}} \Rightarrow \Delta_Y^{\mathcal{C}}$ is a natural transformation. □

Exercise 2.3.3.3. Let $\mathcal{C}, \mathcal{D}$ be categories, $X, Y, Z \in \mathrm{Obj}(\mathcal{D})$, $f \in \mathrm{Hom}_{\mathcal{D}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathcal{D}}(Y, Z)$. Then $\delta_{g \circ f}^{\mathcal{C}} = \delta_g^{\mathcal{C}} \circ \delta_f^{\mathcal{C}}$.

Proof. Let $A \in \mathrm{Obj}(\mathcal{C})$. Then

$$\begin{aligned}
(\delta_{g \circ f}^{\mathcal{C}})_A &= g \circ f \\
&= (\delta_g^{\mathcal{C}})_A \circ (\delta_f^{\mathcal{C}})_A \\
&= (\delta_g^{\mathcal{C}} \circ \delta_f^{\mathcal{C}})_A
\end{aligned}$$

Since $A \in \mathrm{Obj}(\mathcal{C})$ is arbitrary, $\delta_{g \circ f}^{\mathcal{C}} = \delta_g^{\mathcal{C}} \circ \delta_f^{\mathcal{C}}$. □

Exercise 2.3.3.4. Let $\mathcal{C}, \mathcal{D}$ be categories and $X \in \mathrm{Obj}(\mathcal{D})$. Then $\delta^{\mathcal{C}}_{\mathrm{id}_X} = \mathrm{id}_{\Delta^{\mathcal{C}}_X}$.

Proof. Let $A \in \mathrm{Obj}(\mathcal{C})$. Then

$$\begin{aligned}(\delta^{\mathcal{C}}_{\mathrm{id}_X})_A &= \mathrm{id}_X \\ &= \mathrm{id}_{\Delta^{\mathcal{C}}_X(A)} \\ &= (\mathrm{id}_{\Delta^{\mathcal{C}}_X})_A\end{aligned}$$

Since $A \in \mathrm{Obj}(\mathcal{C})$ is arbitrary, $\delta^{\mathcal{C}}_{\mathrm{id}_X} = \mathrm{id}_{\Delta^{\mathcal{C}}_X}$ □

Definition 2.3.3.5. Let $\mathcal{C}$, $\mathcal{D}$ be categories. Suppose that $\mathcal{C}$ is small. We define the **$\mathcal{C}$-ary diagonal functor** on $\mathcal{D}$, denoted by $\Delta^{\mathcal{C}} : \mathcal{D} \to \mathcal{D}^{\mathcal{C}}$, by

- $\Delta^{\mathcal{C}}(X) = \Delta^{\mathcal{C}}_X$
- $\Delta^{\mathcal{C}}(f) = \delta^{\mathcal{C}}_f$

Exercise 2.3.3.6. Let $\mathcal{C}$, $\mathcal{D}$ be categories. Suppose that $\mathcal{C}$ is small. Then $\Delta^{\mathcal{C}} : \mathcal{D} \to \mathcal{D}^{\mathcal{C}}$ is a functor.

Proof.

1. Exercise 2.3.3.2 implies that for each $X, Y \in \mathrm{Obj}(\mathcal{D})$ and $f \in \mathrm{Hom}_{\mathcal{D}}(X, Y)$, $\Delta^{\mathcal{C}}(f) \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(\Delta^{\mathcal{C}}(X), \Delta^{\mathcal{C}}(Y))$
2. Exercise 2.3.3.3 implies that for each $X, Y, Z \in \mathrm{Obj}(\mathcal{D})$, $f \in \mathrm{Hom}_{\mathcal{D}}(X, Y)$ and $g \in \mathrm{Hom}_{\mathcal{D}}(Y, Z)$, $\Delta^{\mathcal{C}}(g \circ f) = \Delta^{\mathcal{C}}(g) \circ \Delta^{\mathcal{C}}(f)$
3. Exercise 2.3.3.4 implies that for each $X \in \mathrm{Obj}(\mathcal{D})$, $\Delta^{\mathcal{C}}(\mathrm{id}_X) = \mathrm{id}_{\Delta^{\mathcal{C}}(X)}$

So $\Delta^{\mathcal{C}} : \mathcal{D} \to \mathcal{D}^{\mathcal{C}}$ is a functor. □

2.4 Isomorphisms

2.4.1 Introduction

Definition 2.4.1.1. Let $\mathcal{C}$ be a category.

- Let $A \in \mathrm{Obj}(\mathcal{C})$
  - Let $f \in \mathrm{Hom}_{\mathcal{C}}(A, A)$. Then f is said to be an **endomorphism**,
  - We define $\mathrm{End}_{\mathcal{C}}(A) := \mathrm{Hom}_{\mathcal{C}}(A, A)$.
- We define $\mathrm{End}_{\mathcal{C}} := \bigcup_{A \in \mathrm{Obj}(\mathcal{C})} \mathrm{End}_{\mathcal{C}}(A)$.

Definition 2.4.1.2. Let $\mathcal{C}$ be a category

- Let $A, B \in \mathrm{Obj}(\mathcal{C})$.
  - Let $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Then f is said to be an **isomorphism** if there exists $g \in \mathrm{Hom}_{\mathcal{C}}(B, A)$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.
  - We define $\mathrm{Iso}_{\mathcal{C}}(A, B) := \{f \in \mathrm{Hom}_{\mathcal{C}}(A, B) : f \text{ is an isomorphism}\}$
- We define $\mathrm{Iso}_{\mathcal{C}} := \bigcup_{(A,B) \in \mathrm{Obj}(\mathcal{C})^2} \mathrm{Iso}_{\mathcal{C}}(A, B)$.

Definition 2.4.1.3. Let $\mathcal{C}$ be a category.

- Let $A \in \mathrm{Obj}(\mathcal{C})$
  - Let $f \in \mathrm{Iso}_{\mathcal{C}}(A, A)$. Then f is said to be an **automorphism**,
  - We define $\mathrm{Aut}_{\mathcal{C}}(A) := \mathrm{Iso}_{\mathcal{C}}(A, A)$.
- We define $\mathrm{Aut}_{\mathcal{C}} := \bigcup_{A \in \mathrm{Obj}(\mathcal{C})} \mathrm{Aut}_{\mathcal{C}}(A)$.

Exercise 2.4.1.4. Uniqueness of Inverses:
Let $\mathcal{C}$ be a category, $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Iso}_{\mathcal{C}}(A, B)$. Then there exists a unique $g \in \mathrm{Iso}_{\mathcal{C}}(B, A)$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.

Proof.

- **Existence:**
  By definition, since f is an isomorphism, there exists $g \in \mathrm{Hom}_{\mathcal{C}}(B, A)$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$. By definition, $g \in \mathrm{Iso}_{\mathcal{C}}(B, A)$.
- **Uniqueness:**
  Let $g' \in \mathrm{Iso}_{\mathcal{C}}(B, A)$. Suppose that $g' \circ f = \mathrm{id}_A$, $f \circ g' = \mathrm{id}_B$. Then

$$\begin{aligned} g' &= g' \circ \mathrm{id}_B \\ &= g' \circ (f \circ g) \\ &= (g' \circ f) \circ g \\ &= \mathrm{id}_A \circ g \\ &= g \end{aligned}$$

□

Definition 2.4.1.5. Let $\mathcal{C}$ be a category, $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Suppose that f is an isomorphism. We define the **inverse of** f, denoted $f^{-1} \in \mathrm{Iso}_{\mathcal{C}}(B, A)$, to be the unique $g \in \mathrm{Iso}_{\mathcal{C}}(B, A)$ such that $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$.

Exercise 2.4.1.6. Let $\mathcal{C}$ be a category and $A \in \mathrm{Obj}(\mathcal{C})$. Then $\mathrm{id}_A \in \mathrm{Aut}_{\mathcal{C}}(A)$ and $(\mathrm{id}_A)^{-1} = \mathrm{id}_A$.

Proof. Since $\mathrm{id}_A \circ \mathrm{id}_A = \mathrm{id}_A$, we have that $\mathrm{id}_A \in \mathrm{Aut}_{\mathcal{C}}(A)$ and $(\mathrm{id}_A)^{-1} = \mathrm{id}_A$. □

Exercise 2.4.1.7. Let $\mathcal{C}$ be a category and $A, B \in \mathrm{Obj}(\mathcal{C})$. Then for each $f \in \mathrm{Iso}_{\mathcal{C}}(A, B)$, $(f^{-1})^{-1} = f$.

Proof. Let $f \in \mathrm{Iso}_{\mathcal{C}}(A, B)$. By definition, $f \circ f^{-1} = \mathrm{id}_B$ and $f^{-1} \circ f = \mathrm{id}_A$. Hence $(f^{-1})^{-1} = f$. □

Exercise 2.4.1.8. Let $\mathcal{C}$ be a category and $A, B, C \in \mathrm{Obj}(\mathcal{C})$. Then for each $f \in \mathrm{Iso}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Iso}_{\mathcal{C}}(B, C)$, $g \circ f \in \mathrm{Iso}_{\mathcal{C}}(A, C)$ and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$.

Proof. Let $f \in \mathrm{Iso}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Iso}_{\mathcal{C}}(B, C)$. Then

$$\begin{aligned}
(f^{-1} \circ g^{-1}) \circ (g \circ f) &= ((f^{-1} \circ g^{-1}) \circ g) \circ f \\
&= (f^{-1} \circ (g^{-1} \circ g)) \circ f \\
&= (f^{-1} \circ \mathrm{id}_B) \circ f \\
&= f^{-1} \circ f \\
&= \mathrm{id}_A
\end{aligned}$$

and

$$\begin{aligned}
(g \circ f) \circ (f^{-1} \circ g^{-1}) &= ((g \circ f) \circ f^{-1}) \circ g^{-1} \\
&= (g \circ (f \circ f^{-1})) \circ g^{-1} \\
&= (g \circ \mathrm{id}_B) \circ g^{-1} \\
&= g \circ g^{-1} \\
&= \mathrm{id}_C\,.
\end{aligned}$$

Hence $g \circ f \in \mathrm{Iso}_{\mathcal{C}}(A, C)$ and $(g \circ f)^{-1} = f^{-1} \circ g^{-1}$. □

Definition 2.4.1.9. Let $\mathcal{C}$ be a category.

- Let $A, B \in \mathrm{Obj}(\mathcal{C})$. Then A is said to be **isomorphic** to B if $\mathrm{Iso}_{\mathcal{C}}(A, B) \neq \varnothing$.
- We define the **isomorphism class relation on** $\mathrm{Obj}(\mathcal{C})$, denoted $\cong_{\mathcal{C}} \in \mathcal{R}(A, B)$, by $A \cong_{\mathcal{C}} B$ iff A is isomorphic to B.

Note 2.4.1.10. When the context is clear, we may write $\cong$ in place of $\cong_{\mathcal{C}}$.

Exercise 2.4.1.11. Let $\mathcal{C}$ be a category. Then $\cong \in \mathrm{Equiv}(\mathrm{Obj}(C))$ does $\mathcal{C}$ neet to be small or just locally small?.

Proof.

1. **reflexivity:**
   Let $A \in \mathrm{Obj}(\mathcal{C})$. Exercise 2.4.1.6 implies that $\mathrm{id}_A \in \mathrm{Aut}(A)$. Thus $A \cong A$. Since $A \in \mathrm{Obj}(\mathcal{C})$ is arbitrary, we have that for each $A \in \mathrm{Obj}(\mathcal{C})$, $A \cong A$ and thus $\cong$ is reflexive.

2. **symmetry:**
   Let $A, B \in \mathrm{Obj}(\mathcal{C})$. Suppose that $A \cong B$. Then there exists $f \in \mathrm{Iso}_{\mathcal{C}}(A, B)$. Exercise 2.4.1.7 implies that $f^{-1} \in \mathrm{Iso}_{\mathcal{C}}(B, A)$. Hence $B \cong A$. Since $A, B \in \mathrm{Obj}(\mathcal{C})$ with $A \cong B$ are arbitrary, we have that for each $A, B \in \mathrm{Obj}(\mathcal{C})$, $A \cong B$ implies that $B \cong A$ and thus $\cong$ is reflexive.

3. **transitivity:** Let $A, B, C \in \mathrm{Obj}(\mathcal{C})$. Suppose that $A \cong B$ and $B \cong C$. Then there exist $f \in \mathrm{Iso}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Iso}_{\mathcal{C}}(B, C)$. Exercise 2.4.1.8 implies that $g \circ f \in \mathrm{Iso}_{\mathcal{C}}(A, C)$. Thus $A \cong C$. Since $A, B, C \in \mathrm{Obj}(\mathcal{C})$ with $A \cong B$ and $B \cong C$ are arbitrary, we have that for each $A, B, C \in \mathrm{Obj}(\mathcal{C})$, $A \cong B$ and $B \cong C$ implies that $A \cong C$ and thus $\cong$ is transitive.

Thus $\cong \in \mathrm{Equiv}(\mathrm{Obj}(C))$. □

2.4.2 Isomorphisms between Categories

Exercise 2.4.2.1. Let $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathbf{Cat})$ and $F \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D})$. Write $F = (F_0, F_1)$. Then $F \in \mathrm{Iso}_{\mathbf{Cat}}$ iff $F_0, F_1 \in \mathrm{Iso}_{\mathbf{Set}}$

Proof.

- ($\Longrightarrow$):
  Suppose that $F \in \mathrm{Iso}_{\mathbf{Cat}}$. Then there exists $G \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{D}, \mathcal{C})$ such that $G \circ_{\mathbf{Cat}} F = \mathrm{id}_{\mathcal{C}}$ and $F \circ_{\mathbf{Cat}} G = \mathrm{id}_{\mathcal{D}}$. Write $G = (G_0, G_1)$. By Definition 2.2.3.1, we have that $G_0 \circ_{\mathbf{Set}} F_0 = \mathrm{id}^{\mathbf{Set}}_{\mathrm{Obj}(\mathcal{C})}$, $F_0 \circ_{\mathbf{Set}} G_0 = \mathrm{id}^{\mathbf{Set}}_{\mathrm{Obj}(\mathcal{D})}$, $G_1 \circ_{\mathbf{Set}} F_1 = \mathrm{id}^{\mathbf{Set}}_{\mathrm{Hom}_{\mathcal{C}}}$ and $F_1 \circ_{\mathbf{Set}} G_1 = \mathrm{id}^{\mathbf{Set}}_{\mathrm{Hom}_{\mathcal{D}}}$. Hence $F_0, G_0 \in \mathrm{Iso}_{\mathbf{Set}}$.

- ($\Longleftarrow$):
  Suppose that $F_0, F_1 \in \mathrm{Iso}_{\mathbf{Set}}$. Define $G_0 := F_0^{-1}$ and $G_1 := F_1^{-1}$.

  1. Let $A', B' \in \mathrm{Obj}(\mathcal{D})$ and $f' \in \mathrm{Hom}_{\mathcal{D}}(A', B')$. Define $f \in \mathrm{Hom}_{\mathcal{C}}$ and $A, B \in \mathrm{Obj}(\mathcal{C})$ by $f := G_1(f')$, $A := \mathrm{dom}(f)$ and $B := \mathrm{cod}(f)$. Since $F \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D})$, we have that
  $$\begin{aligned} f' &= F_1(f) \\ &\in \mathrm{Hom}_{\mathcal{D}}(F_0(A), F_0(B)). \end{aligned}$$
  Since $f' \in \mathrm{Hom}_{\mathcal{D}}(A', B')$, we have that
  $$\begin{aligned} A' &= \mathrm{dom}(f') \\ &= F_0(A). \end{aligned}$$
  Similarly, $B' = F_0(B)$. Hence $G_0(A') = A$, $G_0(B') = B$ and
  $$\begin{aligned} G_1(f') &= G_1(F_1(f)) \\ &= f \\ &\in \mathrm{Hom}_{\mathcal{C}}(A, B) \\ &= \mathrm{Hom}_{\mathcal{C}}(A, B) \\ &= \mathrm{Hom}_{\mathcal{C}}(G_0(A'), G_0(B')). \end{aligned}$$

  2. Let $A', B', C' \in \mathrm{Obj}(\mathcal{D})$, $f' \in \mathrm{Hom}_{\mathcal{D}}(A', B')$ and $g' \in \mathrm{Hom}_{\mathcal{D}}(B', C')$. Define $f, g \in \mathrm{Hom}_{\mathcal{C}}$ by $f := G_1(f')$ and $g := G_1(g')$. Then $F_1(f) = f'$ and $F_1(g) = g'$. Since $F \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D})$, we have that
  $$\begin{aligned} G_1(g' \circ_{\mathcal{D}} f') &= G_1(F_1(g) \circ_{\mathcal{D}} F_1(f)) \\ &= G_1(F_1(g \circ_{\mathcal{C}} f)) \\ &= g \circ_{\mathcal{C}} f \\ &= G_1(g') \circ_{\mathcal{C}} G_1(f'). \end{aligned}$$

  3. Let $A' \in \mathrm{Obj}(\mathcal{D})$. Define $A \in \mathrm{Obj}(\mathcal{C})$ by $A := G_0(A')$. Then $F_0(A) = A'$ and since $F \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D})$, we have that
  $$\begin{aligned} G_1(\mathrm{id}^{\mathcal{D}}_{A'}) &= G_1(\mathrm{id}^{\mathcal{D}}_{F_0(A)}) \\ &= G_1(F_1(\mathrm{id}^{\mathcal{C}}_{A})) \\ &= \mathrm{id}^{\mathcal{C}}_{A} \\ &= \mathrm{id}^{\mathcal{C}}_{G_0(A')}. \end{aligned}$$

  Hence $G \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{D}, \mathcal{C})$. By construction, $G \circ_{\mathbf{Cat}} F = \mathrm{id}_{\mathcal{C}}$ and $F \circ_{\mathbf{Cat}} G = \mathrm{id}_{\mathcal{D}}$. Thus $F \in \mathrm{Iso}_{\mathbf{Cat}}$.

$\square$

2.4.3 Natural Isomorphisms

Definition 2.4.3.1. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, $F, G : \mathcal{C} \to \mathcal{D}$ and $\alpha : F \Rightarrow G$. Then α is said to be a **natural isomorphism** if for each $A \in \mathrm{Obj}(\mathcal{C})$, $\alpha_A \in \mathrm{Iso}_{\mathcal{D}}(F(A), G(A))$.

Definition 2.4.3.2. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, $F, G : \mathcal{C} \to \mathcal{D}$ and $\alpha : F \Rightarrow G$. Suppose that α is a natural isomorphism. We define $\alpha^{-1} : G \Rightarrow F$ by $(\alpha^{-1})_A = \alpha_A^{-1}$.

Exercise 2.4.3.3. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, $F, G : \mathcal{C} \to \mathcal{D}$ and $\alpha : F \Rightarrow G$. Suppose that α is a natural isomorphism. Then $\alpha^{-1} : G \Rightarrow F$ is a natural transformation

Proof.

1. Let $A \in \mathrm{Obj}(\mathcal{C})$. Since $\alpha_A \in \mathrm{Hom}_{\mathcal{D}}(F(A), G(A))$, we have that

$$\begin{aligned}(\alpha^{-1})_A &= \alpha_A^{-1} \\ &\in \mathrm{Hom}_{\mathcal{D}}(G(A), F(A))\end{aligned}$$

2. Let $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Since $G(f) \circ \alpha_A = \alpha_B \circ F(f)$, i.e. the following diagram commutes:

$$\begin{tikzcd} F(A) \arrow[r, "\alpha_A"] \arrow[d, "F(f)"'] & G(A) \arrow[d, "G(f)"] \\ F(B) \arrow[r, "\alpha_B"] & G(B) \end{tikzcd}$$

we have that

$$\begin{aligned}F(f) \circ (\alpha^{-1})_A &= F(f) \circ \alpha_A^{-1} \\ &= \mathrm{id}_{F(B)} \circ (F(f) \circ \alpha_A^{-1}) \\ &= (\alpha_B^{-1} \circ \alpha_B) \circ (F(f) \circ \alpha_A^{-1}) \\ &= \alpha_B^{-1} \circ (\alpha_B \circ (F(f) \circ \alpha_A^{-1})) \\ &= \alpha_B^{-1} \circ ((\alpha_B \circ F(f)) \circ \alpha_A^{-1}) \\ &= \alpha_B^{-1} \circ ((G(f) \circ \alpha_A) \circ \alpha_A^{-1}) \\ &= \alpha_B^{-1} \circ (G(f) \circ (\alpha_A \circ \alpha_A^{-1})) \\ &= \alpha_B^{-1} \circ (G(f) \circ \mathrm{id}_{G(A)}) \\ &= \alpha_B^{-1} \circ G(f) \\ &= (\alpha^{-1})_B \circ G(f)\end{aligned}$$

i.e. the following diagram commutes:

$$\begin{tikzcd} G(A) \arrow[r, "(\alpha^{-1})_A"] \arrow[d, "G(f)"'] & F(A) \arrow[d, "F(f)"] \\ G(B) \arrow[r, "(\alpha^{-1})_B"] & F(B) \end{tikzcd}$$

So $\alpha^{-1} : G \Rightarrow F$.

□

Exercise 2.4.3.4. Let $\mathcal{C}$ and $\mathcal{D}$ be categories, $F, G : \mathcal{C} \to \mathcal{D}$ and $\alpha : F \Rightarrow G$. Suppose that α is a natural isomorphism. Then $\alpha^{-1} \circ \alpha = \mathrm{id}_F$ and $\alpha \circ \alpha^{-1} = \mathrm{id}_G$.

Proof. Let $A \in \mathrm{Obj}(\mathcal{C})$. Then

$$\begin{aligned}(\alpha^{-1} \circ \alpha)_A &= (\alpha^{-1})_A \circ \alpha_A \\ &= \alpha_A^{-1} \circ \alpha_A \\ &= \mathrm{id}_{F(A)} \\ &= (\mathrm{id}_F)_A\end{aligned}$$

and

$$\begin{aligned}(\alpha \circ \alpha^{-1})_A &= \alpha_A \circ (\alpha^{-1})_A \\ &= \alpha_A \circ \alpha_A^{-1} \\ &= \mathrm{id}_{G(A)} \\ &= (\mathrm{id}_G)_A\end{aligned}$$

Since $A \in \mathrm{Obj}(\mathcal{C})$ is arbitrary, $\alpha^{-1} \circ \alpha = \mathrm{id}_F$ and $\alpha \circ \alpha^{-1} = \mathrm{id}_G$. □

Exercise 2.4.3.5. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Suppose that $\mathcal{C}$ is small. Let $F, G \in \mathrm{Obj}(\mathcal{D}^{\mathcal{C}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(F, G)$. Then α is a natural isomorphism iff $\alpha \in \mathrm{Iso}_{\mathcal{D}^{\mathcal{C}}}(F, G)$.

Proof.

- ($\Longrightarrow$):
  Suppose that α is a natural isomorphism. Exercise 2.4.3.4 implies that $\alpha \in \mathrm{Iso}_{\mathcal{D}^{\mathcal{C}}}(F, G)$.
- ($\Longleftarrow$):
  Suppose that $\alpha \in \mathrm{Iso}_{\mathcal{D}^{\mathcal{C}}}(F, G)$. Let $A \in \mathrm{Obj}(\mathcal{C})$. Then

  $$\begin{aligned}\alpha_A \circ (\alpha^{-1})_A &= (\alpha \circ \alpha^{-1})_A \\ &= (\mathrm{id}_G)_A \\ &= \mathrm{id}_{G(A)}\end{aligned}$$

  and similarly, $\alpha_A^{-1} \circ \alpha_A = \mathrm{id}_{F(A)}$. Thus $\alpha_A \in \mathrm{Iso}_{\mathcal{D}}(F(A), G(A))$. Since $A \in \mathrm{Obj}(\mathcal{C})$ is arbitrary, we have that for each $A \in \mathrm{Obj}(\mathcal{C})$, $\alpha_A \in \mathrm{Iso}_{\mathcal{D}}(F(A), G(A))$. By definition, α is a natural isomorphism.

□

Chapter 3

Fundamental Structures

3.1 Initial and Terminal Objects

Definition 3.1.0.1. Let $\mathcal{C}$ be a category and $0 \in \mathrm{Obj}(\mathcal{C})$ and $1 \in \mathrm{Obj}(\mathcal{C})$. Then

- 0 is said to be **$\mathcal{C}$-initial** if for each $A \in \mathrm{Obj}(\mathcal{C})$, there exists $f \in \mathrm{Hom}_{\mathcal{C}}(0, A)$ such that $\mathrm{Hom}_{\mathcal{C}}(0, A) = \{f\}$.
- 1 is said to be **$\mathcal{C}$-terminal** if for each $A \in \mathrm{Obj}(\mathcal{C})$, there exists $f \in \mathrm{Hom}_{\mathcal{C}}(A, 1)$ such that $\mathrm{Hom}_{\mathcal{C}}(A, 1) = \{f\}$. redo, use duality

Note 3.1.0.2. When the context is clear, we say "initial" and "terminal" in place of "$\mathcal{C}$-initial" and "$\mathcal{C}$-terminal".

Exercise 3.1.0.3. Let $\mathcal{C}$ be a category and $0 \in \mathrm{Obj}(\mathcal{C})$ and $1 \in \mathrm{Obj}(\mathcal{C})$. Then

1. 0 is $\mathcal{C}$-initial iff 0 is $\mathcal{C}^{\mathrm{op}}$-terminal
2. 1 is $\mathcal{C}$-terminal iff 1 is $\mathcal{C}^{\mathrm{op}}$-initial

Proof.

1.
   - ($\Longrightarrow$):
     Suppose that 0 is $\mathcal{C}$-initial. Let $A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$. Since $\mathrm{Obj}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Obj}(\mathcal{C})$, we have that $A \in \mathrm{Obj}(\mathcal{C})$. Since 0 is $\mathcal{C}$-initial, there exists $f \in \mathrm{Hom}_{\mathcal{C}}(0, A)$ such that $\mathrm{Hom}_{\mathcal{C}}(0, A) = \{f\}$. Then
     $$\begin{aligned}\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, 0) &= \mathrm{Hom}_{\mathcal{C}}(0, A) \\ &= \{f\}.\end{aligned}$$
     Since A is arbitrary, we have that for each $A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$, there exists $f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, 0)$ such that $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, 0) = \{f\}$. Thus 0 is $\mathcal{C}^{\mathrm{op}}$-terminal.
   - ($\Longleftarrow$):
     Suppose that 0 is $\mathcal{C}^{\mathrm{op}}$-terminal. Let $A \in \mathrm{Obj}(\mathcal{C})$. Since $\mathrm{Obj}(\mathcal{C}) = \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$, we have that $A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$. Since 0 is $\mathcal{C}^{\mathrm{op}}$-terminal, there exists $f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, 0)$ such that $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(0, A) = \{f\}$. Then
     $$\mathrm{Hom}_{\mathcal{C}}(0, A) = \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, 0) \qquad = \{f\}.$$
     Since A is arbitrary, we have that for each $A \in \mathrm{Obj}(\mathcal{C})$, there exists $f \in \mathrm{Hom}_{\mathcal{C}}(0, A)$ such that $\mathrm{Hom}_{\mathcal{C}}(0, A) = \{f\}$. Thus 0 is $\mathcal{C}$-initial.
2. Set $\mathcal{D} := \mathcal{C}^{\mathrm{op}}$. Exercise 2.1.3.3 implies that $\mathcal{D}^{\mathrm{op}} = \mathcal{C}$. Then
   $$\begin{aligned}1 \text{ is } \mathcal{C}\text{-terminal} &\iff 1 \text{ is } \mathcal{C}\text{-terminal} \\ &\iff 1 \text{ is } \mathcal{D}^{\mathrm{op}}\text{-terminal} \\ &\iff 1 \text{ is } \mathcal{D}\text{-initial}, \quad \text{(by (reference) prev part)} \\ &\iff 1 \text{ is } \mathcal{C}^{\mathrm{op}}\text{-initial}.\end{aligned}$$

□

Exercise 3.1.0.4. Let $\mathcal{C}$ be a category and $0 \in \mathrm{Obj}(\mathcal{C})$.

1. If 0 is initial, then $\mathrm{Hom}_{\mathcal{C}}(0,0) = \{\mathrm{id}_0\}$.
2. If 1 is terminal, then $\mathrm{Hom}_{\mathcal{C}}(1,1) = \{\mathrm{id}_1\}$.

Proof.

1. Suppose that 0 is initial. Then there exists a $f \in \mathrm{Hom}_{\mathcal{C}}(0,0)$ such that $\mathrm{Hom}_{\mathcal{C}}(0,0) = \{f\}$. Since $\mathrm{id}_1 \in \mathrm{Hom}_{\mathcal{C}}(0,0)$, $f = \mathrm{id}_1$ and therefore $\mathrm{Hom}_{\mathcal{C}}(0,0) = \{\mathrm{id}_1\}$.
2. use duality.

□

Exercise 3.1.0.5. Let $\mathcal{C}$ be a category and $0, 0', 1, 1' \in \mathrm{Obj}(\mathcal{C})$.

1. If 0 and $0'$ are initial, then
   (a) $0 \cong_{\mathcal{C}} 0'$,
   (b) $\mathrm{Hom}_{\mathcal{C}}(0,0') = \mathrm{Iso}_{\mathcal{C}}(0,0')$.
2. If 1 and $1'$ are terminal, then
   (a) $1 \cong_{\mathcal{C}} 1'$,
   (b) $\mathrm{Hom}_{\mathcal{C}}(1,1') = \mathrm{Iso}_{\mathcal{C}}(1,1')$.

Proof.

1. Suppose that 0 and $0'$ are initial.
   (a) By definition, there exist $f \in \mathrm{Hom}_{\mathcal{C}}(0,0')$ and $f' \in \mathrm{Hom}_{\mathcal{C}}(0',0)$ such that $\mathrm{Hom}_{\mathcal{C}}(0,0') = \{f\}$ and $\mathrm{Hom}_{\mathcal{C}}(0',0) = \{f'\}$, i.e. we have the following commutative diagram:

$$\begin{tikzcd} 0 \arrow[loop left, "f' \circ f"] \arrow[r, bend right, "f"'] & 0' \arrow[l, bend right, "f'"'] \arrow[loop right, "f \circ f'"] \end{tikzcd}$$

   Exercise 3.1.0.4 implies that $f' \circ f = \mathrm{id}_0$ and $f \circ f' = \mathrm{id}_{0'}$. Hence $f \in \mathrm{Iso}_{\mathcal{C}}(0,0')$. Since $\mathrm{Iso}_{\mathcal{C}}(0,0') \neq \varnothing$, we have that $0 \cong_{\mathcal{C}} 0'$.
   (b) Since

$$\begin{aligned} \mathrm{Iso}_{\mathcal{C}}(0,0') &\subset \mathrm{Hom}_{\mathcal{C}}(0,0') \\ &= \{f\} \\ &\subset \mathrm{Iso}_{\mathcal{C}}(0,0'), \end{aligned}$$

   we have that $\mathrm{Hom}_{\mathcal{C}}(0,0') = \mathrm{Iso}_{\mathcal{C}}(0,0')$.
2. use duality

□

Exercise 3.1.0.6. Let $X, Y \in \mathrm{Obj}(\mathbf{Set})$. Then

1. X is initial iff $X = \varnothing$.
2. Y is terminal iff $Y \cong_{\mathbf{Set}} \{\varnothing\}$.

Proof.

1. • ($\Longrightarrow$):
Suppose that X is initial. Since $\mathrm{Hom}_{\mathbf{Set}}(X, \{0,1\}) \cong_{\mathbf{Set}} \mathcal{P}(X)$, we have that

$$\begin{aligned} \#\,\mathrm{Hom}_{\mathbf{Set}}(X, \{0,1\}) = 1 &\iff \#\mathcal{P}(X) = 1 \\ &\iff \mathcal{P}(X) = \{\varnothing\} \\ &\iff X = \varnothing. \end{aligned}$$

Since X is initial, there exists $f \in \mathrm{Hom}_{\mathbf{Set}}(X, \{0,1\})$ such that $\mathrm{Hom}_{\mathbf{Set}}(X, \{0,1\}) = \{f\}$. Thus $X = \varnothing$.

• ($\Longleftarrow$):
Suppose that $X = \varnothing$. Let $A \in \mathrm{Obj}(\mathbf{Set})$.

– We note that for each $f \in \mathrm{Hom}_{\mathbf{Set}}$,

$$\begin{aligned} f \in \mathrm{Hom}_{\mathbf{Set}}(X, A) &\iff f \in \mathrm{Hom}_{\mathbf{Set}}(\varnothing, A) \\ &\Longrightarrow f \subset \varnothing \times A \\ &= \iff f \subset \varnothing \\ &\iff f = \varnothing. \end{aligned}$$

Thus for each $f \in \mathrm{Hom}_{\mathbf{Set}}(X, A)$, $f \in \{\varnothing\}$. Hence $\mathrm{Hom}_{\mathbf{Set}}(X, A) \subset \{\varnothing\}$.

– Since

$$\begin{aligned} \varnothing &\in \mathrm{Hom}_{\mathbf{Set}}(\varnothing, A) \\ &= \mathrm{Hom}_{\mathbf{Set}}(X, A), \end{aligned}$$

we have that $\{\varnothing\} \subset \mathrm{Hom}_{\mathbf{Set}}(X, A)$.

Thus $\mathrm{Hom}_{\mathbf{Set}}(X, A) = \{\varnothing\}$. Therefore X is initial.

2. • ($\Longrightarrow$):
Suppose that Y is terminal. Since $\#\,\mathrm{Hom}_{\mathbf{Set}}(\{\varnothing\}, Y) = \#Y$, we have that

$$\begin{aligned} \#\,\mathrm{Hom}_{\mathbf{Set}}(\{\varnothing\}, Y) = 1 &\iff \#Y = 1 \\ &\iff Y \cong_{\mathbf{Set}} \{\varnothing\}. \end{aligned}$$

Since Y is terminal, there exists $f \in \mathrm{Hom}_{\mathbf{Set}}(\{\varnothing\}, Y)$ such that $\mathrm{Hom}_{\mathbf{Set}}(\{\varnothing\}, Y) = \{f\}$. Thus $Y \cong_{\mathbf{Set}} \{\varnothing\}$.

• ($\Longleftarrow$):
Suppose that $Y \cong_{\mathbf{Set}} \{\varnothing\}$. Then there exists $y \in Y$ such that $Y = \{y\}$. Let $A \in \mathrm{Obj}(\mathbf{Set})$. Define $f_0 \in \mathrm{Hom}_{\mathbf{Set}}(A, Y)$ by $f_0(a) := y$. Let $f \in \mathrm{Hom}_{\mathbf{Set}}(A, Y)$. Let $a \in A$. Since

$$\begin{aligned} f(a) &\in Y \\ &= \{y\}, \end{aligned}$$

we have that $f(a) = y$. Since $a \in A$ is arbitrary, we have that for each $a \in A$,

$$\begin{aligned} f(a) &= y \\ &= f_0(a). \end{aligned}$$

Thus $f = f_0$. Since $f \in \mathrm{Hom}_{\mathbf{Set}}(A, Y)$ is arbitrary, we have that for each $f \in \mathrm{Hom}_{\mathbf{Set}}(A, Y)$, $f = f_0$. Hence $\mathrm{Hom}_{\mathbf{Set}}(A, Y) = \{f_0\}$. Since $A \in \mathrm{Obj}(\mathbf{Set})$ is arbitrary, we have that for each $A \in \mathrm{Obj}(\mathbf{Set})$, there exists $f_0 \in \mathrm{Hom}_{\mathbf{Set}}(A, Y)$ such that $\mathrm{Hom}_{\mathbf{Set}}(A, Y) = \{f_0\}$. Thus Y is terminal.

□

Exercise 3.1.0.7. Let $\mathcal{C}$ be a category. Suppose that $\mathcal{C}$ is locally small. Let $X \in \mathrm{Obj}(\mathcal{C})$. Then

1. X is $\mathcal{C}$-initial iff for each $A \in \mathrm{Obj}(\mathcal{C})$, $\mathrm{Hom}_{\mathcal{C}}(X, A)$ is **Set**-terminal.
2. X is $\mathcal{C}$-terminal iff for each $A \in \mathrm{Obj}(\mathcal{C})$, $\mathrm{Hom}_{\mathcal{C}}(A, X)$ is **Set**-terminal.

Proof.

1. We have that

$$\begin{aligned} X \text{ is } \mathcal{C}\text{-initial} &\iff \text{for each } A \in \mathrm{Obj}(\mathcal{C}),\ \#\,\mathrm{Hom}_{\mathcal{C}}(X, A) = 1 \\ &\iff \text{for each } A \in \mathrm{Obj}(\mathcal{C}),\ \#\,\mathrm{Hom}_{\mathcal{C}}(X, A) \cong_{\mathbf{Set}} \{\varnothing\} \\ &\iff \text{for each } A \in \mathrm{Obj}(\mathcal{C}),\ \#\,\mathrm{Hom}_{\mathcal{C}}(X, A) \text{ is } \mathbf{Set}\text{-terminal}, \quad \text{(by Exercise 3.1.0.6).} \end{aligned}$$

2. We have that

$$\begin{aligned} X \text{ is } \mathcal{C}\text{-terminal} &\iff X \text{ is } \mathcal{C}^{\mathrm{op}}\text{-initial}, \quad \text{(by Exercise 3.1.0.3)} \\ &\iff \text{for each } A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}}),\ \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, A) \text{ is } \mathbf{Set}\text{-terminal}, \quad \text{(by ref prev part)} \\ &\iff \text{for each } A \in \mathrm{Obj}(\mathcal{C}),\ \mathrm{Hom}_{\mathcal{C}}(A, X) \text{ is } \mathbf{Set}\text{-terminal} \end{aligned}$$

□

Exercise 3.1.0.8. We have that

1. $\mathbf{0}$ is **Cat**-initial.
2. $\mathbf{1}$ is **Cat**-terminal.

Proof. 1. Let $\mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$. It is clear that $\mathrm{Hom}_{\mathbf{Cat}}(\mathbf{0}, \mathcal{C}) = \{E_{\mathcal{C}}\}$. Hence $\mathbf{0}$ is **Cat**-initial. Fill out like below ex

2. Let $\mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$. Let $F : \mathcal{C} \to \mathbf{1}$. Since $\mathrm{Obj}(\mathbf{1}) = \{*\}$ and $\mathrm{Hom}_{\mathbf{1}}(*, *) = \{\mathrm{id}_*^{\mathbf{1}}\}$, we have for each $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$, $F(A) = *$ and $F(f) = \mathrm{id}_*^{\mathbf{1}}$. Thus $F = \Delta_*^{\mathcal{C}}$. Since $F : \mathcal{C} \to \mathbf{1}$ is arbitrary, we have that for each $F : \mathcal{C} \to \mathbf{1}$, $F = \Delta_*^{\mathcal{C}}$. Thus $\mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathbf{1}) = \{\Delta_*^{\mathcal{C}}\}$. Since $\mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$ is arbitrary, we have that for each $\mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$, $\mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathbf{1}) = \{\Delta_*^{\mathcal{C}}\}$. Hence $\mathbf{1}$ is **Cat**-terminal.

□

Definition 3.1.0.9. Let $\mathcal{C}$, $\mathcal{D}$ be categories and $0, 1 \in \mathrm{Obj}(\mathcal{D})$ and $F : \mathcal{C} \to \mathcal{D}$. Suppose that 0 is $\mathcal{D}$-initial and 1 is $\mathcal{D}$-terminal. Then for each $A \in \mathrm{Obj}(\mathcal{C})$, there exists $f_A \in \mathrm{Hom}_{\mathcal{D}}(0, F(A))$ and $g_A \in \mathrm{Hom}_{\mathcal{D}}(F(A), 1)$ such that $\mathrm{Hom}_{\mathcal{D}}(0, F(A)) = \{f_A\}$ and $\mathrm{Hom}_{\mathcal{D}}(F(A), 1) = \{g_A\}$. We define the

- **initial natural transformation induced by** 0 **from** $\Delta_0^{\mathcal{C}}$ **to** F, denoted $\eta^{(0,F)} : \Delta_0^{\mathcal{C}} \Rightarrow F$, by $\eta_A^{(0,F)} := f_A$.
- **terminal natural transformation induced by** 1 **from** F **to** $\Delta_1^{\mathcal{C}}$, denoted $\phi^{(F,1)} : F \Rightarrow \Delta_1^{\mathcal{C}}$, by $\phi_A^{(F,1)} := g_A$.

Exercise 3.1.0.10. Let $\mathcal{C}$, $\mathcal{D}$ be categories and $0 \in \mathrm{Obj}(\mathcal{D})$ and $F : \mathcal{C} \to \mathcal{D}$. Suppose that 0 is $\mathcal{D}$-initial and 1 is $\mathcal{D}$-terminal. Then

1. $\eta^{(0,F)} : \Delta_0^{\mathcal{C}} \Rightarrow F$ is a natural transformation.
2. $\phi^{(F,1)} : F \Rightarrow \Delta_1^{\mathcal{C}}$ is a natural transformation.

Proof.

1. (a) By definition, for each $A \in \mathrm{Obj}(\mathcal{C})$, $\eta_A^{(0,F)} \in \mathrm{Hom}_{\mathcal{D}}(\Delta_0^{\mathcal{C}}(A), F(A))$
   (b) Let $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Since

$$\begin{aligned} F(f) \circ \eta_A^{(0,F)} &\in \mathrm{Hom}_{\mathcal{D}}(0, F(B)) \\ &= \{\eta_B^{(0,F)}\} \end{aligned}$$

we have that

$$\begin{aligned} F(f) \circ \eta_A^{(0,F)} &= \eta_B^{(0,F)} \\ &= \eta_B^{(0,F)} \circ \mathrm{id}_0^{\mathcal{C}} \end{aligned}$$

i.e. the following diagram commutes:

$$\begin{tikzcd} \Delta_0^{\mathcal{C}}(A) \arrow[r, "\eta_A^{(0,F)}"] \arrow[d, "\Delta_0^{\mathcal{C}}(f)"'] & F(A) \arrow[d, "F(f)"] \\ \Delta_0^{\mathcal{C}}(B) \arrow[r, "\eta_B^{(0,F)}"'] & F(B) \end{tikzcd} = \begin{tikzcd} 0 \arrow[r, "\eta_A^{(0,F)}"] \arrow[d, "\mathrm{id}_0^{\mathcal{C}}"'] & F(A) \arrow[d, "F(f)"] \\ 0 \arrow[r, "\eta_B^{(0,F)}"'] & F(B) \end{tikzcd}$$

So $\eta^{(0,F)} : \Delta_0^{\mathcal{C}} \Rightarrow F$ is a natural transformation.

2. FINISH!!!

□

Exercise 3.1.0.11. Let $\mathcal{C}$, $\mathcal{D}$ be categories and $0, 1 \in \mathrm{Obj}(\mathcal{D})$. Suppose that $\mathcal{C}$ is small.

1. If 0 is $\mathcal{D}$-initial, then $\Delta_0^{\mathcal{C}}$ is $\mathcal{D}^{\mathcal{C}}$-initial.
2. If 1 is $\mathcal{D}$-terminal, then $\Delta_1^{\mathcal{C}}$ is $\mathcal{D}^{\mathcal{C}}$-terminal.

Proof.

1. Suppose that 0 is $\mathcal{D}$-initial. Let $F \in \mathrm{Obj}(\mathcal{D}^{\mathcal{C}})$, $\alpha \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(\Delta_0^{\mathcal{C}}, F)$ and $A \in \mathrm{Obj}(\mathcal{C})$. Then

$$\begin{aligned} \alpha_A &\in \mathrm{Hom}_{\mathcal{D}}(\Delta_0^{\mathcal{C}}(A), F(A)) \\ &= \mathrm{Hom}_{\mathcal{D}}(0, F(A)) \\ &= \{\eta_A^{(0,F)}\} \end{aligned}$$

Hence $\alpha_A = \eta_A^{(0,F)}$. Since $A \in \mathrm{Obj}(\mathcal{C})$ is arbitrary, we have that for each $A \in \mathrm{Obj}(\mathcal{C})$, $\alpha_A = \eta_A^{(0,F)}$. Thus $\alpha = \eta^{(0,F)}$. Since $\alpha \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(\Delta_0^{\mathcal{C}}, F)$ is arbitrary, we have that for each $\alpha \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(\Delta_0^{\mathcal{C}}, F)$, $\alpha = \eta^{(0,F)}$. This $\mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(\Delta_0^{\mathcal{C}}, F) = \{\eta^{(0,F)}\}$. Therefore $\Delta_0^{\mathcal{C}}$ is $\mathcal{D}^{\mathcal{C}}$-initial.

2. use duality

□

3.2 Products and Coproducts

Definition 3.2.0.1. Let $\mathcal{C}$ be a category, A a set (class?), $(X_\alpha)_{\alpha\in A} \in \mathrm{Obj}(\mathcal{C})^A$ and $X \in \mathrm{Obj}(\mathcal{C})$.

- – Let $(\pi_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(X, X_\alpha)$. Then $(X, (\pi_\alpha)_{\alpha\in A})$ is said to be a **$\mathcal{C}$-product of** $(X_\alpha)_{\alpha\in A}$ if for each $Y \in \mathrm{Obj}(\mathcal{C})$ and $(p_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(Y, X_\alpha)$, there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}}(Y, X)$ such that for each $\alpha \in A$, from $\pi_\alpha \circ_{\mathcal{C}} \phi = p_\alpha$, i.e. the following diagram commutes:

$$\begin{tikzcd} Y \arrow[d, dashed, "\phi"'] \arrow[dr, "p_\alpha"] & \\ X \arrow[r, "\pi_\alpha"'] & X_\alpha \end{tikzcd}$$

  – We define

$$\mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A}) := \left\{ (X, (\pi_\alpha)_{\alpha\in A}) \in \mathrm{Obj}(\mathcal{C}) \times \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(X, X_\alpha) : (X, (\pi_\alpha)_{\alpha\in A}) \text{ is a } \mathcal{C}\text{-product of } (X_\alpha)_{\alpha\in A} \right\}.$$

- – Let $(\iota_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(X_\alpha, X)$. Then $(X, (\iota_\alpha)_{\alpha\in A})$ is said to be a **$\mathcal{C}$-coproduct of** $(X_\alpha)_{\alpha\in A}$ if for each $Y \in \mathrm{Obj}(\mathcal{C})$ and $(i_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(X_\alpha, Y)$, there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$ such that for each $\alpha \in A$, from $\phi \circ_{\mathcal{C}} \iota_\alpha = i_\alpha$, i.e. the following diagram commutes:

$$\begin{tikzcd} & X_\alpha \arrow[dl, "\iota_\alpha"'] \arrow[dd, "i_\alpha"] \\ X \arrow[d, dashed, "\phi"'] & \\ Y & \end{tikzcd}$$

  – We define

$$\mathrm{Coprod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A}) := \left\{ (X, (\iota_\alpha)_{\alpha\in A}) \in \mathrm{Obj}(\mathcal{C}) \times \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(X_\alpha, X) : (X, (\iota_\alpha)_{\alpha\in A}) \text{ is a } \mathcal{C}\text{-coproduct of } (X_\alpha)_{\alpha\in A} \right\}.$$

Exercise 3.2.0.2. Let $\mathcal{C}$ be a category, A a set (class?) and $(X_\alpha)_{\alpha\in A} \in \mathrm{Obj}(\mathcal{C})^A$. Then

1. $\mathrm{Prod}_{\mathcal{C}^{\mathrm{op}}}((X_\alpha)_{\alpha\in A}) = \mathrm{Coprod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$.
2. $\mathrm{Coprod}_{\mathcal{C}^{\mathrm{op}}}((X_\alpha)_{\alpha\in A}) = \mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$.

Proof.

1. - Let $(X, (\pi_\alpha)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}^{\mathrm{op}}}((X_\alpha)_{\alpha\in A})$. We note that $(X_\alpha)_{\alpha\in A} \in \mathrm{Obj}(\mathcal{C})^A$, $X \in \mathrm{Obj}(\mathcal{C})$ and $(\pi_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(X_\alpha, X)$. Let $Y \in \mathrm{Obj}(\mathcal{C})$ and $(i_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(X_\alpha, Y)$.
   - **Existence:**
     Then $Y \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ and $(i_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(Y, X_\alpha)$. Since $(X, (\pi_\alpha)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}^{\mathrm{op}}}((X_\alpha)_{\alpha\in A})$, there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(Y, X)$ such that for each $\alpha \in A$, $\pi_\alpha \circ_{\mathcal{C}^{\mathrm{op}}} \phi = i_\alpha$. Then $\phi \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$ and for each $\alpha \in A$, $\phi \circ_{\mathcal{C}} \pi_\alpha = i_\alpha$.
   - **Uniqueness:**
     Let $\phi' \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$. Suppose that for each $\alpha \in A$, $\phi' \circ_{\mathcal{C}} \pi_\alpha = i_\alpha$. Then $\phi' \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(Y, X)$ and for each $\alpha \in A$, $\pi_\alpha \circ_{\mathcal{C}^{\mathrm{op}}} \phi' = i_\alpha$. Uniqueness of ϕ implies that $\phi' = \phi$.

Since $Y \in \mathrm{Obj}(\mathcal{C})$ and $(i_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(X_\alpha, Y)$ are arbitrary, we have that for each $Y \in \mathrm{Obj}(\mathcal{C})$ and $(i_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(X_\alpha, Y)$, there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$ such that for each $\alpha \in A$, $\phi \circ_{\mathcal{C}} \pi_\alpha = i_\alpha$. Thus $(X, (\pi_\alpha)_{\alpha\in A}) \in \mathrm{Coprod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$. Since $(X, (\pi_\alpha)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}^{\mathrm{op}}}((X_\alpha)_{\alpha\in A})$ is arbitrary, we have that for each $(X, (\pi_\alpha)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}^{\mathrm{op}}}((X_\alpha)_{\alpha\in}$ $(X, (\pi_\alpha)_{\alpha\in A}) \in \mathrm{Coprod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$. Hence $\mathrm{Prod}_{\mathcal{C}^{\mathrm{op}}}((X_\alpha)_{\alpha\in A}) \subset \mathrm{Coprod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$.

- Similar, fill out!!!

Since $\mathrm{Prod}_{\mathcal{C}^{\mathrm{op}}}((X_\alpha)_{\alpha\in A}) \subset \mathrm{Coprod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$ and $\mathrm{Coprod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A}) \subset \mathrm{Prod}_{\mathcal{C}^{\mathrm{op}}}((X_\alpha)_{\alpha\in A})$, we have that $\mathrm{Prod}_{\mathcal{C}^{\mathrm{op}}}((X_\alpha)_{\alpha\in A}) = \mathrm{Coprod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$.

2. The prev part (ref) implies that

$$\begin{aligned}\mathrm{Coprod}_{\mathcal{C}^{\mathrm{op}}}((X_\alpha)_{\alpha\in A}) &= \mathrm{Prod}_{(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}}}((X_\alpha)_{\alpha\in A}) \\ &= \mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A}).\end{aligned}$$

□

Exercise 3.2.0.3. Let $\mathcal{C}$ be a category, A a set (class?) and $(X_\alpha)_{\alpha\in A} \in \mathrm{Obj}(\mathcal{C})^A$. Then

1. for each $(X, (\pi_\alpha^X)_{\alpha\in A}), (Y, (\pi_\alpha^Y)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$, $X \cong_{\mathcal{C}} Y$.
2. for each $(X, (\iota_\alpha^X)_{\alpha\in A}), (Y, (\iota_\alpha^Y)_{\alpha\in A}) \in \mathrm{Coprod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$, $X \cong_{\mathcal{C}} Y$.

Proof.

1. Let $(X, (\pi_\alpha^X)_{\alpha\in A}), (Y, (\pi_\alpha^Y)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$.
   - – By definition, there exists a unique $\phi_{Y,X} \in \mathrm{Hom}_{\mathcal{C}}(Y, X)$ such that for each $\alpha \in A$, $\pi_\alpha^X \circ_{\mathcal{C}} \phi_{Y,X} = \pi_\alpha^Y$. Similarly, there exists a unique $\phi_{X,Y} \in \mathrm{Hom}_{\mathcal{C}}(X, Y)$ such that for each $\alpha \in A$, $\pi_\alpha^Y \circ_{\mathcal{C}} \phi_{X,Y} = \pi_\alpha^X$.

$$\begin{aligned}\pi_\alpha^Y \circ_C (\phi_{X,Y} \circ_{\mathcal{C}} \phi_{Y,X}) &= (\pi_\alpha^Y \circ_{\mathcal{C}} \phi_{X,Y}) \circ_{\mathcal{C}} \phi_{Y,X} \\ &= \pi_\alpha^X \circ_{\mathcal{C}} \phi_{Y,X} \\ &= \pi_\alpha^Y.\end{aligned}$$

     – By definition, there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}}(Y, Y)$ such that $\pi_\alpha^Y \circ \phi = \pi_\alpha^Y$.
     – We note that $\pi_\alpha^Y \circ \mathrm{id}_Y^{\mathcal{C}} = \pi_\alpha^Y$.

     Thus the following diagrams commutes:

$$\begin{tikzcd} Y \arrow[d, dashed, "\phi_{X,Y}\circ_{\mathcal{C}}\phi_{Y,X}"'] \arrow[ddr, "\pi_\alpha"] & \\ Y \arrow[dr, "\pi_\alpha"'] & \\ & X_\alpha \end{tikzcd} \qquad \begin{tikzcd} Y \arrow[d, dashed, "\phi"'] \arrow[ddr, "\pi_\alpha"] & \\ Y \arrow[dr, "\pi_\alpha"'] & \\ & X_\alpha \end{tikzcd} \qquad \begin{tikzcd} Y \arrow[d, dashed, "\mathrm{id}_Y^{\mathcal{C}}"'] \arrow[ddr, "\pi_\alpha"] & \\ Y \arrow[dr, "\pi_\alpha"'] & \\ & X_\alpha \end{tikzcd}$$

     Uniqueness of ϕ then implies that

$$\begin{aligned}\phi_{X,Y} \circ_{\mathcal{C}} \phi_{Y,X} &= \phi \\ &= \mathrm{id}_Y^{\mathcal{C}},\end{aligned}$$

   - Similarly, $\phi_{Y,X} \circ_{\mathcal{C}} \phi_{X,Y} = \mathrm{id}_X^{\mathcal{C}}$.

   Hence $\phi_{Y,X} \in \mathrm{Iso}_{\mathcal{C}}(Y, X)$ and $X \cong_{\mathcal{C}} Y$. Since $(X, (\pi_\alpha^X)_{\alpha\in A}), (Y, (\pi_\alpha^Y)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$ are arbitrary, we have that for each $(X, (\pi_\alpha^X)_{\alpha\in A}), (Y, (\pi_\alpha^Y)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$, $X \cong_{\mathcal{C}} Y$.

2. Let $(X, (\iota_\alpha^X)_{\alpha\in A}), (Y, (\iota_\alpha^Y)_{\alpha\in A}) \in \mathrm{Coprod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$. The prev ex (ref) implies that $(X, (\iota_\alpha^X)_{\alpha\in A}), (Y, (\iota_\alpha^Y)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}^{\mathrm{op}}}((X_\alpha)_{\alpha\in A})$ The prev part (ref) then implies that $X \cong_{\mathcal{C}^{\mathrm{op}}} Y$. An exercise in isomorphism section (ref) implies that $X \cong_{\mathcal{C}} Y$.

□

Note 3.2.0.4.

- – If $(X, (\pi_\alpha)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$, we may write $\prod_{\alpha\in A} X_\alpha$ in place of X.
  – If $(X, (\iota_\alpha)_{\alpha\in A}) \in \mathrm{Coprod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$, we may write $\coprod_{\alpha\in A} X_\alpha$ in place of X.
- If $A = \{1, 2\}$, we may write
  – $\mathrm{Prod}_{\mathcal{C}}(X_1, X_2)$ and $\mathrm{Coprod}_{\mathcal{C}}(X_1, X_2)$ in place of $\mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$ and $\mathrm{Coprod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$ respectively.
  – $X_1 \times X_2$ in place of $\prod_{\alpha\in A} X_\alpha$ and $X_1 \coprod X_2$ in place of $\coprod_{\alpha\in A} X_\alpha$

Exercise 3.2.0.5. Let $\mathcal{C}$ be a category, A a set (class?), $X \in \mathrm{Obj}(\mathcal{C})$, $(X_\alpha)_{\alpha\in A} \in \mathrm{Obj}(\mathcal{C})^A$, $(\prod_{\alpha\in A} X_\alpha, (\pi_\alpha)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$ and $f, g \in \mathrm{Hom}_{\mathcal{C}}(X, \prod_{\alpha\in A} X_\alpha)$. If for each $\alpha \in A$, $\pi_\alpha \circ_{\mathcal{C}} f = \pi_\alpha \circ_{\mathcal{C}} g$, then $f = g$.

Proof. Suppose that for each $\alpha \in A$, $\pi_\alpha \circ_{\mathcal{C}} f = \pi_\alpha \circ_{\mathcal{C}} g$. Since $(\pi_\alpha \circ_{\mathcal{C}} g)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(X, X_\alpha)$ and $(\prod_{\alpha\in A} X_\alpha, (\pi_\alpha)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$, there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}}(X, \prod_{\alpha\in A} X_\alpha)$ such that for each $\alpha \in A$, $\pi_\alpha \circ_{\mathcal{C}} \phi = \pi_\alpha \circ_{\mathcal{C}} g$, i.e. the following diagram commutes:

$$\begin{tikzcd}
X \arrow[d, dashed, "\phi"'] \arrow[ddr, "\pi_\alpha \circ_{\mathcal{C}} g"] & \\
\prod_{\alpha\in A} X_\alpha \arrow[dr, "\pi_\alpha"'] & \\
& X_\alpha
\end{tikzcd}$$

Since for each $\alpha \in A$, $\pi_\alpha \circ_{\mathcal{C}} f = \pi_\alpha \circ_{\mathcal{C}} g$ and $\pi_\alpha \circ_{\mathcal{C}} g = \pi_\alpha \circ_{\mathcal{C}} g$, uniqueness of ϕ implies that

$$\begin{aligned} f &= \phi \\ &= g. \end{aligned}$$

□

Definition 3.2.0.6. Let $\mathcal{C}$ be a category, A a set (class?), $(X_\alpha)_{\alpha\in A}, (Y_\alpha)_{\alpha\in A} \in \mathrm{Obj}(\mathcal{C})^A$, $(f_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(X_\alpha, Y_\alpha)$, $(\prod_{\alpha\in A} X_\alpha, (\pi^X_\alpha)_{\alpha\in A}$ $\mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$ and $(\prod_{\alpha\in A} Y_\alpha, (\pi^Y_\alpha)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}}((Y_\alpha)_{\alpha\in A})$. We define the **$\mathcal{C}$-product of** $(f_\alpha)_{\alpha\in A}$, denoted $\prod_{\alpha\in A} f_\alpha \in \mathrm{Hom}_{\mathcal{C}}(\prod_{\alpha\in A} X_\alpha, \prod_{\alpha\in A} Y$ to be the unique $f \in \mathrm{Hom}_{\mathcal{C}}(\prod_{\alpha\in A} X_\alpha, \prod_{\alpha\in A} Y_\alpha)$ such that for each $\alpha \in A$, $\pi^Y_\alpha \circ_{\mathcal{C}} f = f_\alpha \circ \pi^X_\alpha$, i.e the following diagram commutes:

$$\begin{tikzcd}
\prod_{\alpha\in A} X_\alpha \arrow[d, dashed, "f"'] \arrow[ddr, "f_\alpha \circ_{\mathcal{C}} \pi^X_\alpha"] & \\
\prod_{\alpha\in A} Y_\alpha \arrow[dr, "\pi^Y_\alpha"'] & \\
& Y_\alpha
\end{tikzcd}$$

Exercise 3.2.0.7. Let $\mathcal{C}$ be a category, A a set (class?), $(X_\alpha)_{\alpha\in A}, (Y_\alpha)_{\alpha\in A} \in \mathrm{Obj}(\mathcal{C})^A$, $(f_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathcal{C}}(X_\alpha, Y_\alpha)$, $(\prod_{\alpha\in A} X_\alpha, (\pi^X_\alpha)_{\alpha\in A})$ $\mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha\in A})$ and $(\prod_{\alpha\in A} Y_\alpha, (\pi^Y_\alpha)_{\alpha\in A}) \in \mathrm{Prod}_{\mathcal{C}}((Y_\alpha)_{\alpha\in A})$. Then $(\pi^X_\alpha, \pi^Y_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(\prod_{\alpha\in A} f_\alpha, f_\alpha)$.

Proof. By construction, for each $\alpha \in A$,

$$f_\alpha \circ_{\mathcal{C}} \pi_\alpha^X = \pi_\alpha^Y \circ_{\mathcal{C}} \prod_{\alpha \in A} f_\alpha,$$

i.e. the following diagram commutes:

$$\begin{tikzcd}
\prod_{\alpha \in A} X_\alpha \arrow[r, "\pi_\alpha^X"] \arrow[d, "\prod_{\alpha \in A} f_\alpha"'] & X_\alpha \arrow[d, "f_\alpha"] \\
\prod_{\alpha \in A} Y_\alpha \arrow[r, "\pi_\alpha^Y"'] & Y_\alpha
\end{tikzcd}$$

Thus $(\pi_\alpha^X, \pi_\alpha^Y)_{\alpha \in A} \in \prod_{\alpha \in A} \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(\prod_{\alpha \in A} f_\alpha, f_\alpha)$. □

Next exercise shows that the product of morphisms is in fact a product object in the arrow category

Exercise 3.2.0.8. Let $\mathcal{C}$ be a category, A a set (class?), $(X_\alpha)_{\alpha \in A}, (Y_\alpha)_{\alpha \in A} \in \mathrm{Obj}(\mathcal{C})^A$, $(f_\alpha)_{\alpha \in A} \in \prod_{\alpha \in A} \mathrm{Hom}_{\mathcal{C}}(X_\alpha, Y_\alpha)$, $(\prod_{\alpha \in A} X_\alpha, (\pi_\alpha^X)_{\alpha \in A}) \in \mathrm{Prod}_{\mathcal{C}}((X_\alpha)_{\alpha \in A})$ and $(\prod_{\alpha \in A} Y_\alpha, (\pi_\alpha^Y)_{\alpha \in A}) \in \mathrm{Prod}_{\mathcal{C}}((Y_\alpha)_{\alpha \in A})$. Define $(\pi_\alpha)_{\alpha \in A} \in \prod_{\alpha \in A} \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(\prod_{\alpha \in A} f_\alpha, f_\alpha)$ by $\pi_\alpha := (\pi_\alpha^X, \pi_\alpha^Y)$. Then $(\prod_{\alpha \in A} f_\alpha, \pi_\alpha) \in \mathrm{Prod}_{\mathrm{Arr}(\mathcal{C})}((f_\alpha)_{\alpha \in A})$.

Proof. Let $g \in \mathrm{Obj}(\mathrm{Arr}(\mathcal{C}))$ and $(p_\alpha)_{\alpha \in A} \in \prod_{\alpha \in A} \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(g, f_\alpha)$. Define $A, B \in \mathrm{Obj}(\mathcal{C})$ by $A := \mathrm{dom}_{\mathcal{C}}(g)$ and $B := \mathrm{cod}_{\mathcal{C}}(g)$. Write $p_\alpha = (p_\alpha^A, p_\alpha^A)$. We note that

- By definition, for each $\alpha \in A$, $f_\alpha \circ_{\mathcal{C}} p_\alpha^A = p_\alpha^B \circ_{\mathcal{C}} g$, i.e. the following diagram commutes:

$$\begin{tikzcd}
A \arrow[r, "p_\alpha^A"] \arrow[d, "g"'] & X_\alpha \arrow[d, "f_\alpha"] \\
B \arrow[r, "p_\alpha^B"'] & Y_\alpha
\end{tikzcd}$$

- By construction, for each $\alpha \in A$, $f_\alpha \circ_{\mathcal{C}} \pi_\alpha^X = \pi_\alpha^Y \circ_{\mathcal{C}} \prod_{\alpha \in A} f_\alpha$, i.e. the following diagram commutes:

$$\begin{tikzcd}
\prod_{\alpha \in A} X_\alpha \arrow[r, "\pi_\alpha^X"] \arrow[d, "\prod_{\alpha \in A} f_\alpha"'] & X_\alpha \arrow[d, "f_\alpha"] \\
\prod_{\alpha \in A} Y_\alpha \arrow[r, "\pi_\alpha^Y"'] & Y_\alpha
\end{tikzcd}$$

- **Existence:**
  By definition, there exists a unique $\phi_X \in \mathrm{Hom}_{\mathcal{C}}(A, \prod_{\alpha \in A} X_\alpha)$ such that for each $\alpha \in A$, $\pi_\alpha^X \circ_{\mathcal{C}} \phi_X = p_\alpha^A$, i.e. the following diagram commutes:

$$\begin{tikzcd}
A \arrow[d, dashed, "\phi_X"'] \arrow[ddr, "p_\alpha^A"] & \\
\prod_{\alpha \in A} X_\alpha \arrow[dr, "\pi_\alpha^X"'] & \\
& X_\alpha
\end{tikzcd}$$

  Similiarly, there exists a unique $\phi_Y \in \mathrm{Hom}_{\mathcal{C}}(B, \prod_{\alpha \in A} Y_\alpha)$ such that for each $\alpha \in A$,

$\pi_\alpha^Y \circ_{\mathcal{C}} \phi_Y = p_\alpha^B$, i.e. the following diagram commutes:

$$\begin{tikzcd}
B \arrow[d, dashed, "\phi_Y"'] \arrow[ddr, "p_\alpha^B"] & \\
\prod_{\alpha\in A} Y_\alpha \arrow[dr, "\pi_\alpha^Y"'] & \\
& Y_\alpha
\end{tikzcd}$$

Then for each $\alpha \in A$,

$$\begin{aligned}
\pi_\alpha^Y \circ_{\mathcal{C}} (\prod_{\alpha\in A} f_\alpha \circ_{\mathcal{C}} \phi_X) &= (\pi_\alpha^Y \circ_{\mathcal{C}} \prod_{\alpha\in A} f_\alpha) \circ_{\mathcal{C}} \phi_X \\
&= (f_\alpha \circ_{\mathcal{C}} \pi_\alpha^X) \circ_{\mathcal{C}} \phi_X \\
&= f_\alpha \circ_{\mathcal{C}} (\pi_\alpha^X \circ_{\mathcal{C}} \phi_X) \\
&= f_\alpha \circ_{\mathcal{C}} p_\alpha^A \\
&= p_\alpha^B \circ_{\mathcal{C}} g \\
&= (\pi_\alpha^Y \circ_{\mathcal{C}} \phi_Y) \circ_{\mathcal{C}} g \\
&= \pi_\alpha^Y \circ_{\mathcal{C}} (\phi_Y \circ_{\mathcal{C}} g).
\end{aligned}$$

the prev ex (ref) then implies that $\prod_{\alpha\in A} f_\alpha \circ_{\mathcal{C}} \phi_X = \phi_Y \circ_{\mathcal{C}} g$, i.e. the following diagram commutes:

$$\begin{tikzcd}
A \arrow[r, "\phi_X"] \arrow[d, "g"'] & \prod_{\alpha\in A} X_\alpha \arrow[d, "\prod_{\alpha\in A} f_\alpha"] \\
B \arrow[r, "\phi_Y"'] & \prod_{\alpha\in A} Y_\alpha
\end{tikzcd}$$

Thus $(\phi_X, \phi_Y) \in \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(g, \prod_{\alpha\in A} f_\alpha)$. Define $\phi \in \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(g, \prod_{\alpha\in A} f_\alpha)$ by $\phi := (\phi_X, \phi_Y)$. Let $\alpha \in A$. Since $\pi_\alpha^X \circ_{\mathcal{C}} \phi_X = p_\alpha^A$ and $\pi_\alpha^Y \circ_{\mathcal{C}} \phi_Y = p_\alpha^B$, we have that

$$\begin{aligned}
\pi_\alpha \circ_{\mathrm{Arr}(\mathcal{C})} \phi &= (\pi_\alpha^X, \pi_\alpha^Y) \circ_{\mathrm{Arr}(\mathcal{C})} (\phi_X, \phi_Y) \\
&= (\pi_\alpha^X \circ_{\mathcal{C}} \phi_X, \pi_\alpha^Y \circ_{\mathcal{C}} \phi_Y) \\
&= (p_\alpha^A, p_\alpha^B) \\
&= p_\alpha.
\end{aligned}$$

Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $\pi_\alpha \circ_{\mathrm{Arr}(\mathcal{C})} \phi = p_\alpha$, i.e. the following diagram commutes:

$$\begin{tikzcd}
g \arrow[d, dashed, "\phi"'] \arrow[ddr, "p_\alpha"] & \\
\prod_{\alpha\in A} f_\alpha \arrow[dr, "\pi_\alpha"'] & \\
& f_\alpha
\end{tikzcd}$$

- **Uniqueness:**
  Let $\phi' \in \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(g, \prod_{\alpha\in A} f_\alpha)$. Write $\phi' = (\phi'_X, \phi'_Y)$. Suppose that for each $\alpha \in A$, $\pi_\alpha \circ_{\mathrm{Arr}(\mathcal{C})} \phi' = p_\alpha$, i.e. the following diagram

commutes:

$$\begin{tikzcd}
g \arrow[d, dashed, "\phi'"'] \arrow[ddr, "p_\alpha"] & \\
\prod_{\alpha\in A} f_\alpha \arrow[dr, "\pi_\alpha"'] & \\
& f_\alpha
\end{tikzcd}$$

Let $\alpha \in A$. Then

$$\begin{aligned}
(\pi_\alpha^X \circ_{\mathcal{C}} \phi'_X, \pi_\alpha^Y \circ_{\mathcal{C}} \phi'_Y) &= (\pi_\alpha^X, \pi_\alpha^Y) \circ_{\mathrm{Arr}(\mathcal{C})} (\phi'_X, \phi'_Y) \\
&= \pi_\alpha \circ_{\mathrm{Arr}(\mathcal{C})} \phi' \\
&= p_\alpha \\
&= (p_\alpha^A, p_\alpha^B).
\end{aligned}$$

Therefore $\pi_\alpha^X \circ_{\mathcal{C}} \phi'_X = p_\alpha^A$ and $\pi_\alpha^Y \circ_{\mathcal{C}} \phi'_Y = p_\alpha^B$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $\pi_\alpha^X \circ_{\mathcal{C}} \phi'_X = p_\alpha^A$ and $\pi_\alpha^Y \circ_{\mathcal{C}} \phi'_Y = p_\alpha^B$, i.e. the following diagrams commute:

$$\begin{tikzcd}
A \arrow[d, dashed, "\phi'_X"'] \arrow[ddr, "p_\alpha^A"] & \\
\prod_{\alpha\in A} X_\alpha \arrow[dr, "\pi_\alpha^X"'] & \\
& X_\alpha
\end{tikzcd}
\qquad
\begin{tikzcd}
B \arrow[d, dashed, "\phi'_Y"'] \arrow[ddr, "p_\alpha^B"] & \\
\prod_{\alpha\in A} Y_\alpha \arrow[dr, "\pi_\alpha^Y"'] & \\
& Y_\alpha
\end{tikzcd}$$

Uniqueness of ϕ_X and ϕ_Y imply that $\phi'_X = \phi_X$ and $\phi'_Y = \phi_Y$. Thus

$$\begin{aligned}
\phi' &= (\phi'_X, \phi'_Y) \\
&= (\phi_X, \phi_Y) \\
&= \phi.
\end{aligned}$$

Since $\phi' \in \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(g, \prod_{\alpha\in A} f_\alpha)$ is arbitrary, we have that for each $\phi' \in \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(g, \prod_{\alpha\in A} f_\alpha)$, $\phi' = \phi$.

Thus there exists a unique $\phi \in \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(g, \prod_{\alpha\in A} f_\alpha)$ such that for each $\alpha \in A$, $\pi_\alpha \circ_{\mathrm{Arr}(\mathcal{C})} \phi = p_\alpha$. Since $g \in \mathrm{Obj}(\mathrm{Arr}(\mathcal{C}))$ and $(p_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(g, f_\alpha)$ are arbitrary, we have that for each $g \in \mathrm{Obj}(\mathrm{Arr}(\mathcal{C}))$ and $(p_\alpha)_{\alpha\in A} \in \prod_{\alpha\in A} \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(g, f_\alpha)$, there exists a unique $\phi \in \mathrm{Hom}_{\mathrm{Arr}(\mathcal{C})}(g, \prod_{\alpha\in A} f_\alpha)$ such that for each $\alpha \in A$, $\pi_\alpha \circ_{\mathrm{Arr}(\mathcal{C})} \phi = p_\alpha$, i.e. the following diagram commutes:

$$\begin{tikzcd}
g \arrow[d, dashed, "\phi"'] \arrow[ddr, "p_\alpha"] & \\
\prod_{\alpha\in A} f_\alpha \arrow[dr, "\pi_\alpha"'] & \\
& f_\alpha
\end{tikzcd}$$

Hence $(\prod_{\alpha\in A} f_\alpha, (\pi_\alpha)_{\alpha\in A}) \in \mathrm{Prod}_{\mathrm{Arr}(\mathcal{C})}((f_\alpha)_{\alpha\in A})$. □

Definition 3.2.0.9. Let $\mathcal{C}$ be a category. Then $\mathcal{C}$ is said to have all

- **products** if for each $A \in \mathrm{Obj}(\mathbf{Set})$ and $(X_\alpha)_{\alpha\in A} \in \mathrm{Obj}(\mathcal{C})^A$, $\mathrm{Prod}_{\mathcal{C}}((A_\alpha)_{\alpha\in A}) \neq \varnothing$,

- **coproducts** if for each $A \in \mathrm{Obj}(\mathbf{Set})$ and $(X_\alpha)_{\alpha \in A} \in \mathrm{Obj}(\mathcal{C})^A$, $\mathrm{Prod}_{\mathcal{C}}((A_\alpha)_{\alpha \in A}) \neq \varnothing$,
- **finite products** if for each $A, B \in \mathrm{Obj}(\mathcal{C})$, $\mathrm{Prod}_{\mathcal{C}}(A, B) \neq \varnothing$,
- **finite coproducts** if for each $A, B \in \mathrm{Obj}(\mathcal{C})$, $\mathrm{Coprod}_{\mathcal{C}}(A, B) \neq \varnothing$,

3.3 Equalizers and Coequalizers

Definition 3.3.0.1. Let $\mathcal{C}$ be a category. Let $A, B \in \mathrm{Obj}(\mathcal{C})$ and $g, h \in \mathrm{Hom}_{\mathcal{C}}(A, B)$.

- Let $C \in \mathrm{Obj}(\mathcal{C})$.
  - Let $f \in \mathrm{Hom}_{\mathcal{C}}(C, A)$. Then (C, f) is said to be a **$\mathcal{C}$-equalizer of g and h** if
    1. $g \circ_{\mathcal{C}} f = h \circ_{\mathcal{C}} f$, i.e. the following diagram commutes:

$$\begin{tikzcd} & & B \\ C \arrow[r, "f"] & A \arrow[ur, "g"] \arrow[dr, "h"'] & \\ & & B \end{tikzcd}$$

    2. for each $C' \in \mathrm{Obj}(\mathcal{C})$ and $f' \in \mathrm{Hom}_{\mathcal{C}}(C', A)$, if $g \circ_{\mathcal{C}} f' = h \circ_{\mathcal{C}} f'$, then there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}}(C', C)$ such that $f \circ_{\mathcal{C}} \phi = f'$, i.e. the following diagram commutes:

$$\begin{tikzcd} C' \arrow[dr, "f'"] \arrow[dd, dashed, "\phi"'] & & B \\ & A \arrow[ur, "g"] \arrow[dr, "h"] & \\ C \arrow[ur, "f"] & & B \end{tikzcd}$$

  - Let $f \in \mathrm{Hom}_{\mathcal{C}}(B, C)$. Then (C, f) is said to be a **$\mathcal{C}$-coequalizer** of g and h if
    1. $f \circ_{\mathcal{C}} g = f \circ_{\mathcal{C}} h$, i.e. the following diagram commutes:

$$\begin{tikzcd} A \arrow[dr, "g"] & & \\ & B \arrow[r, "f"] & C \\ A \arrow[ur, "h"] & & \end{tikzcd}$$

    2. for each $C' \in \mathrm{Obj}(\mathcal{C})$ and $f' \in \mathrm{Hom}_{\mathcal{C}}(B, C')$, if $f' \circ_{\mathcal{C}} g = f' \circ_{\mathcal{C}} h$, then there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}}(C, C')$ such that $\phi \circ_{\mathcal{C}} f = f'$, i.e. the following diagram commutes:

$$\begin{tikzcd} A \arrow[dr, "g"] & & C \arrow[dd, dashed, "\phi"] \\ & B \arrow[ur, "f"] \arrow[dr, "f'"] & \\ A \arrow[ur, "h"] & & C' \end{tikzcd}$$

- We define
  - $\mathrm{Equi}_{\mathcal{C}}(g, h) := \{(C, f) \in \mathrm{Obj}(\mathcal{C}) \times \mathrm{Hom}_{\mathcal{C}} : \mathrm{cod}_{\mathcal{C}}(f) = A \text{ and } f \text{ is a } \mathcal{C}\text{-equalizer of } g \text{ and } h\}$

- $\mathrm{Coequi}_{\mathcal{C}}(g,h) := \{(C,f) \in \mathrm{Obj}(\mathcal{C}) \times \mathrm{Hom}_{\mathcal{C}} : \mathrm{dom}_{\mathcal{C}}(f) = B$ and f is a $\mathcal{C}$-coequalizer of g and $h\}$

Definition 3.3.0.2. Let $\mathcal{C}$ be a category.

- Let $C \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}$. Then (C,f) is said to be a
  1. $\mathcal{C}$-equalizer if there exist $A, B \in \mathrm{Obj}(\mathcal{C})$ and $g, h \in \mathrm{Hom}_{\mathcal{C}}(A,B)$ such that $f \in \mathrm{Hom}_{\mathcal{C}}(C,A)$ and f is a $\mathcal{C}$-equalizer of g and h.
  2. $\mathcal{C}$-coequalizer if there exist $A, B \in \mathrm{Obj}(\mathcal{C})$ and $g, h \in \mathrm{Hom}_{\mathcal{C}}(A,B)$ such that $f \in \mathrm{Hom}_{\mathcal{C}}(B,C)$ and f is a $\mathcal{C}$-coequalizer of g and h.
- We define
  - $\mathrm{Equi}_{\mathcal{C}} := \{(C,f) \in \mathrm{Obj}(\mathcal{C}) \times \mathrm{Hom}_{\mathcal{C}} : (C,f)$ is a $\mathcal{C}$-equalizer$\}$
  - $\mathrm{Coequi}_{\mathcal{C}} := \{(C,f) \in \mathrm{Obj}(\mathcal{C}) \times \mathrm{Hom}_{\mathcal{C}} : (C,f)$ is a $\mathcal{C}$-coequalizer$\}$

Exercise 3.3.0.3. Let $\mathcal{C}$ be a category. Then

1. for each $A, B \in \mathrm{Obj}(\mathcal{C})$ and $g, h \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A,B)$,
   (a) $\mathrm{Equi}_{\mathcal{C}^{\mathrm{op}}}(g,h) = \mathrm{Coequi}_{\mathcal{C}}(g,h)$
   (b) $\mathrm{Coequi}_{\mathcal{C}^{\mathrm{op}}}(g,h) = \mathrm{Equi}_{\mathcal{C}}(g,h)$
2. (a) $\mathrm{Equi}_{\mathcal{C}^{\mathrm{op}}} = \mathrm{Coequi}_{\mathcal{C}}$
   (b) $\mathrm{Coequi}_{\mathcal{C}^{\mathrm{op}}} = \mathrm{Equi}_{\mathcal{C}}$

Proof.

1. (a)
   - Let $(C,f) \in \mathrm{Equi}_{\mathcal{C}^{\mathrm{op}}}(f,g)$. By definition (ref def), $C \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$, $f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(C,A)$ and
     - $g \circ_{\mathcal{C}^{\mathrm{op}}} f = h \circ_{\mathcal{C}^{\mathrm{op}}} f$,
     - for each $C' \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ and $f' \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(C',A)$, if $g \circ_{\mathcal{C}^{\mathrm{op}}} f' = h \circ_{\mathcal{C}^{\mathrm{op}}} f'$, then there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(C',C)$ such that $f \circ_{\mathcal{C}^{\mathrm{op}}} \phi = f'$.

     We note that $g, h \in \mathrm{Hom}_{\mathcal{C}}(B,A)$ and $f \in \mathrm{Hom}_{\mathcal{C}}(B,C)$.
     i. Thus
     $$\begin{aligned} f \circ_{\mathcal{C}} g &= g \circ_{\mathcal{C}^{\mathrm{op}}} f \\ &= h \circ_{\mathcal{C}^{\mathrm{op}}} f \\ &= f \circ_{\mathcal{C}} h. \end{aligned}$$
     ii. Let $C' \in \mathrm{Obj}(\mathcal{C})$ and $f' \in \mathrm{Hom}_{\mathcal{C}}(B,C')$. Suppose that $f' \circ_{\mathcal{C}} g = f' \circ_{\mathcal{C}} h$. Then $f' \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(C',B)$ and
     $$\begin{aligned} g \circ_{\mathcal{C}^{\mathrm{op}}} f' &= f' \circ_{\mathcal{C}} g \\ &= f' \circ_{\mathcal{C}} h \\ &= h \circ_{\mathcal{C}^{\mathrm{op}}} f'. \end{aligned}$$
     Thus there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(C',C)$ such that $f \circ_{\mathcal{C}^{\mathrm{op}}} \phi = f'$. Therefore there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}}(C,C')$ such that
     $$\begin{aligned} \phi \circ_{\mathcal{C}} f &= f \circ_{\mathcal{C}^{\mathrm{op}}} \phi \\ &= f'. \end{aligned}$$
     Since $C' \in \mathrm{Obj}(\mathcal{C})$ and $f' \in \mathrm{Hom}_{\mathcal{C}}(B,C')$ are arbitrary, we have that for each $C' \in \mathrm{Obj}(\mathcal{C})$ and $f' \in \mathrm{Hom}_{\mathcal{C}}(B,C')$, there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}}(C,C')$ such that $\phi \circ_{\mathcal{C}} f = f'$.

   Thus $(C,f) \in \mathrm{Coequi}_{\mathcal{C}}(f,g)$. Since $(C,f) \in \mathrm{Equi}_{\mathcal{C}^{\mathrm{op}}}(f,g)$ is arbitrary, we have that for each $(C,f) \in \mathrm{Equi}_{\mathcal{C}^{\mathrm{op}}}(f,g)$, $(C,f) \in \mathrm{Coequi}_{\mathcal{C}}(f,g)$. Thus $\mathrm{Equi}_{\mathcal{C}^{\mathrm{op}}}(f,g) \subset \mathrm{Coequi}_{\mathcal{C}}(f,g)$.

- Similarly, $\mathrm{Coequi}_{\mathcal{C}}(f,g) \subset \mathrm{Equi}_{\mathcal{C}^{\mathrm{op}}}(f,g)$. show details

Since $\mathrm{Equi}_{\mathcal{C}^{\mathrm{op}}}(f,g) \subset \mathrm{Coequi}_{\mathcal{C}}(f,g)$ and $\mathrm{Coequi}_{\mathcal{C}}(f,g) \subset \mathrm{Equi}_{\mathcal{C}^{\mathrm{op}}}(f,g)$, we have that $\mathrm{Equi}_{\mathcal{C}^{\mathrm{op}}}(f,g) = \mathrm{Coequi}_{\mathcal{C}}(f,g)$

(b) Similar

2. FINISH!!!

FINISH!!! □

3.4 Pullbacks and Pushouts

3.5 Exponentials

Definition 3.5.0.1. Let $\mathcal{C}$ be a category. Suppose that $\mathcal{C}$ has finite products. Let $B, C \in \mathrm{Obj}(\mathcal{C})$.

- – Let $D \in \mathrm{Obj}(\mathcal{C})$ and $\mathrm{eval} : D \times B \to C$. Then (D, eval) is said to be a **$\mathcal{C}$-exponential object of B and C** if for each $A \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A \times B, C)$, there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}}(A, D)$ such that $\mathrm{eval} \circ_{\mathcal{C}} (\phi \times \mathrm{id}_B^{\mathcal{C}}) = f$, i.e. the following diagram commutes:

$$\begin{tikzcd} A \arrow[d, dashed, "\phi"'] & A \times B \arrow[d, dashed, "\phi \times \mathrm{id}_B^{\mathcal{C}}"'] \arrow[dr, dashed, "f"] & \\ D & D \times B \arrow[r, "\mathrm{eval}"'] & C \end{tikzcd}$$

  – We define

$$\mathrm{Exp}_{\mathcal{C}}(B, C) := \left\{ (D, \mathrm{eval}) \in \coprod_{D \in \mathrm{Obj}(\mathcal{C})} \mathrm{Hom}_{\mathcal{C}}(D \times B, C) : (D, \mathrm{eval}) \text{ is a } \mathcal{C}\text{-exponential of } B \text{ and } C \right\}.$$

- We define define coexponential objects

Exercise 3.5.0.2. Let $\mathcal{C}$ be a category, A a set (class?) and $(X_\alpha)_{\alpha \in A} \in \mathrm{Obj}(\mathcal{C})^A$. Then

1. $\mathrm{Exp}_{\mathcal{C}^{\mathrm{op}}}(B, C) = \mathrm{Coexp}_{\mathcal{C}}(B, C)$.
2. $\mathrm{Coexp}_{\mathcal{C}^{\mathrm{op}}}(B, C) = \mathrm{Exp}_{\mathcal{C}}(B, C)$.

Proof. □

Exercise 3.5.0.3.

3.6 Monomorphisms and Epimorphisms

3.6.1 Introduction

Definition 3.6.1.1. Let $\mathcal{C}$ be a category and $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Then f is said to be

- a **monomorphism** if for each $C \in \mathrm{Obj}(C)$ and $g, h \in \mathrm{Hom}_{\mathcal{C}}(C, A)$, $f \circ g = f \circ h$ implies that $g = h$, i.e. we have the following implication of commutative diagrams:

$$\begin{tikzcd} C \arrow[dr, "g"] & & \\ & A \arrow[r, "f"] & B \\ C \arrow[ur, "h"'] & & \end{tikzcd} \implies \begin{tikzcd} C \arrow[r, bend left, "g"] \arrow[r, bend right, "h"'] & A \end{tikzcd}$$

- an **epimorphism** if for each $C \in \mathrm{Obj}(C)$ and $g, h \in \mathrm{Hom}_{\mathcal{C}}(B, C)$, $g \circ f = h \circ f$ implies that $g = h$, i.e. we have the following implication of commutative diagrams:

$$\begin{tikzcd} & & C \\ A \arrow[r, "f"] & B \arrow[ur, "g"] \arrow[dr, "h"'] & \\ & & C \end{tikzcd} \implies \begin{tikzcd} B \arrow[r, bend left, "g"] \arrow[r, bend right, "h"'] & C \end{tikzcd}$$

make note about common short-hand using parallel arrows

Definition 3.6.1.2. Let $\mathcal{C}$ be a category.

- We define
  - $\mathrm{Mono}_{\mathcal{C}} := \{f \in \mathrm{Hom}_{\mathcal{C}} : f \text{ is a monomorphism}\}$
  - $\mathrm{Epi}_{\mathcal{C}} := \{f \in \mathrm{Hom}_{\mathcal{C}} : f \text{ is a epimorphism}\}$
- Let $A, B \in \mathrm{Obj}(\mathcal{C})$. We define
  - $\mathrm{Mono}_{\mathcal{C}}(A, B) := \mathrm{Mono}_{\mathcal{C}} \cap \mathrm{Hom}_{\mathcal{C}}(A, B)$.
  - $\mathrm{Epi}_{\mathcal{C}}(A, B) := \mathrm{Epi}_{\mathcal{C}} \cap \mathrm{Hom}_{\mathcal{C}}(A, B)$.

Exercise 3.6.1.3. Let $\mathcal{C}$ be a category. Then

1. for each $A, B \in \mathrm{Obj}(\mathcal{C})$,
   (a) $\mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Epi}_{\mathcal{C}}(B, A)$
   (b) $\mathrm{Epi}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Mono}_{\mathcal{C}}(B, A)$
2. (a) $\mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}} = \mathrm{Epi}_{\mathcal{C}}$
   (b) $\mathrm{Epi}_{\mathcal{C}^{\mathrm{op}}} = \mathrm{Mono}_{\mathcal{C}}$

Proof. FIX!!!

1. (a) Let $A, B \in \mathrm{Obj}(\mathcal{C})$.

- Let $m \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$. Then $m \in \mathrm{Hom}_{\mathcal{C}}(B, A)$. Let $C \in \mathrm{Obj}(\mathcal{C})$ and $f, g \in \mathrm{Hom}_{\mathcal{C}}(A, C)$. Suppose that $f \circ_{\mathcal{C}} m = g \circ_{\mathcal{C}} m$. Then $m \circ_{\mathcal{C}^{\mathrm{op}}} f = m \circ_{\mathcal{C}^{\mathrm{op}}} g$. Since $m \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}$, we have that $f = g$. Since $C \in \mathrm{Obj}(\mathcal{C})$ and $f, g \in \mathrm{Hom}_{\mathcal{C}}(A, C)$ are arbitrary, we have that for each $C \in \mathrm{Obj}(\mathcal{C})$ and $f, g \in \mathrm{Hom}_{\mathcal{C}}(A, C)$, $f \circ_{\mathcal{C}} m = g \circ_{\mathcal{C}} m$ implies that $f = g$. Thus $m \in \mathrm{Epi}_{\mathcal{C}}(B, A)$. Since $m \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$ is arbitrary, we have that for each $m \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$, $m \in \mathrm{Epi}_{\mathcal{C}}(B, A)$. Hence $\mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B) \subset \mathrm{Epi}_{\mathcal{C}}(B, A)$.
- Let $e \in \mathrm{Epi}_{\mathcal{C}}(B, A)$. Then $e \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B)$. Let $C \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ and $f, g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(C, A)$. Suppose that $e \circ_{\mathcal{C}^{\mathrm{op}}} f = e \circ_{\mathcal{C}^{\mathrm{op}}} g$. Then $f \circ_{\mathcal{C}} e = g \circ_{\mathcal{C}} e$. Since $e \in \mathrm{Epi}_{\mathcal{C}}$, we have that $f = g$. Since $C \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ and $f, g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(C, A)$ are arbitrary, we have that for each $C \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ and $f, g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(C, A)$, $e \circ_{\mathcal{C}^{\mathrm{op}}} f = e \circ_{\mathcal{C}^{\mathrm{op}}} g$ implies that $f = g$. Thus $e \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$. Since $e \in \mathrm{Epi}_{\mathcal{C}}(B, A)$ is arbitrary, we have that for each $e \in \mathrm{Epi}_{\mathcal{C}}(B, A)$, $e \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$. Hence $\mathrm{Epi}_{\mathcal{C}}(B, A) \subset \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$.

Since $\mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B) \subset \mathrm{Epi}_{\mathcal{C}}(B, A)$ and $\mathrm{Epi}_{\mathcal{C}}(B, A) \subset \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$, we have that $\mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Epi}_{\mathcal{C}}(B, A)$.

(b)

2. (a) Since $A, B \in \mathrm{Obj}(\mathcal{C})$ are arbitrary, we have that for each $A, B \in \mathrm{Obj}(\mathcal{C})$, $\mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Epi}_{\mathcal{C}}(B, A)$. Hence

$$\begin{aligned}
\mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}} &= \bigcup_{(A,B) \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})^2} \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B) \\
&= \bigcup_{(A,B) \in \mathrm{Obj}(\mathcal{C})^2} \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B) \\
&= \bigcup_{(B,A) \in \mathrm{Obj}(\mathcal{C})^2} \mathrm{Epi}_{\mathcal{C}}(B, A) \\
&= \mathrm{Epi}_{\mathcal{C}}\,.
\end{aligned}$$

(b) The previous part (ref needed) implies that

$$\begin{aligned}
\mathrm{Mono}_{\mathcal{C}} &= \mathrm{Mono}_{(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}}} \\
&= \mathrm{Epi}_{\mathcal{C}^{\mathrm{op}}}\,.
\end{aligned}$$

□

Exercise 3.6.1.4. Let $\mathcal{C}$ be a category and $A, B, C \in \mathrm{Obj}(\mathcal{C})$. Then

1. for each $f \in \mathrm{Mono}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Mono}_{\mathcal{C}}(B, C)$, $g \circ_{\mathcal{C}} f \in \mathrm{Mono}_{\mathcal{C}}(A, C)$,
2. for each $f \in \mathrm{Epi}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Epi}_{\mathcal{C}}(B, C)$, $g \circ_{\mathcal{C}} f \in \mathrm{Epi}_{\mathcal{C}}(A, C)$.

Proof.

1. Let $f \in \mathrm{Mono}_{\mathcal{C}}(A, B)$, $g \in \mathrm{Mono}_{\mathcal{C}}(B, C)$, $X \in \mathrm{Obj}(\mathcal{C})$ and $\alpha, \beta \in \mathrm{Hom}_{\mathcal{C}}(X, A)$. We note that since

$$\begin{aligned}
(g \circ_{\mathcal{C}} f) \circ_{\mathcal{C}} \alpha = (g \circ_{\mathcal{C}} f) \circ_{\mathcal{C}} \beta &\iff g \circ_{\mathcal{C}} (f \circ_{\mathcal{C}} \alpha) = g \circ_{\mathcal{C}} (f \circ_{\mathcal{C}} \beta) \\
&\implies f \circ_{\mathcal{C}} \alpha = f \circ_{\mathcal{C}} \beta \quad (g \in \mathrm{Mono}_{\mathcal{C}}) \\
&\implies \alpha = \beta \quad (f \in \mathrm{Mono}_{\mathcal{C}}).
\end{aligned}$$

Since $X \in \mathrm{Obj}(\mathcal{C})$ and $\alpha, \beta \in \mathrm{Hom}_{\mathcal{C}}(X, A)$ with $(g \circ_{\mathcal{C}} f) \circ_{\mathcal{C}} \alpha = (g \circ_{\mathcal{C}} f) \circ_{\mathcal{C}} \beta$ are arbitrary, we have that for each $X \in \mathrm{Obj}(\mathcal{C})$ and $\alpha, \beta \in \mathrm{Hom}_{\mathcal{C}}(X, A)$, $(g \circ_{\mathcal{C}} f) \circ_{\mathcal{C}} \alpha = (g \circ_{\mathcal{C}} f) \circ_{\mathcal{C}} \beta$ implies that $\alpha = \beta$. Hence $g \circ_{\mathcal{C}} f \in \mathrm{Mono}_{\mathcal{C}}(A, C)$. Since $f \in \mathrm{Mono}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Mono}_{\mathcal{C}}(B, C)$ are arbitrary, we have that for each $f \in \mathrm{Mono}_{\mathcal{C}}(A, B)$, $g \in \mathrm{Mono}_{\mathcal{C}}(B, C)$, $g \circ_{\mathcal{C}} f \in \mathrm{Mono}_{\mathcal{C}}(A, C)$.

2. Let $f \in \mathrm{Epi}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Epi}_{\mathcal{C}}(B, C)$. Exercise 6.1.0.10 (ref prev ex) implies that $f \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(B, A)$ and $g \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(C, B)$, (ref prev part) then implies that

$$\begin{aligned}
g \circ_{\mathcal{C}} f &= f \circ_{\mathcal{C}^{\mathrm{op}}} g \\
&\in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}} \\
&= \mathrm{Epi}_{\mathcal{C}}(A, C).
\end{aligned}$$

Since $f \in \mathrm{Epi}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Epi}_{\mathcal{C}}(B, C)$ are arbitrary, we have that for each $f \in \mathrm{Epi}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Epi}_{\mathcal{C}}(B, C)$, $g \circ_{\mathcal{C}} f \in \mathrm{Epi}_{\mathcal{C}}(A, C)$.

□

Exercise 3.6.1.5. Let $\mathcal{C}$ be a category, $A, B, C \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(B, C)$. Then

1. $g \in \mathrm{Mono}_{\mathcal{C}}$ implies that $g \circ_{\mathcal{C}} f \in \mathrm{Mono}_{\mathcal{C}}$ iff $f \in \mathrm{Mono}_{\mathcal{C}}$.
2. $f \in \mathrm{Epi}_{\mathcal{C}}$ implies that $g \circ_{\mathcal{C}} f \in \mathrm{Epi}_{\mathcal{C}}$ iff $g \in \mathrm{Epi}_{\mathcal{C}}$.

Proof.

1. Suppose that $g \in \mathrm{Mono}_{\mathcal{C}}$.
   - ($\Longrightarrow$):
     Suppose that $g \circ_{\mathcal{C}} f \in \mathrm{Mono}_{\mathcal{C}}$. Let $D \in \mathrm{Obj}(\mathcal{C})$ and $\alpha, \beta \in \mathrm{Hom}_{\mathcal{C}}(D, A)$. Suppose that $f \circ_{\mathcal{C}} \alpha = f \circ_{\mathcal{C}} \beta$. Then
     $$\begin{aligned}(g \circ_{\mathcal{C}} f) \circ_{\mathcal{C}} \alpha \\ &= g \circ_{\mathcal{C}} (f \circ_{\mathcal{C}} \alpha) \\ &= g \circ_{\mathcal{C}} (f \circ_{\mathcal{C}} \beta) \\ &= (g \circ_{\mathcal{C}} f) \circ_{\mathcal{C}} \beta.\end{aligned}$$
     Since $g \circ_{\mathcal{C}} f \in \mathrm{Mono}_{\mathcal{C}}$, $\alpha = \beta$. Since $D \in \mathrm{Obj}(\mathcal{C})$ and $\alpha, \beta \in \mathrm{Hom}_{\mathcal{C}}(D, A)$ with $f \circ_{\mathcal{C}} \alpha = f \circ_{\mathcal{C}} \beta$ are arbitrary, we have that for each $D \in \mathrm{Obj}(\mathcal{C})$ and $\alpha, \beta \in \mathrm{Hom}_{\mathcal{C}}(D, A)$, $f \circ_{\mathcal{C}} \alpha = f \circ_{\mathcal{C}} \beta$ implies that $\alpha = \beta$. Thus $f \in \mathrm{Mono}_{\mathcal{C}}$.
   - ($\Longleftarrow$):
     Suppose that $f \in \mathrm{Mono}_{\mathcal{C}}$. Since $g \in \mathrm{Mono}_{\mathcal{C}}$, Exercise 6.1.0.10 (ref prev ex) implies that $g \circ_{\mathcal{C}} f \in \mathrm{Mono}_{\mathcal{C}}$.
2. Suppose that $f \in \mathrm{Epi}_{\mathcal{C}}$. Exercise 6.1.0.10 (ref prev ex abotu duality) implies that $f \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}$. Then
   $$\begin{aligned} g \circ_{\mathcal{C}} f \in \mathrm{Epi}_{\mathcal{C}} &\Longleftrightarrow g \circ_{\mathcal{C}} f \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}} \quad \text{(ref prev ex about duality)} \\ &\Longleftrightarrow f \circ_{\mathcal{C}^{\mathrm{op}}} g \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}} \\ &\Longleftrightarrow g \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}} \quad \text{(ref prev part)} \\ &\Longleftrightarrow g \in \mathrm{Epi}_{\mathcal{C}} \quad \text{(ref prev ex about duality)}. \end{aligned}$$

□

Exercise 3.6.1.6. Let $\mathcal{C}$ be a category and $A, B \in \mathrm{Obj}(\mathcal{C})$. Then $\mathrm{Iso}_{\mathcal{C}}(A, B) \subset \mathrm{Mono}_{\mathcal{C}}(A, B) \cap \mathrm{Epi}_{\mathcal{C}}(A, B)$.

Proof. Let $f \in \mathrm{Iso}_{\mathcal{C}}(A, B)$.

- (monomorphism):
  Let $C \in \mathrm{Obj}(\mathcal{C})$ and $g, h \in \mathrm{Hom}_{\mathcal{C}}(C, A)$. Suppose that $f \circ_{\mathcal{C}} g = f \circ_{\mathcal{C}} h$. Then
  $$\begin{aligned} g &= \mathrm{id}_A \circ_{\mathcal{C}} g \\ &= (f^{-1} \circ_{\mathcal{C}} f) \circ_{\mathcal{C}} g \\ &= f^{-1} \circ_{\mathcal{C}} (f \circ_{\mathcal{C}} g) \\ &= f^{-1} \circ_{\mathcal{C}} (f \circ_{\mathcal{C}} h) \\ &= (f^{-1} \circ_{\mathcal{C}} f) \circ_{\mathcal{C}} h \\ &= \mathrm{id}_A \circ_{\mathcal{C}} h \\ &= h \end{aligned}$$
  Since $g, h \in \mathrm{Hom}_{\mathcal{C}}(C, A)$ with $f \circ_{\mathcal{C}} g = f \circ_{\mathcal{C}} h$ are arbitrary, we have that for each $g, h \in \mathrm{Hom}_{\mathcal{C}}(C, A)$, $f \circ_{\mathcal{C}} g = f \circ_{\mathcal{C}} h$ implies that $g = h$. Therefore $f \in \mathrm{Mono}_{\mathcal{C}}(A, B)$.

- (epimorphism):
  Let $C \in \mathrm{Obj}(\mathcal{C})$ and $g, h \in \mathrm{Hom}_{\mathcal{C}}(B, C)$. Suppose that $g \circ_{\mathcal{C}} f = h \circ_{\mathcal{C}} f$. Then

$$\begin{aligned} g &= g \circ_{\mathcal{C}} \mathrm{id}_B \\ &= g \circ_{\mathcal{C}} (f \circ_{\mathcal{C}} f^{-1}) \\ &= (g \circ_{\mathcal{C}} f) \circ_{\mathcal{C}} f^{-1} \\ &= (h \circ_{\mathcal{C}} f) \circ_{\mathcal{C}} f^{-1} \\ &= h \circ_{\mathcal{C}} (f \circ_{\mathcal{C}} f^{-1}) \\ &= h \circ_{\mathcal{C}} \mathrm{id}_B \\ &= h \end{aligned}$$

  Since $g, h \in \mathrm{Hom}_{\mathcal{C}}(B, C)$ with $g \circ_{\mathcal{C}} f = h \circ_{\mathcal{C}} f$ are arbitrary, we have that for each $g, h \in \mathrm{Hom}_{\mathcal{C}}(B, C)$, $g \circ_{\mathcal{C}} f = h \circ_{\mathcal{C}} f$ implies that $g = h$. Therefore $f \in \mathrm{Epi}_{\mathcal{C}}(A, B)$.

Thus $f \in \mathrm{Mono}_{\mathcal{C}}(A, B) \cap \mathrm{Epi}_{\mathcal{C}}(A, B)$. Since $f \in \mathrm{Iso}_{\mathcal{C}}(A, B)$ is arbitrary, we have that for each $f \in \mathrm{Iso}_{\mathcal{C}}(A, B)$, $f \in \mathrm{Mono}_{\mathcal{C}}(A, B) \cap \mathrm{Epi}_{\mathcal{C}}(A, B)$. Hence $\mathrm{Iso}_{\mathcal{C}}(A, B) \subset \mathrm{Mono}_{\mathcal{C}}(A, B) \cap \mathrm{Epi}_{\mathcal{C}}(A, B)$. □

Exercise 3.6.1.7. Let $\mathcal{C}$ be a category and $X \in \mathrm{Obj}(\mathcal{C})$. Then $\mathrm{id}_X \in \mathrm{Mono}_{\mathcal{C}} \cap \mathrm{Epi}_{\mathcal{C}}$.

Proof. ref prev ex in isomorphism section implies that $\mathrm{id}_X \in \mathrm{Aut}_{\mathcal{C}}(X)$. ref prev ex implies that $\mathrm{id}_X \in \mathrm{Mono}_{\mathcal{C}} \cap \mathrm{Epi}_{\mathcal{C}}$. □

Exercise 3.6.1.8. mono/epi in **Set** Let $A, B \in \mathrm{Obj}(\mathbf{Set})$ and $f \in \mathrm{Hom}_{\mathbf{Set}}(A, B)$. Then

1. f is a monomorphism iff f is injective
2. f is an epimorphism iff f is surjective

Hint: consider $C := \{0\}$ and $C := \{0, 1\}$.

Proof.

1. - ($\Longrightarrow$):
     Suppose that f is injective. Let $C \in \mathrm{Obj}(\mathbf{Set})$ and $g, h \in \mathrm{Hom}_{\mathbf{Set}}(C, A)$. Suppose that $f \circ_{\mathbf{Set}} g = f \circ_{\mathbf{Set}} h$. Let $x \in C$. Then $f(g(x)) = f(h(x))$. Injectivity of f implies that $g(x) = h(x)$. Since $x \in C$ is arbitrary, $g = h$. Hence $f \in \mathrm{Mono}_{\mathbf{Set}}$.

   - ($\Longleftarrow$):
     Conversely, suppose that $f \in \mathrm{Mono}_{\mathbf{Set}}$. Let $a, b \in A$. Suppose that $f(a) = f(b)$. Set $C = \{0\}$ and define $g, h : C \to A$ by $g(0) = a$ and $h(0) = b$. Then

$$\begin{aligned} f \circ_{\mathbf{Set}} g(0) &= f(g(0)) \\ &= f(a) \\ &= f(b) \\ &= f(h(0)) \\ &= f \circ_{\mathbf{Set}} h(0) \end{aligned}$$

     Therefore $f \circ_{\mathbf{Set}} g = f \circ_{\mathbf{Set}} h$. Since $f \in \mathrm{Mono}_{\mathbf{Set}}$, we have that $g = h$. Hence

$$\begin{aligned} a &= g(0) \\ &= h(0) \\ &= b \end{aligned}$$

     Since $a, b \in A$ with $f(a) = f(b)$ are arbitrary, we have that for each $a, b \in A$, $f(a) = f(b)$ implies that $a = b$. Thus f is injective.

2. • ($\Longrightarrow$):
Suppose that f is surjective. Let $C \in \mathrm{Obj}(\mathcal{C})$ and $g, h \in \mathrm{Hom}_{\mathbf{Set}}(B, C)$. Suppose that $g \circ_{\mathbf{Set}} f = h \circ_{\mathbf{Set}} f$. Let $y \in B$. Surjective of f implies that there exists $x \in A$ such that $y = f(x)$. Then

$$\begin{aligned} g(y) &= g(f(x)) \\ &= g \circ_{\mathbf{Set}} f(x) \\ &= h \circ_{\mathbf{Set}} f(x) \\ &= h(f(x)) \\ &= h(y) \end{aligned}$$

Since $y \in B$ is arbitrary, we have that for each $y \in B$, $g(y) = h(y)$. Hence $g = h$. Since $C \in \mathrm{Obj}(\mathcal{C})$ and $g, h \in \mathrm{Hom}_{\mathbf{Set}}(B, C)$ with $g \circ_{\mathbf{Set}} f = h \circ_{\mathbf{Set}} f$ are arbitrary, we have that for each $C \in \mathrm{Obj}(\mathcal{C})$ and $g, h \in \mathrm{Hom}_{\mathbf{Set}}(B, C)$, $g \circ_{\mathbf{Set}} f = h \circ_{\mathbf{Set}} f$ implies that $g = h$. Hence f is an epimorphism.

• ($\Longleftarrow$):
Conversely, suppose that f is an epimorphism. Define $C \in \mathrm{Obj}(\mathbf{Set})$ and $g, h : B \to C$ by $C := \{0, 1\}$, $g := \chi_{f(A)}$ and $h := \chi_B$. Then $g \circ_{\mathbf{Set}} f = h \circ_{\mathbf{Set}} f$. Since $f \in \mathrm{Epi}_{\mathcal{C}}$, $g = h$ and $f(A) = B$. Hence f is surjective.

□

3.6.2 Extremal Monomorphisms and Epimorphisms

Definition 3.6.2.1. Let $\mathcal{C}$ be a category, $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Then f is said to be an

- **extremal monomorphism** if
  - $f \in \mathrm{Mono}_{\mathcal{C}}$
  - for each $C \in \mathrm{Obj}(\mathcal{C})$, $e \in \mathrm{Epi}_{\mathcal{C}}(A, C)$ and $g \in \mathrm{Hom}_{\mathcal{C}}(C, B)$, $f = g \circ_{\mathcal{C}} e$ implies that $e \in \mathrm{Iso}_{\mathcal{C}}(A, C)$.
- **extremal epimorphism** if
  - $f \in \mathrm{Epi}_{\mathcal{C}}$
  - for each $C \in \mathrm{Obj}(\mathcal{C})$, $g \in \mathrm{Hom}_{\mathcal{C}}(A, C)$ and $m \in \mathrm{Mono}_{\mathcal{C}}(C, B)$, $f = m \circ_{\mathcal{C}} g$ implies that $m \in \mathrm{Iso}_{\mathcal{C}}(C, B)$.

Definition 3.6.2.2. Let $\mathcal{C}$ be a category.

- We define
  - $\mathrm{ExtMono}_{\mathcal{C}} := \{f \in \mathrm{Hom}_{\mathcal{C}} : f \text{ is an extremal monomorphism}\}$,
  - $\mathrm{ExtEpi}_{\mathcal{C}} := \{f \in \mathrm{Hom}_{\mathcal{C}} : f \text{ is an extremal epiorphism}\}$.
- Let $A, B \in \mathrm{Obj}(\mathcal{C})$. We define
  - $\mathrm{ExtMono}_{\mathcal{C}}(A, B) := \mathrm{ExtMono}_{\mathcal{C}} \cap \mathrm{Hom}_{\mathcal{C}}(A, B)$,
  - $\mathrm{ExtEpi}_{\mathcal{C}}(A, B) := \mathrm{ExtEpi}_{\mathcal{C}} \cap \mathrm{Hom}_{\mathcal{C}}(A, B)$.

Exercise 3.6.2.3. Let $\mathcal{C}$ be a category. Then

1. for each $A, B \in \mathrm{Obj}(\mathcal{C})$,
   (a) $\mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$
   (b) $\mathrm{ExtEpi}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{ExtMono}_{\mathcal{C}}(B, A)$
2. (a) $\mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}} = \mathrm{ExtEpi}_{\mathcal{C}}$
   (b) $\mathrm{ExtEpi}_{\mathcal{C}^{\mathrm{op}}} = \mathrm{ExtMono}_{\mathcal{C}}$

Proof. FIX!!!

1. Let $A, B \in \mathrm{Obj}(\mathcal{C})$.

   - Let $f \in \mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$.
     - Then $f \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$. ref prev ex in section on monomorphisms implies that $f \in \mathrm{Epi}_{\mathcal{C}}(B, A)$.
     - Let $C \in \mathrm{Obj}(\mathcal{C})$, $g \in \mathrm{Hom}_{\mathcal{C}}(B, C)$ and $m \in \mathrm{Mono}_{\mathcal{C}}(C, A)$. Suppose that $f = m \circ_{\mathcal{C}} g$. Then $g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(C, B)$, a prev ex in section on monomorphs implies that $m \in \mathrm{Epi}_{\mathcal{C}^{\mathrm{op}}}(A, C)$ and $f = g \circ_{\mathcal{C}^{\mathrm{op}}} m$. Since $f \in \mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$, we have that $m \in \mathrm{Iso}_{\mathcal{C}^{\mathrm{op}}}(B, C)$. Exercise 6.1.0.10 (make and ref prev ex that $\mathrm{Iso}_{\mathcal{C}} = \mathrm{Iso}_{\mathcal{C}^{\mathrm{op}}}$) implies that $m \in \mathrm{Iso}_{\mathcal{C}}(C, B)$. Since $C \in \mathrm{Obj}(\mathcal{C})$, $g \in \mathrm{Hom}_{\mathcal{C}}(B, C)$ and $m \in \mathrm{Mono}_{\mathcal{C}}(C, A)$ are arbitrary, we have that for each $C \in \mathrm{Obj}(\mathcal{C})$, $g \in \mathrm{Hom}_{\mathcal{C}}(B, C)$ and $m \in \mathrm{Mono}_{\mathcal{C}}(C, A)$, $f = m \circ_{\mathcal{C}} g$ implies that $m \in \mathrm{Iso}_{\mathcal{C}}(C, A)$.

   Thus $f \in \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$. Since $f \in \mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$ is arbitrary, we have that for each $f \in \mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$, $f \in \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$. Hence $\mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B) \subset \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$.

   - Let $f \in \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$.
     - Then $f \in \mathrm{Epi}_{\mathcal{C}}(B, A)$. ref prev ex in section on monomorphisms implies that $f \in \mathrm{Mono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$.
     - Let $C \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$, $e \in \mathrm{Epi}_{\mathcal{C}^{\mathrm{op}}}(A, C)$ and $g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(C, B)$. Suppose that $f = g \circ_{\mathcal{C}^{\mathrm{op}}} e$. Then $g \in \mathrm{Hom}_{\mathcal{C}}(B, C)$, a prev ex in section on monomorphs implies that $e \in \mathrm{Mono}_{\mathcal{C}}(C, A)$ and $f = e \circ_{\mathcal{C}} g$. Since $f \in \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$, we have that $e \in \mathrm{Iso}_{\mathcal{C}}(C, A)$. Exercise 6.1.0.10 (make and ref prev ex that $\mathrm{Iso}_{\mathcal{C}} = \mathrm{Iso}_{\mathcal{C}^{\mathrm{op}}}$) implies that $e \in \mathrm{Iso}_{\mathcal{C}^{\mathrm{op}}}(A, C)$. Since $C \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$, $e \in \mathrm{Epi}_{\mathcal{C}^{\mathrm{op}}}(A, C)$ and $g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(C, B)$ are arbitrary, we have that for each $C \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$, $e \in \mathrm{Epi}_{\mathcal{C}^{\mathrm{op}}}(A, C)$ and $g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(C, B)$, $f = g \circ_{\mathcal{C}^{\mathrm{op}}} e$ implies that $e \in \mathrm{Iso}_{\mathcal{C}^{\mathrm{op}}}(A, C)$.

   Thus $f \in \mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$. Since $f \in \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$ is arbitrary, we have that for each $f \in \mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B)$, $f \in \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$. Hence $\mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B) \subset \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$.

   Since $\mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B) \subset \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$ and $\mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B) \subset \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$, we have that $\mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$. Since $A, B \in \mathrm{Obj}(\mathcal{C})$ are arbitrary, we have that for each $A, B \in \mathrm{Obj}(\mathcal{C})$, $\mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{ExtEpi}_{\mathcal{C}}(B, A)$. Therefore

$$\begin{aligned}
\mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}} &= \bigcup_{(A,B) \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})^2} \mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B) \\
&= \bigcup_{(A,B) \in \mathrm{Obj}(\mathcal{C})^2} \mathrm{ExtMono}_{\mathcal{C}^{\mathrm{op}}}(A, B) \\
&= \bigcup_{(B,A) \in \mathrm{Obj}(\mathcal{C})^2} \mathrm{ExtEpi}_{\mathcal{C}}(B, A) \\
&= \mathrm{ExtEpi}_{\mathcal{C}}\,.
\end{aligned}$$

2. The previous part (ref needed) implies that

$$\begin{aligned}
\mathrm{ExtMono}_{\mathcal{C}} &= \mathrm{ExtMono}_{(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}}} \\
&= \mathrm{ExtEpi}_{\mathcal{C}^{\mathrm{op}}}\,.
\end{aligned}$$

□

3.6.3 Regular Monomorphisms and Epimorphisms

Exercise 3.6.3.1. Let $\mathcal{C}$ be a category, $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f, g \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Rewrite using notation $\mathrm{Equi}_{\mathcal{C}}$

1. Let $M \in \mathrm{Obj}(\mathcal{C})$ and $m \in \mathrm{Hom}_{\mathcal{C}}(M, A)$. If (M, m) is an equalizer of f and g, then $m \in \mathrm{Mono}_{\mathcal{C}}(M, A)$.

2. Let $E \in \mathrm{Obj}(\mathcal{C})$ and $e \in \mathrm{Hom}_{\mathcal{C}}(B, E)$. If (E, e) is a coequalizer of f and g, then $e \in \mathrm{Epi}_{\mathcal{C}}(B, E)$.

Proof.

1. Suppose that (M, m) is an equalizer of f and g. Let $C \in \mathrm{Obj}(\mathcal{C})$ and $\alpha, \beta \in \mathrm{Hom}_{\mathcal{C}}(C, M)$. Suppose that $m \circ_{\mathcal{C}} \alpha = m \circ_{\mathcal{C}} \beta$. Since (M, m) is an equalizer of f and g, we have that $f \circ_{\mathcal{C}} m = g \circ_{\mathcal{C}} m$, i.e. the following diagram commutes:

$$\begin{tikzcd} & & B \\ M \arrow[r, "m"] & A \arrow[ur, "f"] \arrow[dr, "g"'] & \\ & & B \end{tikzcd}$$

Define $M' \in \mathrm{Obj}(\mathcal{C})$ and $m' \in \mathrm{Hom}_{\mathcal{C}}(M', A)$ by $M' := C$ and $m' := m \circ_{\mathcal{C}} \alpha$. Then

$$\begin{aligned} f \circ_{\mathcal{C}} m' &= f \circ_{\mathcal{C}} (m \circ_{\mathcal{C}} \alpha) \\ &= (f \circ_{\mathcal{C}} m) \circ_{\mathcal{C}} \alpha \\ &= (g \circ_{\mathcal{C}} m) \circ_{\mathcal{C}} \alpha \\ &= g \circ_{\mathcal{C}} (m \circ_{\mathcal{C}} \alpha) \\ &= g \circ_{\mathcal{C}} m'. \end{aligned}$$

Since (E, m) is an equalizer of f and g, there exists a unique $\phi \in \mathrm{Hom}_{\mathcal{C}}(E', E)$ such that $m \circ_{\mathcal{C}} \phi = m'$, i.e. the following diagram commutes:

$$\begin{tikzcd} E' \arrow[dr, "m'"] \arrow[dd, dashed, "\phi"'] & & B \\ & A \arrow[ur, "f"] \arrow[dr, "g"'] & \\ E \arrow[ur, "m"'] & & B \end{tikzcd}$$

Since $m \circ_{\mathcal{C}} \alpha = m'$, uniqueness of ϕ implies that $\phi = \alpha$. Since

$$\begin{aligned} m \circ_{\mathcal{C}} \beta &= m \circ_{\mathcal{C}} \alpha \\ &= m', \end{aligned}$$

uniqueness of ϕ implies that $\phi = \beta$. Hence $\alpha = \beta$. Since $C \in \mathrm{Obj}(\mathcal{C})$ and $\alpha, \beta \mathrm{Hom}_{\mathcal{C}}(C, E)$ with $m \circ_{\mathcal{C}} \alpha = m \circ_{\mathcal{C}} \beta$ are arbitrary, we have that for each $C \in \mathrm{Obj}(\mathcal{C})$ and $\alpha, \beta \in \mathrm{Hom}_{\mathcal{C}}(C, E)$, $m \circ_{\mathcal{C}} \alpha = m \circ_{\mathcal{C}} \beta$ implies that $\alpha = \beta$, i.e. we have the following implication of commutative diagrams:

$$\begin{tikzcd} C \arrow[dr, "\alpha"] & & \\ & E \arrow[r, "m"] & A \\ C \arrow[ur, "\beta"'] & & \end{tikzcd} \implies \begin{tikzcd} C \arrow[r, bend left, "\alpha"] \arrow[r, bend right, "\beta"'] & E \end{tikzcd}$$

Thus $m \in \mathrm{Mono}_{\mathcal{C}}(E, A)$.

2. FINISH!!!

$\square$

Definition 3.6.3.2. Let $\mathcal{C}$ be a category, $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$. Then f is said to be a

- **regular monomorphism** if there exists $C \in \mathrm{Obj}(\mathcal{C})$ and $g, h \in \mathrm{Hom}_{\mathcal{C}}(B, C)$ such that $(A, f) \in \mathrm{Equi}_{\mathcal{C}}(g, h)$.

- **regular epimorphism** if there exists $C \in \mathrm{Obj}(\mathcal{C})$ and $g, h \in \mathrm{Hom}_{\mathcal{C}}(C, A)$ such that $(B, f) \in \mathrm{Coequi}_{\mathcal{C}}(g, h)$.

Definition 3.6.3.3. Let $\mathcal{C}$ be a category.

- We define
  - $\mathrm{RegMono}_{\mathcal{C}} := \{f \in \mathrm{Hom}_{\mathcal{C}} : f \text{ is a regular monomorphism}\}$,
  - $\mathrm{RegEpi}_{\mathcal{C}} := \{f \in \mathrm{Hom}_{\mathcal{C}} : f \text{ is a regular epiorphism}\}$.
- Let $A, B \in \mathrm{Obj}(\mathcal{C})$. We define
  - $\mathrm{RegMono}_{\mathcal{C}}(A, B) := \mathrm{RegMono}_{\mathcal{C}} \cap \mathrm{Hom}_{\mathcal{C}}(A, B)$,
  - $\mathrm{RegEpi}_{\mathcal{C}}(A, B) := \mathrm{RegEpi}_{\mathcal{C}} \cap \mathrm{Hom}_{\mathcal{C}}(A, B)$.

Exercise 3.6.3.4. Let $\mathcal{C}$ be a category. Then

1. for each $A, B \in \mathrm{Obj}(\mathcal{C})$,
   (a) $\mathrm{RegMono}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{RegEpi}_{\mathcal{C}}(B, A)$
   (b) $\mathrm{RegEpi}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{RegMono}_{\mathcal{C}}(B, A)$
2. (a) $\mathrm{RegMono}_{\mathcal{C}^{\mathrm{op}}} = \mathrm{RegEpi}_{\mathcal{C}}$
   (b) $\mathrm{RegEpi}_{\mathcal{C}^{\mathrm{op}}} = \mathrm{RegMono}_{\mathcal{C}}$

Proof. FINISH!!! □

Exercise 3.6.3.5. Let $\mathcal{C}$ be a category. Then for each $A, B \in \mathrm{Obj}(\mathcal{C})$,

1. $\mathrm{RegMono}_{\mathcal{C}}(A, B) \subset \mathrm{ExtMono}_{\mathcal{C}}(A, B)$,
2. $\mathrm{RegEpi}_{\mathcal{C}}(A, B) \subset \mathrm{ExtEpi}_{\mathcal{C}}(A, B)$.

Proof. Let $A, B \in \mathrm{Obj}(\mathcal{C})$. FINISH!!!

1.
2.

□

Exercise 3.6.3.6. Let $A, B \in \mathrm{Obj}(\mathbf{Set})$ and $f, g \in \mathrm{Hom}_{\mathbf{Set}}(A, B)$. Then $(\mathrm{Equi}(f, g), \iota_{\mathrm{Equi}(f,g)}) \in \mathrm{Equi}_{\mathbf{Set}}(f, g)$.

Proof. Define $E \in \mathrm{Obj}(\mathbf{Set})$ by $E := \mathrm{Equi}(f, g)$.

1. By definition, $E = \{x \in A : f(x) = g(x)\}$. Therefore for each $x \in E$,

$$\begin{aligned} f \circ_{\mathbf{Set}} \iota_E(x) &= f(\iota_E(x)) \\ &= f(x) \\ &= g(x) \\ &= g(\iota_E(x)) \\ &= g \circ_{\mathbf{Set}} \iota_E(x). \end{aligned}$$

   Thus $f \circ_{\mathbf{Set}} \iota_E = g \circ_{\mathbf{Set}} \iota_E$.

2. Let $E' \in \mathrm{Obj}(\mathbf{Set})$ and $e' \in \mathrm{Hom}_{\mathbf{Set}}(E', A)$. Suppose that $f \circ_{\mathbf{Set}} e' = g \circ_{\mathbf{Set}} e'$. Let $x \in E'$. Define $a \in A$ by $a := e'(x)$. Then

$$\begin{aligned} f(a) &= f(e'(x)) \\ &= f \circ_{\mathbf{Set}} e'(x) \\ &= f \circ_{\mathbf{Set}} e'(x) \\ &= g(e'(x)) \\ &= g(a). \end{aligned}$$

Thus $a \in \mathrm{Equi}(f, g)$. Therefore

$$\begin{aligned} e'(x) &= a \\ &\in \mathrm{Equi}(f, g) \\ &= E. \end{aligned}$$

Since $x \in E'$ is arbitrary, we have that for each $x \in E'$, $e'(x) \in E$. Define $\phi \in \mathrm{Hom}_{\mathbf{Set}}(E', E)$ by $\phi(x) := e'(x)$.

- For each $x \in E'$,

$$\begin{aligned} \iota_E \circ_{\mathbf{Set}} \phi(x) \\ &= \iota_E(\phi(x)) \\ &= \iota_E(e'(x)) \\ &= e'(x). \end{aligned}$$

  Thus $\iota_E \circ_{\mathbf{Set}} \phi = e'$.
- Let $\phi' \in \mathrm{Hom}_{\mathbf{Set}}(E', E)$. Suppose that $\iota_E \circ_{\mathbf{Set}} \phi' = e'$. Then for each $x \in E'$,

$$\begin{aligned} \phi'(x) &= \iota_E(\phi'(x)) \\ &= \iota_E \circ_{\mathbf{Set}} \phi'(x) \\ &= e'(x) \\ &= \iota_E \circ_{\mathbf{Set}} \phi(x) \\ &= \iota_E(\phi(x)) && = \phi(x). \end{aligned}$$

  Hence $\phi' = \phi$. Since $\phi' \in \mathrm{Hom}_{\mathbf{Set}}(E', E)$ with $\iota_E \circ_{\mathbf{Set}} \phi' = e'$ is arbitrary, we have that for each $\phi' \in \mathrm{Hom}_{\mathbf{Set}}(E', E)$, $\iota_E \circ_{\mathbf{Set}} \phi' = e'$ implies that $\phi' = \phi$.

Thus there exists a unique $\phi \in \mathrm{Hom}_{\mathbf{Set}}(E', E)$ such that $\iota_E \circ_{\mathbf{Set}} \phi = e'$. Since $E' \in \mathrm{Obj}(\mathbf{Set})$ and $e' \in \mathrm{Hom}_{\mathbf{Set}}(E', A)$ with $f \circ_{\mathbf{Set}} e' = g \circ_{\mathbf{Set}} e'$ are arbitrary, we have that for each $E' \in \mathrm{Obj}(\mathbf{Set})$ and $e' \in \mathrm{Hom}_{\mathbf{Set}}(E', A)$, $f \circ_{\mathbf{Set}} e' = g \circ_{\mathbf{Set}} e'$ implies that there exists a unique $\phi \in \mathrm{Hom}_{\mathbf{Set}}(E', E)$ such that $\iota_E \circ_{\mathbf{Set}} \phi = e'$.

Thus $(E, \iota_E) \in \mathrm{Equi}_{\mathbf{Set}}(f, g)$. □

Exercise 3.6.3.7. We have that $\mathrm{RegMono}_{\mathbf{Set}} = \mathrm{Mono}_{\mathbf{Set}}$.

Proof.

- a prev ex implies that $\mathrm{RegMono}_{\mathbf{Set}} \subset \mathrm{Mono}_{\mathbf{Set}}$.
- Let $m \in \mathrm{Mono}_{\mathbf{Set}}$. Define $M, A \in \mathrm{Obj}(\mathbf{Set})$ by $M := \mathrm{dom}_{\mathbf{Set}}(m)$ and $A := \mathrm{cod}_{\mathbf{Set}}(m)$. a prev ex implies that m is injective. Define $B \in \mathrm{Obj}(\mathbf{Set})$ and $f, g \in \mathrm{Hom}_{\mathbf{Set}}(A, B)$ by $B := m(M) \cup \{0, 1\}$ and

$$f(x) := \begin{cases} x, & x \in m(M) \\ 0, & x \in m(M)^c, \end{cases} \quad g(x) := \begin{cases} x, & x \in m(M) \\ 1, & x \in m(M)^c. \end{cases}$$

  Define $E \in \mathrm{Obj}(\mathbf{Set})$ by $E := \mathrm{Equi}(f, g)$. Then

$$\begin{aligned} E &= \mathrm{Equi}(f, g) \\ &= \{x \in A : f(x) = g(x)\} \\ &= m(M). \end{aligned}$$

1. By construction, for each $x \in M$,

$$\begin{aligned} f \circ_{\mathbf{Set}} m(x) &= f(m(x)) \\ &= g(m(x)) \quad (\text{since } m(x) \in E) \\ &= g \circ_{\mathbf{Set}} m(x). \end{aligned}$$

Thus $f \circ_{\mathbf{Set}} m = g \circ_{\mathbf{Set}} m$, i.e. the following diagram commutes:

$$\begin{tikzcd} & & B \\ M \arrow[r, "m"] & A \arrow[ur, "f"] \arrow[dr, "g"'] & \\ & & B \end{tikzcd}$$

2. Let $M' \in \mathrm{Obj}(\mathbf{Set})$ and $m' \in \mathrm{Hom}_{\mathbf{Set}}(M', A)$. Suppose that $f \circ_{\mathbf{Set}} m' = g \circ_{\mathbf{Set}} m'$. Exercise 6.1.0.10 ref prev ex implies that $(E, \iota_E) \in \mathrm{Equi}_{\mathbf{Set}}(f, g)$. Thus
   - there exists a unique $\alpha \in \mathrm{Hom}_{\mathbf{Set}}(M, E)$ such that $\iota_E \circ_{\mathbf{Set}} \alpha = m$,
   - there exists a unique $\alpha' \in \mathrm{Hom}_{\mathbf{Set}}(M', E)$ such that $\iota_E \circ_{\mathbf{Set}} \alpha' = m'$,

   i.e. the following diagram commutes:

$$\begin{tikzcd} M' \arrow[dr, "m'"] \arrow[ddr, bend right, dashed, "\alpha'"'] & & B \\ & A \arrow[ur, "f"] \arrow[dr, "g"'] & \\ M \arrow[r, "m"] \arrow[d, dashed, "\alpha"'] & & B \\ E \arrow[uur, "\iota_E"'] & & \end{tikzcd}$$

Since $\iota_E \circ m|^E = m$, uniqueness of α implies that $\alpha = m|^E$. Since m is injective, $m|^E \in \mathrm{Iso}_{\mathbf{Set}}(M, E)$.

- Define $\phi \in \mathrm{Hom}_{\mathbf{Set}}(M', M)$ by $\phi' := \alpha^{-1} \circ_{\mathbf{Set}} \alpha'$. Then

$$\begin{aligned} m \circ_{\mathbf{Set}} \phi &= (\iota_E \circ_{\mathbf{Set}} m|^E) \circ_{\mathbf{Set}} (\alpha^{-1} \circ_{\mathbf{Set}} \alpha') \\ &= \iota_E \circ_{\mathbf{Set}} (m|^E \circ_{\mathbf{Set}} \alpha^{-1}) \circ_{\mathbf{Set}} \alpha' \\ &= \iota_E \circ_{\mathbf{Set}} (\alpha \circ_{\mathbf{Set}} \alpha^{-1}) \circ_{\mathbf{Set}} \alpha' \\ &= \iota_E \circ_{\mathbf{Set}} \mathrm{id}_E \circ_{\mathbf{Set}} \alpha' \\ &= \iota_E \circ_{\mathbf{Set}} \alpha' \\ &= m'. \end{aligned}$$

- * Let $\phi_0 \in \mathrm{Hom}_{\mathbf{Set}}(M', M)$. Suppose that $m \circ \phi_0 = m'$. Then

$$\begin{aligned} \iota_E \circ_{\mathbf{Set}} (\alpha \circ_{\mathbf{Set}} \phi_0) &= (\iota_E \circ_{\mathbf{Set}} \alpha) \circ_{\mathbf{Set}} \phi_0 \\ &= (\iota_E \circ_{\mathbf{Set}} m|^E) \circ_{\mathbf{Set}} \phi_0 \\ &= m \circ_{\mathbf{Set}} \phi_0 \\ &= m'. \end{aligned}$$

  Uniqueness of α' implies that $\alpha \circ_{\mathbf{Set}} \phi_0 = \alpha'$. Since $\phi_0 \in \mathrm{Hom}_{\mathbf{Set}}(M', M)$ such that $m \circ \phi_0 = m'$ is arbitrary, we have that for each $\phi_0 \in \mathrm{Hom}_{\mathbf{Set}}(M', M)$, $m \circ \phi_0 = m'$ implies that $\alpha \circ_{\mathbf{Set}} \phi_0 = \alpha'$.
  * In particular, ref prev part (above) implies that $\alpha \circ_{\mathbf{Set}} \phi = \alpha'$.

* Let $\phi' \in \mathrm{Hom}_{\mathbf{Set}}(M', M)$. Suppose that $m \circ \phi' = m'$. ref above prev parts implies that

$$\begin{aligned} \alpha \circ_{\mathbf{Set}} \phi' &= \alpha' \\ &= \alpha \circ_{\mathbf{Set}} \phi. \end{aligned}$$

Since

$$\begin{aligned} \alpha &= m|^E \\ &\in \mathrm{Iso}_{\mathbf{Set}}(M, E) \\ &\subset \mathrm{Mono}_{\mathbf{Set}}(M, E), \end{aligned}$$

we have that $\phi' = \phi$. Since $\phi' \in \mathrm{Hom}_{\mathbf{Set}}(M', M)$ with $m \circ \phi' = m'$ is arbitrary, we have that for each $\phi' \in \mathrm{Hom}_{\mathbf{Set}}(M', M)$, $m \circ \phi' = m'$ implies that $\phi' = \phi$.

Thus there exists a unique $\phi \in \mathrm{Hom}_{\mathbf{Set}}(M', M)$ such that $m \circ_{\mathbf{Set}} \phi = m'$. Since $M' \in \mathrm{Obj}(\mathbf{Set})$ and $m' \in \mathrm{Hom}_{\mathbf{Set}}(M', A)$ with $f \circ_{\mathbf{Set}} m' = g \circ_{\mathbf{Set}} m'$ are arbitrary, we have that for each $M' \in \mathrm{Obj}(\mathbf{Set})$ and $m' \in \mathrm{Hom}_{\mathbf{Set}}(M', A)$, $f \circ_{\mathbf{Set}} m' = g \circ_{\mathbf{Set}} m'$ implies that there exists a unique $\phi \in \mathrm{Hom}_{\mathbf{Set}}(M', M)$ such that $m \circ_{\mathbf{Set}} \phi = m'$, i.e. the following diagram commutes:

$$\begin{tikzcd} M' \arrow[dd, dashed, "\phi"'] \arrow[dr, "m'"] & & B \\ & A \arrow[ur, "f"] \arrow[dr, "g"] & \\ M \arrow[ur, "m"] & & B \end{tikzcd}$$

Therefore $(M, m) \in \mathrm{Equi}_{\mathbf{Set}}(f, g)$. Hence $m \in \mathrm{RegMono}_{\mathbf{Set}}$. Since $m \in \mathrm{Mono}_{\mathbf{Set}}$ is arbitrary, we have that for each $m \in \mathrm{Mono}_{\mathbf{Set}}$, $m \in \mathrm{RegMono}_{\mathbf{Set}}$. Thus $\mathrm{Mono}_{\mathbf{Set}} \subset \mathrm{RegMono}_{\mathbf{Set}}$.

Since $\mathrm{RegMono}_{\mathbf{Set}} \subset \mathrm{Mono}_{\mathbf{Set}}$ and $\mathrm{Mono}_{\mathbf{Set}} \subset \mathrm{RegMono}_{\mathbf{Set}}$, we have that $\mathrm{RegMono}_{\mathbf{Set}} = \mathrm{Mono}_{\mathbf{Set}}$. $\square$

3.7 Subobjects and Quotient Objects

Definition 3.7.0.1. maybe move this to subsection on subcategories, define wide category generated by hom closed morphisms Let $\mathcal{C}$ be a category and $H \subset \mathrm{Hom}_{\mathcal{C}}$. Then H is said to be **hom closed** if

1.

2.

Definition 3.7.0.2. define wide category with all morphisms mono, resp epi Let $\mathcal{C}$ be a category. We define

- **the restriction of $\mathcal{C}$ to $\mathrm{Mono}_{\mathcal{C}}$,** denoted $\mathbf{Mono}(\mathcal{C}, M)$, by
  - $\mathrm{Obj}(\mathbf{Mono}(\mathcal{C}, M)) := \mathrm{Obj}(\mathcal{C})$
  - For $A, B \in \mathrm{Obj}(\mathcal{C})$, $\mathrm{Hom}_{\mathbf{Mono}(\mathcal{C},M)}(A, B) := \mathrm{Mono}_{\mathcal{C}}(A, B)$
  - For $A, B, C \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathbf{Mono}(\mathcal{C},M)}(A, B)$ and $g \in \mathrm{Hom}_{\mathbf{Mono}(\mathcal{C},M)}(B, C)$, $g \circ_{\mathbf{Mono}(\mathcal{C},M)} f := g \circ_{\mathcal{C}} f$.
- **the restriction of $\mathcal{C}$ to $\mathrm{Epi}_{\mathcal{C}}$,** denoted $\mathbf{Epi}(\mathcal{C})$, by
  - $\mathrm{Obj}(\mathbf{Epi}(\mathcal{C})) := \mathrm{Obj}(\mathcal{C})$
  - For $A, B \in \mathrm{Obj}(\mathcal{C})$, $\mathrm{Hom}_{\mathbf{Epi}(\mathcal{C})}(A, B) := \mathrm{Epi}_{\mathcal{C}}(A, B)$
  - For $A, B, C \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathbf{Epi}(\mathcal{C})}(A, B)$ and $g \in \mathrm{Hom}_{\mathbf{Epi}(\mathcal{C})}(B, C)$, $g \circ_{\mathbf{Epi}(\mathcal{C})} f := g \circ_{\mathcal{C}} f$.

Exercise 3.7.0.3. Let $\mathcal{C}$ be a category. Then

1. (a) $\mathbf{Mono}(\mathcal{C}, M)$ is a category ,
   (b) $\mathbf{Mono}(\mathcal{C}, M)$ is a wide subcategory of $\mathcal{C}$,
2. (a) $\mathbf{Epi}(\mathcal{C})$ is a category,
   (b) $\mathbf{Epi}(\mathcal{C})$ is a wide subcategory of $\mathcal{C}$.

Proof. clear. fill out, ref exercies showing that composition of mono is mono, id_X is mono □

Exercise 3.7.0.4. Let $\mathcal{C}$ be a category and $X \in \mathrm{Obj}(\mathcal{C})$. Then

1. $\mathbf{Mono}(\mathcal{C}, M)/X$ is a proset category.
2.

Proof. 1. Let $(A, f), (B, g) \in \mathbf{Mono}(\mathcal{C})/X$ and $\phi \in \mathrm{Hom}_{\mathbf{Mono}(\mathcal{C})/X}((A, f), (B, g))$. Then

2.

□

Definition 3.7.0.5. Let $\mathcal{C}$ be a category, $M \subset \mathrm{Mono}_{\mathcal{C}}$, $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Mono}_{\mathcal{C}}(A, B)$. Then (A, f) is said to be an **M-subobject** of B if $f \in M$.

Definition 3.7.0.6. Let $\mathcal{C}$ be a category, $M \subset \mathrm{Mono}_{\mathcal{C}}$ and $X \in \mathrm{Obj}(\mathcal{C})$. We define the **category of M-subobjects of X**, denoted $\mathrm{Sub}_{\mathcal{C}}^{M}(X)$, by

- $\mathrm{Obj}(\mathrm{Sub}_{\mathcal{C}}^{M}(X)) := \{(A, \alpha) \in \mathrm{Obj}(\mathcal{C}/X) : \alpha \text{ is an } M\text{-subobject of } X\}$
- for $(A, f), (B, g) \in \mathrm{Obj}(\mathrm{Sub}_{\mathcal{C}}^{M}(X))$, $\mathrm{Hom}_{(\mathrm{Sub}_{\mathcal{C}}^{M}(X))}((A, f), (B, g)) := \mathrm{Hom}_{\mathcal{C}/X}((A, f), (B, g))$
-

is there a notion of full subcategory generated by a set of objects? can we say that $\mathrm{Sub}_{\mathcal{C}}^{M}(X)$ is the full subcategory of $\mathcal{C}/X$ generated by the M-subobjects of X?

Exercise 3.7.0.7. show $\mathrm{Sub}_{\mathcal{C}}^{M}(X)$ is a preodered set/category, i.e. if (A, f) and (B, g) are M-subobjects of X, then $\#\,\mathrm{Hom}_{\mathrm{Sub}_{\mathcal{C}}^{M}(X)}((A, f),$
1.

Chapter 4

Constructions on Categories

4.1 Subcategories

are subcategories subobjects in **Cat** where subobjects are defined as in 'Elementary categories, elementary toposes' by mcLarty?, how to place this subsection?

TODO: move this subsection until after functors and natural transformations, maybe even after adjunctions. We definitely want to to be able to characterize subcategories as subobjects of categories and in terms of the inclusion functor, so maybe should come after monomorphisms, subobjects and functors. We also want to describe subcategories generated by a subgraph, so we need to talk about the graph of a category and the adjoint of the forgetful functor from **Cat** to Graph, so maybe put after adjoints, then this will come after limits. Would like to cover the order of subcat, prodcat, coprodcat? quocat? projlimcat?, so need to check of the product cat can come after limits and adjoints and all that. also subobjects should come after slice categories, since we turn slice categories of wide categories generated by graphs whose edges belong to a class of monomorphisms into a proset, but if subobjects come after generated categories, they need to come after adjoints, so maybe we should later give examples of subobjects in Cat as subcats. This sounds good, so we can characterize subcats in terms of the inclusion functor and then then return to subcats as an example in the context of subobjects.

The subcategories generated by a graph as an adjunction should also be applicable to describing the topology generated by a subset I think, not sure if there is a lattice formulation of this topic as well and if the adjunct description generalizes this lattice description. Maybe there is just a straight lattice based characteriszation of generated subcateogries when we order by inclusion functors.

idea: see https://math.stackexchange.com/questions/596071/subcategory-generated-by-a-graph

all we need to do is talk about galois connections between the posets of subcats and subgraphs. so we ccan just pull from the lattice notes here. With this, we can define subcategories generated by a subgraph at any time, so no need to discuss subcats after adjoints. so just talk about subcats/prodcats/etc after functors/nat trans/ but before subobjects?.

So: cats (and their graphs) functors natty transf subcats (with gal conn between posets of subcats and graphs and then subcats generated by subgraphs) iso/mono/epi init/terminal prod(including) coprod equal/coequal pull/push-subobj? END!!!!!!!!!!!!!!!!!!!!!END!!!!!!!!!!! END!!!!!!!!!!!!!!!!!!!!END!!!!!!!!!!!!!!!!!!!!END!!!!!!!!!!!!!!!!!!!!END!!!!!!!!!!!!!!!!!!!!

Definition 4.1.0.1. Let $\mathcal{C}$ be a category.

- Let $\mathcal{D}$ be categories. Then $\mathcal{D}$ is said to be a **subcategory of** $\mathcal{C}$, denoted $\mathcal{D} \subset \mathcal{C}$, if
  1. $\mathrm{Obj}(\mathcal{D}) \subset \mathrm{Obj}(\mathcal{C})$
  2. for each $A, B \in \mathrm{Obj}(\mathcal{D})$, $\mathrm{Hom}_{\mathcal{D}}(A, B) \subset \mathrm{Hom}_{\mathcal{C}}(A, B)$
  3. for each $A, B, C \in \mathrm{Obj}(\mathcal{D})$, $f \in \mathrm{Hom}_{\mathcal{D}}(A, B)$ and $g \in \mathrm{Hom}_{\mathcal{D}}(B, C)$, $g \circ_{\mathcal{D}} f = g \circ_{\mathcal{C}} f$
  4. for each $A \in \mathrm{Obj}(\mathcal{D})$, $\mathrm{id}_A^{\mathcal{D}} = \mathrm{id}_A^{\mathcal{C}}$
- We define $\mathrm{Subcat}(\mathcal{C}) := \{\mathcal{D} \in \mathbf{Cat} : \mathcal{L} \text{ is a subcategory of } \mathcal{C}\}$. need to give more justification for why we restrict to **Cat**, with universes, this should not be an issue, but need to formalize it

- We define the **subcategory relation on** $\mathrm{Subcat}(\mathcal{C})$, denoted $\leq_{\mathrm{Cat}} \subset \mathrm{Subcat}(\mathcal{C}) \times \mathrm{Subcat}(\mathcal{C})$, by $\mathcal{D} \leq_{\mathrm{Cat}} \mathcal{C}$ iff $\mathcal{D}$ is a subcat of $\mathcal{C}$.

Definition 4.1.0.2. Let $\mathcal{C}$ be a category and $\mathcal{D} \in \mathrm{Subcat}(\mathcal{C})$. We define the **inclusion functor** from $\mathcal{D}$ to $\mathcal{C}$, denoted $\iota_{\mathcal{D}}^{\mathcal{C}} : \mathcal{D} \to \mathcal{C}$, by

- $\iota_{\mathcal{D}}^{\mathcal{C}}(X) := X, \quad X \in \mathrm{Obj}(\mathcal{D})$
- $\iota_{\mathcal{D}}^{\mathcal{C}}(f) := f \in \mathrm{Hom}_{\mathcal{D}}$

Note 4.1.0.3. When the context is clear, we may write $\iota_{\mathcal{D}}$ in place of $\iota_{\mathcal{D}}^{\mathcal{C}}$.

Exercise 4.1.0.4. Let $\mathcal{C}$ be a category and $\mathcal{D} \in \mathrm{Subcat}(\mathcal{C})$. Then $\iota_{\mathcal{D}} : \mathcal{D} \to \mathcal{C}$ is a functor.

Proof.

- Let $X, Y \in \mathrm{Obj}(\mathcal{D})$ and $f \in \mathrm{Hom}_{\mathcal{D}}(X, Y)$. By Definition 4.1.0.1,

$$\begin{aligned}\iota_{\mathcal{D}}(f) &= f \\ &\in \mathrm{Hom}_{\mathcal{D}}(X, Y) \\ &\subset \mathrm{Hom}_{\mathcal{C}}(X, Y) \\ &= \mathrm{Hom}_{\mathcal{C}}(\iota_{\mathcal{D}}(X), \iota_{\mathcal{D}}(Y)).\end{aligned}$$

- Let $X, Y, Z \in \mathrm{Obj}(\mathcal{D})$, $g \in \mathrm{Hom}_{\mathcal{D}}(X, Y)$ and $h \in\in \mathrm{Hom}_{\mathcal{D}}(Y, Z)$. By Definition 4.1.0.1,

$$\begin{aligned}\iota_{\mathcal{D}}(g \circ_{\mathcal{D}} f) &= g \circ_{\mathcal{D}} f \\ &= g \circ_{\mathcal{C}} f \\ &= \iota_{\mathcal{D}}(g) \circ_{\mathcal{C}} \iota_{\mathcal{D}}(f).\end{aligned}$$

- Let $X \in \mathrm{Obj}(\mathcal{D})$. By Definition 4.1.0.1,

$$\begin{aligned}\iota_{\mathcal{D}}(\mathrm{id}_X^{\mathcal{D}}) &= \mathrm{id}_X^{\mathcal{D}} \\ &= \mathrm{id}_X^{\mathcal{C}} \\ &= \mathrm{id}_{\iota_{\mathcal{D}}(X)}^{\mathcal{C}}.\end{aligned}$$

Hence $\iota_{\mathcal{D}} : \mathcal{D} \to \mathcal{C}$ is a functor. □

Definition 4.1.0.5. We define the **forgetful functor from Cat to Quiv**, denoted $\mathrm{Forget}_{\mathbf{Quiv}}^{\mathbf{Cat}} : \mathbf{Cat} \to \mathbf{Quiv}$ finish

Definition 4.1.0.6. Let $\mathcal{C}$ be a category and $H \subset \mathrm{Hom}_{\mathcal{C}}$. We define the **subcategory of $\mathcal{C}$ generated by** H, denoted $\mathrm{SubcatGen}(H, \mathcal{C})$, by $\mathrm{SubcatGen}(H, \mathcal{C}) :=$

Exercise 4.1.0.7. content...

Definition 4.1.0.8.

Definition 4.1.0.9. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Suppose that $\mathcal{D}$ is a subcategory of $\mathcal{C}$. Then $\mathcal{C}$ is said to be

- **full** if for each $A, B \in \mathrm{Obj}(\mathcal{D})$, $\mathrm{Hom}_{\mathcal{D}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(A, B)$,
- **wide** or **lluf** if for each $\mathrm{Obj}(\mathcal{D}) = \mathrm{Obj}(\mathcal{C})$.

Note 4.1.0.10. The motivation for the adjective "lluf" seems to be that it is just "full" backwards.

Exercise 4.1.0.11. content...

Exercise 4.1.0.12. erase after making iff Let $\mathcal{C}$ be a category, $\mathcal{D}_0 \subset \mathrm{Obj}(\mathcal{C})$ and $\mathcal{D}_1 \subset \mathrm{Hom}_{\mathcal{C}}$. Define $\mathcal{D}_1^{(2)} \subset \mathcal{D}_1 \times \mathcal{D}_1$ by $\mathcal{D}_1^{(2)} := \{(g, f) \in \mathcal{D}_1 \times \mathcal{D}_1 : \mathrm{dom}_{\mathcal{C}}(g) = \mathrm{cod}_{\mathcal{C}}(f)\}$. If

- $\mathrm{dom}_{\mathcal{C}}(\mathcal{D}_1), \mathrm{cod}_{\mathcal{C}}(\mathcal{D}_1) \subset \mathcal{D}_0$,
- $\circ_{\mathcal{C}}(\mathcal{D}_1^{(2)}) \subset \mathcal{D}_1$,
- for each $X \in \mathcal{D}_0$, $\mathrm{id}_X^{\mathcal{C}} \in \mathcal{D}_1$,

then

1. $(\mathcal{D}_0, \mathcal{D}_1, \mathrm{dom}_{\mathcal{C}}|_{\mathcal{D}_1}, \mathrm{cod}_{\mathcal{C}}|_{\mathcal{D}_1}, \circ_{\mathcal{C}}|_{\mathcal{D}_1^{(2)}})$ is a category,
2. $(\mathcal{D}_0, \mathcal{D}_1, \mathrm{dom}_{\mathcal{C}}|_{\mathcal{D}_1}, \mathrm{cod}_{\mathcal{C}}|_{\mathcal{D}_1}, \circ_{\mathcal{C}}|_{\mathcal{D}_1^{(2)}})$ is a subcategory of $\mathcal{C}$.

Proof. Set $\mathrm{dom}_{\mathcal{D}} := \mathrm{dom}_{\mathcal{C}}|_{\mathcal{D}_1}$, $\mathrm{cod}_{\mathcal{D}} := \mathrm{cod}_{\mathcal{C}}|_{\mathcal{D}_1}$, $\circ_{\mathcal{D}} := \circ_{\mathcal{C}}|_{\mathcal{D}_1^{(2)}}$ and $\mathcal{D} := (\mathcal{D}_0, \mathcal{D}_1, \mathrm{dom}_{\mathcal{C}}|_{\mathcal{D}_1}, \mathrm{cod}_{\mathcal{C}}|_{\mathcal{D}_1}, \circ_{\mathcal{C}}|_{\mathcal{D}_1^{(2)}})$.

1. (a) **(Composition:)**
   For each $(g, f) \in \mathcal{D}_1^{(2)}$,
   $$\begin{aligned}\mathrm{dom}_{\mathcal{D}}(g \circ_{\mathcal{D}} f) &= \mathrm{dom}_{\mathcal{C}}(g \circ_{\mathcal{C}}) \\ &= \mathrm{dom}_{\mathcal{C}}(f) \\ &= \mathrm{dom}_{\mathcal{D}}(f).\end{aligned}$$
   and
   $$\begin{aligned}\mathrm{cod}_{\mathcal{D}}(g \circ_{\mathcal{D}} f) &= \mathrm{cod}_{\mathcal{C}}(g \circ_{\mathcal{C}}) \\ &= \mathrm{cod}_{\mathcal{C}}(g) \\ &= \mathrm{cod}_{\mathcal{D}}(g).\end{aligned}$$

   (b) **(Associativity:)**
   For each $f, g, h \in \mathcal{D}_1$, if $(g, f), (h, g) \in \mathcal{D}_1^{(2)}$, then
   $$\begin{aligned}(h \circ_{\mathcal{D}} g) \circ_{\mathcal{D}} f &= (h \circ_{\mathcal{C}} g) \circ_{\mathcal{C}} f \\ &= h \circ_{\mathcal{C}} (g \circ_{\mathcal{C}} f) \\ &= h \circ_{\mathcal{D}} (g \circ_{\mathcal{D}} f).\end{aligned}$$

   (c) **(Identity:)**
   Let $X \in \mathcal{D}_0$. Since $\mathrm{id}_X^{\mathcal{C}} \in \mathcal{D}_1$,
   $$\begin{aligned}\mathrm{dom}_{\mathcal{D}}(\mathrm{id}_X^{\mathcal{C}}) &= \mathrm{dom}_{\mathcal{C}}(\mathrm{id}_X^{\mathcal{C}}) \\ &= X,\end{aligned}$$
   $$\begin{aligned}\mathrm{cod}_{\mathcal{D}}(\mathrm{id}_X^{\mathcal{C}}) &= \mathrm{cod}_{\mathcal{C}}(\mathrm{id}_X^{\mathcal{C}}) \\ &= X,\end{aligned}$$
   and for each $f, g \in \mathcal{D}_1$, if $\mathrm{dom}_{\mathcal{D}}(f) = X$ and $\mathrm{cod}_{\mathcal{D}}(g) = X$, then
   $$\begin{aligned}f \circ_{\mathcal{D}} \mathrm{id}_X^{\mathcal{C}} &= f \circ_{\mathcal{C}} \mathrm{id}_X^{\mathcal{C}} \\ &= f\end{aligned}$$
   and
   $$\begin{aligned}\mathrm{id}_X^{\mathcal{C}} \circ_{\mathcal{D}} g &= id_X^{\mathcal{C}} \circ_{\mathcal{C}} g \\ &= g.\end{aligned}$$

   Hence $\mathcal{D}$ is a category.

2. Clear by construction.

□

4.2 Product Categories

4.2.1 Introduction

Definition 4.2.1.1. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. We define the **product category of $\mathcal{C}$ and $\mathcal{D}$**, denoted $\mathcal{C} \times \mathcal{D}$ by

- $\mathrm{Obj}(\mathcal{C} \times \mathcal{D}) := \mathrm{Obj}(\mathcal{C}) \times \mathrm{Obj}(\mathcal{D})$
- for each $(A, A'), (B, B') \in \mathrm{Obj}(\mathcal{C} \times \mathcal{D})$,
$$\mathrm{Hom}_{\mathcal{C}\times\mathcal{D}}((A, A'), (B, B')) := \mathrm{Hom}_{\mathcal{C}}(A, B) \times \mathrm{Hom}_{\mathcal{C}}(A', B').$$
- for each $(A, A'), (B, B'), (C, C') \in \mathrm{Obj}(\mathcal{C} \times \mathcal{D})$, $(f, f') \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}}((A, A'), (B, B'))$ and $(g, g') \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}}((B, B'), (C, C'))$,
$$(g, g') \circ_{\mathcal{C}\times\mathcal{D}} (f, f') := (g \circ_{\mathcal{C}} f, g' \circ_{\mathcal{D}} f')$$

Exercise 4.2.1.2. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Then $\mathcal{C} \times \mathcal{D}$ is a category.

Proof.

- **well-definedness of composition:**
Let $(A, A'), (B, B'), (C, C') \in \mathrm{Obj}(\mathcal{C} \times \mathcal{D})$, $(f, f') \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}}((A, A'), (B, B'))$ and $(g, g') \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}}((B, B'), (C, C'))$. Then $f \in \mathrm{Hom}_{\mathcal{C}}(A, B)$, $g \in \mathrm{Hom}_{\mathcal{C}}(B, C)$, $f' \in \mathrm{Hom}_{\mathcal{D}}(A', B')$, and $g' \in \mathrm{Hom}_{\mathcal{D}}(B', C')$. Hence $g \circ_{\mathcal{C}} f \in \mathrm{Hom}_{\mathcal{C}}(A, C)$ and $g' \circ_{\mathcal{D}} f' \in \mathrm{Hom}_{\mathcal{D}}(A', C')$. Thus
$$\begin{aligned}(g, g') \circ_{\mathcal{C}\times\mathcal{D}} (f, f') &= (g \circ_{\mathcal{C}} f, g' \circ_{\mathcal{D}} f') \\ &\in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}}((A, A'), (C, C'))\end{aligned}$$
Thus, composition is well defined.

- **associativity of composition:**
Let $(A, A'), (B, B'), (C, C'), (D, D') \in \mathrm{Obj}(\mathcal{C} \times \mathcal{D})$, $(f, f') \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}}((A, A'), (B, B'))$, $(g, g') \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}}((B, B'), (C, C'))$ and $(h, h') \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}}((C, C'), (D, D'))$. Then
$$\begin{aligned}[(h, h') \circ_{\mathcal{C}\times\mathcal{D}} (g, g')] \circ_{\mathcal{C}\times\mathcal{D}} (f, f') &= (h \circ_{\mathcal{C}} g, h' \circ_{\mathcal{D}} g') \circ_{\mathcal{C}\times\mathcal{D}} (f, f') \\ &= ((h \circ_{\mathcal{C}} g) \circ_{\mathcal{C}} f, (h' \circ_{\mathcal{D}} g') \circ_{\mathcal{D}} f') \\ &= (h \circ_{\mathcal{C}} (g \circ_{\mathcal{C}} f), h' \circ_{\mathcal{D}} (g' \circ_{\mathcal{D}} f')) \\ &= (h, h') \circ_{\mathcal{C}\times\mathcal{D}} (g \circ_{\mathcal{C}} f, g' \circ_{\mathcal{D}} f') \\ &= (h, h') \circ_{\mathcal{C}\times\mathcal{D}} [(g, g') \circ_{\mathcal{C}\times\mathcal{D}} (f, f')]\end{aligned}$$
Thus composition is associative.

- **existence of identities:**
Let $(A, B) \in \mathrm{Obj}(\mathcal{C} \times \mathcal{D})$, $(f, f'), (g, g') \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}}$. Suppose that $\mathrm{dom}_{\mathcal{C}\times\mathcal{D}}(f, f') = (A, B)$ and $\mathrm{cod}_{\mathcal{C}\times\mathcal{D}}(g, g') = (A, B)$. Then $\mathrm{dom}_{\mathcal{C}}(f) = A$, $\mathrm{dom}_{\mathcal{D}}(f') = B$, $\mathrm{cod}_{\mathcal{C}}(g) = A$ and $\mathrm{cod}_{\mathcal{D}}(g') = B$. Hence
$$\begin{aligned}(f, f') \circ_{\mathcal{C}\times\mathcal{D}} (\mathrm{id}_A^{\mathcal{C}}, \mathrm{id}_B^{\mathcal{D}}) &= (f \circ_{\mathcal{C}} \mathrm{id}_A^{\mathcal{C}}, f' \circ_{\mathcal{D}} \mathrm{id}_B^{\mathcal{D}}) \\ &= (f, f)\end{aligned}$$
and
$$\begin{aligned}(\mathrm{id}_A^{\mathcal{C}}, \mathrm{id}_B^{\mathcal{D}}) \circ_{\mathcal{C}\times\mathcal{D}} (g, g') &= (\mathrm{id}_A^{\mathcal{C}} \circ_{\mathcal{C}} g, \mathrm{id}_B^{\mathcal{D}} \circ_{\mathcal{D}} g') \\ &= (g, g')\end{aligned}$$
Therefore $\mathrm{id}_{(A,B)}^{\mathcal{C}\times\mathcal{D}} = (\mathrm{id}_A^{\mathcal{C}}, \mathrm{id}_B^{\mathcal{D}})$.

□

Exercise 4.2.1.3. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. Then $\mathcal{C}$ and $\mathcal{D}$ are small iff $\mathcal{C} \times \mathcal{D}$ is small.

Proof. We have that

$$\begin{aligned}
\mathcal{C} \text{ and } \mathcal{D} \text{ are small} &\iff \mathrm{Obj}(\mathcal{C}), \mathrm{Hom}_{\mathcal{C}}, \mathrm{Obj}(\mathcal{D}), \mathrm{Hom}_{\mathcal{D}} \in \mathrm{Obj}(\mathbf{Set}) \\
&\iff \mathrm{Obj}(\mathcal{C}) \times \mathrm{Obj}(\mathcal{D}) \in \mathrm{Obj}(\mathbf{Set}) \text{ and } \mathrm{Hom}_{\mathcal{C}} \times \mathrm{Hom}_{\mathcal{D}} \in \mathrm{Obj}(\mathbf{Set}) \\
&\iff (\mathrm{Obj}(\mathcal{C} \times \mathcal{D}) \in \mathrm{Obj}(\mathbf{Set}) \text{ and } \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}} \in \mathrm{Obj}(\mathbf{Set}) \\
&\iff \mathcal{C} \times \mathcal{D} \text{ is small.}
\end{aligned}$$

□

Definition 4.2.1.4. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$ be categories and $F : \mathcal{A} \to \mathcal{C}$, $G : \mathcal{B} \to \mathcal{D}$. We define the **product functor of** F **and** G, denoted $F \times G : \mathcal{A} \times \mathcal{B} \to \mathcal{C} \times \mathcal{D}$, by

- $F \times G(A, B) := (F(A), G(B)), \quad (A, B) \in \mathrm{Obj}(\mathcal{A} \times \mathcal{B})$,
- $F \times G(f, g) := (F(f), G(g)), \quad (f, g) \in \mathrm{Hom}_{\mathcal{A}\times\mathcal{B}}$

Exercise 4.2.1.5. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}, \mathcal{D}$ be categories and $F : \mathcal{A} \to \mathcal{C}$, $G : \mathcal{B} \to \mathcal{D}$. Then $F \times G$ is a functor.

Proof.

1. Let $X, Y \in \mathrm{Obj}(\mathcal{A} \times \mathcal{B})$ and $h \in \mathrm{Hom}_{\mathcal{A}\times\mathcal{B}}(X, Y)$. Then there exist $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$, $B_1, B_2 \in \mathrm{Obj}(\mathcal{B})$, $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$ and $g \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$ such that $X = (A_1, B_1)$, $Y = (A_2, B_2)$ and $h = (f, g)$. Since $F(f) \in \mathrm{Hom}_{\mathcal{C}}(F(A_1), F(A_2))$ and $G(g) \in \mathrm{Hom}_{\mathcal{D}}(G(B_1), G(B_2))$, we have that
$$\begin{aligned}
F \times G(h) &= F \times G(f, g) \\
&= (F(f), G(g)) \\
&\in \mathrm{Hom}_{\mathcal{A}\times\mathcal{B}}((F(A_1), G(B_1)), (F(A_2), G(B_2))) \\
&= \mathrm{Hom}_{\mathcal{A}\times\mathcal{B}}(F \times G(A_1, B_1), F \times G(A_2, B_2)) \\
&= \mathrm{Hom}_{\mathcal{A}\times\mathcal{B}}(F \times G(X), F \times G(Y)).
\end{aligned}$$

2. Let $X_1, X_2, X_3 \in \mathrm{Obj}(\mathcal{A} \times \mathcal{B})$, $h_{1,2} \in \mathrm{Hom}_{\mathcal{A}\times\mathcal{B}}(X_1, X_2)$ and $h_{2,3} \in \mathrm{Hom}_{\mathcal{A}\times\mathcal{B}}(X_2, X_3)$. Then there exist $A_1, A_2, A_3 \in \mathrm{Obj}(\mathcal{A})$, $B_1, B_2, B_3 \in \mathrm{Obj}(\mathcal{B})$, $f_{1,2} \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$, $g_{1,2} \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$, $f_{2,3} \in \mathrm{Hom}_{\mathcal{A}}(A_2, A_3)$ and $g_{2,3} \in \mathrm{Hom}_{\mathcal{B}}(B_2, B_3)$, such that $X_1 = (A_1, B_1)$, $X_2 = (A_2, B_2)$, $X_3 = (A_3, B_3)$, $h_{1,2} = (f_{1,2}, g_{1,2})$ and $h_{2,3} = (f_{2,3}, g_{2,3})$. Then
$$\begin{aligned}
F \times G(h_{2,3} \circ_{\mathcal{A}\times\mathcal{B}} h_{1,2}) &= F \times G((f_{2,3}, g_{2,3}) \circ_{\mathcal{A}\times\mathcal{B}}, (f_{1,2}, g_{1,2})) \\
&= F \times G(f_{2,3} \circ_{\mathcal{A}} f_{1,2}, g_{2,3} \circ_{\mathcal{B}} g_{1,2}) \\
&= (F(f_{2,3} \circ_{\mathcal{A}} f_{1,2}), G(g_{2,3} \circ_{\mathcal{B}} g_{1,2})) \\
&= (F(f_{2,3}) \circ_{\mathcal{C}} F(f_{1,2}), G(g_{2,3}) \circ_{\mathcal{D}} G(g_{1,2})) \\
&= (F(f_{2,3}), G(g_{2,3})) \circ_{\mathcal{C}\times\mathcal{D}} (F(f_{1,2}), G(g_{1,2})) \\
&= F \times G(f_{2,3}, g_{2,3}) \circ_{\mathcal{C}\times\mathcal{D}} F \times G(f_{1,2}, g_{1,2}) \\
&= F \times G(h_{2,3}) \circ_{\mathcal{C}\times\mathcal{D}} F \times G(h_{1,2}).
\end{aligned}$$

3. Let $X \in \mathrm{Obj}(\mathcal{A} \times \mathcal{B})$. Then there exists $A \in \mathrm{Obj}(\mathcal{A})$ and $B \in \mathrm{Obj}(\mathcal{B})$ such that $X = (A, B)$. Then
$$\begin{aligned}
F \times G(\mathrm{id}_X^{\mathcal{A}\times\mathcal{B}}) &= F \times G(\mathrm{id}_{(A,B)}^{\mathcal{A}\times\mathcal{B}}) \\
&= F \times G(\mathrm{id}_A^{\mathcal{A}}, \mathrm{id}_B^{\mathcal{B}}) \\
&= (F(\mathrm{id}_A^{\mathcal{A}}), G(\mathrm{id}_B^{\mathcal{B}})) \\
&= (\mathrm{id}_{F(A)}^{\mathcal{C}}, \mathrm{id}_{G(B)}^{\mathcal{D}}) \\
&= \mathrm{id}_{(F(A),G(B))}^{\mathcal{C}\times\mathcal{D}} \\
&= \mathrm{id}_{F\times G(A,B)}^{\mathcal{C}\times\mathcal{D}} \\
&= \mathrm{id}_{F\times G(X)}^{\mathcal{C}\times\mathcal{D}}.
\end{aligned}$$

Hence $F \times G$ is a functor. $\square$

Exercise 4.2.1.6. Let $\mathcal{C}_1, \mathcal{C}_2, \mathcal{C}_3 \mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3 \in \mathrm{Obj}(\mathbf{Cat})$, $F_{1,2} \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}_1, \mathcal{C}_2)$, $F_{2,3} \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}_2, \mathcal{C}_3)$, $G_{1,2} \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{D}_1, \mathcal{D}_2)$ and $G_{2,3} \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{D}_2, \mathcal{D}_3)$. Then

$$(F_{2,3} \circ_{\mathbf{Cat}} F_{1,2}) \times (G_{2,3} \circ_{\mathbf{Cat}} G_{1,2}) = (F_{2,3} \times G_{2,3}) \circ_{\mathbf{Cat}} (F_{1,2} \times G_{1,2}).$$

Proof.

- For each $A \in \mathrm{Obj}(\mathcal{C}_1)$ and $B \in \mathrm{Obj}(\mathcal{D}_1)$, we have that

$$\begin{aligned}
[(F_{2,3} \circ_{\mathbf{Cat}} F_{1,2}) \times (G_{2,3} \circ_{\mathbf{Cat}} G_{1,2})](A, B) &= (F_{2,3} \circ_{\mathbf{Cat}} F_{1,2}(A), (G_{2,3} \circ_{\mathbf{Cat}} G_{1,2}(B)) \\
&= (F_{2,3}(F_{1,2}(A)), G_{2,3}(G_{1,2}(B))) \\
&= (F_{2,3} \times G_{2,3})(F_{1,2}(A), G_{1,2}(B)) \\
&= (F_{2,3} \times G_{2,3})((F_{1,2} \times G_{1,2})(A, B)) \\
&= [(F_{2,3} \times G_{2,3}) \circ_{\mathbf{Cat}} (F_{1,2} \times G_{1,2})](A, B).
\end{aligned}$$

- For each $A_1, A_2 \in \mathrm{Obj}(\mathcal{C}_1)$, $B_1, B_2 \in \mathrm{Obj}(\mathcal{D}_1)$ Then

$$\begin{aligned}
(F_{2,3} \circ_{\mathbf{Cat}} F_{1,2}) \times (G_{2,3} \circ_{\mathbf{Cat}} G_{1,2})(f, g) &= (F_{2,3} \circ_{\mathbf{Cat}} F_{1,2}(f), G_{2,3} \circ_{\mathbf{Cat}} G_{1,2}(g)) \\
&= (F_{2,3}(F_{1,2}(f)), G_{2,3}(G_{1,2}(g)) \\
&= (F_{2,3} \times G_{2,3})(F_{1,2}(f), G_{1,2}(g)) \\
&= (F_{2,3} \times G_{2,3})((F_{1,2} \times G_{1,2})(f, g)) \\
&= (F_{2,3} \times G_{2,3}) \circ_{\mathbf{Cat}} (f, g).
\end{aligned}$$

Hence $(F_{2,3} \circ_{\mathbf{Cat}} F_{1,2}) \times (G_{2,3} \circ_{\mathbf{Cat}} G_{1,2}) = (F_{2,3} \times G_{2,3}) \circ_{\mathbf{Cat}} (F_{1,2} \times G_{1,2})$. $\square$

Exercise 4.2.1.7. Let $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathbf{Cat})$. Then $\mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}} \times \mathrm{id}_{\mathcal{D}}^{\mathbf{Cat}} = \mathrm{id}_{\mathcal{C} \times \mathcal{D}}^{\mathbf{Cat}}$.

Proof.

- For each $(A, B) \in \mathrm{Obj}(\mathcal{C} \times \mathcal{D})$,

$$\begin{aligned}
\mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}} \times \mathrm{id}_{\mathcal{D}}^{\mathbf{Cat}}(A, B) &= (\mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}}(A), \mathrm{id}_{\mathcal{D}}^{\mathbf{Cat}}(B)) \\
&= (A, B) \\
&= \mathrm{id}_{\mathcal{C} \times \mathcal{D}}^{\mathbf{Cat}}(A, B).
\end{aligned}$$

- For each $(f, g) \in \mathrm{Hom}_{\mathcal{C} \times \mathcal{D}}$,

$$\begin{aligned}
&\mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}} \times \mathrm{id}_{\mathcal{D}}^{\mathbf{Cat}}(f, g) \\
&\qquad = (\mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}}(f), \mathrm{id}_{\mathcal{D}}^{\mathbf{Cat}}(g)) \\
&\qquad = (f, g) \\
&\qquad = \mathrm{id}_{\mathcal{C} \times \mathcal{D}}^{\mathbf{Cat}}(f, g).
\end{aligned}$$

Hence $\mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}} \times \mathrm{id}_{\mathcal{D}}^{\mathbf{Cat}} = \mathrm{id}_{\mathcal{C} \times \mathcal{D}}^{\mathbf{Cat}}$. $\square$

Definition 4.2.1.8. We define the **product functor on Cat**, denoted $(- \times -) : \mathbf{Cat} \times \mathbf{Cat} \to \mathbf{Cat}$, by

- $(- \times -)(\mathcal{A}, \mathcal{B}) := \mathcal{A} \times \mathcal{B}, \quad (\mathcal{A}, \mathcal{B}) \in \mathrm{Obj}(\mathbf{Cat} \times \mathbf{Cat})$,
- $(- \times -)(F, G) := F \times G, \quad (\mathcal{C}_1, \mathcal{D}_1), (\mathcal{C}_1, \mathcal{D}_2) \in \mathrm{Obj}(\mathbf{Cat} \times \mathbf{Cat})$ and $(F, G) \in \mathrm{Hom}_{\mathbf{Cat} \times \mathbf{Cat}}((\mathcal{C}_1, \mathcal{D}_1), (\mathcal{C}_2, \mathcal{D}_2))$.

Exercise 4.2.1.9. We have that $(- \times -) : \mathbf{Cat} \times \mathbf{Cat} \to \mathbf{Cat}$ is a functor.

Proof.

1. Let $(\mathcal{C}_1, \mathcal{D}_1), (\mathcal{C}_2, \mathcal{D}_2) \in \mathrm{Obj}(\mathbf{Cat} \times \mathbf{Cat})$ and $(F, G) \in \mathrm{Hom}_{\mathbf{Cat} \times \mathbf{Cat}}((\mathcal{C}_1, \mathcal{D}_1), (\mathcal{C}_2, \mathcal{D}_2))$. Exercise 4.2.1.5 implies that

$$\begin{aligned}(- \times -)(F, G) &= F \times G \\ &\in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}_1 \times \mathcal{D}_1, \mathcal{C}_2 \times \mathcal{D}_2) \\ &= \mathrm{Hom}_{\mathbf{Cat}}((- \times -)(\mathcal{C}_1, \mathcal{D}_1), (- \times -)(\mathcal{C}_2, \mathcal{D}_2)).\end{aligned}$$

2. Let $(\mathcal{C}_1, \mathcal{D}_1), (\mathcal{C}_2, \mathcal{D}_2), (\mathcal{C}_3, \mathcal{D}_3) \in \mathrm{Obj}(\mathbf{Cat} \times \mathbf{Cat})$, $(F_{1,2}, G_{1,2}) \in \mathrm{Hom}_{\mathbf{Cat} \times \mathbf{Cat}}((\mathcal{C}_1, \mathcal{D}_1), (\mathcal{C}_2, \mathcal{D}_2))$ and $(F_{2,3}, G_{2,3}) \in \mathrm{Hom}_{\mathbf{Cat} \times \mathbf{Cat}}((\mathcal{C}$ Exercise 4.2.1.6 implies that

$$\begin{aligned}(- \times -)((F_{2,3}, G_{2,3}) \circ_{\mathbf{Cat} \times \mathbf{Cat}} (F_{1,2}, G_{1,2})) &= (- \times -)(F_{2,3} \circ_{\mathbf{Cat}} F_{1,2}, G_{2,3} \circ_{\mathbf{Cat}} G_{1,2}) \\ &= (F_{2,3} \circ_{\mathbf{Cat}} F_{1,2}) \times (G_{2,3} \circ_{\mathbf{Cat}} G_{1,2}) \\ &= (F_{2,3} \times G_{2,3}) \circ_{\mathbf{Cat}} (F_{1,2} \times G_{1,2}) \\ &= (- \times -)(F_{2,3}, G_{2,3}) \circ_{\mathbf{Cat}} (- \times -)(F_{1,2}, G_{1,2}).\end{aligned}$$

3. Exercise 4.2.1.7 implies that for each $(\mathcal{C}, \mathcal{D}) \in \mathrm{Obj}(\mathbf{Cat} \times \mathbf{Cat})$,

$$\begin{aligned}(- \times -)(\mathrm{id}^{\mathbf{Cat} \times \mathbf{Cat}}_{(\mathcal{C}, \mathcal{D})}) &= (- \times -)(\mathrm{id}^{\mathbf{Cat}}_{\mathcal{C}}, \mathrm{id}^{\mathbf{Cat}}_{\mathcal{D}}) \\ &= \mathrm{id}^{\mathbf{Cat}}_{\mathcal{C}} \times \mathrm{id}^{\mathbf{Cat}}_{\mathcal{D}} \\ &= \mathrm{id}^{\mathbf{Cat}}_{\mathcal{C} \times \mathcal{D}} \\ &= \mathrm{id}^{\mathbf{Cat}}_{(- \times -)(\mathcal{C}, \mathcal{D})}.\end{aligned}$$

Hence $(- \times -) : \mathbf{Cat} \times \mathbf{Cat} \to \mathbf{Cat}$ is a functor. □

Exercise 4.2.1.10. maybe switch order of 'basic construction and operations' and 'fundamental structures' and redo the previous exercises by showing that if $\mathcal{C}$ has finite products, then $\times : \mathcal{C} \times \mathcal{C} \mapsto \mathcal{C}$ given by $\times(A, B) := A \times B$ and $\times(f, g) := f \times g$ is a functor and then the previous exercise would be a corollary with $\mathcal{C} = \mathbf{Cat}$ (if we show that $(\mathcal{C} \times \mathcal{D}, (\pi_{\mathcal{C}}, \pi_{\mathcal{D}})) \in \mathrm{Prod}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D})$ and $F \times G$ is the product of morphisms in **Cat** as defined in the section on products in the basic).

4.2.2 Slicing

Definition 4.2.2.1. Let $\mathcal{A}, \mathcal{B}, \mathcal{D}$ be categories, $F : \mathcal{A} \times \mathcal{B} \to \mathcal{D}$ and $A_0 \in \mathrm{Obj}(\mathcal{A})$, $B_0 \in \mathrm{Obj}(\mathcal{B})$. We define the

- **slice functor of** F **at** B_0, denoted $F(\cdot, B_0) : \mathcal{A} \to \mathcal{D}$ by
  - $F(\cdot, B_0)(A) := F(A, B_0), \quad A \in \mathrm{Obj}(\mathcal{A})$
  - $F(\cdot, B_0)(f) := F(f, \mathrm{id}^{\mathcal{B}}_{B_0}), \quad f \in \mathrm{Hom}_{\mathcal{A}}$
- **slice functor of** F **at** A_0, denoted $F(A_0, \cdot) : \mathcal{B} \to \mathcal{D}$
  - $F(A_0, \cdot)(B) := F(A, B), \quad B \in \mathcal{B}$
  - $F(A_0, \cdot)(g) := F(\mathrm{id}^{\mathcal{A}}_{A_0}, g), \quad g \in \mathrm{Hom}_{\mathcal{B}}$

Exercise 4.2.2.2. Let $\mathcal{A}, \mathcal{B}, \mathcal{D}$ be categories, $F : \mathcal{A} \times \mathcal{B} \to \mathcal{D}$. Then

1. for each $B_0 \in \mathrm{Obj}(\mathcal{B})$, $F(\cdot, B_0) : \mathcal{A} \to \mathcal{D}$ is a functor,
2. for each $A_0 \in \mathrm{Obj}(\mathcal{A})$, $F(A_0, \cdot) : \mathcal{B} \to \mathcal{D}$ is a functor.

Proof.

1. Let $B_0 \in \mathrm{Obj}(\mathcal{B})$.

(a) Let $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$ and $f_{12} \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$. Then $(f_{12}, \mathrm{id}^{\mathcal{B}}_{B_0}) \in \mathrm{Hom}_{\mathcal{A}\times\mathcal{B}}((A_1, B_0), (A_2, B_0))$. Since $F : \mathcal{A} \times \mathcal{B} \to \mathcal{D}$, we have that

$$\begin{aligned} F(\cdot, B_0)(f_{12}) &= F(f_{12}, \mathrm{id}^{\mathcal{B}}_{B_0}) \\ &\in \mathrm{Hom}_{\mathcal{D}}(F(A_1, B_0), F(A_2, B_0)) \\ &= \mathrm{Hom}_{\mathcal{D}}(F(\cdot, B_0)(A_1), F(\cdot, B_0)(A_2)) \end{aligned}$$

(b) Let $A_1, A_2, A_3 \in \mathrm{Obj}(\mathcal{A})$, $f_{12} \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$ and $f_{23} \in \mathrm{Hom}_{\mathcal{A}}(A_2, A_3)$. Since $F : \mathcal{A} \times \mathcal{B} \to \mathcal{D}$, we have that

$$\begin{aligned} F(\cdot, B_0)(f_{23} \circ_{\mathcal{A}} f_{12}) &= F(f_{23} \circ_{\mathcal{A}} f_{12}, \mathrm{id}^{\mathcal{B}}_{B_0}) \\ &= F(f_{23} \circ_{\mathcal{A}} f_{12}, \mathrm{id}^{\mathcal{B}}_{B_0} \circ_{\mathcal{B}} \mathrm{id}^{\mathcal{B}}_{B_0}) \\ &= F[(f_{23}, \mathrm{id}^{\mathcal{B}}_{B_0}) \circ_{\mathcal{A}\times\mathcal{B}} (f_{12}, \mathrm{id}^{\mathcal{B}}_{B_0})] \\ &= F(f_{23}, \mathrm{id}^{\mathcal{B}}_{B_0}) \circ_{\mathcal{D}} F(f_{12}, \mathrm{id}^{\mathcal{B}}_{B_0}) \\ &= F(\cdot, B_0)(f_{23}) \circ_{\mathcal{D}} F(\cdot, B_0)(f_{12}). \end{aligned}$$

(c) Let $A \in \mathrm{Obj}(\mathcal{A})$. Then

$$\begin{aligned} F(\cdot, B_0)(\mathrm{id}^{\mathcal{A}}_A) &= F(\mathrm{id}^{\mathcal{A}}_A, \mathrm{id}^{\mathcal{B}}_{B_0}) \\ &= F(\mathrm{id}^{\mathcal{A}\times\mathcal{B}}_{(A,B_0)}) \\ &= \mathrm{id}^{\mathcal{D}}_{F(A,B_0)} \\ &= \mathrm{id}^{\mathcal{D}}_{F(\cdot,B_0)(A)}. \end{aligned}$$

Thus $F(\cdot, B_0) : \mathcal{A} \to \mathcal{D}$ is a functor.

2. Similar to previous part ref prev part.

□

Definition 4.2.2.3. Let $\mathcal{A}, \mathcal{B}$ be categories, $A_0 \in \mathrm{Obj}(\mathcal{A})$ and $B_0 \in \mathrm{Obj}(\mathcal{B})$. We define the

- **embedding functor of $\mathcal{A}$ into $\mathcal{A} \times \mathcal{B}$ at** B_0, denoted $\iota^{B_0}_{\mathcal{A}} : \mathcal{A} \to \mathcal{A} \times \mathcal{B}$, by $\iota^{B_0}_{\mathcal{A}} := \mathrm{id}_{\mathcal{A}} \times \mathrm{id}_{\mathcal{B}}(\cdot, B_0)$
- **embedding functor of $\mathcal{B}$ into $\mathcal{A} \times \mathcal{B}$ at** A_0, denoted $\iota^{A_0}_{\mathcal{B}} : \mathcal{B} \to \mathcal{A} \times \mathcal{B}$, by $\iota^{A_0}_{\mathcal{B}} := \mathrm{id}_{\mathcal{A}} \times \mathrm{id}_{\mathcal{B}}(A_0, \cdot)$.

Exercise 4.2.2.4. Let $\mathcal{A}, \mathcal{B}$ be categories. Then

1. for each $B_0 \in \mathrm{Obj}(\mathcal{A})$, $\iota^{B_0}_{\mathcal{A}} : \mathcal{A} \to \mathcal{A} \times \mathcal{B}$ is a functor and
   - $\iota^{B_0}_{\mathcal{A}}(A) := (A, B_0), \quad A \in \mathrm{Obj}(\mathcal{A})$,
   - $\iota^{B_0}_{\mathcal{A}}(f) := (f, \mathrm{id}^{\mathcal{B}}_{B_0}), \quad f \in \mathrm{Hom}_{\mathcal{A}}$,
2. for each $A_0 \in \mathrm{Obj}(\mathcal{B})$, $\iota^{A_0}_{\mathcal{B}} : \mathcal{B} \to \mathcal{A} \times \mathcal{B}$ is a functor and
   - $\iota^{A_0}_{\mathcal{B}}(B) := (A_0, B), \quad B \in \mathrm{Obj}(\mathcal{B})$,
   - $\iota^{A_0}_{\mathcal{B}}(g) := (\mathrm{id}^{\mathcal{A}}_{A_0}, g), \quad g \in \mathrm{Hom}_{\mathcal{B}}$.

Proof.

1. Let $B_0 \in \mathrm{Obj}(\mathcal{B})$. By prev exercises, $\iota^{B_0}_{\mathcal{A}}$ is a functor. By definition, we have that
   - for each $A \in \mathrm{Obj}(\mathcal{A})$,

$$\begin{aligned} \iota^{B_0}_{\mathcal{A}}(A) &= \mathrm{id}_{\mathcal{A}} \times \mathrm{id}_{\mathcal{B}}(A, B_0) \\ &= (\mathrm{id}_{\mathcal{A}}(A), \mathrm{id}_{\mathcal{B}}(B_0)) \\ &= (A, B_0), \end{aligned}$$

- for each $f \in \mathrm{Hom}_{\mathcal{A}}$,

$$\begin{aligned}\iota_{\mathcal{A}}^{B_0}(f) &= \mathrm{id}_{\mathcal{A}} \times \mathrm{id}_{\mathcal{B}}(f, \mathrm{id}_{B_0}^{\mathcal{B}}) \\ &= (\mathrm{id}_{\mathcal{A}}(f), \mathrm{id}_{\mathcal{B}}(\mathrm{id}_{B_0}^{\mathcal{B}})) \\ &= (f, \mathrm{id}_{B_0}^{\mathcal{B}}).\end{aligned}$$

2. Similar to prev part (ref prev part)

□

Definition 4.2.2.5. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be categories, $F, G : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$, $\alpha : F \Rightarrow G$, $A_0 \in \mathrm{Obj}(\mathcal{A})$ and $B_0 \in \mathrm{Obj}(\mathcal{B})$. We define the

- **slice transformation of** α **at** A_0, denoted $\alpha_{(A_0,\cdot)} : F(A_0, \cdot) \Rightarrow G(A_0, \cdot)$, by $(\alpha_{(A_0,\cdot)})_B := \alpha_{(A_0,B)}$
- **slice transformation of** α **at** B_0, denoted $\alpha_{(\cdot,B_0)} : F(\cdot, B_0) \Rightarrow G(\cdot, B_0)$, by $(\alpha_{(\cdot,B_0)})_A := \alpha_{(A,B_0)}$.

Exercise 4.2.2.6. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be categories, $F, G : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$ and $\alpha : F \Rightarrow G$. Then

1. for each $A_0 \in \mathrm{Obj}(\mathcal{A})$, $\alpha_{(A_0,\cdot)} : F(A_0, \cdot) \Rightarrow G(A_0, \cdot)$ is a natural transformation,
2. for each $B_0 \in \mathrm{Obj}(\mathcal{B})$, $\alpha_{(\cdot,B_0)} : F(\cdot, B_0) \Rightarrow G(\cdot, B_0)$ is a natural transformation.

Proof.

1. Let $A_0 \in \mathrm{Obj}(\mathcal{A})$.
   - By construction, for each $B \in \mathrm{Obj}(\mathcal{B})$,

$$\begin{aligned}(\alpha_{(A_0,\cdot)})_B &= \alpha_{(A_0,B)} \\ &\in \mathrm{Hom}_{\mathcal{C}}(F(A_0, B), G(A_0, B)) \\ &= \mathrm{Hom}_{\mathcal{C}}(F(A_0, \cdot)(B), G(A_0, \cdot)(B)).\end{aligned}$$

   - Let $B_1, B_2 \in \mathrm{Obj}(\mathcal{B})$ and $g \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$. Since $\alpha : F \Rightarrow G$, we have that for each $g \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$, $G(\mathrm{id}_{A_0}^{\mathcal{A}}, g) \circ_{\mathcal{C}} \alpha_{(A_0,B_1)} = \alpha_{(A_0,B_2)} \circ_{\mathcal{C}} F(\mathrm{id}_{A_0}^{\mathcal{A}}, g)$, i.e. the following diagram commutes:

$$\begin{tikzcd} F(A_0, B_1) \arrow[r, "\alpha_{(A_0,B_1)}"] \arrow[d, "F(\mathrm{id}_{A_0}^{\mathcal{A}}, g)"'] & G(A_0, B_1) \arrow[d, "G(\mathrm{id}_{A_0}^{\mathcal{A}}, g)"] \\ F(A_0, B_2) \arrow[r, "\alpha_{(A_0,B_2)}"] & G(A_0, B_2). \end{tikzcd}$$

   Therefore

$$\begin{aligned}G(A_0, \cdot)(g) \circ_{\mathcal{C}} (\alpha_{(A_0,\cdot)})_{B_1} &= G(\mathrm{id}_{A_0}^{\mathcal{A}}, g) \circ_{\mathcal{C}} \alpha_{(A_0,B_1)} \\ &= \alpha_{(A_0,B_2)} \circ_{\mathcal{C}} F(\mathrm{id}_{A_0}^{\mathcal{A}}, g) \\ &= (\alpha_{(A_0,\cdot)})_{B_2} \circ_{\mathcal{C}} F(A_0, \cdot)(g),\end{aligned}$$

   i.e. the following diagram commutes:

$$\begin{tikzcd} F(A_0, \cdot)(B_1) \arrow[r, "(\alpha_{(A_0,\cdot)})_{B_1}"] \arrow[d, "F(A_0,\cdot)(g)"'] & G(A_0, \cdot)(B_1) \arrow[d, "G(A_0,\cdot)(g)"] \\ F(A_0, \cdot)(B_2) \arrow[r, "(\alpha_{(A_0,\cdot)})_{B_2}"] & G(A_0, \cdot)(B_2). \end{tikzcd}$$

   Since $B_1, B_2 \in \mathrm{Obj}(\mathcal{B})$ and $g \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$ are arbitrary, we have that for each $B_1, B_2 \in \mathrm{Obj}(\mathcal{B})$, $G(A_0, \cdot)(g) \circ_{\mathcal{C}} (\alpha_{(A_0,\cdot)})_{B_1} = (\alpha_{(A_0,\cdot)})_{B_2} \circ_{\mathcal{C}} F(A_0, \cdot)(g)$.

Thus $\alpha_{(A_0,\cdot)} : F(A_0,\cdot) \Rightarrow G(A_0,\cdot)$ is a natural transformation. Since $A_0 \in \mathrm{Obj}(\mathcal{A})$ is arbitrary, we have that for each $A_0 \in \mathrm{Obj}(\mathcal{A})$, $\alpha_{(A_0,\cdot)} : F(A_0,\cdot) \Rightarrow G(A_0,\cdot)$ is a natural transformation.

2. Similar to previous part (ref prev part)

□

Exercise 4.2.2.7. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be categories, $F_1, F_2, F_3 : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$, $\alpha_{1,2} : F_1 \Rightarrow F_2$ and $\alpha_{2,3} : F_2 \Rightarrow F_3$. Then

1. for each $A_0 \in \mathrm{Obj}(\mathcal{A})$, $(\alpha_{2,3} \circ \alpha_{1,2})_{(A_0,\cdot)} = (\alpha_{2,3})_{(A_0,\cdot)} \circ (\alpha_{1,2})_{(A_0,\cdot)}$.
2. for each $B_0 \in \mathrm{Obj}(\mathcal{B})$, $(\alpha_{2,3} \circ \alpha_{1,2})_{(\cdot,B_0)} = (\alpha_{2,3})_{(\cdot,B_0)} \circ (\alpha_{1,2})_{(\cdot,B_0)}$.

Proof.

1. Let $A_0 \in \mathrm{Obj}(\mathcal{A})$ and $B \in \mathrm{Obj}(\mathcal{B})$. Then

$$\begin{aligned}
[(\alpha_{2,3} \circ \alpha_{1,2})_{(A_0,\cdot)}]_B &= (\alpha_{2,3} \circ \alpha_{1,2})_{(A_0,B)} \\
&= (\alpha_{2,3})_{(A_0,B)} \circ (\alpha_{1,2})_{(A_0,B)} \\
&= [(\alpha_{2,3})_{(A_0,\cdot)}]_B \circ [(\alpha_{1,2})_{(A_0,\cdot)}]_B \\
&= [(\alpha_{2,3})_{(A_0,\cdot)} \circ (\alpha_{1,2})_{(A_0,\cdot)}]_B.
\end{aligned}$$

Since $B \in \mathrm{Obj}(\mathcal{B})$ is arbitrary, we have that for each $B \in \mathrm{Obj}(\mathcal{B})$, $[(\alpha_{2,3} \circ \alpha_{1,2})_{(A_0,\cdot)}]_B = [(\alpha_{2,3})_{(A_0,\cdot)} \circ (\alpha_{1,2})_{(A_0,\cdot)}]_B$. Thus $(\alpha_{2,3} \circ \alpha_{1,2})_{(A_0,\cdot)} = (\alpha_{2,3})_{(A_0,\cdot)} \circ (\alpha_{1,2})_{(A_0,\cdot)}$. Since $A_0 \in \mathrm{Obj}$ is arbitrary, we have that for each $A_0 \in \mathrm{Obj}(\mathcal{A})$, $(\alpha_{2,3} \circ \alpha_{1,2})_{(A_0,\cdot)} = (\alpha_{2,3})_{(A_0,\cdot)} \circ (\alpha_{1,2})_{(A_0,\cdot)}$.

2. Similar to prev part (ref prev part)

□

Exercise 4.2.2.8. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be categories and $F \in \mathcal{A} \times \mathcal{B} \to \mathcal{C}$. Then

1. for each $A_0 \in \mathrm{Obj}(\mathcal{A})$, $(\mathrm{id}_F)_{(A_0,\cdot)} = \mathrm{id}_{F(A_0,\cdot)}$,
2. for each $B_0 \in \mathrm{Obj}(\mathcal{A})$, $(\mathrm{id}_F)_{(\cdot,B_0)} = \mathrm{id}_{F(\cdot,B_0)}$.

Proof.

1. Let $A_0 \in \mathrm{Obj}(\mathcal{A})$. Then for each $B \in \mathrm{Obj}(\mathcal{B})$,

$$\begin{aligned}
[(\mathrm{id}_F)_{(A_0,\cdot)}]_B &= (\mathrm{id}_F)_{(A_0,B)} \\
&= \mathrm{id}_{F(A_0,B)} \\
&= \mathrm{id}_{F(A_0,\cdot)(B)} \\
&= (\mathrm{id}_{F(A_0,\cdot)})_B.
\end{aligned}$$

Hence $(\mathrm{id}_F)_{(A_0,\cdot)} = \mathrm{id}_{F(A_0,\cdot)}$. Since $A_0 \in \mathrm{Obj}(\mathcal{A})$ is arbitrary, we have that for each $A_0 \in \mathrm{Obj}(\mathcal{A})$, $(\mathrm{id}_F)_{(A_0,\cdot)} = \mathrm{id}_{F(A_0,\cdot)}$.

2. Similart to (ref) prev part

□

4.3 Functor Categories

Definition 4.3.0.1. Let $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathbf{Cat})$. We define the **evaluation functor from $\mathcal{C} \times \mathcal{D}^{\mathcal{C}}$ to $\mathcal{D}$**, denoted $\mathrm{eval}_{(\mathcal{C},\mathcal{D})} : \mathcal{C} \times \mathcal{D}^{\mathcal{C}} \to \mathcal{D}$, by

- $\mathrm{eval}_{(\mathcal{C},\mathcal{D})}(A, F) := F(A), \quad (A, F) \in \mathrm{Obj}(\mathcal{C} \times \mathcal{D}^{\mathcal{C}})$
- $\mathrm{eval}_{(\mathcal{C},\mathcal{D})}(f, \alpha) := G(f) \circ_{\mathcal{D}} \alpha_A, \quad (f, \alpha) \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}^{\mathcal{C}}}$

Exercise 4.3.0.2. Let $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathbf{Cat})$. Then $\mathrm{eval}_{(\mathcal{C},\mathcal{D})} : \mathrm{eval} : \mathcal{C} \times \mathcal{D}^{\mathcal{C}} \to \mathcal{D}$ is a functor.

Proof.

1. Let $(A, F), (B, G) \in \mathrm{Obj}(\mathcal{C} \times \mathcal{D}^{\mathcal{C}})$ and $(f, \alpha) \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}^{\mathcal{C}}}((A, F), (B, G))$. Then $G(f) \in \mathrm{Hom}_{\mathcal{D}}(G(A), G(B))$ and $\alpha_A \in \mathrm{Hom}_{\mathcal{D}}(F(A), G(A))$. Hence

$$\begin{aligned}\mathrm{eval}_{(\mathcal{C},\mathcal{D})}(f, \alpha) &= G(f) \circ_{\mathcal{D}} \alpha_A \\ &\in \mathrm{Hom}_{\mathcal{D}}(F(A), G(B)) \\ &= \mathrm{Hom}_{\mathcal{D}}(\mathrm{eval}_{(\mathcal{C},\mathcal{D})}(A, F), \mathrm{eval}_{(\mathcal{C},\mathcal{D})}(B, G)).\end{aligned}$$

2. Let $(A, F), (B, G), (C, H) \in \mathrm{Obj}(\mathcal{C} \times \mathcal{D}^{\mathcal{C}})$, $(f, \alpha) \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}^{\mathcal{C}}}((A, F), (B, G))$ and $(g, \beta) \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{D}^{\mathcal{C}}}((B, G), (C, H))$. Since $\beta \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(G, H)$, we have that $H(f) \circ_{\mathcal{D}} \beta_A = \beta_B \circ_{\mathcal{D}} G(f)$, i.e. the following diagram commutes:

$$\begin{tikzcd} G(A) \arrow[r, "\beta_A"] \arrow[d, "G(f)"'] & H(A) \arrow[d, "H(f)"] \\ G(B) \arrow[r, "\beta_B"] & H(B). \end{tikzcd}$$

$$\begin{aligned}\mathrm{eval}_{(\mathcal{C},\mathcal{D})}((g, \beta) \circ_{\mathcal{C}\times\mathcal{D}^{\mathcal{C}}} (f, \alpha)) &= \mathrm{eval}_{(\mathcal{C},\mathcal{D})}(g \circ_{\mathcal{C}} f, \beta \circ_{\mathcal{D}^{\mathcal{C}}} \alpha) \\ &= H(g \circ_{\mathcal{C}} f) \circ_{\mathcal{D}} (\beta \circ_{\mathcal{D}^{\mathcal{C}}} \alpha)_A \\ &= H(g) \circ_{\mathcal{D}} (H(f) \circ_{\mathcal{D}} \beta_A) \circ_{\mathcal{D}} \alpha_A \\ &= H(g) \circ_{\mathcal{D}} (\beta_B \circ_{\mathcal{D}} G(f)) \circ_{\mathcal{D}} \alpha_A \\ &= (H(g) \circ_{\mathcal{D}} \beta_B) \circ_{\mathcal{D}} (G(f) \circ_{\mathcal{D}} \alpha_A) \\ &= \mathrm{eval}_{(\mathcal{C},\mathcal{D})}(g, \beta) \circ_D \mathrm{eval}_{(\mathcal{C},\mathcal{D})}(f, \alpha)).\end{aligned}$$

3. For each $(A, F) \in \mathrm{Obj}(\mathcal{C} \times \mathcal{D}^{\mathcal{C}})$, we have that

$$\begin{aligned}\mathrm{eval}_{(\mathcal{C},\mathcal{D})}(\mathrm{id}^{\mathcal{C}\times\mathcal{D}^{\mathcal{C}}}_{(A,F)}) &= \mathrm{eval}_{(\mathcal{C},\mathcal{D})}(\mathrm{id}^{\mathcal{C}}_A, \mathrm{id}^{\mathcal{D}^{\mathcal{C}}}_F) \\ &= \mathrm{eval}_{(\mathcal{C},\mathcal{D})}(\mathrm{id}^{\mathcal{C}}_A, \mathrm{id}^{\mathcal{D}^{\mathcal{C}}}_F) \\ &= F(\mathrm{id}^{\mathcal{C}}_A) \circ_{\mathcal{D}} (\mathrm{id}^{\mathcal{D}^{\mathcal{C}}}_F)_A \\ &= \mathrm{id}^{\mathcal{D}}_{F(A)} \circ_{\mathcal{D}} \mathrm{id}^{\mathcal{D}}_{F(A)} \\ &= \mathrm{id}^{\mathcal{D}}_{F(A)} \\ &= \mathrm{id}^{\mathcal{D}}_{\mathrm{eval}_{(\mathcal{C},\mathcal{D})}(A,F)}\end{aligned}$$

Hence $\mathrm{eval}_{(\mathcal{C},\mathcal{D})} : \mathcal{C} \times \mathcal{D}^{\mathcal{C}} \to \mathcal{D}$ is a functor. □

Exercise 4.3.0.3. Find two functors $F, G : \mathbf{Cat} \times \mathbf{Cat} \to \mathbf{Cat}$ such that $\mathrm{eval} \mapsto \mathrm{eval}_{(\mathcal{C},\mathcal{D})}$ is a natural isomorphism $F \Rightarrow G$.

Definition 4.3.0.4. Let $(\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2) \in \mathrm{Obj}(\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat})$, $(F, G) \in \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}}\times\mathbf{Cat}}((\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2))$, $H, J \in \mathrm{Obj}(\mathcal{B}_1^{\mathcal{A}_1})$ and $\alpha \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}(H, J)$.

- For each $H \in \mathrm{Obj}(\mathcal{B}_1^{\mathcal{A}_1})$, we define the **internal Hom functor object induced by F and G at H**, denoted $G^F(H) \in \mathrm{Obj}(\mathcal{B}_2^{\mathcal{A}_2})$, by $G^F(H) := G \circ_{\mathbf{Cat}} H \circ_{\mathbf{Cat}} F$
- **internal Hom functor natural transformation induced by F and G at α**, denoted $G^F(\alpha) : G^F(H) \Rightarrow G^F(J)$, by $G^F(\alpha)_A := G(\alpha_{F(A)})$.

Exercise 4.3.0.5. Let $(\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2) \in \mathrm{Obj}(\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat})$, $(F, G) \in \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}}((\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2))$, $H, J \in \mathrm{Obj}(\mathcal{B}_1^{\mathcal{A}_1})$ and $\alpha \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}(H, J)$. Then $G^F(\alpha) : G^F(H) \Rightarrow G^F(J)$ is a natural transformation.

Proof.

- Let $A \in \mathrm{Obj}(\mathcal{A}_2)$. Since $\alpha \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}(H, J)$, we have that $\alpha_{F(A} \in \mathrm{Hom}_{\mathcal{B}_1}(H(F(A)), J(F(A)))$. Then

$$\begin{aligned} G^F(\alpha)_A &= G(\alpha_{F(A)}) \\ &\in \mathrm{Hom}_{\mathcal{B}_2}(G(H(F(A))), G(J(F(A)))) \\ &= \mathrm{Hom}_{\mathcal{B}_2}(G^F(H)(A)), G^F(J)(A)). \end{aligned}$$

  Since $A \in \mathrm{Obj}(\mathcal{A}_2)$ is arbitrary, we have that for each $A \in \mathrm{Obj}(\mathcal{A}_2)$, $G^F(\alpha)_A \in \mathrm{Hom}_{\mathcal{B}_2}(G^F(H)(A)), G^F(J)(A))$.

- Let $A_1, A_2 \in \mathrm{Obj}(\mathcal{A}_2)$ and $f \in \mathrm{Hom}_{\mathcal{A}_2}(A_1, A_2)$. Define $A_1', A_2' \in \mathrm{Obj}(\mathcal{A}_1)$ and $f' \in \mathrm{Hom}_{\mathcal{A}_1}(A_1', A_2')$ by $A_k' := F(A_k)$ and $f' := F(f)$. Since $\alpha \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}(H, J)$, we have that

$$J(f') \circ_{\mathcal{B}_1} \alpha_{A_1'} = \alpha_{A_2'} \circ_{\mathcal{B}_1} H(f'),$$

  i.e. the following diagram commutes:

$$\begin{tikzcd} H(A_1') \arrow[r, "\alpha_{A_1'}"] \arrow[d, "H(f')"'] & J(A_1') \arrow[d, "J(f')"] \\ H(A_2') \arrow[r, "\alpha_{A_2'}"] & J(A_2'). \end{tikzcd}$$

  Therefore,

$$\begin{aligned} G^F(J)(f) \circ_{\mathcal{B}_2} G^F(\alpha)_{A_1} &= [G \circ_{\mathbf{Cat}} J \circ_{\mathbf{Cat}} F](f) \circ_{\mathcal{B}_2} G(\alpha_{F(A_1)}) \\ &= G(J(F(f))) \circ_{\mathcal{B}_2} G(\alpha_{F(A_1)}) \\ &= G(J(f')) \circ_{\mathcal{B}_1} G(\alpha_{A_2'}) \\ &= G(J(f') \circ_{\mathcal{B}_1} \alpha_{A_1'}) \\ &= G(\alpha_{A_2'} \circ_{\mathcal{B}_1} H(f')) \\ &= G(\alpha_{A_2'}) \circ_{\mathcal{B}_2} G(H(f')) \\ &= G(\alpha_{F(A_2)}) \circ_{\mathcal{B}_2} G(H(F(f))) \\ &= G(\alpha_{F(A_2)}) \circ_{\mathcal{B}_2} [G \circ_{\mathbf{Cat}} H \circ_{\mathbf{Cat}} F](f) \\ &= G^F(\alpha)_{A_2} \circ_{\mathcal{B}_2} G^F(H)(f), \end{aligned}$$

  i.e. the following diagram commutes:

$$\begin{tikzcd} G^F(H)(A_1) \arrow[r, "(G^F_\alpha)_{A_1}"] \arrow[d, "G^F(H)(f)"'] & G^F(J)(A_1) \arrow[d, "G^F(J)(f)"] \\ G^F(H)(A_2) \arrow[r, "(G^F_\alpha)_{A_2}"] & G^F(J)(A_2). \end{tikzcd}$$

  Since $A_1, A_2 \in \mathrm{Obj}(\mathcal{A}_2)$ and $f \in \mathrm{Hom}_{\mathcal{A}_2}(A_1, A_2)$ are arbitrary, we have that for each $A_1, A_2 \in \mathrm{Obj}(\mathcal{A}_2)$ and $f \in \mathrm{Hom}_{\mathcal{A}_2}(A_1, A_2)$, $G^F(J)(f) \circ_{\mathcal{B}_2} G^F(\alpha)_{A_1} = G^F(\alpha)_{A_2} \circ_{\mathcal{B}_2} G^F(H)(f)$.

Thus $G^F(\alpha) : G^F(H) \Rightarrow G^F(J)$ is a natural transformation. □

Exercise 4.3.0.6. Let $(\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2) \in \mathrm{Obj}(\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat})$, $(F, G) \in \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}}((\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2))$, $H, J, K \in \mathrm{Obj}(\mathcal{B}_1^{\mathcal{A}_1})$, $\alpha \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}(H, J)$ and $\beta \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}(J, K)$. Then $G^F(\beta \circ_{\mathcal{B}_1^{\mathcal{A}_1}} \alpha) = G^F(\beta) \circ_{\mathcal{B}_2^{\mathcal{A}_2}} G^F(\alpha)$.

Proof. For each $A \in \mathrm{Obj}(\mathcal{A}_2)$,

$$\begin{aligned} G^F(\beta \circ_{\mathcal{B}_1^{\mathcal{A}_1}} \alpha)_A &= G([\beta \circ_{\mathcal{B}_1^{\mathcal{A}_1}} \alpha]_{F(A)}) \\ &= G(\beta_{F(A)} \circ_{\mathcal{B}_1} \alpha_{F(A)}) \\ &= G(\beta_{F(A)}) \circ_{\mathcal{B}_2} G(\alpha_{F(A)}) \\ &= G^F(\beta)_A \circ_{\mathcal{B}_2} G^F(\alpha)_A \\ &= (G^F(\beta) \circ_{\mathcal{B}_2^{\mathcal{A}_2}} G^F(\alpha))_A. \end{aligned}$$

Hence $G^F(\beta \circ_{\mathcal{B}_1^{\mathcal{A}_1}} \alpha) = G^F(\beta) \circ_{\mathcal{B}_2^{\mathcal{A}_2}} G^F(\alpha)$. □

Exercise 4.3.0.7. Let $(\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2) \in \mathrm{Obj}(\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat})$, $(F, G) \in \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}}((\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2))$, $H \in \mathrm{Obj}(\mathcal{B}_1^{\mathcal{A}_1})$, Then $G^F(\mathrm{id}_H^{\mathcal{B}_1^{\mathcal{A}_1}}) = \mathrm{id}_{G^F(H)}^{\mathcal{B}_2^{\mathcal{A}_2}}$.

Proof. For each $A \in \mathcal{A}_2$,

$$\begin{aligned} G^F(\mathrm{id}_H^{\mathcal{B}_1^{\mathcal{A}_1}})_A &= G((\mathrm{id}_H^{\mathcal{B}_1^{\mathcal{A}_1}})_{F(A)}) \\ &= G(\mathrm{id}_{H(F(A))}^{\mathcal{B}_1}) \\ &= \mathrm{id}_{G(H(F(A)))}^{\mathcal{B}_1} \\ &= \mathrm{id}_{G \circ_{\mathbf{Cat}} H \circ_{\mathbf{Cat}} F(A)}^{\mathcal{B}_1} \\ &= \mathrm{id}_{G^F(H)(A)}^{\mathcal{B}_1} \\ &= (\mathrm{id}_{G^F(H)}^{\mathcal{B}_2^{\mathcal{A}_2}})_A. \end{aligned}$$

Hence $G^F(\mathrm{id}_H^{\mathcal{B}_1^{\mathcal{A}_1}}) = \mathrm{id}_{G^F(H)}^{\mathcal{B}_2^{\mathcal{A}_2}}$. □

Definition 4.3.0.8. Let $(\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2) \in \mathrm{Obj}(\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat})$ and $(F, G) \in \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}}((\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2))$. We define the **internal Hom functor of Cat at** (F, G), denoted $[F, G]_{\mathbf{Cat}} : \mathcal{B}_1^{\mathcal{A}_1} \to \mathcal{B}_2^{\mathcal{A}_2}$ or $G^F : \mathcal{B}_1^{\mathcal{A}_1} \to \mathcal{B}_2^{\mathcal{A}_2}$, by

- $[F, G]_{\mathbf{Cat}}(H) := G^F(H), \quad H \in \mathrm{Obj}(\mathcal{B}_1^{\mathcal{A}_1})$
- $[F, G]_{\mathbf{Cat}}(\alpha) := G^F(\alpha), \quad \alpha \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}$

Exercise 4.3.0.9. Let $(\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2) \in \mathrm{Obj}(\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat})$ and $(F, G) \in \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}}((\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2))$. Then $G^F : \mathcal{B}_1^{\mathcal{A}_1} \to \mathcal{B}_2^{\mathcal{A}_2}$ is a functor.

Proof.

1. Exercise 4.3.0.5 implies that for each $H, J \in \mathrm{Obj}(\mathcal{B}_1^{\mathcal{A}_1})$ and $\alpha \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}(H, J)$, $G^F(\alpha) \in \mathrm{Hom}_{\mathcal{B}_2^{\mathcal{A}_2}}(G^F(H), G^F(J))$.
2. Exercise 4.3.0.6 implies that for each $H, J, K \in \mathrm{Obj}(\mathcal{B}_1^{\mathcal{A}_1})$, $\alpha \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}(H, J)$ and $\beta \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}(J, K)$, $G^F(\beta \circ_{\mathcal{B}_1^{\mathcal{A}_1}} \alpha) = G^F(\beta) \circ_{\mathcal{B}_2^{\mathcal{A}_2}} G^F(\alpha)$.
3. Exercise 4.3.0.7 implies that for each $A \in \mathrm{Obj}(\mathcal{A}_2)$, $G^F(\mathrm{id}_H^{\mathcal{B}_1^{\mathcal{A}_1}}) = \mathrm{id}_{G^F(H)}^{\mathcal{B}_2^{\mathcal{A}_2}}$.

Hence $G^F : \mathcal{B}_1^{\mathcal{A}_1} \to \mathcal{B}_2^{\mathcal{A}_2}$ is a functor. □

Exercise 4.3.0.10. Let $(\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2), (\mathcal{A}_3, \mathcal{B}_3) \in \mathrm{Obj}(\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat})$, $(F_{1,2}, G_{1,2}) \in \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}}((\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2))$ and $(F_{2,3}, G_{2,3}) \in \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}}((\mathcal{A}_2, \mathcal{B}_2), (\mathcal{A}_3, \mathcal{B}_3))$. Then

$$(G_{2,3} \circ_{\mathbf{Cat}} G_{1,2})^{F_{2,3} \circ_{\mathbf{Cat}^{\mathrm{op}}} F_{1,2}} = G_{2,3}^{F_{2,3}} \circ_{\mathbf{Cat}} G_{1,2}^{F_{1,2}}.$$

Proof.

- For each $H \in \mathrm{Obj}(\mathcal{B}_1^{\mathcal{A}_1})$,

$$\begin{aligned}(G_{2,3} \circ_{\mathbf{Cat}} G_{1,2})^{F_{2,3} \circ_{\mathbf{Cat}^{\mathrm{op}}} F_{1,2}}(H) &= G_{2,3} \circ_{\mathbf{Cat}} G_{1,2} \circ_{\mathbf{Cat}} H \circ_{\mathbf{Cat}} (F_{2,3} \circ_{\mathbf{Cat}^{\mathrm{op}}} F_{1,2}) \\ &= G_{2,3} \circ_{\mathbf{Cat}} G_{1,2} \circ_{\mathbf{Cat}} H \circ_{\mathbf{Cat}} (F_{1,2} \circ_{\mathbf{Cat}} F_{2,3}) \\ &= G_{2,3} \circ_{\mathbf{Cat}} (G_{1,2} \circ_{\mathbf{Cat}} H \circ_{\mathbf{Cat}} F_{1,2}) \circ_{\mathbf{Cat}} F_{2,3} \\ &= G_{2,3} \circ_{\mathbf{Cat}} (G_{1,2}^{F_{1,2}}(H)) \circ_{\mathbf{Cat}} F_{2,3} \\ &= G_{2,3}^{F_{2,3}}(G_{1,2}^{F_{1,2}}(H)) \\ &= G_{2,3}^{F_{2,3}} \circ_{\mathbf{Cat}} G_{1,2}^{F_{1,2}}(H).\end{aligned}$$

- Let $H, K \in \mathrm{Obj}(\mathcal{B}_1^{\mathcal{A}_1})$ and $\alpha \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}(H, K)$. Then for each $A \in \mathrm{Obj}(\mathcal{A}_3)$,

$$\begin{aligned}[(G_{2,3} \circ_{\mathbf{Cat}} G_{1,2})^{F_{2,3} \circ_{\mathbf{Cat}^{\mathrm{op}}} F_{1,2}}(\alpha)]_A &= [(G_{2,3} \circ_{\mathbf{Cat}} G_{1,2})^{F_{1,2} \circ_{\mathbf{Cat}} F_{2,3}}(\alpha)]_A \\ &= G_{2,3} \circ_{\mathbf{Cat}} G_{1,2}(\alpha_{F_{1,2} \circ_{\mathbf{Cat}} F_{2,3}(A)}) \\ &= G_{2,3}(G_{1,2}(\alpha_{F_{1,2}(F_{2,3}(A))})) \\ &= G_{2,3}(G_{1,2}^{F_{1,2}}(\alpha)_{F_{2,3}(A)}) \\ &= [G_{2,3}^{F_{2,3}}(G_{1,2}^{F_{1,2}}(\alpha))]_A \\ &= [G_{2,3}^{F_{2,3}} \circ_{\mathbf{Cat}} G_{1,2}^{F_{1,2}}(\alpha)]_A\end{aligned}$$

  Thus $(G_{2,3} \circ_{\mathbf{Cat}} G_{1,2})^{F_{2,3} \circ_{\mathbf{Cat}^{\mathrm{op}}} F_{1,2}}(\alpha) = G_{2,3}^{F_{2,3}} \circ_{\mathbf{Cat}} G_{1,2}^{F_{1,2}}(\alpha)$. Since $H, K \in \mathrm{Obj}(\mathcal{B}_1^{\mathcal{A}_1})$ and $\alpha \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}(H, K)$ are arbitrary, we have that for each $H, K \in \mathrm{Obj}(\mathcal{B}_1^{\mathcal{A}_1})$ and $\alpha \in \mathrm{Hom}_{\mathcal{B}_1^{\mathcal{A}_1}}(H, K)$, $(G_{2,3} \circ_{\mathbf{Cat}} G_{1,2})^{F_{2,3} \circ_{\mathbf{Cat}^{\mathrm{op}}} F_{1,2}}(\alpha) = G_{2,3}^{F_{2,3}} \circ_{\mathbf{Cat}} G_{1,2}^{F_{1,2}}(\alpha)$.

Therefore $(G_{2,3} \circ_{\mathbf{Cat}} G_{1,2})^{F_{2,3} \circ_{\mathbf{Cat}^{\mathrm{op}}} F_{1,2}} = G_{2,3}^{F_{2,3}} \circ_{\mathbf{Cat}} G_{1,2}^{F_{1,2}}$. □

Exercise 4.3.0.11. Let $\mathcal{A}, \mathcal{B} \in \mathrm{Obj}(\mathbf{Cat})$. Then

$$(\mathrm{id}_{\mathcal{B}}^{\mathbf{Cat}})^{\mathrm{id}_{\mathcal{A}}^{\mathbf{Cat}}} = \mathrm{id}_{\mathcal{B}^{\mathcal{A}}}^{\mathbf{Cat}}.$$

Proof. Let $\mathcal{A}, \mathcal{B} \in \mathrm{Obj}(\mathbf{Cat})$.

- For each $H \in \mathrm{Obj}(\mathcal{B}^{\mathcal{A}})$,

$$\begin{aligned}(\mathrm{id}_{\mathcal{B}}^{\mathbf{Cat}})^{\mathrm{id}_{\mathcal{A}}^{\mathbf{Cat}}}(H) &= \mathrm{id}_{\mathcal{B}}^{\mathbf{Cat}} \circ_{\mathbf{Cat}} H \circ_{\mathbf{Cat}} \mathrm{id}_{\mathcal{A}}^{\mathbf{Cat}} \\ &= H \\ &= \mathrm{id}_{\mathcal{B}^{\mathcal{A}}}^{\mathbf{Cat}}(H).\end{aligned}$$

- Let $H, K \in \mathrm{Obj}(\mathcal{B}^{\mathcal{A}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{B}^{\mathcal{A}}}(H, K)$. Then for each $A \in \mathrm{Obj}(\mathcal{A})$,

$$\begin{aligned}[(\mathrm{id}_{\mathcal{B}}^{\mathbf{Cat}})^{\mathrm{id}_{\mathcal{A}}^{\mathbf{Cat}}}(\alpha)]_A &= \mathrm{id}_{\mathcal{B}}^{\mathbf{Cat}}(\alpha_{\mathrm{id}_{\mathcal{A}}^{\mathbf{Cat}}(A)}) \\ &= \mathrm{id}_{\mathcal{B}}^{\mathbf{Cat}}(\alpha_A) \\ &= \alpha_A \\ &= [\mathrm{id}_{\mathcal{B}^{\mathcal{A}}}^{\mathbf{Cat}}(\alpha)]_A\end{aligned}$$

  Hence $(\mathrm{id}_{\mathcal{B}}^{\mathbf{Cat}})^{\mathrm{id}_{\mathcal{A}}^{\mathbf{Cat}}}(\alpha) = \mathrm{id}_{\mathcal{B}^{\mathcal{A}}}^{\mathbf{Cat}}(\alpha)$.

□

Definition 4.3.0.12. We define the **internal Hom functor of Cat**, denoted $[\cdot, \cdot]_{\mathbf{Cat}} : \mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat} \to \mathbf{Cat}$, by

- $[\mathcal{C}, \mathcal{D}]_{\mathbf{Cat}} := \mathcal{D}^{\mathcal{C}}, \quad (\mathcal{C}, \mathcal{D}) \in \mathrm{Obj}(\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat})$,

- $[F, G]_{\mathbf{Cat}} := G^F, \quad (F, G) \in \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}}$.

Exercise 4.3.0.13. We have that $[\cdot, \cdot]_{\mathbf{Cat}} : \mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat} \to \mathbf{Cat}$ is a functor.

Note 4.3.0.14. When the context is clear, we may write $[\cdot, \cdot]$ in place of $[\cdot, \cdot]_{\mathbf{Cat}}$.

Proof.

1. Let $(\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2) \in \mathrm{Obj}(\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat})$ $(F, G) \in \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}}((\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2))$. Exercise 4.3.0.9 implies that

$$\begin{aligned}
[\cdot, \cdot](F, G) &= G^F \\
&\in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{B}_1^{\mathcal{A}_1}, \mathcal{B}_2^{\mathcal{A}_2}) \\
&= \mathrm{Hom}_{\mathbf{Cat}}([\mathcal{A}_1, \mathcal{B}_1], [\mathcal{A}_2, \mathcal{B}_2]) \\
&= \mathrm{Hom}_{\mathbf{Cat}}([\cdot, \cdot](\mathcal{A}_1, \mathcal{B}_1), [\cdot, \cdot](\mathcal{A}_2, \mathcal{B}_2)).
\end{aligned}$$

2. $(\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2), (\mathcal{A}_2, \mathcal{B}_3) \in \mathrm{Obj}(\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat})$ $(F_{1,2}, G_{1,2}) \in \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}}((\mathcal{A}_1, \mathcal{B}_1), (\mathcal{A}_2, \mathcal{B}_2))$ and $(F_{2,3}, G_{2,3}) \in \mathrm{Hom}_{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}}((\mathcal{A}$
Exercise 4.3.0.10 then implies that

$$\begin{aligned}
[\cdot, \cdot]((F_{2,3}, G_{2,3}) \circ_{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}} (F_{1,2}, G_{1,2})) &= [\cdot, \cdot](F_{2,3} \circ_{\mathbf{Cat}^{\mathrm{op}}} F_{1,2}, G_{2,3} \circ_{\mathbf{Cat}} G_{1,2}) \\
&= (G_{2,3} \circ_{\mathbf{Cat}} G_{1,2})^{F_{2,3} \circ_{\mathbf{Cat}^{\mathrm{op}}} F_{1,2}} \\
&= G_{2,3}^{F_{2,3}} \circ_{\mathbf{Cat}} G_{1,2}^{F_{1,2}} \\
&= [\cdot, \cdot](F_{2,3}, G_{2,3}) \circ_{\mathbf{Cat}} [\cdot, \cdot](F_{1,2}, G_{1,2}).
\end{aligned}$$

3. Exercise 4.3.0.11 implies that for each $\mathcal{A}, \mathcal{B} \in \mathrm{Obj}(\mathbf{Cat})$,

$$\begin{aligned}
[\cdot, \cdot](\mathrm{id}^{\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}}_{(\mathcal{A}, \mathcal{B})}) &= [\cdot, \cdot](\mathrm{id}^{\mathbf{Cat}}_{\mathcal{A}}, \mathrm{id}^{\mathbf{Cat}}_{\mathcal{B}}) \\
&= (\mathrm{id}^{\mathbf{Cat}}_{\mathcal{B}})^{\mathrm{id}^{\mathbf{Cat}}_{\mathcal{A}}} \\
&= \mathrm{id}^{\mathbf{Cat}}_{\mathcal{B}^{\mathcal{A}}} \\
&= \mathrm{id}^{\mathbf{Cat}}_{[\cdot, \cdot](\mathcal{A}, \mathcal{B})}.
\end{aligned}$$

Hence $[\cdot, \cdot] : \mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat} \to \mathbf{Cat}$ is a functor. $\square$

4.4 Opposition

relabel and organize

Definition 4.4.0.1. Let $\mathcal{C}, \mathcal{D}$ be a categories and $F : \mathcal{C} \to \mathcal{D}$. We define the **opposite of** F, denoted $F^{\mathrm{op}} : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}^{\mathrm{op}}$, by

- $F^{\mathrm{op}}(A) := F(A), \quad A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$
- $F^{\mathrm{op}}(f) := F(f), \quad f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}$

Exercise 4.4.0.2. Let $\mathcal{C}, \mathcal{D}$ be a categories and $F : \mathcal{C} \to \mathcal{D}$. Then $F^{\mathrm{op}} : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}^{\mathrm{op}}$ is a functor.

Proof.

1. For each $A, B \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ and $f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B)$,
$$\begin{aligned} F^{\mathrm{op}}(f) &= f \\ &\in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) \\ &= \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(F^{\mathrm{op}}(A), F^{\mathrm{op}}(B)). \end{aligned}$$
2. For each $A, B, C \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$, $f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B)$ and $g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(B, C)$,
$$\begin{aligned} F^{\mathrm{op}}(g \circ_{\mathcal{C}^{\mathrm{op}}} f) &= g \circ_{\mathcal{C}^{\mathrm{op}}} f \\ &= F^{\mathrm{op}}(g) \circ_{\mathcal{C}^{\mathrm{op}}} F^{\mathrm{op}}(f). \end{aligned}$$
3. For each $A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$,
$$\begin{aligned} F^{\mathrm{op}}(\mathrm{id}_A^{\mathcal{C}^{\mathrm{op}}}) &= \mathrm{id}_A^{\mathcal{C}^{\mathrm{op}}} \\ &= \mathrm{id}_{F^{\mathrm{op}}(A)}^{\mathcal{C}^{\mathrm{op}}}. \end{aligned}$$

Hence $F^{\mathrm{op}} : \mathcal{C}^{\mathrm{op}} \to \mathcal{D}^{\mathrm{op}}$ is a functor. □

Exercise 4.4.0.3. Let $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$, $F \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{A}, \mathcal{B})$ and $G \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{B}, \mathcal{C})$. Then $G \circ_{\mathbf{Cat}} F^{\mathrm{op}} = G^{\mathrm{op}} \circ_{\mathbf{Cat}} F^{\mathrm{op}}$.

Proof.

- For each $A \in \mathrm{Obj}(A^{\mathrm{op}})$,
$$\begin{aligned} G \circ_{\mathbf{Cat}} F^{\mathrm{op}}(A) &= G \circ_{\mathbf{Cat}} F(A) \\ &= G(F(A)) \\ &= G^{\mathrm{op}}(F^{\mathrm{op}}(A)) \\ &= G^{\mathrm{op}} \circ_{\mathbf{Cat}} F^{\mathrm{op}}(A). \end{aligned}$$
- For each $A, B, C \in \mathrm{Obj}(\mathcal{A}^{\mathrm{op}})$ and $f \in \mathrm{Hom}_{\mathcal{A}^{\mathrm{op}}}(A, B)$,
$$\begin{aligned} G \circ_{\mathbf{Cat}} F^{\mathrm{op}}(f) &= G \circ_{\mathbf{Cat}} F(f) \\ &= G(F(f)) \\ &= G^{\mathrm{op}}(F^{\mathrm{op}}(f)) \\ &= G^{\mathrm{op}} \circ_{\mathbf{Cat}} F^{\mathrm{op}}(f). \end{aligned}$$

Thus $G \circ_{\mathbf{Cat}} F^{\mathrm{op}} = G^{\mathrm{op}} \circ_{\mathbf{Cat}} F^{\mathrm{op}}$. □

Exercise 4.4.0.4. Let $\mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$. Then $\mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}^{\mathrm{op}}} = \mathrm{id}_{\mathcal{C}^{\mathrm{op}}}^{\mathbf{Cat}}$.

Proof.

- For each $A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$,

$$\begin{aligned}\mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}^{\mathrm{op}}}(A) &= \mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}}(A)\\ &= A\\ &= \mathrm{id}_{\mathcal{C}^{\mathrm{op}}}^{\mathbf{Cat}}(A).\end{aligned}$$

- For each $A, B \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ and $f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B)$,

$$\begin{aligned}\mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}^{\mathrm{op}}}(f) &= \mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}}(f)\\ &= f\\ &= \mathrm{id}_{\mathcal{C}^{\mathrm{op}}}^{\mathbf{Cat}}(f).\end{aligned}$$

Thus $\mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}^{\mathrm{op}}} = \mathrm{id}_{\mathcal{C}^{\mathrm{op}}}^{\mathbf{Cat}}$. □

Definition 4.4.0.5. We define the **opposite functor on Cat**, denoted $\mathrm{op}^{\mathbf{Cat}} : \mathbf{Cat} \to \mathbf{Cat}$, by

- $\mathrm{op}^{\mathbf{Cat}}(\mathcal{C}) := \mathcal{C}^{\mathrm{op}}$
- $\mathrm{op}^{\mathbf{Cat}}(F) := F^{\mathrm{op}}$

Note 4.4.0.6. When the context is clear, we may write op in place of $\mathrm{op}^{\mathbf{Cat}}$.

Exercise 4.4.0.7. We have that $\mathrm{op}^{\mathbf{Cat}} : \mathbf{Cat} \to \mathbf{Cat}$ is a functor.

Proof.

1. Exercise 4.4.0.2 implies that for each $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathbf{Cat})$ and $F \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}, \mathcal{D})$,

$$\begin{aligned}\mathrm{op}(F) &= F^{\mathrm{op}}\\ &\in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{C}^{\mathrm{op}}, \mathcal{D}^{\mathrm{op}})\\ &= \mathrm{Hom}_{\mathbf{Cat}}(\mathrm{op}(\mathcal{C}), \mathrm{op}(\mathcal{D})).\end{aligned}$$

2. Exercise 4.4.0.3 implies that for each $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$, $F \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{A}, \mathcal{B})$ and $G \in \mathrm{Hom}_{\mathbf{Cat}}(\mathcal{B}, \mathcal{C})$,

$$\begin{aligned}\mathrm{op}(G \circ_{\mathbf{Cat}} F) &= G \circ_{\mathbf{Cat}} F^{\mathrm{op}}\\ &= G^{\mathrm{op}} \circ_{\mathbf{Cat}} F^{\mathrm{op}}\\ &= \mathrm{op}((G) \circ_{\mathbf{Cat}} \mathrm{op}(F).\end{aligned}$$

3. Exercise 4.4.0.4 implies that for each $\mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$,

$$\begin{aligned}\mathrm{op}(\mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}}) &= \mathrm{id}_{\mathcal{C}}^{\mathbf{Cat}^{\mathrm{op}}} &&= \mathrm{id}_{\mathcal{C}^{\mathrm{op}}}^{\mathbf{Cat}}\\ &= \mathrm{id}_{\mathrm{op}(\mathcal{C})}^{\mathbf{Cat}}.\end{aligned}$$

Thus $\mathrm{op} : \mathbf{Cat} \to \mathbf{Cat}$ is a functor. □

Definition 4.4.0.8. Let $\mathcal{C}, \mathcal{D}$ be categories, $F, G : \mathcal{C} \to \mathcal{D}$ functors and $\alpha : F \Rightarrow G$ a natural transformation. We define the **opposite of** α, denoted $\alpha^{\mathrm{op}} : G^{\mathrm{op}} \Rightarrow F^{\mathrm{op}}$, by

$$\alpha_A^{\mathrm{op}} := \alpha_A, \quad A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$$

Exercise 4.4.0.9. Let $\mathcal{C}, \mathcal{D}$ be categories, $F, G : \mathcal{C} \to \mathcal{D}$ functors and $\alpha : F \Rightarrow G$ a natural transformation. Then $\alpha^{\mathrm{op}} : G^{\mathrm{op}} \Rightarrow F^{\mathrm{op}}$ is a natural transformation.

Proof.

- Let $A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$. Then $A \in \mathrm{Obj}(\mathcal{C})$ and

$$\begin{aligned}\alpha_A^{\mathrm{op}} &= \alpha_A \\ &\in \mathrm{Hom}_{\mathcal{D}}(F(A), G(A)) \\ &= \mathrm{Hom}_{\mathcal{D}^{\mathrm{op}}}(G(A), F(A)) \\ &= \mathrm{Hom}_{\mathcal{D}^{\mathrm{op}}}(G^{\mathrm{op}}(A), F^{\mathrm{op}}(A)).\end{aligned}$$

  Since $A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ is arbitrary, we have that for each $A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$, $\alpha_A^{\mathrm{op}} \in \mathrm{Hom}_{\mathcal{D}^{\mathrm{op}}}(G^{\mathrm{op}}(A), F^{\mathrm{op}}(A))$.

- Let $A, B \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ and $f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B)$. Then $A, B \in \mathrm{Obj}(\mathcal{C})$ and $f \in \mathrm{Hom}_{\mathcal{C}}(B, A)$. Since $\alpha : F \Rightarrow G$ is a natural transformation, $G(f) \circ_{\mathcal{D}} \alpha_B = \alpha_A \circ_{\mathcal{D}} F(f)$, i.e. the following diagram commutes:

$$\begin{tikzcd} F(B) \arrow[r, "\alpha_B"] \arrow[d, "F(f)"'] & G(B) \arrow[d, "G(f)"] \\ F(A) \arrow[r, "\alpha_A"'] & G(A). \end{tikzcd}$$

  Therefore

$$\begin{aligned}F^{\mathrm{op}}(f) \circ_{\mathcal{D}^{\mathrm{op}}} \alpha_A^{\mathrm{op}} &= F(f) \circ_{\mathcal{D}^{\mathrm{op}}} \alpha_A \\ &= \alpha_A \circ_{\mathcal{D}} F(f) \\ &= G(f) \circ_{\mathcal{D}} \alpha_B \\ &= \alpha_B \circ_{\mathcal{D}^{\mathrm{op}}} G(f) \\ &= \alpha_B^{\mathrm{op}} \circ_{\mathcal{D}^{\mathrm{op}}} G^{\mathrm{op}}(f),\end{aligned}$$

  i.e. the following diagram commutes:

$$\begin{tikzcd} G^{\mathrm{op}}(A) \arrow[r, "\alpha_A^{\mathrm{op}}"] \arrow[d, "G^{\mathrm{op}}(f)"'] & F^{\mathrm{op}}(A) \arrow[d, "F^{\mathrm{op}}(f)"] \\ G^{\mathrm{op}}(B) \arrow[r, "\alpha_B^{\mathrm{op}}"'] & F^{\mathrm{op}}(B). \end{tikzcd}$$

  Since $A, B \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ and $f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B)$ are arbitrary, we have that for each $A, B \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ and $f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B)$, $F^{\mathrm{op}}(f) \circ_{\mathcal{D}^{\mathrm{op}}} \alpha_A^{\mathrm{op}} = \alpha_B^{\mathrm{op}} \circ_{\mathcal{D}^{\mathrm{op}}} G^{\mathrm{op}}(f)$.

Thus $\alpha^{\mathrm{op}} : G^{\mathrm{op}} \Rightarrow F^{\mathrm{op}}$ is a natural transformation. □

Exercise 4.4.0.10. Let $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathbf{Cat})$, $F, G, H \in \mathrm{Obj}(\mathcal{D}^{\mathcal{C}})$, $\alpha \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(F, G)$ and $\beta \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(G, H)$. Then $(\beta \circ_{\mathcal{D}^{\mathcal{C}}} \alpha)^{\mathrm{op}} = \alpha^{\mathrm{op}} \circ_{(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}} \beta^{\mathrm{op}}$.

Proof. For each $A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$,

$$\begin{aligned}(\beta \circ_{\mathcal{D}^{\mathcal{C}}} \alpha)_A^{\mathrm{op}} &= (\beta \circ_{\mathcal{D}^{\mathcal{C}}} \alpha)_A \\ &= \beta_A \circ_{\mathcal{D}} \alpha_A \\ &= \alpha_A \circ_{\mathcal{D}^{\mathrm{op}}} \beta_A \\ &= \alpha_A^{\mathrm{op}} \circ_{\mathcal{D}^{\mathrm{op}}} \beta_A^{\mathrm{op}} \\ &= (\alpha^{\mathrm{op}} \circ_{(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}} \beta^{\mathrm{op}})_A.\end{aligned}$$

Thus $(\beta \circ_{\mathcal{D}^{\mathcal{C}}} \alpha)^{\mathrm{op}} = \alpha^{\mathrm{op}} \circ_{(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}} \beta^{\mathrm{op}}$. □

Exercise 4.4.0.11. Let $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathbf{Cat})$ and $F \in \mathrm{Obj}(\mathcal{D}^{\mathcal{C}})$. Then $(\mathrm{id}_F^{\mathcal{D}^{\mathcal{C}}})^{\mathrm{op}} = \mathrm{id}_{F^{\mathrm{op}}}^{(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}}$.

Proof. For each $A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$,

$$\begin{aligned}(\mathrm{id}_F^{\mathcal{D}^{\mathcal{C}\,\mathrm{op}}})_A &= (\mathrm{id}_F^{\mathcal{D}^{\mathcal{C}}})_A \\ &= \mathrm{id}_{F(A)}^{\mathcal{D}} \\ &= \mathrm{id}_{F(A)}^{\mathcal{D}^{\mathrm{op}}} \\ &= \mathrm{id}_{F^{\mathrm{op}}(A)}^{\mathcal{D}^{\mathrm{op}}} \\ &= (\mathrm{id}_{F^{\mathrm{op}}}^{(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}})_A.\end{aligned}$$

Thus $(\mathrm{id}_F^{\mathcal{D}^{\mathcal{C}}})^{\mathrm{op}} = \mathrm{id}_{F^{\mathrm{op}}}^{(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}}$. □

Definition 4.4.0.12. Let $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathbf{Cat})$. We define the **opposite functor on** $\mathcal{D}^{\mathcal{C}}$, denoted $\mathrm{op}_{(\mathcal{C},\mathcal{D})} : \mathcal{D}^{\mathcal{C}} \to [(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}]^{\mathrm{op}}$, by

- $\mathrm{op}_{(\mathcal{C},\mathcal{D})}(F) := F^{\mathrm{op}}$
- $\mathrm{op}_{(\mathcal{C},\mathcal{D})}(\alpha) := \alpha^{\mathrm{op}}$

Exercise 4.4.0.13. Let $\mathcal{C}, \mathcal{D} \in \mathrm{Obj}(\mathbf{Cat})$. Then $\mathrm{op}_{(\mathcal{C},\mathcal{D})} : \mathcal{D}^{\mathcal{C}} \to [(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}]^{\mathrm{op}}$ is a functor.

Proof.

1. Let $F, G \in \mathrm{Obj}(\mathcal{D}^{\mathcal{C}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(F, G)$. Exercise 4.4.0.2 implies that

$$\begin{aligned}\mathrm{op}_{(\mathcal{C},\mathcal{D})}(F), \mathrm{op}_{(\mathcal{C},\mathcal{D})}(G) &\in \mathrm{Obj}((\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}) \\ &= \mathrm{Obj}([(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}]^{\mathrm{op}})\end{aligned}$$

   and Exercise 4.4.0.9 implies that

$$\begin{aligned}\mathrm{op}_{(\mathcal{C},\mathcal{D})}(\alpha) &\in \mathrm{Hom}_{(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}}(G^{\mathrm{op}}, F^{\mathrm{op}}) \\ &= \mathrm{Hom}_{[(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}]^{\mathrm{op}}}(F^{\mathrm{op}}, G^{\mathrm{op}}) \\ &= \mathrm{Hom}_{[(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}]^{\mathrm{op}}}(\mathrm{op}_{(\mathcal{C},\mathcal{D})}(F), \mathrm{op}_{(\mathcal{C},\mathcal{D})}(G)).\end{aligned}$$

   Since $F, G \in \mathrm{Obj}(\mathcal{D}^{\mathcal{C}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(F, G)$ are arbitrary, we have that for each $F, G \in \mathrm{Obj}(\mathcal{D}^{\mathcal{C}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(F, G)$, $\mathrm{op}_{(\mathcal{C},\mathcal{D})}(\alpha) \in \mathrm{Hom}_{[(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}]^{\mathrm{op}}}(\mathrm{op}_{(\mathcal{C},\mathcal{D})}(F), \mathrm{op}_{(\mathcal{C},\mathcal{D})}(G))$.

2. Let $F, G, H \in \mathrm{Obj}(\mathcal{D}^{\mathcal{C}})$, $\alpha \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(F, G)$ and $\beta \in \mathrm{Hom}_{\mathcal{D}^{\mathcal{C}}}(G, H)$. Exercise 4.4.0.10 implies that

$$\begin{aligned}\mathrm{op}_{(\mathcal{C},\mathcal{D})}(\beta \circ_{\mathcal{D}^{\mathcal{C}}} \alpha) &= \beta \circ_{\mathcal{D}^{\mathcal{C}}} \alpha^{\mathrm{op}} \\ &= \alpha^{\mathrm{op}} \circ_{(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}} \beta^{\mathrm{op}} \\ &= \beta^{\mathrm{op}} \circ_{[(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}]^{\mathrm{op}}} \alpha^{\mathrm{op}} \\ &= \mathrm{op}_{(\mathcal{C},\mathcal{D})}(\beta) \circ_{[(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}]^{\mathrm{op}}} \mathrm{op}_{(\mathcal{C},\mathcal{D})}(\alpha).\end{aligned}$$

3. Exercise 4.4.0.11 implies that for each $F \in \mathrm{Obj}(\mathcal{D}^{\mathcal{C}})$,

$$\begin{aligned}\mathrm{op}_{(\mathcal{C},\mathcal{D})}(\mathrm{id}_F^{\mathcal{D}^{\mathcal{C}}}) &= (\mathrm{id}_F^{\mathcal{D}^{\mathcal{C}}})^{\mathrm{op}} \\ &= \mathrm{id}_{F^{\mathrm{op}}}^{(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}} \\ &= \mathrm{id}_{\mathrm{op}_{(\mathcal{C},\mathcal{D})}(F)}^{(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}}.\end{aligned}$$

Thus $\mathrm{op}_{(\mathcal{C},\mathcal{D})} : \mathcal{D}^{\mathcal{C}} \to [(\mathcal{D}^{\mathrm{op}})^{\mathcal{C}^{\mathrm{op}}}]^{\mathrm{op}}$ is a functor. □

Definition 4.4.0.14. We define $\mathrm{op} : [\cdot,\cdot]_{\mathbf{Cat}} \Rightarrow \mathrm{op}^{\mathbf{Cat}} \circ_{\mathbf{Cat}} [\cdot,\cdot] \circ (\mathrm{op}^{\mathbf{Cat}}, \mathrm{op}^{\mathbf{Cat}})$ by $\mathrm{op}_{(\mathcal{C},\mathcal{D})} := \mathrm{op}_{(\mathcal{C},\mathcal{D})}$ rename stuff and reorganize the opposite stuff

Exercise 4.4.0.15. $\emptyset$

4.5 Currying

Definition 4.5.0.1. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be categories, $F : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$, $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$, $B_1, B_2 \in \mathrm{Obj}(\mathcal{B})$, $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$ and $g \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$. We define the

- **natural transformation from** $F(A_1, \cdot)$ **to** $F(A_2, \cdot)$ **induced by** f, denoted $\alpha(F)^f : F(A_1, \cdot) \Rightarrow F(A_2, \cdot)$, by $\alpha(F)^f_B := F(f, \mathrm{id}^{\mathcal{B}}_B)$,
- **natural transformation from** $F(\cdot, B_1)$ **to** $F(\cdot, B_2)$ **induced by** g, denoted $\beta(F)^g : F(\cdot, B_1) \Rightarrow F(\cdot, B_2)$, by $\beta(F)^g_A := F(\mathrm{id}^{\mathcal{A}}_A, g)$,

Exercise 4.5.0.2. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be categories and $F : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$. Then

1. for each $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$ and $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$, $\alpha(F)^f : F(A_1, \cdot) \Rightarrow F(A_2, \cdot)$ is a natural transformation,
2. for each $B_1, B_2 \in \mathrm{Obj}(\mathcal{B})$ and $g_0 \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$, $\beta(F)^g : F(\cdot, B_1) \Rightarrow F(\cdot, B_2)$ is a natural transformation.

Proof.

1. Let $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$.
   - Let $B \in \mathrm{Obj}(\mathcal{B})$. Then $(f, \mathrm{id}^{\mathcal{B}}_B) \in \mathrm{Hom}_{\mathcal{A}\times\mathcal{B}}((A_1, B), (A_2, B))$. Therefore
   $$\begin{aligned}\alpha(F)^f_B &= F(f, \mathrm{id}^{\mathcal{B}}_B)\\ &\in \mathrm{Hom}_{\mathcal{A}\times\mathcal{B}}(F(A_1, B), F(A_2, B))\\ &= \mathrm{Hom}_{\mathcal{A}\times\mathcal{B}}(F(A_1, \cdot)(B), F(A_2, \cdot)(B)).\end{aligned}$$
   Since $B \in \mathrm{Obj}(\mathcal{B})$ is arbitrary, we have that for each $B \in \mathrm{Obj}(\mathcal{B})$, $\alpha(F)^f_B \in \mathrm{Hom}_{\mathcal{A}\times\mathcal{B}}(F(A_1, \cdot)(B), F(A_2, \cdot)(B))$.
   - Let $B_1, B_2 \in \mathrm{Obj}(\mathcal{B})$, $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$ and $g \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$. Then
   $$\begin{aligned}F(A_2, \cdot)(g) \circ_{\mathcal{C}} \alpha(F)^f_{B_1} &= F(\mathrm{id}^{\mathcal{A}}_{A_2}, g) \circ_{\mathcal{C}} F(f, \mathrm{id}^{\mathcal{B}}_{B_1})\\ &= F((\mathrm{id}^{\mathcal{A}}_{A_2}, g) \circ_{\mathcal{A}\times\mathcal{B}} (f, \mathrm{id}^{\mathcal{B}}_{B_1}))\\ &= F(\mathrm{id}^{\mathcal{A}}_{A_2} \circ_{\mathcal{A}} f, g \circ_{\mathcal{B}} \mathrm{id}^{\mathcal{B}}_{B_1})\\ &= F(f, g)\\ &= F(f \circ_{\mathcal{A}} \circ_{\mathcal{A}} \mathrm{id}^{\mathcal{A}}_{A_1}, \mathrm{id}^{\mathcal{B}}_{B_2} \circ_{\mathcal{B}} g)\\ &= F((f, \mathrm{id}^{\mathcal{B}}_{B_2}) \circ_{\mathcal{A}\times\mathcal{B}} (\mathrm{id}^{\mathcal{A}}_{A_1}, g))\\ &= F(f, \mathrm{id}^{\mathcal{B}}_{B_2}) \circ_{\mathcal{C}} F(\mathrm{id}^{\mathcal{A}}_{A_1}, g)\\ &= \alpha(F)^f_{B_2} \circ_{\mathcal{C}} F(A_1, \cdot)(g),\end{aligned}$$
   i.e. the following diagram commutes:
   $$\begin{tikzcd} F(A_1, \cdot)(B_1) \arrow[r, "\alpha(F)^f_{B_1}"] \arrow[d, "F(A_1,\cdot)(g)"'] & F(A_2, \cdot)(B_1) \arrow[d, "F(A_2,\cdot)(g)"] \\ F(A_1, \cdot)(B_2) \arrow[r, "\alpha(F)^f_{B_2}"] & F(A_2, \cdot)(B_2). \end{tikzcd}$$
   Since $B_1, B_2 \in \mathrm{Obj}(\mathcal{B})$ and $g \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$ are arbitrary, we have that for each $B_1, B_2 \in \mathrm{Obj}(\mathcal{B})$ and $g \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$, $F(A_2, \cdot)(g) \circ_{\mathcal{C}} \alpha(F)^f_{B_1} = \alpha(F)^f_{B_2} \circ_{\mathcal{C}} F(A_1, \cdot)(g)$.

   Thus $\alpha(F)^f : F(A_1, \cdot) \Rightarrow F(A_2, \cdot)$ is a natural transformation. Since $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$ and $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$ are arbitrary, we have that for each $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$ and $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$, $\alpha(F)^f : F(A_1, \cdot) \Rightarrow F(A_2, \cdot)$ is a natural transformation.
2. Similar to prev part (ref part).

□

Exercise 4.5.0.3. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be categories and $F : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$. Then

1. for each $A_1, A_2, A_3 \in \mathrm{Obj}(\mathcal{A})$, $f_{1,2} \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$ and $f_{2,3} \in \mathrm{Hom}_{\mathcal{A}}(A_2, A_3)$, $\alpha(F)^{f_{2,3} \circ_{\mathcal{A}} f_{1,2}} = \alpha(F)^{f_{2,3}} \circ_{\mathcal{C}^{\mathcal{B}}} \alpha(F)^{f_{1,2}}$
2. for each $B_1, B_2, B_3 \in \mathrm{Obj}(\mathcal{B})$, $g_{1,2} \in \mathrm{Hom}_{\mathcal{B}}(B_1, B_2)$ and $g_{2,3} \in \mathrm{Hom}_{\mathcal{B}}(B_2, B_3)$, $\beta(F)^{g_{2,3} \circ_{\mathcal{B}} g_{1,2}} = \beta(F)^{g_{2,3}} \circ_{\mathcal{C}^{\mathcal{A}}} \beta(F)^{g_{1,2}}$

Proof.

1. Let $A_1, A_2, A_3 \in \mathrm{Obj}(\mathcal{A})$, $f_{1,2} \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$ and $f_{2,3} \in \mathrm{Hom}_{\mathcal{A}}(A_2, A_3)$. Then for each $B \in \mathrm{Obj}(\mathcal{B})$,

$$\begin{aligned}
(\alpha(F)^{f_{2,3} \circ_{\mathcal{A}} f_{1,2}})_B &= F(f_{2,3} \circ_{\mathcal{A}} f_{1,2}, \mathrm{id}_B^{\mathcal{B}}) \quad \text{(by Definition 4.5.0.1)} \\
&= F(f_{2,3} \circ_{\mathcal{A}} f_{1,2}, \mathrm{id}_B^{\mathcal{B}} \circ_{\mathcal{B}} \mathrm{id}_B^{\mathcal{B}}) \\
&= F((f_{2,3}, \mathrm{id}_B^{\mathcal{B}}) \circ_{\mathcal{A} \times \mathcal{B}} (f_{1,2}, \mathrm{id}_B^{\mathcal{B}})) \\
&= F(f_{2,3}, \mathrm{id}_B^{\mathcal{B}}) \circ_{\mathcal{C}} F(f_{1,2}, \mathrm{id}_B^{\mathcal{B}}) \\
&= \alpha(F)_B^{f_{2,3}} \circ_{\mathcal{C}} \alpha(F)_B^{f_{1,2}} \\
&= (\alpha(F)^{f_{2,3}} \circ_{\mathcal{C}^{\mathcal{B}}} \alpha(F)^{f_{1,2}})_B.
\end{aligned}$$

   Thus $\alpha(F)^{f_{2,3} \circ_{\mathcal{A}} f_{1,2}} = \alpha(F)^{f_{2,3}} \circ_{\mathcal{C}^{\mathcal{B}}} \alpha(F)^{f_{1,2}}$.

2. similar to prev part (ref prev part)

□

Exercise 4.5.0.4. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be categories and $F : \mathcal{A} \times \mathcal{B} \to \mathcal{C}$. Then

1. For each $A \in \mathrm{Obj}(\mathcal{A})$, $\alpha(F)^{\mathrm{id}_A^{\mathcal{A}}} = \mathrm{id}_{F(A,\cdot)}^{\mathcal{C}^{\mathcal{B}}}$.
2. For each $B \in \mathrm{Obj}(\mathcal{B})$, $\beta(F)^{\mathrm{id}_B^{\mathcal{B}}} = \mathrm{id}_{F(\cdot,B)}^{\mathcal{C}^{\mathcal{A}}}$.

Proof.

1. For each $B \in \mathrm{Obj}(\mathcal{B})$,

$$\begin{aligned}
\alpha(F)_B^{\mathrm{id}_A^{\mathcal{A}}} &= F(\mathrm{id}_A^{\mathcal{A}}, \mathrm{id}_B^{\mathcal{B}}) \\
&= F(\mathrm{id}_{(A,B)}^{\mathcal{A} \times \mathcal{B}}) \\
&= \mathrm{id}_{F(A,B)}^{\mathcal{C}} \\
&= \mathrm{id}_{F(A,\cdot)(B)}^{\mathcal{C}} \\
&= (\mathrm{id}_{F(A,\cdot)})_B \quad \text{(by Definition 2.3.2.4).}
\end{aligned}$$

   Since $B \in \mathrm{Obj}(\mathcal{B})$ is arbitrary, we have that for each $B \in \mathrm{Obj}(\mathcal{B})$, $\alpha(F)_B^{\mathrm{id}_A^{\mathcal{A}}} = (\mathrm{id}_{F(A,\cdot)})_B$. Thus $\alpha(F)^{\mathrm{id}_A^{\mathcal{A}}} = \mathrm{id}_{F(A,\cdot)}$.

2. Similar to prev part (ref prev part).

□

Definition 4.5.0.5. Let $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$ and $F \in \mathrm{Obj}(\mathcal{C}^{\mathcal{A} \times \mathcal{B}})$. We define the **Curry functor induced by** F, denoted $\mathrm{cur}(F) : \mathcal{A} \to \mathcal{C}^{\mathcal{B}}$ by

- $\mathrm{cur}(F)(A) := F(A, \cdot), \quad (A \in \mathrm{Obj}(\mathcal{A}))$,
- $\mathrm{cur}(F)(f) := \alpha(F)^f, \quad (f \in \mathrm{Hom}_{\mathcal{A}})$,

Exercise 4.5.0.6. Let $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$ and $F \in \mathrm{Obj}(\mathcal{C}^{\mathcal{A} \times \mathcal{B}})$. Then $\mathrm{cur}(F) : \mathcal{A} \to \mathcal{C}^{\mathcal{B}}$ is a functor.

Proof.

1. Let $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$ and $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$. Exercise 4.5.0.2 implies that

$$\begin{aligned}\mathrm{cur}(F)(f) &= \alpha(F)^f \\ &\in \mathrm{Hom}_{\mathcal{C}^{\mathcal{B}}}(F(A_1, \cdot), F(A_2, \cdot)) \\ &= \mathrm{Hom}_{\mathcal{C}^{\mathcal{B}}}(\mathrm{cur}(F)(A_1), \mathrm{cur}(F)(A_1)).\end{aligned}$$

2. Let $A_1, A_2, A_3 \in \mathrm{Obj}(\mathcal{A})$, $f_{1,2} \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$ and $f_{2,3} \in \mathrm{Hom}_{\mathcal{A}}(A_2, A_3)$. Exercise 4.5.0.3 implies that

$$\begin{aligned}\mathrm{cur}(F)(f_{2,3} \circ_{\mathcal{A}} f_{1,2}) &= \alpha(F)^{f_{2,3} \circ_{\mathcal{A}} f_{1,2}} \\ &= \alpha(F)^{f_{2,3}} \circ_{\mathcal{C}^{\mathcal{B}}} \alpha(F)^{f_{1,2}} \\ &= \mathrm{cur}(F)(f_{2,3}) \circ_{\mathcal{C}^{\mathcal{B}}} \mathrm{cur}(F)(f_{1,2}).\end{aligned}$$

3. Let $A \in \mathrm{Obj}(\mathcal{A})$. Exercise 4.5.0.4 implies that,

$$\begin{aligned}\mathrm{cur}(F)(\mathrm{id}_A^{\mathcal{A}}) &= \alpha(F)^{\mathrm{id}_A^{\mathcal{A}}} \\ &= \mathrm{id}_{F(A,\cdot)}^{\mathcal{C}^{\mathcal{B}}} \\ &= \mathrm{id}_{\mathrm{cur}(F)(A)}^{\mathcal{C}^{\mathcal{B}}}.\end{aligned}$$

Hence $\mathrm{cur}(F) : \mathcal{A} \to \mathcal{C}^{\mathcal{B}}$ is a functor. □

Definition 4.5.0.7. Let $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$ and $F, G \in \mathrm{Obj}(\mathcal{C}^{\mathcal{A}\times\mathcal{B}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}(F, G)$. We define the **Curry transformation induced by** α, denoted $\mathrm{cur}(\alpha) : \mathrm{cur}(F) \Rightarrow \mathrm{cur}(G)$ by

$$\mathrm{cur}(\alpha)_A := \alpha_{(A,\cdot)}, \quad A \in \mathrm{Obj}(\mathcal{A}).$$

Exercise 4.5.0.8. Let $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$, $F, G \in \mathrm{Obj}(\mathcal{C}^{\mathcal{A}\times\mathcal{B}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}(F, G)$. Then $\mathrm{cur}(\alpha) : \mathrm{cur}(F) \Rightarrow \mathrm{cur}(G)$ is a natural transformation.

Proof.

- Exercise 4.2.2.6 implies that for each $A \in \mathrm{Obj}(\mathcal{A})$,

$$\begin{aligned}\mathrm{cur}(\alpha)_A &= \alpha_{(A,\cdot)} \\ &\in \mathrm{Hom}_{\mathcal{C}^{\mathcal{B}}}(F(A, \cdot), G(A, \cdot)) \\ &= \mathrm{Hom}_{\mathcal{C}^{\mathcal{B}}}(\mathrm{cur}(F)(A), \mathrm{cur}(G)(A)).\end{aligned}$$

- Let $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$ and $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$. Since $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}(F, G)$, we have that

$$G(f, \mathrm{id}_B^{\mathcal{B}}) \circ_{\mathcal{C}} \alpha_{(A_1,B)} = \alpha_{(A_2,B)} \circ_{\mathcal{C}} F(f, \mathrm{id}_B^{\mathcal{B}}),$$

  i.e. the following diagram commutes:

$$\begin{tikzcd} F(A_1, B) \arrow[r, "\alpha_{(A_1,B)}"] \arrow[d, "F(f,\mathrm{id}_B^{\mathcal{B}})"'] & G(A_1, B) \arrow[d, "G(f,\mathrm{id}_B^{\mathcal{B}})"] \\ F(A_2, B) \arrow[r, "\alpha_{(A_2,B)}"] & G(A_2, B). \end{tikzcd}$$

  Therefore for each $B \in \mathrm{Obj}(\mathcal{B})$,

$$\begin{aligned}(\mathrm{cur}(G)(f) \circ_{\mathcal{C}^{\mathcal{B}}} \mathrm{cur}(\alpha)_{A_1})_B &= \mathrm{cur}(G)(f)_B \circ_{\mathcal{C}} (\mathrm{cur}(\alpha)_{A_1})_B \\ &= G(f, \mathrm{id}_B^{\mathcal{B}}) \circ_{\mathcal{C}} \alpha_{(A_1,B)} \\ &= \alpha_{(A_2,B)} \circ_{\mathcal{C}} F(f, \mathrm{id}_B^{\mathcal{B}}) \\ &= \mathrm{cur}(\alpha)_{A_1})_B \circ_{\mathcal{C}} \mathrm{cur}(F)(f)_B \\ &= (\mathrm{cur}(\alpha)_{A_1} \circ_{\mathcal{C}} \mathrm{cur}(F)(f))_B.\end{aligned}$$

Hence $\mathrm{cur}(G)(f) \circ_{\mathcal{C}^{\mathcal{B}}} \mathrm{cur}(\alpha)_{A_1} = \mathrm{cur}(\alpha)_{A_2} \circ_{\mathcal{C}} \mathrm{cur}(F)(f)$, i.e. the following diagram commutes:

$$\begin{tikzcd} \mathrm{cur}(F)(A_1) \arrow[r, Rightarrow, "\mathrm{cur}(\alpha)_{A_1}"] \arrow[d, Rightarrow, "\mathrm{cur}(F)(f)"'] & \mathrm{cur}(G)(A_1) \arrow[d, Rightarrow, "\mathrm{cur}(G)(f)"] \\ \mathrm{cur}(F)(A_2) \arrow[r, Rightarrow, "\mathrm{cur}(\alpha)_{A_2}"] & \mathrm{cur}(G)(A_2). \end{tikzcd}$$

Since $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$ and $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$ are arbitrary, we have that for each $A_1, A_2 \in \mathrm{Obj}(\mathcal{A})$ and $f \in \mathrm{Hom}_{\mathcal{A}}(A_1, A_2)$, $\mathrm{cur}(G)(f) \circ_{\mathcal{C}^{\mathcal{B}}} \mathrm{cur}(\alpha)_{A_1} = \mathrm{cur}(\alpha)_{A_2} \circ_{\mathcal{C}} \mathrm{cur}(F)(f)$.

Thus $\mathrm{cur}(\alpha) : \mathrm{cur}(F) \Rightarrow \mathrm{cur}(G)$ is a natural transformation. □

Exercise 4.5.0.9. Let $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$, $F_1, F_2, F_3 \in \mathrm{Obj}(\mathcal{C}^{\mathcal{A}\times\mathcal{B}})$, $\alpha_{1,2} \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}(F_1, F_2)$ and $\alpha_{2,3} \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}(F_2, F_3)$. Then $\mathrm{cur}(\alpha_{2,3} \circ_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}} \alpha_{1,2}) = \mathrm{cur}(\alpha_{2,3}) \circ_{(\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}} \mathrm{cur}(\alpha_{1,2})$.

Proof. Exercise 4.2.2.7 implies that for each $A \in \mathrm{Obj}(\mathcal{A})$,

$$\begin{aligned} \mathrm{cur}(\alpha_{2,3} \circ_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}} \alpha_{1,2})_A &= (\alpha_{2,3} \circ_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}} \alpha_{1,2})_{(A,\cdot)} \\ &= (\alpha_{2,3})_{(A,\cdot)} \circ_{\mathcal{C}^{\mathcal{B}}} (\alpha_{1,2})_{(A,\cdot)} \\ &= \mathrm{cur}(\alpha_{2,3})_A \circ_{\mathcal{C}^{\mathcal{B}}} \mathrm{cur}(\alpha_{1,2})_A \\ &= [\mathrm{cur}(\alpha_{2,3}) \circ_{(\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}} \mathrm{cur}(\alpha_{1,2})]_A. \end{aligned}$$

Therefore $\mathrm{cur}(\alpha_{2,3} \circ_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}} \alpha_{1,2}) = \mathrm{cur}(\alpha_{2,3}) \circ_{(\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}} \mathrm{cur}(\alpha_{1,2})$. □

Exercise 4.5.0.10. Let $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$ and $F \in \mathrm{Obj}(\mathcal{C}^{\mathcal{A}\times\mathcal{B}})$. Then $\mathrm{cur}(\mathrm{id}_F^{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}) = \mathrm{id}_{\mathrm{cur}(F)^{(\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}}}$.

Proof. Exercise 4.2.2.8 implies that for each $A \in \mathrm{Obj}(\mathcal{A})$,

$$\begin{aligned} \mathrm{cur}(\mathrm{id}_F^{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}})_A &= (\mathrm{id}_F^{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}})_{(A,\cdot)} \\ &= \mathrm{id}_{F(A,\cdot)}^{\mathcal{C}^{\mathcal{B}})} \\ &= \mathrm{id}_{\mathrm{cur}(F)(A)}^{\mathcal{C}^{\mathcal{B}}} \\ &= (\mathrm{id}_{\mathrm{cur}(F)}^{(\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}})_A. \end{aligned}$$

Hence $\mathrm{cur}(\mathrm{id}_F^{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}) = \mathrm{id}_{\mathrm{cur}(F)}^{(\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}}$. □

Definition 4.5.0.11. Let $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$. We define the **Curry functor from $\mathcal{C}^{\mathcal{A}\times\mathcal{B}}$ to $(\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}$**, denoted $\mathrm{cur}_{(\mathcal{A},\mathcal{B},\mathcal{C})}^{\mathbf{Cat}} : \mathcal{C}^{\mathcal{A}\times\mathcal{B}} \to (\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}$ by

- $\mathrm{cur}_{(\mathcal{A},\mathcal{B},\mathcal{C})}^{\mathbf{Cat}}(F) := \mathrm{cur}(F)$
- $\mathrm{cur}_{(\mathcal{A},\mathcal{B},\mathcal{C})}^{\mathbf{Cat}}(\alpha) := \mathrm{cur}(\alpha)$.

Exercise 4.5.0.12. Let $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$. Then $\mathrm{cur}_{(\mathcal{A},\mathcal{B},\mathcal{C})}^{\mathbf{Cat}} : \mathcal{C}^{\mathcal{A}\times\mathcal{B}} \to (\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}$ is a functor.

Proof.

1. Let $F, G \in \mathrm{Obj}(\mathcal{C}^{\mathcal{A}\times\mathcal{B}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}(F, G)$. Exercise 4.5.0.8 implies that

$$\begin{aligned} \mathrm{cur}_{(\mathcal{A},\mathcal{B},\mathcal{C})}^{\mathbf{Cat}}(\alpha) &= \mathrm{cur}(\alpha) \\ &\in \mathrm{Hom}_{\mathcal{C}^{\mathcal{B}}}(\mathrm{cur}(F), \mathrm{cur}(G)) \\ &= \mathrm{Hom}_{\mathcal{C}^{\mathcal{B}}}(\mathrm{cur}_{(\mathcal{A},\mathcal{B},\mathcal{C})}^{\mathbf{Cat}}(F), \mathrm{cur}_{(\mathcal{A},\mathcal{B},\mathcal{C})}^{\mathbf{Cat}}(G)). \end{aligned}$$

2. Let $F_1, F_2, F_3 \in \mathrm{Obj}(\mathcal{C}^{\mathcal{A}\times\mathcal{B}})$, $\alpha_{1,2} \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}(F_1, F_2)$ and $\alpha_{2,3} \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}(F_2, F_3)$. Exercise 4.5.0.9 implies that
$$\begin{aligned}\mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})}(\alpha_{2,3} \circ_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}} \alpha_{1,2}) &= \mathrm{cur}((\alpha_{2,3} \circ_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}} \alpha_{1,2})) \\ &= \mathrm{cur}(\alpha_{2,3}) \circ_{(\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}} \mathrm{cur}(\alpha_{1,2}) \\ &= \mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})}(\alpha_{2,3}) \circ_{(\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}} \mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})}(\alpha_{1,2}).\end{aligned}$$

3. Exercise 4.5.0.10 implies that
$$\begin{aligned}\mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})}(\mathrm{id}_F^{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}) &= \mathrm{cur}(\mathrm{id}_F^{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}) \\ &= \mathrm{id}^{(\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}}_{\mathrm{cur}(F)} \\ &= \mathrm{id}^{(\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}}_{\mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})}(F)}.\end{aligned}$$

Thus $\mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})} : \mathcal{C}^{\mathcal{A}\times\mathcal{B}} \to (\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}$ is a functor. □

Definition 4.5.0.13. We define the **Currying natural transformation on Cat**, denoted $\mathrm{cur}^{\mathbf{Cat}} : [(- \times -)_{\mathbf{Cat}}, -]_{\mathbf{Cat}} \Rightarrow [-, [-, -]_{\mathbf{Cat}}]_{\mathbf{Cat}}$, by $(\mathrm{cur}^{\mathbf{Cat}})_{(\mathcal{A},\mathcal{B},\mathcal{C})} := \mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})}$

Exercise 4.5.0.14. We have that $\mathrm{cur}^{\mathbf{Cat}} : [(- \times -)_{\mathbf{Cat}}, -]_{\mathbf{Cat}} \Rightarrow [-, [-, -]_{\mathbf{Cat}}]_{\mathbf{Cat}}$ is a natural transformation.

Proof.

show it is in correct Hom set

Show diagram commutes □

To Do:

- make a note that $[-, [-, -]] : \mathbf{Cat}^{\mathrm{op}} \times (\mathbf{Cat}^{\mathrm{op}} \times \mathbf{Cat}) \to \mathbf{Cat}$ means $[-, -] \circ \mathrm{id}_{\mathbf{Cat}} \times [-, -]$
- define opposite functor $OP : \mathcal{C} \to \mathcal{C}^{\mathrm{op}}$
- show that $\mathrm{cur}^{\mathbf{Cat}} : F \Rightarrow G$ is a natural transformation (and natural isomorphism)

Definition 4.5.0.15.

Exercise 4.5.0.16. content...

Proof. content... □

Definition 4.5.0.17. Let $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$. We define the **Currying functor from** $(\mathcal{C})^{\mathcal{A}\times\mathcal{B}}$ **to** $(\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}$, denoted $\mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})} : (\mathcal{C})^{\mathcal{A}\times\mathcal{B}} \to (\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}$, by

- $\mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})}(F) := \mathrm{cur}(F), \quad (A \in \mathrm{Obj}(\mathcal{A}))$
- $\mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})}(\alpha)(A) := \alpha_{(A,\cdot)}$.

(need exercise showing that if $\mathcal{A}, \mathcal{B}$ are small, then $\mathcal{A} \times \mathcal{B}$ is small)

Exercise 4.5.0.18. Let $\mathcal{A}, \mathcal{B}, \mathcal{C} \in \mathrm{Obj}(\mathbf{Cat})$. Then $\mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})} : \mathcal{C}^{\mathcal{A}\times\mathcal{B}} \to (\mathcal{C}^{\mathcal{B}})^{\mathcal{A}}$ is a functor.

Proof.

1. Let $F, G \in \mathrm{Obj}(\mathcal{C}^{\mathcal{A}\times\mathcal{B}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{A}\times\mathcal{B}}}(F, G)$. Then for each $A \in \mathrm{Obj}(\mathcal{A})$,
$$\begin{aligned}\mathrm{cur}^{\mathbf{Cat}}_{\mathcal{A},\mathcal{B},\mathcal{C}}(\alpha)(A) &= \alpha_{(A,\cdot)} \\ &\in \mathrm{Hom}_{\mathcal{C}^{\mathcal{B}}}(F(A,\cdot), G(A,\cdot)) \quad \text{(by ref prev ex)} \\ &= \mathrm{Hom}_{\mathcal{C}^{\mathcal{B}}}(\mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})}(F)(A), \mathrm{cur}^{\mathbf{Cat}}_{(\mathcal{A},\mathcal{B},\mathcal{C})}(G)(A))\end{aligned}$$

2.

3.

□

4.6 Equalizer and Coequalizer Categories

Chapter 5

Limits

5.1 Cones and Cocones

Definition 5.1.0.1. Let $\mathcal{J}$, $\mathcal{C}$ be categories and $D : \mathcal{J} \to \mathcal{C}$. Then D is said to be a **diagram of type $\mathcal{J}$ in $\mathcal{C}$**.

Note 5.1.0.2. We are usually interested in the case that $\mathcal{J}$ is small. We will identify a diagram D with its image and we visualize a diagram by the generating morphisms in $\mathcal{J}$.

Example 5.1.0.3. create example reference Define $\mathcal{J}$ by

- $\mathrm{Obj}(\mathcal{J}) = \{1, 2, 3, 4\}$,
- for $i, j \in \mathrm{Obj}(\mathcal{J})$, $\mathrm{Hom}_{\mathcal{J}}(i, j) = \{a_{ij}\}$.

Let $\mathcal{C}$ be a category and $D : \mathcal{J} \to \mathcal{C}$. Without including the identity morphisms or compositions, we can visualize D by its set of generating morphisms (i.e. not including the identity morphisms) as follows:

$$\begin{tikzcd}
 & 1 \arrow[r, "a_{1,2}"] \arrow[dl, "a_{1,3}"'] & 2 \arrow[dl, "a_{2,4}"] \\
3 \arrow[r, "a_{3,4}"'] & 4 &
\end{tikzcd}
\xrightarrow{D}
\begin{tikzcd}
 & D_1 \arrow[r, "D_{1,2}"] \arrow[dl, "D_{1,3}"'] & D_2 \arrow[dl, "D_{2,4}"] \\
D_3 \arrow[r, "D_{3,4}"'] & D_4 &
\end{tikzcd}$$

Definition 5.1.0.4. Let $\mathcal{J}$, $\mathcal{C}$ be categories. Suppose that $\mathcal{J}$ is small. Let $D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$.

- We define the **category of cones over** D, denoted $\mathbf{Cone}(D)$, by $\mathbf{Cone}(D) = (\Delta^{\mathcal{J}} \downarrow D)$.
- We define the **category of cocones under** D, denoted $\mathbf{Cocone}(D)$, by $\mathbf{Cocone}(D) = (D \downarrow \Delta^{\mathcal{J}})$.

Note 5.1.0.5.

- By defintion,

$$\begin{aligned}
\mathrm{Obj}(\mathbf{Cone}(D)) &= \{(X, \phi) : X \in \mathrm{Obj}(\mathcal{C}) \text{ and } \phi \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(\Delta^{\mathcal{J}}(X), D)\} \\
&= \{(X, \phi) : X \in \mathrm{Obj}(\mathcal{C}) \text{ and } \phi : \Delta^{\mathcal{J}}_X \Rightarrow D\}
\end{aligned}$$

  and for $(X, \phi), (Y, \psi) \in \mathrm{Obj}(\mathbf{Cone}(D))$,

$$\begin{aligned}
\mathrm{Hom}_{\mathbf{Cone}(D)}((X, \phi), (Y, \psi)) &= \{\alpha \in \mathrm{Hom}_{\mathcal{C}}(X, Y) : \psi \circ_{\mathcal{C}^{\mathcal{J}}} \Delta^{\mathcal{J}}(\alpha) = \phi\} \\
&= \{\alpha \in \mathrm{Hom}_{\mathcal{C}}(X, Y) : \psi \circ_{\mathcal{C}^{\mathcal{J}}} \delta^{\mathcal{J}}_{\alpha} = \phi\} \\
&= \{\alpha \in \mathrm{Hom}_{\mathcal{C}}(X, Y) : \text{for each } j \in \mathcal{J},\ \psi_j \circ_{\mathcal{C}} \alpha = \phi_j\}.
\end{aligned}$$

make ex in subcategory section and rephrase this in terms of the commutative diagram on only the generating morphisms. This will enable us to more precisely cover limits like products and projective systems where the identity maorphisms are not typically included in the definition of a diagram, but rather only the set of generating morphisms. Therefore, $(X, \phi) \in \mathrm{Obj}(\mathbf{Cone}(D))$ iff for each $i, j \in \mathrm{Obj}(\mathcal{J})$ and $f_{i,j} \in \mathrm{Hom}_{\mathcal{J}}(i, j)$, the following diagram commutes:

$$\begin{tikzcd}
 & X \arrow[dl, "\phi_i"'] \arrow[dr, "\phi_j"] & \\
D_i \arrow[rr, "D_{f_{i,j}}"'] & & D_j
\end{tikzcd}$$

and $\alpha \in \mathrm{Hom}_{\mathbf{Cone}(D)}((X, \phi), (Y, \psi))$ iff for each $i, j \in \mathrm{Obj}(\mathcal{J})$ and $(i, j) \in \mathrm{Hom}_{\mathcal{J}}(i, j)$ the following diagram commutes:

$$\begin{tikzcd}
 & X \arrow[d, "\alpha"'] \arrow[ddl, bend right, "\phi_i"'] \arrow[ddr, bend left, "\phi_j"] & \\
 & Y \arrow[dl, "\psi_i"'] \arrow[dr, "\psi_j"] & \\
D_i \arrow[rr, "D_{i,j}"'] & & D_j
\end{tikzcd}$$

- By defintion,

$$\begin{aligned}
\mathrm{Obj}(\mathbf{Cocone}(D)) &= \{(X, \phi) : X \in \mathrm{Obj}(\mathcal{C}) \text{ and } \phi \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D, \Delta^{\mathcal{J}}(X))\} \\
&= \{(X, \phi) : X \in \mathrm{Obj}(\mathcal{C}) \text{ and } \phi : D \Rightarrow \Delta^{\mathcal{J}}_X\}
\end{aligned}$$

and for $(X, \phi), (Y, \psi) \in \mathrm{Obj}(\mathbf{Cocone}(D))$,

$$\begin{aligned}
\mathrm{Hom}_{\mathbf{Cocone}(D)}((X, \phi), (Y, \psi)) &= \{\alpha \in \mathrm{Hom}_{\mathcal{C}}(X, Y) : \Delta^{\mathcal{J}}(\alpha) \circ_{\mathcal{C}^{\mathcal{J}}} \phi = \psi\} \\
&= \{\alpha \in \mathrm{Hom}_{\mathcal{C}}(X, Y) : \delta^{\mathcal{J}}_{\alpha} \circ_{\mathcal{C}^{\mathcal{J}}} \phi = \psi\} \\
&= \{\alpha \in \mathrm{Hom}_{\mathcal{C}}(X, Y) : \text{for each } j \in \mathcal{J},\ \alpha \circ_{\mathcal{C}} \phi_j = \psi_j\}.
\end{aligned}$$

make ex in subcategory section and rephrase this in terms of the commutative diagram on only the generating morphisms. This will enable us to more precisely cover limits like products and projective systems where the identity maorphisms are not typically included in the definition of a diagram, but rather only the set of generating morphisms. Therefore, $(X, \phi) \in \mathrm{Obj}(\mathbf{Cone}(D))$ iff for each $i, j \in \mathrm{Obj}(\mathcal{J})$ and $f_{i,j} \in \mathrm{Hom}_{\mathcal{J}}(i, j)$, the following diagram commutes:

$$\begin{tikzcd}
D_i \arrow[rr, "D_{f_{i,j}}"] \arrow[dr, "\phi_i"'] & & D_j \arrow[dl, "\phi_j"] \\
 & X &
\end{tikzcd}$$

and $\alpha \in \mathrm{Hom}_{\mathbf{Cone}(D)}((X, \phi), (Y, \psi))$ iff for each $i, j \in \mathrm{Obj}(\mathcal{J})$ and $(i, j) \in \mathrm{Hom}_{\mathcal{J}}(i, j)$ the following diagram commutes:

$$\begin{tikzcd}
D_i \arrow[rr, "D_{f_{i,j}}"] \arrow[dr, "\phi_i"'] \arrow[ddr, bend right, "\psi_i"'] & & D_j \arrow[dl, "\phi_j"] \arrow[ddl, bend left, "\psi_j"] \\
 & X \arrow[d, "\alpha"'] & \\
 & Y &
\end{tikzcd}$$

note that in the case of say a product of projective system, the category $\mathcal{J}$ does not include the identity morphisms in the Hom sets, to avoid having to specify the category with all morphisms, rephrase this definition in terms of the subcategory generated by a set of morphisms. This will need some exercises made in section on subcategories about natural transformations on a generated subcategory needing only to satisfy the composition rules in the def natl transf's on the generating morphisms of that category. maybe other similar exercises needed as well. For example in a product, when we write the diagram, we include the objects X, Y but we never include the morphisms $\mathrm{id}_X, \mathrm{id}_Y$ in the diagram.

Example 5.1.0.6. ref example Define $\mathcal{J}$ and $\mathcal{D}$ as in previous example.

- Let $(X, \phi) \in \mathrm{Obj}(\mathbf{Cone}(D))$. We can visualize (X, ϕ) as follows:

$$\begin{tikzcd}
 & X, \arrow[d, "\phi_1"] \arrow[ddl, "\phi_3"'] \arrow[ddr, "\phi_4"] \arrow[drr, "\phi_2"] & & \\
 & D_1 \arrow[dl, "D_{1,3}"] \arrow[rr, "D_{1,2}"] & & D2 \arrow[dl, "D_{2,4}"] \\
D_3 \arrow[rr, "D_{3,4}"'] & & D_4 &
\end{tikzcd}$$

- Let $(X, \phi) \in \mathrm{Obj}(\mathbf{Cocone}(D))$. We can visualize (X, ϕ) as follows:

$$\begin{tikzcd}
 & X, \arrow[d, "\phi_1"] \arrow[ddl, "\phi_3"'] \arrow[ddr, "\phi_4"] \arrow[drr, "\phi_2"] & & \\
 & D_1 \arrow[dl, "D_{1,3}"] \arrow[rr, "D_{1,2}"] & & D2 \arrow[dl, "D_{2,4}"] \\
D_3 \arrow[rr, "D_{3,4}"'] & & D_4 &
\end{tikzcd}$$

FIX!!!!

Exercise 5.1.0.7. show $\mathbf{Cocone}(D) \cong \mathbf{Cone}(D)^{\mathrm{op}}$ (ref ex about comma categories and opposite categories)

Proof. □

Exercise 5.1.0.8. Let $\mathcal{C}$ be a category, $\mathcal{J} \in \mathrm{Obj}(\mathbf{Cat})$ and $D_1, D_2 \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$.

1. Let $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_1, D_2)$.
   (a) For each $(X, \phi) \in \mathrm{Obj}(\mathbf{Cone}(D_1))$,
   $$(X, \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi) \in \mathrm{Obj}(\mathbf{Cone}(D_2)).$$
   (b) For each $(X, \phi), (X', \phi') \in \mathrm{Obj}(\mathbf{Cone}(D_1))$ and $\eta \in \mathrm{Hom}_{\mathbf{Cone}(D_1)}((X, \phi), (X', \phi'))$,
   $$\eta \in \mathrm{Hom}_{\mathbf{Cone}(D_2)}((X, \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi), (X', \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi')).$$
2. Let $\beta \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_2, D_1)$.
   (a) FIX for cocones

(b)

Proof.

1. (a) Let $(X, \phi) \in \mathrm{Obj}(\mathbf{Cone}(D_1))$. Then $X \in \mathrm{Obj}(\mathcal{C})$ and $\phi \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(\Delta_X^{\mathcal{J}}, D_1)$. Therefore $\alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(\Delta_X^{\mathcal{J}}, D_2)$. Hence $(X, \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi) \in \mathrm{Obj}(\mathbf{Cone}(D_2))$.

   (b) Let $(X, \phi), (X', \phi') \in \mathrm{Obj}(\mathbf{Cone}(D_1))$ and $\eta \in \mathrm{Hom}_{\mathbf{Cone}(D_1)}((X, \phi), (X', \phi'))$. Then $\eta \in \mathrm{Hom}_{\mathcal{C}}(X, X')$ and $\phi' \circ_{\mathcal{C}^{\mathcal{J}}} \Delta^{\mathcal{J}}(\eta) = \phi$. Therefore $\alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi' \circ_{\mathcal{C}^{\mathcal{J}}} \Delta^{\mathcal{J}}(\eta) = \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi$. Hence $\eta \in \mathrm{Hom}_{\mathbf{Cone}(D_2)}((X, \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi), (X', \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi'))$.

2. use duality

□

Definition 5.1.0.9. Let $\mathcal{C}$ be a category, $\mathcal{J} \in \mathrm{Obj}(\mathbf{Cat})$ and $D_1, D_2 \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$.

1. Let $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_1, D_2)$. We define the **cone functor induced by** α, denoted $\mathrm{Cn}_\alpha : \mathbf{Cone}(D_1) \to \mathbf{Cone}(D_2)$, by
   - $\mathrm{Cn}_\alpha(X, \phi) := (X, \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi), \quad (X, \phi) \in \mathrm{Obj}(\mathbf{Cone}(D_1))$,
   - $\mathrm{Cn}_\alpha(\eta) := \eta, \quad \eta \in \mathrm{Hom}_{\mathbf{Cone}(D_1)}$.
2. similar for cocone functor

Exercise 5.1.0.10. Let $\mathcal{C}$ be a category, $\mathcal{J} \in \mathrm{Obj}(\mathbf{Cat})$ and $D_1, D_2 \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$.

1. Let $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_1, D_2)$. Then $\mathrm{Cn}_\alpha : \mathbf{Cone}(D_1) \to \mathbf{Cone}(D_2)$ is a functor.
2. similar for cocone functor

Proof.

1. We note that ref prev ex implies that Cn_α is well-defined.

   (a) ref prev ex implies that for each $(X, \phi), (X', \phi') \in \mathrm{Obj}(\mathbf{Cone}(D_1))$, and $\eta \in \mathrm{Hom}_{\mathbf{Cone}(D_1)}((X, \phi), (X', \phi'))$,

$$\begin{aligned}\mathrm{Cn}_\alpha(\eta) &= \eta \\ &\in \mathrm{Hom}_{\mathbf{Cone}(D_2)}((X, \alpha \circ \mathcal{C}^{\mathcal{J}} \phi), (X', \alpha \circ \mathcal{C}^{\mathcal{J}} \phi')) \\ &= \mathrm{Hom}_{\mathbf{Cone}(D_2)}(\mathrm{Cn}_\alpha(X, \phi), \mathrm{Cn}_\alpha(X', \phi')).\end{aligned}$$

   (b) Let $(X, \phi), (Y, \psi), (Z, \chi) \in \mathrm{Obj}(\mathbf{Cone}(D_1))$ and $\mu \in \mathrm{Hom}_{\mathbf{Cone}(D_1)}((X, \phi), (Y, \psi))$ and $\nu \in \mathrm{Hom}_{\mathbf{Cone}(D_1)}((Y, \psi), (Z, \chi))$. Since $\mathbf{Cone}(D_1) = (\Delta^{\mathcal{J}} \downarrow D_1)$ and $\mathbf{Cone}(D_2) = (\Delta^{\mathcal{J}} \downarrow D_2)$, Definition 2.2.4.7 implies that

$$\begin{aligned}\mathrm{Cn}_\alpha(\nu \circ_{\mathbf{Cone}(D_1)} \mu) &= \nu \circ_{\mathbf{Cone}(D_1)} \mu \\ &= \nu \circ_{(\Delta^{\mathcal{J}} \downarrow D_1)} \mu \\ &= \nu \circ_{\mathcal{C}} \mu \\ &= \nu \circ_{(\Delta^{\mathcal{J}} \downarrow D_2)} \mu \\ &= \nu \circ_{\mathbf{Cone}(D_2)} \mu \\ &= \mathrm{Cn}_\alpha(\nu) \circ_{\mathbf{Cone}(D_2)} \mathrm{Cn}_\alpha(\mu).\end{aligned}$$

   (c) By Definition 5.1.0.4,

$$\begin{aligned}\mathrm{id}_{(X,\phi)}^{\mathbf{Cone}(D_1)} &= \mathrm{id}_{(X,\phi)}^{(\Delta^{\mathcal{J}} \downarrow D_1)} \\ &= \mathrm{id}_X^{\mathcal{C}} \\ &= \mathrm{id}_{(X, \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi)}^{(\Delta^{\mathcal{J}} \downarrow D_2)} \\ &= \mathrm{id}_{(X, \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi)}^{\mathbf{Cone}(D_2)}.\end{aligned}$$

Hence for each $(X, \phi) \in \mathrm{Obj}(\mathbf{Cone}(D_1))$,

$$\begin{aligned}\mathrm{Cn}_\alpha(\mathrm{id}^{\mathbf{Cone}(D_1)}_{(X,\phi)}) &= \mathrm{id}^{\mathbf{Cone}(D_1)}_{(X,\phi)} \\ &= \mathrm{id}^{\mathbf{Cone}(D_2)}_{(X,\alpha\circ_{\mathcal{C}^{\mathcal{J}}}\phi)} \\ &= \mathrm{id}^{\mathbf{Cone}(D_2)}_{\mathrm{Cn}_\alpha(X,\phi)}.\end{aligned}$$

Hence $\mathrm{Cn}_\alpha : \mathbf{Cone}(D_1) \to \mathbf{Cone}(D_2)$ is a functor.

2. use duality

□

Definition 5.1.0.11. Let $\mathcal{C}$ be a category, $\mathcal{J} \in \mathrm{Obj}(\mathbf{Cat})$.

1. We define the **Cone functor**, denoted $\mathbf{Cone}(\cdot) : \mathcal{C}^{\mathcal{J}} \to \mathbf{Cat}$, by
   - $\mathbf{Cone}(\cdot)(D) := \mathbf{Cone}(D), \quad D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$
   - $\mathbf{Cone}(\cdot)(\alpha) := \mathrm{Cn}_\alpha, \quad \alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}$.
2. define cocone functor

Exercise 5.1.0.12. Let $\mathcal{C}$ be a category, $\mathcal{J} \in \mathrm{Obj}(\mathbf{Cat})$.

1. We have that $\mathbf{Cone}(\cdot) : \mathcal{C}^{\mathcal{J}} \to \mathbf{Cat}$ is a functor.
2. show cocone functor is functor

Proof.

1. Set $C := \mathbf{Cone}(\cdot)$.
   (a) Let $D_1, D_2 \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_1, D_2)$. Then

$$\begin{aligned}C(\alpha) &= \mathrm{Cn}_\alpha \\ &\in \mathrm{Hom}_{\mathbf{Cat}}(\mathbf{Cone}(D_1), \mathbf{Cone}(D_2)) \\ &= \mathrm{Hom}_{\mathbf{Cat}}(C(D_1), C(D_2)).\end{aligned}$$

   (b) Let $D_1, D_2, D_3 \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$, $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_1, D_2)$ and $\beta \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_2, D_3)$. Then for each $(X, \phi) \in \mathrm{Obj}(\mathbf{Cone}(D))$,

$$\begin{aligned}[C(\beta \circ_{\mathcal{C}^{\mathcal{J}}} \alpha)](X, \phi) &= \mathrm{Cn}_{\beta\circ_{\mathcal{C}^{\mathcal{J}}}\alpha}(X, \phi) \\ &= (X, [\beta \circ_{\mathcal{C}^{\mathcal{J}}} \alpha] \circ_{\mathcal{C}^{\mathcal{J}}} \phi) \\ &= (X, \beta \circ_{\mathcal{C}^{\mathcal{J}}} [\alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi]) \\ &= \mathrm{Cn}_\beta(X, \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi) \\ &= \mathrm{Cn}_\beta(\mathrm{Cn}_\alpha(X, \phi)) \\ &= C(\beta)[C(\alpha)(X, \phi)] \\ &= [C(\beta) \circ_{\mathbf{Cat}} C(\alpha)](X, \phi)\end{aligned}$$

   Therefore $C(\beta \circ_{\mathcal{C}^{\mathcal{J}}} \alpha) = C(\beta) \circ_{\mathbf{Cat}} C(\alpha)$.
   (c) Let $D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$. Then for each $(X, \phi), (X', \phi') \in \mathrm{Obj}(\mathbf{Cone}(D))$ and $\eta \in \mathrm{Hom}_{\mathbf{Cone}(D)}((X, \phi), (X', \phi'))$,

$$\begin{aligned}C(\mathrm{id}^{\mathcal{C}^{\mathcal{J}}}_D)(X, \phi) &= \mathrm{Cn}_{\mathrm{id}^{\mathcal{C}^{\mathcal{J}}}_D}(X, \phi) \\ &= (X, \mathrm{id}^{\mathcal{C}^{\mathcal{J}}}_D \circ_{\mathcal{C}^{\mathcal{J}}} \phi) \\ &= (X, \phi) \\ &= \mathrm{id}^{\mathbf{Cat}}_{\mathbf{Cone}(D)}(X, \phi) \\ &= \mathrm{id}^{\mathbf{Cat}}_{C(D)}(X, \phi)\end{aligned}$$

and

$$\begin{aligned} C(\mathrm{id}_D^{\mathcal{C}^{\mathcal{J}}})(\eta) &= \mathrm{Cn}_{\mathrm{id}_D^{\mathcal{C}^{\mathcal{J}}}}(\eta) \\ &= \eta \\ &= \mathrm{id}^{\mathbf{Cat}}_{\mathbf{Cone}(D)}(\eta) \\ &= \mathrm{id}^{\mathbf{Cat}}_{C(D)}(\eta). \end{aligned}$$

Thus $C(\mathrm{id}_D^{\mathcal{C}^{\mathcal{J}}}) = \mathrm{id}^{\mathbf{Cat}}_{C(D)}$.

Hence $\mathbf{Cone}(\cdot) : \mathcal{C}^{\mathcal{J}} \to \mathbf{Cat}$ is a functor.

2. FINISH for cocones!!!!

□

5.2 Limits and Colimits

5.2.1 Universal Morphisms

make subsection instead of section and put it in section on limits after section on cones and cocones

Definition 5.2.1.1. Let $\mathcal{C}, \mathcal{D}$ be a categories, $X \in \mathrm{Obj}(\mathcal{D})$, $F : \mathcal{C} \to \mathcal{D}$ and $A \in \mathrm{Obj}(\mathcal{C})$.

- Let $f \in \mathrm{Hom}_{\mathcal{D}}(X, F(A))$. Then (A, f) is said to be a **universal morphism from** X **to** F if for each $A' \in \mathrm{Obj}(\mathcal{C})$ and $f' \in \mathrm{Hom}_{\mathcal{D}}(X, F(A'))$, there exists a unique $\alpha \in \mathrm{Hom}_{\mathcal{C}}(A, A')$ such that $F(\alpha) \circ_{\mathcal{C}} f = f'$, i.e. the following diagram commutes:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[dr, "f'"'] & F(A) \arrow[d, dashed, "F(\alpha)"] & A \arrow[d, dashed, "\alpha"] \\ & F(A') & A' \end{tikzcd}$$

- Let $f \in \mathrm{Hom}_{\mathcal{D}}(F(A), X)$. Then (A, f) is said to be a **universal morphism from** F **to** X if for each $A' \in \mathrm{Obj}(\mathcal{C})$ and $f' \in \mathrm{Hom}_{\mathcal{D}}(F(A'), X)$, there exists a unique $\alpha \in \mathrm{Hom}_{\mathcal{C}}(A', A)$ such that $f \circ_{\mathcal{D}} F(\alpha) = f'$, i.e. the following diagram commutes:

$$\begin{tikzcd} X & F(A) \arrow[l, "f"'] & A \\ & F(A') \arrow[u, dashed, "F(\alpha)"'] \arrow[ul, "f'"] & A' \arrow[u, dashed, "\alpha"'] \end{tikzcd}$$

Exercise 5.2.1.2. Let $\mathcal{C}, \mathcal{D}$ be a categories, $X \in \mathrm{Obj}(\mathcal{D})$, $F : \mathcal{C} \to \mathcal{D}$ and $A \in \mathrm{Obj}(\mathcal{C})$.

1. Let $f \in \mathrm{Hom}_{\mathcal{D}}(X, F(A))$. Then (A, f) is a universal morphism from X to F iff (A, f) is $(X \downarrow F)$-initial.
2. Let $f \in \mathrm{Hom}_{\mathcal{D}}(F(A), X)$. Then (A, f) is a universal morphism from F to X iff (A, f) is $(F \downarrow X)$-terminal.

Proof.

1.
2.

FINISH!!! □

Note 5.2.1.3. make a comment on how if (A, f) is universal from X to F, then for each (A', f'), f' is a post-processing of f and (A, f) is universal from F to X, then for each (A', f'), f' is a pre-processing of f

5.2.2 Limits and Colimits

Definition 5.2.2.1. Let $\mathcal{J}, \mathcal{C}$ be categories. Suppose that $\mathcal{J}$ is small. Let $D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$.

- – Let $(X, \phi) \in \mathrm{Obj}(\mathbf{Cone}(D))$. Then (X, ϕ) is said to be a D**-limit**, denoted $(X, \phi) = \lim D$, if (X, ϕ) is a universal morphism from $\Delta^{\mathcal{J}}$ to D.
  – We define the **limit set of** D, denoted $\mathrm{limSet}(D)$, by

$$\mathrm{limSet}\, D := \{(X, \phi) \in \mathrm{Obj}(\mathbf{Cone}(D)) : \mathrm{Obj}(\mathbf{Cone}(D)) \text{ is a } D\text{-limit}\}.$$

- – Let $(X, \phi) \in \mathrm{Obj}(\mathbf{Cocone}(D))$. Then (X, ϕ) is said to be a D**-colimit**, denoted $(X, \phi) = \mathrm{colim}\, D$, if (X, ϕ) is a universal morphism from D to $\Delta^{\mathcal{J}}$.
  – We define the **colimit set of** D, denoted $\mathrm{colimSet}(D)$, by

$$\mathrm{colimSet}\, D := \{(X, \phi) \in \mathrm{Obj}(\mathbf{Cocone}(D)) : \mathrm{Obj}(\mathbf{Cocone}(D)) \text{ is a } D\text{-colimit}\}.$$

maybe define lim set, but maybe not, we want to be able to write $(X, \phi) = \lim D$, but we need some guarantee that anything we would possibly want to say is invariant under isomorphism, other wise different limits of D may always be interchangeable, ask someone if we need to be careful here and define limit sets. For instance commutativity

Note 5.2.2.2. Let $\mathcal{J}$, $\mathcal{C}$ be categories. Suppose that $\mathcal{J}$ is small. Let $D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$.

- Let $(X, \phi) \in \mathrm{Obj}(\mathbf{Cone}(D))$. Then

$$\begin{aligned}(X, \phi) \text{ is a limit of } D &\iff (X, \phi) \text{ is } \mathbf{Cone}(D)\text{-terminal} \\ &\iff \text{for each } (X', \phi') \in \mathrm{Obj}(\mathbf{Cone}(D)), \text{ there exists a unique} \\ &\qquad \alpha \in \mathrm{Hom}_{\mathcal{C}}(X', X) \text{ such that for each } j \in \mathcal{J},\ \phi_j \circ_{\mathcal{C}} \alpha = \phi'_j\end{aligned}$$

- Let $(X, \phi) \in \mathrm{Obj}(\mathbf{Cocone}(D))$. Then

$$\begin{aligned}(X, \phi) \text{ is a colimit of } D &\iff (X, \phi) \text{ is } \mathbf{Cocone}(D)\text{-initial} \\ &\iff \text{for each } (X', \phi') \in \mathrm{Obj}(\mathbf{Cocone}(D)), \text{ there exists a unique} \\ &\qquad \alpha \in \mathrm{Hom}_{\mathcal{C}}(X, X') \text{ such that for each } j \in \mathcal{J},\ \alpha \circ_{\mathcal{C}} \phi_j = \phi'_j\end{aligned}$$

Exercise 5.2.2.3. limits and colimits are unique up to isomorphism

1. ?Let $(X, \phi) \in \mathrm{limSet}\, D$ and $Y \in \mathrm{Obj}(\mathcal{C})$, then $(Y, \psi) \in \mathrm{limSet}\, D$ iff there exists $\alpha \in \mathrm{Iso}_{\mathcal{C}}(X, Y)$ such that $\psi \circ_{\mathcal{C}^{\mathcal{J}}} \delta^{\mathcal{J}}_{\alpha} = \phi$?
2. colimits

Proof.

1.
2.

□

Definition 5.2.2.4. Let $\mathcal{J}$, $\mathcal{C}$ be categories. Suppose that $\mathcal{J}$ is small. Let $L \in \prod_{\mathrm{Obj}(\mathcal{C}^{\mathcal{J}})} \mathrm{limSet}(D)$.

5.3 Fundamental Examples

5.3.1 Products and Coproducts

Definition 5.3.1.1. Let $\mathcal{J}$ be a discrete category, $\mathcal{C}$ a category, $D \in \mathcal{C}^{\mathcal{J}}$ and $(X, \pi) \in \mathbf{Cone}(D)$. Then (X, π) is said to be a **product of** D.

Note 5.3.1.2. We may think of a product of D as a tuple $(X,)$

Exercise 5.3.1.3. show that if $\underline{1}$ is terminal, then $X \times \underline{1} \cong X$.

Exercise 5.3.1.4. show that if $\mathcal{C}$ has products, then $\mathcal{C}$ has a copy morphism (define a copy morphism) see page 32 in 'elementary categories, elementary toposes' by McLarty.

5.3.2 Equalizers and Coequalizers

5.3.3 Projective and Inductive Limits

Definition 5.3.3.1. Let $\mathcal{J}$ be a directed poset, $\mathcal{C}$ a category and $D \in \mathcal{C}^{\mathcal{J}}$. Then D is said to be a **$\mathcal{C}$-projective system**.

Note 5.3.3.2. We may think of

- a $\mathcal{C}$-projective system as a tuple $((X_j)_{j \in J}, (\pi_{j,k})_{(j,k) \in \leq})$ where $(J, \leq)$ is a directed poset, $(X_j)_{j \in J} \subset \mathrm{Obj}(\mathcal{C})$ and $(\pi_{j,k})_{(j,k) \in \leq} \subset \mathrm{Hom}_{\mathcal{C}}$ satisfy that for each $j, k, l \in J$,

1. $j \leq k$ implies that $\pi_{j,k} \in \mathrm{Hom}_{\mathcal{C}}(X_k, X_j)$,
2. $\pi_{j,j} = \mathrm{id}_{X_j}$,
3. $j \leq k$ and $k \leq l$ implies that $\pi_{j,k} \circ_{\mathcal{C}} \pi_{k,l} = \pi_{j,l}$.

- a cone over D as a tuple $(X, (\pi_j)_{j \in J})$ where $X \in \mathrm{Obj}(\mathcal{C})$ and $(\pi_j)_{j \in J} \subset \mathrm{Hom}_{\mathcal{C}}$ satisfy that for each $j, k \in J$, $j \leq k$ implies that $\pi_{j,k} \circ_{\mathcal{C}} \pi_k = \pi_j$.

make some diagrams

Exercise 5.3.3.3. Let $\mathcal{C}$ be a category and $(X_j)_{j \in \mathbb{N}} \subset \mathrm{Obj}(\mathcal{C})$. Define $(Y_n)_{n \in \mathbb{N}} \subset \mathrm{Obj}(\mathcal{C})$ and $(\pi_{n,k})_{n \leq k} \in \prod_{n \leq k} \mathrm{Hom}_{\mathcal{C}}(Y_k, Y_n)$ by $Y_n := \prod_{j=1}^{n} X_j$ and $\pi_{n,k}$. Suppose that for each $(\prod_{n \in \mathbb{N}}, (\pi_n)_{n \in \mathbb{N}})$ arg1.

Definition 5.3.3.4. Let $(\mathcal{J}, \mathcal{C}, D)$ be a $\mathcal{C}$-projective system and $(X, \phi) \in \mathrm{Obj}(\mathbf{Cone}(D))$. Then (X, ϕ) is said to be a $\mathcal{C}$-projective limit of $(\mathcal{J}, \mathcal{C}, D)$ if (X, ϕ) is a limit of D.

Note 5.3.3.5. We may think of a projective limit if $(\mathcal{J}, \mathcal{C}, D)$ as a cone $(X, (\pi_j)_{j \in J})$ over D satisfying that for each cone $(Y, (\tau_j)_{j \in J})$ over D, there exists a unique $\alpha \in \mathrm{Hom}_{\mathcal{C}}(Y, X)$ such that $\tau_j = \pi_j \circ_{\mathcal{C}} \alpha$.

make some diagrams

5.4 Completeness

Definition 5.4.0.1. Let $\mathcal{C}$ be a category and $\mathcal{J} \in \mathrm{Obj}(\mathbf{Cat})$. Then $\mathcal{C}$ is said to

- **have all $\mathcal{J}$-limits** if for each $D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$, $\mathrm{limSet}\, D \neq \varnothing$.
- **have all $\mathcal{J}$-colimits** if for each $D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$, $\mathrm{colimSet}\, D \neq \varnothing$.

Definition 5.4.0.2. Let $\mathcal{C}$ be a category and $\mathcal{J} \in \mathrm{Obj}(\mathbf{Cat})$.

- – Suppose that $\mathcal{C}$ has all $\mathcal{J}$-limits. Let $L \in \prod_{D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})} \mathrm{limSet}\, D$. Then L is said to be a **$\mathcal{C}^{\mathcal{J}}$-limit choice function**.
  - – We define $\mathrm{limChoice}(\mathcal{C}^{\mathcal{J}}) := \prod_{D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})} \mathrm{limSet}\, D$.
- – Suppose that $\mathcal{C}$ has all $\mathcal{J}$-colimits. Let $C \in \prod_{D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})} \mathrm{colimSet}\, D$. Then C is said to be a **$\mathcal{C}^{\mathcal{J}}$-colimit choice function**.
  - – We define $\mathrm{colimChoice}(\mathcal{C}^{\mathcal{J}}) := \prod_{D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})} \mathrm{colimSet}\, D$.

Exercise 5.4.0.3. Let $\mathcal{C}$ be a category and $\mathcal{J} \in \mathrm{Obj}(\mathbf{Cat})$.

1. Suppose that $\mathcal{C}$ has all $\mathcal{J}$-limits. Let $L \in \mathrm{limChoice}(\mathcal{C}^{\mathcal{J}})$. Then for each $D_1, D_2 \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_1, D_2)$, there exists $\eta_\alpha^L \in \mathrm{Hom}_{\mathbf{Cone}(D_2)}(\mathrm{Cn}_\alpha(L_{D_1}), L_{D_2})$ such that $\mathrm{Hom}_{\mathbf{Cone}(D_2)}(\mathrm{Cn}_\alpha(L_{D_1}), L_{D_2}) = \{\eta_\alpha^L\}$.
2. do somethign similar for colimtis

Proof.

1. Let $D_1, D_2 \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_1, D_2)$. Since L_{D_2} is a D_2-limit, L_{D_2} is $\mathbf{Cone}(D_2)$-terminal. Thus there exists $\eta_\alpha^L \in \mathrm{Hom}_{\mathbf{Cone}(D_2)}(\mathrm{Cn}_\alpha(L_{D_1}), L_{D_2})$ such that $\mathrm{Hom}_{\mathbf{Cone}(D_2)}(\mathrm{Cn}_\alpha(L_{D_1}), L_{D_2}) = \{\eta_\alpha^L\}$.
2. FINISH!!!!

□

Definition 5.4.0.4. Let $\mathcal{C}$ be a category and $\mathcal{J} \in \mathrm{Obj}(\mathbf{Cat})$.

- Suppose that $\mathcal{C}$ has all $\mathcal{J}$-limits. Let $L \in \mathrm{limChoice}(\mathcal{C}^{\mathcal{J}})$.
  - – For $D_1, D_2 \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_1, D_2)$, we define $\eta_\alpha^L \in \mathrm{Hom}_{\mathbf{Cone}(D_2)}(\mathrm{Cn}_\alpha(L_{D_1}), L_{D_2})$ as in Exercise 5.4.0.3.
  - – We define the **L-limit functor**, denoted $\lim_L : \mathcal{C}^{\mathcal{J}} \to \mathcal{C}$, by
    - $\lim_L(D) := \mathrm{proj}_{\mathrm{Obj}(\mathcal{C})}(L_D), \quad D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$
    - $\lim_L(\alpha) := \eta_\alpha^L, \quad \alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}$.
- define something similar for colimits, might need to change notation from η_α^L to something else

Exercise 5.4.0.5. Let $\mathcal{C}$ be a category and $\mathcal{J} \in \mathrm{Obj}(\mathbf{Cat})$.

1. Suppose that $\mathcal{C}$ has all $\mathcal{J}$-limits. Let $L \in \mathrm{limChoice}(\mathcal{C}^{\mathcal{J}})$. Then $\lim_L : \mathcal{C}^{\mathcal{J}} \to \mathcal{C}$ is a functor.
2. do something similar for colimits

Proof.

1. (a) Let $D_1, D_2 \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$ and $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_1, D_2)$. Write $L_{D_1} = (X_1, \phi_1)$ and $L_{D_2} = (X_2, \phi_2)$. Then $\lim_L(D_1) = X_1$ and $\lim_L(D_2) = X_2$. By Definition 5.1.0.4,

$$\begin{aligned}\mathrm{Hom}_{\mathbf{Cone}(D_2)}(\mathrm{Cn}_\alpha(L_{D_1}), L_{D_2}) &= \mathrm{Hom}_{\mathbf{Cone}(D_2)}((X_1, \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi_1), (X_2, \phi_2)) \\ &\subset \mathrm{Hom}_{\mathcal{C}}(X_1, X_2).\end{aligned}$$

Therefore

$$\begin{aligned}\lim_L(\alpha) &= \eta^L_\alpha \\ &\in \mathrm{Hom}_{\mathbf{Cone}(D_2)}(\mathrm{Cn}_\alpha(L_{D_1}), L_{D_2}) \\ &\subset \mathrm{Hom}_{\mathcal{C}}(X_1, X_2) \\ &= \mathrm{Hom}_{\mathcal{C}}(\lim_L(D_1), \lim_L(D_2)).\end{aligned}$$

(b) Let $D_1, D_2, D_3 \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$, $\alpha \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_1, D_2)$ and $\beta \in \mathrm{Hom}_{\mathcal{C}^{\mathcal{J}}}(D_2, D_3)$. By Exercise 5.4.0.3,

- $\mathrm{Hom}_{\mathbf{Cone}(D_2)}(\mathrm{Cn}_\alpha(L_{D_1}), L_{D_2}) = \{\eta^L_\alpha\}$,
- $\mathrm{Hom}_{\mathbf{Cone}(D_3)}(\mathrm{Cn}_\beta(L_{D_2}), L_{D_3}) = \{\eta^L_\beta\}$,
- $\mathrm{Hom}_{\mathbf{Cone}(D_3)}(\mathrm{Cn}_{\beta \circ_{\mathcal{C}^{\mathcal{J}}} \alpha}(L_{D_1}), L_{D_3}) = \{\eta^L_{\beta \circ_{\mathcal{C}^{\mathcal{J}}} \alpha}\}$.

By definition Definition 5.1.0.4,

- $\eta^L_\alpha \in \mathrm{Hom}_{\mathcal{C}}(X_1, X_2)$ and $\phi_2 \circ_{\mathcal{C}^{\mathcal{J}}} \Delta^{\mathcal{J}}(\eta^L_\alpha) = \alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi_1$
- $\eta^L_\beta \in \mathrm{Hom}_{\mathcal{C}}(X_2, X_3)$ and $\phi_3 \circ_{\mathcal{C}^{\mathcal{J}}} \Delta^{\mathcal{J}}(\eta^L_\beta) = \beta \circ_{\mathcal{C}^{\mathcal{J}}} \phi_2$.
-

Therefore $\eta^L_\beta \circ_{\mathcal{C}} \eta^L_\alpha \in \mathrm{Hom}_{\mathcal{C}}(X_1, X_3)$ and

$$\begin{aligned}\phi_3 \circ_{\mathcal{C}^{\mathcal{J}}} \Delta^{\mathcal{J}}(\eta^L_\beta \circ_{\mathcal{C}} \eta^L_\alpha) &= \phi_3 \circ_{\mathcal{C}^{\mathcal{J}}} [\Delta^{\mathcal{J}}(\eta^L_\beta) \circ_{\mathcal{C}^{\mathcal{J}}} \Delta^{\mathcal{J}}(\eta^L_\alpha)] \\ &= [\phi_3 \circ_{\mathcal{C}^{\mathcal{J}}} \Delta^{\mathcal{J}}(\eta^L_\beta)] \circ_{\mathcal{C}^{\mathcal{J}}} \Delta^{\mathcal{J}}(\eta^L_\alpha) \\ &= [\beta \circ_{\mathcal{C}^{\mathcal{J}}} \phi_2] \circ_{\mathcal{C}^{\mathcal{J}}} \Delta^{\mathcal{J}}(\eta^L_\alpha) \\ &= \beta \circ_{\mathcal{C}^{\mathcal{J}}} [\phi_2 \circ_{\mathcal{C}^{\mathcal{J}}} \Delta^{\mathcal{J}}(\eta^L_\alpha)] \\ &= \beta \circ_{\mathcal{C}^{\mathcal{J}}} [\alpha \circ_{\mathcal{C}^{\mathcal{J}}} \phi_1] \\ &= [\beta \circ_{\mathcal{C}^{\mathcal{J}}} \alpha] \circ_{\mathcal{C}^{\mathcal{J}}} \phi_1.\end{aligned}$$

Hence

$$\begin{aligned}\eta^L_\beta \circ_{\mathcal{C}} \eta^L_\alpha &\in \mathrm{Hom}_{\mathbf{Cone}(D_3)}(X_1, (\beta \circ_{\mathcal{C}^{\mathcal{J}}} \alpha) \circ_{\mathcal{C}^{\mathcal{J}}} \phi_1, (X_3, \phi_3)) \\ &= \mathrm{Hom}_{\mathbf{Cone}(D_3)}(\mathrm{Cn}_{\beta \circ_{\mathcal{C}^{\mathcal{J}}} \alpha}(L_{D_1}), L_{D_3}) \\ &= \{\eta^L_{\beta \circ_{\mathcal{C}^{\mathcal{J}}} \alpha}\}.\end{aligned}$$

Thus

$$\begin{aligned}\lim_L(\beta \circ_{\mathcal{C}^{\mathcal{J}}} \alpha) &= \eta^L_{\beta \circ_{\mathcal{C}^{\mathcal{J}}} \alpha} \\ &= \eta^L_\beta \circ_{\mathcal{C}} \eta^L_\alpha \\ &= \lim_L(\beta) \circ_{\mathcal{C}} \lim_L(\alpha).\end{aligned}$$

(c) Let $D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$. Write $L_D = (X, \phi)$. Exercise 6.1.0.10 ref prev ex implies that $\mathbf{Cone}(\cdot) : \mathcal{C}^{\mathcal{J}} \to \mathbf{Cat}$ and

$$\begin{aligned}\mathrm{Cn}_{\mathrm{id}^{\mathcal{C}^{\mathcal{J}}}_D} &= \mathbf{Cone}(\cdot)(\mathrm{id}^{\mathcal{C}^{\mathcal{J}}}_D) \\ &= \mathrm{id}^{\mathbf{Cat}}_{\mathbf{Cone}(D)}.\end{aligned}$$

Then

$$\begin{aligned}\mathrm{Cn}_{\mathrm{id}_D^{\mathcal{C}^{\mathcal{J}}}}(L_D) &= \mathrm{id}^{\mathbf{Cat}}_{\mathbf{Cone}(D)}(L_D)\\ &= L_D.\end{aligned}$$

Since L_D is $\mathbf{Cone}(D)$-terminal,

$$\begin{aligned}\lim_L(\mathrm{id}_D^{\mathcal{C}^{\mathcal{J}}}) &= \eta^L_{\mathrm{id}_D^{\mathcal{C}^{\mathcal{J}}}}\\ &\in \mathrm{Hom}_{\mathbf{Cone}(D)}(\mathrm{Cn}_{\mathrm{id}_D^{\mathcal{C}^{\mathcal{J}}}}(L_D), L_D)\\ &= \mathrm{Hom}_{\mathbf{Cone}(D)}(L_D, L_D)\\ &= \{\mathrm{id}^{\mathbf{Cone}(D)}_{L_D}\}.\end{aligned}$$

By Definition **??** ref cone def,

$$\begin{aligned}\mathrm{id}^{\mathbf{Cone}(D)}_{L_D} &= \mathrm{id}^{(\Delta^{\mathcal{J}}\downarrow D)}_{L_D}\\ &= \mathrm{id}^{(\Delta^{\mathcal{J}}\downarrow D)}_{(X,\phi)}\\ &= \mathrm{id}^{\mathcal{C}}_X.\end{aligned}$$

Therefore

$$\begin{aligned}\lim_L(\mathrm{id}_D^{\mathcal{C}^{\mathcal{J}}}) &= \mathrm{id}^{\mathbf{Cone}(D)}_{L_D}\\ &= \mathrm{id}^{\mathcal{C}}_X\\ &= \mathrm{id}^{\mathcal{C}}_{\lim_L(D)}.\end{aligned}$$

Hence $\lim_L : \mathcal{C}^{\mathcal{J}} \to \mathcal{C}$ is a functor.

2. colimit functor

□

Exercise 5.4.0.6. For a choice of L, we get a functor $\lim_L$. Rephrasing this, for each object (a set $L \in \mathrm{limChoice}(\mathcal{C}^{\mathcal{J}})$) we get a an object (a functor $\lim_L : \mathcal{C}^{\mathcal{J}} \to \mathcal{C}$). Maybe we can consider $\mathrm{limChoice}(\mathcal{C}^{\mathcal{J}})$ as a category and consider an arrow $f : L \to M$ as a function $f : \bigcup_{D\in\mathrm{Obj}(\mathcal{C}^{\mathcal{J}})} \mathbf{Cone}(D) \to \bigcup_{D\in\mathrm{Obj}(\mathcal{C}^{\mathcal{J}})} \mathbf{Cone}(D)$ such that $f \circ_{\mathbf{Set}} L = M$, i.e. for each $D \in \mathrm{Obj}(\mathcal{C}^{\mathcal{J}})$ $f(L_D) = M_D$, idk, think about

Definition 5.4.0.7. Let $\mathcal{C}$ be a category. Then $\mathcal{C}$ is said to be

- **complete** if for each $J \in \mathrm{Obj}(\mathbf{Cat})$, $\mathcal{C}$ has all $\mathcal{J}$-limits.
- **cocomplete** if for each $J \in \mathrm{Obj}(\mathbf{Cat})$, $\mathcal{C}$ has all $\mathcal{J}$-colimits.

Chapter 6

Yoneda Lemma

6.1 Representable Functors

Exercise 6.1.0.1. Let $\mathcal{C}$ be a category, $(A_1, A_2), (B_1, B_2) \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}} \times \mathcal{C})$ and $(f_1, f_2) \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}}((A_1, A_2), (B_1, B_2))$. Then for each $h \in \mathrm{Hom}_{\mathcal{C}}(A_1, A_2)$, $f_2 \circ_{\mathcal{C}} h \circ_{\mathcal{C}} f_1 \in \mathrm{Hom}_{\mathcal{C}}(B_1, B_2)$.

Proof. By definition of $\mathcal{C}^{\mathrm{op}} \times \mathcal{C}$, we have that

- $$\begin{aligned} A_1, B_1 &\in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}}) \\ &= \mathrm{Obj}(\mathcal{C}), \end{aligned}$$

  $$\begin{aligned} f_1 &\in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A_1, B_1) \\ &= \mathrm{Hom}_{\mathcal{C}}(B_1, A_1), \end{aligned}$$

- $A_2, B_2 \in \mathrm{Obj}(\mathcal{C})$ and $f_2 \in \mathrm{Hom}_{\mathcal{C}}(A_2, B_2)$.

Therefore, for each $h \in \mathrm{Hom}_{\mathcal{C}}(A_1, A_2)$, $f_2 \circ_{\mathcal{C}} h \circ_{\mathcal{C}} f_1 \in \mathrm{Hom}_{\mathcal{C}}(B_1, B_2)$. □

Note 6.1.0.2. If $\mathcal{C}$ is locally small, then for each $(A_1, A_2) \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}} \times \mathcal{C})$, $\mathrm{Hom}_{\mathcal{C}}(A_1, A_2) \in \mathrm{Obj}(\mathbf{Set})$.

Definition 6.1.0.3. Let $\mathcal{C}$ be a locally small category. We define the **Hom-functor of** $\mathcal{C}$, denoted $\mathrm{Hom}_{\mathcal{C}}(\cdot, \cdot) : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathbf{Set}$, by

- $\mathrm{Hom}_{\mathcal{C}}(\cdot, \cdot)(A_1, A_2) := \mathrm{Hom}_{\mathcal{C}}(A_1, A_2), \quad (A_1, A_2) \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}} \times \mathcal{C})$
- $\mathrm{Hom}_{\mathcal{C}}(\cdot, \cdot)(f_1, f_2)(h) := f_2 \circ_{\mathcal{C}} h \circ_{\mathcal{C}} f_1, \quad (f_1, f_2) \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}}((A_1, A_2), (B_1, B_2)), h \in \mathrm{Hom}_{\mathcal{C}}(A_1, A_2)$

Exercise 6.1.0.4. Let $\mathcal{C}$ be a locally small category. Then $\mathrm{Hom}_{\mathcal{C}} : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathbf{Set}$ is a functor.

Proof. Set $H := \mathrm{Hom}_{\mathcal{C}}(\cdot, \cdot)$.

1. The previous ex implies that for each $(A_1, A_2), (B_1, B_2) \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}} \times \mathcal{C})$ and $(f_1, f_2) \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}}((A_1, A_2), (B_1, B_2))$,

$$\begin{aligned} H(f_1, f_2) &\in \mathrm{Hom}_{\mathbf{Set}}(\mathrm{Hom}_{\mathcal{C}}(A_1, A_2), \mathrm{Hom}_{\mathcal{C}}(B_1, B_2)) \\ &= \mathrm{Hom}_{\mathbf{Set}}(H(A_1, A_2), H(B_1, B_2)) \end{aligned}$$

2. Let $(A_1, A_2), (B_1, B_2), (C_1, C_2) \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}} \times \mathcal{C})$, $(f_1, f_2) \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}}((A_1, A_2), (B_1, B_2))$ and $(g_1, g_2) \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}}((B_1, B_2), (C_1, C_2))$. Then for each $h \in \mathrm{Hom}_{\mathcal{C}}(A_1, A_2)$, we have that

$$\begin{aligned} H((g_1, g_2) \circ_{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}} (f_1, f_2))(h) &= H(g_1 \circ_{\mathcal{C}^{\mathrm{op}}} f_1, g_2 \circ_{\mathcal{C}} f_2)(h) \\ &= H(f_1 \circ_{\mathcal{C}} g_1, g_2 \circ_{\mathcal{C}} f_2)(h) \\ &= (g_2 \circ_{\mathcal{C}} f_2) \circ_{\mathcal{C}} h \circ_{\mathcal{C}} (f_1 \circ_{\mathcal{C}} g_1) \\ &= g_2 \circ_{\mathcal{C}} (f_2 \circ_{\mathcal{C}} h \circ_{\mathcal{C}} f_1) \circ_{\mathcal{C}} g_1 \\ &= g_2 \circ_{\mathcal{C}} [H(f_1, f_2)(h)] \circ_{\mathcal{C}} g_1 \\ &= H(g_1, g_2)(H(f_1, f_2)(h)) \\ &= H(g_1, g_2) \circ_{\mathbf{Set}} H(f_1, f_2)(h). \end{aligned}$$

Thus $H((g_1, g_2) \circ_{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}} (f_1, f_2)) = H(g_1, g_2) \circ_{\mathbf{Set}} H(f_1, f_2)$. Since $(A_1, A_2), (B_1, B_2), (C_1, C_2) \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}} \times \mathcal{C})$, $(f_1, f_2) \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}}((A_1, A_2), (B_1, B_2))$ and $(g_1, g_2) \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}}((B_1, B_2), (C_1, C_2))$ are arbitrary, we have that for each $(A_1, A_2), (B_1, B_2), (C_1$ $\mathrm{Obj}(\mathcal{C}^{\mathrm{op}} \times \mathcal{C})$, $(f_1, f_2) \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}}((A_1, A_2), (B_1, B_2))$ and $(g_1, g_2) \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}}((B_1, B_2), (C_1, C_2))$, $H((g_1, g_2) \circ_{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}} (f_1, f_2)) = H(g_1, g_2) \circ_{\mathbf{Set}} H(f_1, f_2)$.

3. Let $(A_1, A_2) \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}} \times \mathcal{C})$. Then for each $h \in \mathrm{Hom}_{\mathcal{C}}(A_1, A_2)$,

$$\begin{aligned} H(\mathrm{id}^{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}}_{(A_1, A_2)})(h) &= H(\mathrm{id}^{\mathcal{C}^{\mathrm{op}}}_{A_1}, \mathrm{id}^{\mathcal{C}}_{A_2})(h) \\ &= H(\mathrm{id}^{\mathcal{C}}_{A_1}, \mathrm{id}^{\mathcal{C}}_{A_2})(h) \\ &= \mathrm{id}^{\mathcal{C}}_{A_2} \circ_{\mathcal{C}} h \circ_{\mathcal{C}} \mathrm{id}^{\mathcal{C}}_{A_1} \\ &= h \\ &= \mathrm{id}^{\mathbf{Set}}_{\mathrm{Hom}_{\mathcal{C}}(A_1, A_2)}(h) \\ &= \mathrm{id}^{\mathbf{Set}}_{H(A_1, A_2)}(h). \end{aligned}$$

Thus $H(\mathrm{id}^{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}}_{(A_1, A_2)}) = \mathrm{id}^{\mathbf{Set}}_{H(A_1, A_2)}$. Since $(A_1, A_2) \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}} \times \mathcal{C})$ is arbitrary, we have that for each $(A_1, A_2) \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}} \times \mathcal{C})$, $\mathrm{Hom}_{\mathcal{C}}(\mathrm{id}^{\mathcal{C}^{\mathrm{op}} \times \mathcal{C}}_{(A_1, A_2)}) = \mathrm{id}^{\mathbf{Set}}_{H(A_1, A_2)}$.

Hence $\mathrm{Hom}_{\mathcal{C}} : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathbf{Set}$ is a functor. □

Note 6.1.0.5. If $\mathcal{C}$ is locally small, Exercise 4.2.2.2 implies that for each $X \in \mathrm{Obj}(\mathcal{C})$, $\mathrm{Hom}_{\mathcal{C}}(X, \cdot) : \mathcal{C} \to \mathbf{Set}$ and $\mathrm{Hom}_{\mathcal{C}}(\cdot, X) : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$ are functors.

Exercise 6.1.0.6. Let $\mathcal{C}$ be a category. Suppose that $\mathcal{C}$ is locally small. Then

1. for each $X \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$, $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, \cdot) = \mathrm{Hom}_{\mathcal{C}}(\cdot, X)$

2. for each $X \in \mathrm{Obj}(\mathcal{C})$, $\mathrm{Hom}_{\mathcal{C}}(X, \cdot) = \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(\cdot, X)$

Proof. We note that Exercise 6.1.0.10 (ref ex) implies that $(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} = \mathcal{C}$. So $(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} \times \mathcal{C}^{\mathrm{op}} = \mathcal{C} \times \mathcal{C}^{\mathrm{op}}$.

1. Let $X \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$. Then $X \in \mathrm{Obj}(\mathcal{C})$.

   - Let $A \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$. Then $A \in \mathrm{Obj}(\mathcal{C})$ and

$$\begin{aligned} \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, \cdot)(A) &= \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(\cdot, \cdot)(X, A) \\ &= \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, A) \\ &= \mathrm{Hom}_{\mathcal{C}}(A, X) \\ &= \mathrm{Hom}_{\mathcal{C}}(\cdot, \cdot)(A, X) \\ &= \mathrm{Hom}_{\mathcal{C}}(\cdot, X)(A). \end{aligned}$$

- Let $A, B \in \mathrm{Obj}(\mathcal{C}^{\mathrm{op}})$ and $f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B)$ and $h \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, A)$. Then $A, B \in \mathrm{Obj}(\mathcal{C})$, $f \in \mathrm{Hom}_{\mathcal{C}}(B, A)$ and $h \in \mathrm{Hom}_{\mathcal{C}}(A, X)$. Thus $(\mathrm{id}_X^{\mathcal{C}}, f) \in \mathrm{Hom}_{\mathcal{C}\times\mathcal{C}^{\mathrm{op}}}((X, A), (X, B))$ and

$$\begin{aligned}
\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, \cdot)(f)(h) &= \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(\cdot, \cdot)(\mathrm{id}_X^{\mathcal{C}}, f)(h) \\
&= f \circ_{\mathcal{C}^{\mathrm{op}}} h \circ_{\mathcal{C}^{\mathrm{op}}} \mathrm{id}_X^{\mathcal{C}} \\
&= \mathrm{id}_X^{\mathcal{C}} \circ_{\mathcal{C}} h \circ_{\mathcal{C}} f \\
&= \mathrm{Hom}_{\mathcal{C}}(\cdot, \cdot)(f, \mathrm{id}_X^{\mathcal{C}})(h) \\
&= \mathrm{Hom}_{\mathcal{C}}(\cdot, X)(f)(h)
\end{aligned}$$

  Since $h \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, A)$ is arbitrary, we have that for each $h \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, A)$, $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, \cdot)(f)(h) = \mathrm{Hom}_{\mathcal{C}}(\cdot, X)(f)(h)$. Thus $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, \cdot)(f) = \mathrm{Hom}_{\mathcal{C}}(\cdot, X)(f)$.

Hence $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, \cdot) = \mathrm{Hom}_{\mathcal{C}}(\cdot, X)$.

2. FINISH!!! (similar)

□

Definition 6.1.0.7. Let $\mathcal{C}$ be a category. Suppose that $\mathcal{C}$ is locally small. Let $F : \mathcal{C} \to \mathbf{Set}$.

- Let $X \in \mathrm{Obj}(\mathcal{C})$ and $\phi : F \Rightarrow \mathrm{Hom}_{\mathcal{C}}(X, \cdot)$. Then (X, ϕ) is said to be a **representation of** F if ϕ is a natural isomorphism.
- Then F is said to be **representable** if there exists $X \in \mathrm{Obj}(\mathcal{C})$ and $\phi : F \Rightarrow \mathrm{Hom}_{\mathcal{C}}(X, \cdot)$ such that (X, ϕ) is a representation of F.

Exercise 6.1.0.8. Let $\mathcal{C}$ be a category. Suppose that $\mathcal{C}$ is small. Let $X \in \mathrm{Obj}(\mathcal{C})$. Let $1 \in \mathrm{Obj}(\mathbf{Set})$. Suppose that 1 is **Set**-terminal. Then

1. X is $\mathcal{C}$-initial iff $\mathrm{Hom}_{\mathcal{C}}(X, \cdot) \cong_{\mathbf{Set}^{\mathcal{C}}} \Delta_1^{\mathcal{C}}$.
2. X is $\mathcal{C}$-terminal iff $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(\cdot, X) \cong_{\mathbf{Set}^{\mathcal{C}^{\mathrm{op}}}} \Delta_1^{\mathcal{C}^{\mathrm{op}}}$.

Proof. Define $F \in \mathbf{Set}^{\mathcal{C}}$ by $F := \mathrm{Hom}_{\mathcal{C}}(X, \cdot)$.

1. Exercise 3.1.0.11 implies that $\phi^{(F,1)} \in \mathrm{Hom}_{\mathbf{Set}^{\mathcal{C}}}(F, \Delta_1^{\mathcal{C}})$. Exercise 3.1.0.6 implies that there exists $x \in 1$ such that $1 = \{x\}$.

   - ($\Longrightarrow$):
     Suppose that X is $\mathcal{C}$-initial. Let $A \in \mathrm{Obj}(\mathcal{C})$. Since $F(A) = \mathrm{Hom}_{\mathcal{C}}(X, A)$, Exercise 3.1.0.7 implies that $F(A)$ is **Set**-terminal. Since 1 is **Set**-terminal, Exercise 3.1.0.5 implies that $F(A) \cong_{\mathbf{Set}} 1$ and $\mathrm{Hom}_{\mathbf{Set}}(F(A), 1) = \mathrm{Iso}_{\mathbf{Set}}(F(A), 1)$. Therefore

$$\begin{aligned}
\phi_A^{(F,1)} &\in \mathrm{Hom}_{\mathbf{Set}}(F(A), 1) \\
&= \mathrm{Iso}_{\mathbf{Set}}(F(A), 1) \\
&= \mathrm{Iso}_{\mathbf{Set}}(F(A), \Delta_1^{\mathcal{C}}(A)).
\end{aligned}$$

     Since $A \in \mathrm{Obj}(\mathcal{C})$ is arbitrary, we have that for each $A \in \mathrm{Obj}(\mathcal{C})$, $\phi_A^{(F,1)} \in \mathrm{Iso}_{\mathbf{Set}}(F(A), \Delta_1^{\mathcal{C}}(A))$. Thus $\phi^{(F,1)} : F \Rightarrow \Delta_1^{\mathcal{C}}$ is a natural isomorphism. Exercise 2.4.3.5 then implies that $\phi^{(F,1)} \in \mathrm{Iso}_{\mathbf{Set}^{\mathcal{C}}}(F, \Delta_1^{\mathcal{C}})$. Since $\mathrm{Iso}_{\mathbf{Set}^{\mathcal{C}}}(F, \Delta_1^{\mathcal{C}}) \neq \varnothing$, we have that $F \cong_{\mathbf{Set}^{\mathcal{C}}} \Delta_1^{\mathcal{C}}$.
   - ($\Longleftarrow$):
     Suppose that $F \cong_{\mathbf{Set}^{\mathcal{C}}} \Delta_1^{\mathcal{C}}$. Then there exists $\phi \in \mathrm{Iso}_{\mathbf{Set}^{\mathcal{C}}}(F, \Delta_1^{\mathcal{C}})$. Exercise 2.4.3.5 then implies that $\phi : F \Rightarrow \Delta_1^{\mathcal{C}}$ is a natural isomorphism. Let $A \in \mathrm{Obj}(\mathcal{C})$. Then

$$\begin{aligned}
\phi_A &\in \mathrm{Iso}_{\mathbf{Set}}(F(A), \Delta_1^{\mathcal{C}}(A)) \\
&= \mathrm{Iso}_{\mathbf{Set}}(F(A), 1).
\end{aligned}$$

Since $\mathrm{Iso}_{\mathbf{Set}}(F(A), 1) \neq \varnothing$ and 1 is **Set**-terminal, we have that

$$\begin{aligned}\mathrm{Hom}_{\mathcal{C}}(X, A) &= F(A) \\ &\cong_{\mathbf{Set}} 1 \\ &\cong_{\mathbf{Set}} \{\varnothing\}, \quad \text{(by Exercise 3.1.0.6).}\end{aligned}$$

Thus there exists $f \in \mathrm{Hom}_{\mathcal{C}}(X, A)$ such that $\mathrm{Hom}_{\mathcal{C}}(X, A) = \{f\}$. Since $A \in \mathrm{Obj}(\mathcal{X})$ is arbitrary, we have that for each $A \in \mathrm{Obj}(\mathcal{C})$, there exists $f \in \mathrm{Hom}_{\mathcal{C}}(X, A)$ such that $\mathrm{Hom}_{\mathcal{C}}(X, A) = \{f\}$. Hence X is $\mathcal{C}$-initial.

2. We have that

$$\begin{aligned}X \text{ is } \mathcal{C}\text{-terminal} &\iff X \text{ is } \mathcal{C}^{\mathrm{op}}\text{-initial}, \quad \text{(by Exercise 3.1.0.3)} \\ &\iff \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, \cdot) \cong_{\mathbf{Set}^{\mathcal{C}^{\mathrm{op}}}} \Delta_1^{\mathcal{C}^{\mathrm{op}}}, \quad \text{(by (ref) prev part)} \\ &\iff \mathrm{Hom}_{\mathcal{C}}(\cdot, X) \cong_{\mathbf{Set}^{\mathcal{C}^{\mathrm{op}}}} \Delta_1^{\mathcal{C}^{\mathrm{op}}}, \quad \text{(by (ref) prev ex about } \mathrm{Hom}_{\mathcal{C}}(\cdot, X) = \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(X, \cdot))\end{aligned}$$

□

Exercise 6.1.0.9. Let $\mathcal{C}$ be a category, $F : \mathcal{C} \to \mathbf{Set}$, $X \in \mathrm{Obj}(\mathbf{Set})$ and $(A, f) \in (X \downarrow F)$. Suppose that $\mathcal{C}$ is locally small, X is **Set**-terminal and (A, f) is a universal morphism from X to F. Then

relevant exercises
Exercise 4.3.0.2
Exercise 4.5.0.12

Exercise 6.1.0.10. Yoneda Lemma: Let $\mathcal{C}$ be a category. Suppose that $\mathcal{C}$ is locally small. Then

$$\Rightarrow \mathrm{eval}_{(\mathcal{C}, \mathbf{Set})}$$

6.2 Fundamental Examples

look up applications of yoneda lemma to analysis. For example, see Tom LaGatta's answer on stackexchange and look up yoneda lemma construction of dedikind cuts for posets.

Chapter 7

Adjunctions

Exercise 7.0.0.1. Let

Definition 7.0.0.2. Let $\mathcal{C}$, $\mathcal{D}$ be categories, $F : \mathcal{C} \to \mathcal{D}$ and $G : \mathcal{D} \to \mathcal{C}$. Then (F, G) is said to be an **adjunction** if

7.1 TO DO

- Define subcategories and full subcategories and show that if $\mathrm{Obj}(D) \subset \mathrm{Obj}(C)$ and for each $X, Y \in \mathrm{Obj}(D)$, $\mathrm{Hom}_D(X, Y) = \mathrm{Hom}_C(X, Y)$, then D is a full subcategory of C. I used this in differential
- discuss projective/inductive systems and the projective/inductive limits and applications to topology and measure theory

Chapter 8

Monoidal Categories

Definition 8.0.0.1. Let $\mathcal{C}$ be a category, $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$, $\rho : (- \otimes -) \otimes \cdot \Rightarrow - \otimes (- \otimes -)$.

Chapter 9

To Do

- look into the schroder bernstein property wikipedia

Appendix A

App

A.1 Reading Diagrams and associated digraphs of diagrams

Definition A.1.0.1. Let

$$\begin{tikzcd} C \arrow[r, "g"] \arrow[d, "h"'] & A \arrow[d, "f"] \\ A \arrow[r, "f"'] & B \end{tikzcd} \implies \begin{tikzcd} C \arrow[r, bend left, "g"] \arrow[r, bend right, "h"'] & A \end{tikzcd}$$

see an intro to the language of category theory by roman for description

Definition A.1.0.2. A diagram is said to be **commutative** if for each path of length ≥ 2, in the associated digraph gives the same morphism.